



MANUAL FOR MAINTENANCE OF TRACK



**WDFC CTP 1 & 2
REWARI – IQBALGARH**

MANUAL FOR MAINTENANCE OF TRACK



Foreword

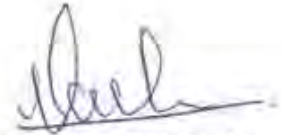
*As DFCCIL embarks upon the operation and maintenance phase, It is a matter of immense pleasure that Sojitz-L&T JV Consortium have come up with the "**Manual for Maintenance of Track**" as part of its obligation under CTP 1&2 Contract.*

DFC will be a game changer in the transport logistics scenario of the country. For the first time in Indian Railways, the earthwork and bridges have been designed for 32.5 T axle load and the track designed for 25 T axle load in DFCCIL. Besides, the Design build philosophy adopted by DFC has enabled use of many new Heavy Haul components/practices hitherto not being used by Indian Railways such as relating to maintenance of CWRs continuing through Turnouts, Anti Creep Devices at Turnouts, improved inspection system through rail based portable track geometry recording system etc. Besides, DFC has planned totally mechanised maintenance system over its network which will bring about a paradigm shift in the maintenance practices over DFC and Indian Railways as well. As this Manual is to be read with the Indian Railway P Way Manual, the provisions of this Manual are supplementary guidelines to help maintainers in understanding the Heavy Haul Track components and its maintenance procedures.

I appreciate the pioneering effort by SOJITZ-L&T Consortium and I am sure that this will motivate and guide other Contractors following suite for its further improvement, updation and development incorporating the best of maintenance policies.

This would also pave way for the DFCCIL's own Manual incorporating the specific designs of all the D&B Contracts of both the corridors i.e. EDFC and WDFC.

With best wishes,



(Anurag Sachan)
Managing Director, DFCCIL

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SECTION 1 - GENERAL

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REFERENCES

- Guidelines to Best Practices for Heavy Haul Railway Operations-International Heavy Haul Association (IHHA)
- Continuous Welded Rail (CWR) Track (RT/CE/S/011)
- Australian Rail Track Corporation Limited (ARTC) Maintenance Manuals
- FRA Track Safety Standards
- IRPWM
- IR LWR manual

MANUAL FOR MAINTENANCE OF TRACK

ABBREVIATIONS

A

ACD	Anti- Creep Device
AC	Alternating Current
AM	Assistant Manager
AT weld	Alumino Thermic Weld
ATS	Actual Toe of Switch

B

BCM	Ballast Cleaning Machine
BDT	Ballast Distribution Team
BG	Broad Gauge
BTC	Beginning of Transition Curve

C

CGRSP	Composite Grooved Rubber Sole Plates
CMS	Cast Manganese Steel Crossing
CMT	Civil Maintenance Team
CSR	Clear Standing Room
CTE	Chief Track Engineer
CTR	Complete Track Renewal
CTTT	Conventional Track Tamping Team
CWR	Continuous Welded Rail

D

DFCCIL	Dedicated Freight Corridor Corporation of India Limited
DS	Derailing Switch

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E

EN	European Norms
ER	Employer's Requirements
ERC	Elastic Rail Clip
ETC	End of Transition Curve

F

FBW	Flash Butt Weld
FRA	Failure Rate Analysis

G

GFN	Glass Filled Nylon liners
GM	General Manager
GMT	Gross Metric Ton
GPS	Global Positioning System
GRC	Geometric Rail Car

H

HH	Head Hardened
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I

IMD	Integrated Maintenance Depot
IMSD	Integrated Maintenance Sub-Depot
IRLWR Manual	Indian Railway Long Welded Rail Manual
IRM	Inner Reference Mark
IRPWM	Indian Railway Permanent Way Manual

K

KPI	Key Performance Indices
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L

LC	Level Crossing
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LWR Long Welded Rail

M

MDT Material Distribution Teams

MIS Management Information System

MMIS Maintenance Management Information System

MT Metric Ton

N

NTC New Track Construction machine

O

OEM Original Equipment Manufacturer

OHE Over Head Equipment

ORM Outer Reference Mark

P

P. Way Permanent Way

P&C Points and Crossings

PSC Pre-Stressed Concrete Sleeper

R

RCR Rail cum Road Vehicle

RDSO Research Designs and Standards Organisation

RFD Rail Flaw Detection

RMMU Regional Mechanized Maintenance Unit

RMT Rail Maintenance Teams

ROB Road Over Bridge

RQA Ride Quality Accelerometers

RUB Road Under Bridge

S

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S&T	Signalling and Telecom
SEJ	Switch Expansion Joint
SFT	Stress Free Temperature
SPURT	Self-Propelled Ultrasonic Rail Testing
SSD	Spring Setting Device

T

Td	Destressing Temperature
TMT	Track Maintenance Teams
TQI	Track Quality Indices
TSR	Through Sleeper Renewal
TTR	Through Track Renewal

U

UIC	International Union of Railways
USFD	Ultra-Sonic Flaw Detection
UTS	Ultimate Tensile Strength

V

VERSE	Vertical Rail Stressing Equipment
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W

WDFC	Western Dedicated Freight Corridor
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MANUAL FOR MAINTENANCE OF TRACK

A. INTRODUCTION

The Western Dedicated Freight Corridor consists of approximately 1500 km of double rail line, running along the existing Indian Railway Tracks, is an electrified track with 2x25 kV AC, 50 Hz. Overhead catenary system capable of operating at a maximum train speed of 100 KMPH. Formation and bridge structures are provided for 32.5t axle load and the track structure of Continuous Welded Rail (CWR) for 25t axle load.

The WDFCC project starts at Dadri-Khurja near Delhi and passes through Phulera, Ajmer, Beawer, Marwar, Iqbalgarh, Sanand, Vadodara and ends at JNPT near Mumbai. The total length of Alignment is around 1500kms and is passing through Haryana, Rajasthan, Gujarat and Maharashtra states. The proposed alignment is passing generally parallel to the existing IR network. The alignment takes a detour at few places. Around 90% of the proposed alignment passes parallel to the existing IR network and 10% of the alignment passes through detours.

All track maintenance and engineering activities shall be guided and managed from integrated maintenance depots of DFCCIL. The intention is that resources for most maintenance activities will be based at both IMD's and IMSD's. Such Integrated maintenance depots are set up at appx 160 kms distance, throughout the DFC, with sub depots appx 45 kms apart from the main depots. These depots and sub depots shall act as common hubs for Civil, Track, Electrical, Signal and Telecom engineering.

Considering the need of timely inspection, the DFCC has planned to use Rail-cum Road vehicles for daily inspection of track. The conventional method of push trolley inspection has been completely dispensed with. DFCCIL will use portable track geometry measurement devices to test track for geometry correctness/defects. They are to be strapped to the rear bumper of an inspection rail-cum-road vehicle so as to provide track geometry measurements to supplement visual observations.

On-Track machines will be deployed for maintaining the Continuous Welded Track. The machines that are capable of maintaining the plain track as well as the turnouts are adopted by DFCC for holistic maintenance of the track.

The Manual for Rail Track maintenance has been compiled in line with the Indian Railway Permanent Way Manual. **The manual provisions are in addition to the IRPWM**, which helps the maintainers in understanding the Heavy Haul track components and its maintenance procedures. The timely and economic maintenance are required for improving the life cycle of the assets.

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B. DIFFERENTIATING FEATURES OF DFCCIL OVER INDIAN RAILWAYS

DFCCIL freight corridors are different in few aspects from the present running mixed (passenger & freight) lines of Indian Railways. The special features of the Western Dedicated Freight Corridor (WDFC) are listed below:

i) TRAIN TYPE/WAGON

- Present commissioning with 25t axle loads (Ultimately being 32.5 t axle loads). (as per Clause 3.2 para 26 of ER Vol II of Functional part)
- High Speed freight trains running at 100kmph. (as per Clause 3.2 para 26 of ER Vol II of Functional part)
- Mass of Loaded Train of 12000 T capacity. (as per Clause 16.6.6 of ER Vol II of Design part)
- Wagons of 3660 mm width and 7.1 metre height. (as per Clause 16.5 of ER Vol II of Design part)
- Trains with double stack containers. (as per Clause 16.1 of ER Vol II of Design part)

ii) ROUTE DESIGN STANDARDS

- Stations at 14-48km interval
- Minimum curve radius of 700m on Main Line.
- Ruling gradient in general is 1in200 except in few small stretches as per topographic requirement. Details of such stretches shall be maintained by field units and Depots

iii) ROAD BED

- Formation width of 13.5 m minimum for double track on embankment and 12.9m minimum for double track in cutting.
- Formation and Bridges are designed and constructed for an axle load of 32.5 T
- Track structure has been designed and installed for 25T axle load.
- Ballast cushion of 350mm in main lines, connecting line & 250mm in Loop Lines is provided.

iv) MODERN TRACK TECHNOLOGY

- Rail Standard specification IRS T-12:
 - Main lines and Points and Crossings- 60E1 1080 HH (as per IRS T-12)
 - Other than Main lines & Points and Crossings- UIC 60kg 880grade
 - For Guard rail- 880 grades rails UIC 60kg
- Continuous Welded Rails without any SEJ.
- Flash butt welded panels in all Lines and Turnouts with AT welds.

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- Upgraded Fastening system (10mm Composite Grooved Rubber Sole Plates (CGRSP), ERC-Mark V) are used.
- New track is laid using fully mechanized system (NTC).
- Nowhere track gauge is permitted less than “Zero” in the WDFCC.
- Destressing using Super Puller system has been adopted primarily.

v) POINTS AND CROSSINGS

- Turnouts of 1 in 12 with thick web switches.
- Turnouts of 1 in 8.5 with thick web switches.
- CWR through Points & Crossings with weldable CMS crossings.
- Turnout design speed 1 in 12 (50kmph on diverging track)
- Turnout design speed 1 in 8.5 (25kmph on diverging track)
- Canted Rails (1 in 20) are used for turnouts to avoid change in rail plane in turnout
 - For switch portion in turnout- 60 E1A1 1080 HH RAIL (as per IRS T-12)
 - For stock rail, crossing portion, lead rail & Intermediate Portion- 60E1, 1080 HH RAIL (as per IRS T-12)
 - For check rail- EN33C1
- Anti-Creep Device (ACD)
- Drive mechanism – Spring Setting Device (SSD)

PIROLL System (Plate Integrated ROLLER system): - PIROLLs are provided (Rewari-Phulera Section) in the switch device of a turnout of 1 in 12 only. The PIROLL mounted on the sides of the slide plates or slide chair plates allows an easy setting process, as the switches are rolling on the rollers only during the setting operation. With the use of PIROLLs, the open switch is lifted about 1mm to 2mm (max) off the slide chair plates and therefore, the turnout works properly without lubrication of the slide chairs and slide plates even in polluted track areas. The mounting is made on approximately every second or third plate. The rollers are made from PA6 with slide bearing.

vi) LEVEL CROSSINGS

No surface crossings are provided. All road crossings are with grade separation either ROB or RUBs / Limited Head Subways. However, few temporary LCs may be there till the ROB/RUB is completed.

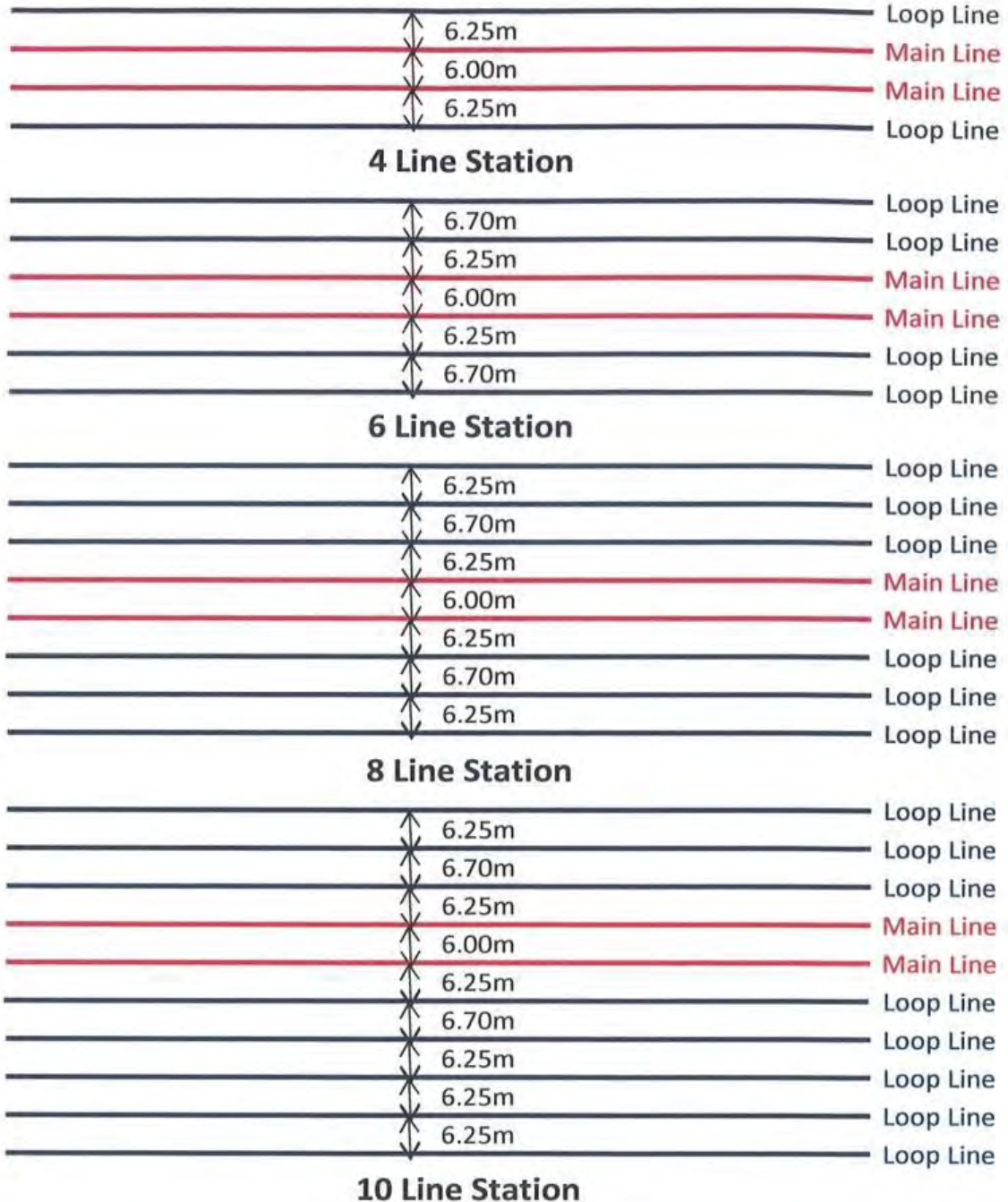
vii) FRICTION BUFFER STOPS

- Buffer stop for design parameters train mass of 2000MT & Impact speed of 5 kmph for use on siding track other than overrun loop lines, ballast sidings, accident relief sidings, etc

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- Buffer stop for design parameters train mass of 6500mt & Impact speed of 10 kmph for use on overrun of loop lines, etc having minimum CSR of 750m.

viii) TRACK CENTER DISTANCE IN YARDS



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Figure 1: Key Plan

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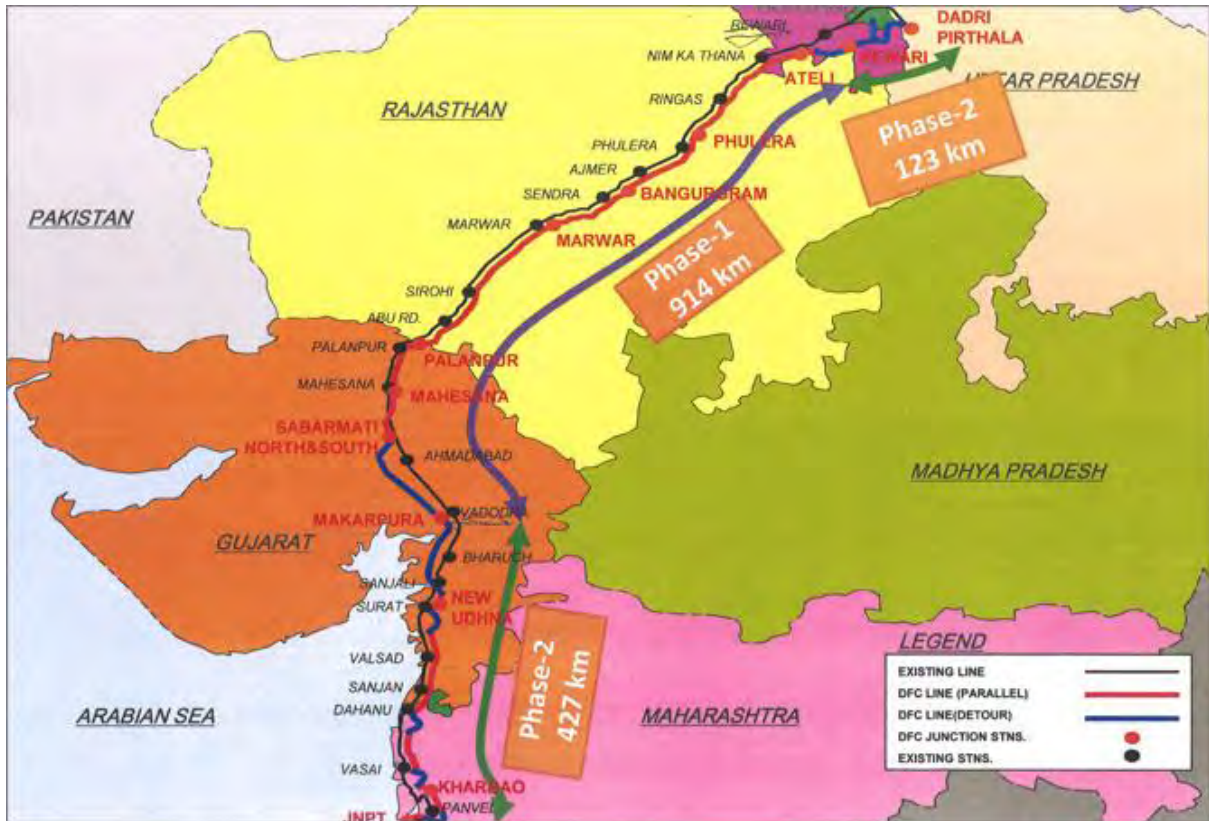


Figure 2 WDFC Route Map

SECTION 2 – MANUAL PROVISIONS

CHAPTER I - DUTIES OF PERMANENT WAY OFFICIALS/MEN

1.1 ROLES, RESPONSIBILITIES AND PROCESSES

1.1.1 DUTIES AND RESPONSIBILITIES

1.1.1.1 DEPOT ENGINEER (MANAGER/APM)

The Depot Engineer shall be responsible for the maintenance and safety of all Pway/Track in his charge, for the accuracy, quality and progress of new works and control over all expenditure in relation to budget allotment. The duties of the Depot Engineer are detailed in various chapters of this track maintenance manual and other reference manuals including Indian Railways Permanent Way Manual, LWR Manual, Track machine Manual and other relevant Manuals. He shall be specifically responsible for:

- A) Mainline track inspection and maintenance in a satisfactory and safe condition, subject to KPIs and requirements as may be prescribed by this manual or DFCCIL or its constituent office, for traffic. Depot Engineer shall be responsible for all Track maintenance and inspection activities at IMD's and associated IMSD's.
- B) Sr Executive/Executive (Tracks), track maintenance foremen, welder foremen and bridge foremen will report to him or her.
- C) Reporting of requirement of all works incidental to track maintenance, including track relaying works. He/she should ensure that all works are carried out according to plans and specifications laid down
- D) Planning and execution of inspection and maintenance programs of track infrastructure; including Track Machines deployment, Upkeep and Overhauls.
- E) Civil budget, safety and infrastructure performance on territory.
- F) Monitoring of performance of IMD civil track Teams and take measures to correct problems with productivity, quality and safety.
- G) Undertaking one inspection of territory per month by RCR vehicle or train along with concerned Sr Executive/Executive (Tracks), reviewing their observations, reports and recordings. He/she shall provide feedback to Sr Executive/Executive (Tracks) on their recent inspections and submitted reports. In addition, he shall carry out all other inspections as per the given schedule by DFCCIL.

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- H) Meeting at least weekly with direct reports.
- I) Accountable and periodical verification of such works as may be reported to be required in track area.
- J) Maintenance of land boundaries between DFCCIL track and IR track as may be specified by the DFCCIL administration.
- K) Submission of proposals for inclusion in the track renewal programme, revenue budget and the works programme.
- L) Maintaining up-to-date knowledge of all codes and manuals, with all amendments, updates and corrections, as may be prescribed by DFCCIL or its constituent office. S/He shall be well acquainted with the rules, regulations and procedures concerning his work and duties as enjoined in these codes and manuals. S/He shall keep himself in touch with the orders and circulars issued by higher authorities from time to time and efficiently act upon them. He shall ensure that all staff working under him are well acquainted with the relevant rules and working methods and efficiently perform their duties.
- M) Keeping of Materials - The Depot Engineer shall see to the security of RCR's and other equipment including rails, chairs, sleepers and other materials in his charge and ensure that unused materials are stacked properly clear of the line, so as not to interfere with the safe running of trains.
- N) Keeping close coordination with officers of other departments of DFCCIL, Indian Railway, State Government and other organisations having frequent dealing with DFCCIL
- O) Ensure proper field training of supervisors and officers sent to his depot for training.
- P) Accompanying on Inspections of Higher Officials - When the Depot Engineer accompanies a periodical or special inspection by the higher officials, he should have with him the following registers and documents pertaining to his section, other than the codes and manuals mentioned above:
 - I) Latest MIS reports generated from Maintenance management system including results of track recording and monitoring, all recent and relevant inspection reports and logs pertaining to track, curves, points and crossings, etc.
 - II) Latest applicable Time Table.
 - III) Permanent way diagrams of section and yards.

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- IV) Any other document as may be mandated by DFCCIL
- V) List of Permanent and Temporary speed restrictions, if any
- VI) List of works and other details.
- Q) Action in case of Emergency – When alerted to an accident (including breaches) affecting any part of track, restricting free passage of trains, the Depot Engineer should proceed to site by the quickest available means. On the way he should collect information regarding the damage, the men and material requirement at site for restoration and arrange for movement of men and materials and thereafter the restoration
- R) He shall keep himself updated with the safety and Incident management manual of DFCCIL and shall play all roles or tasks as may be assigned to him by DFCCIL, in accordance with such manuals.
- S) It shall be responsibility of the Depot Engineer to ensure that all his tasks, duties and responsibilities are undertaken in accordance and absolute compliance with the DFCCIL's safety management system.
- T) Undertake any other responsibility as may be assigned by DFCCIL or under this manual.

1.1.1.2 Sr. EXECUTIVE/EXECUTIVE (TRACK)

The Sr Executive/Executive (Track) shall be responsible for:

- A) Mainline track inspection in a satisfactory and safe condition, subject to KPIs and requirements as may be prescribed by this manual or DFCCIL or its constituent office, for traffic.
- B) Reporting of requirement of all works incidental to track maintenance, including track relaying works.
- C) Shall hold CWR Track & CWR Turnout Maintenance Competency Certificate, keep abreast of CWR maintenance techniques and renew CWR Competency Certificate at intervals fixed by DFCCIL administration.
- D) Accountable and periodical verification of such works as may be reported to be required in track area.
- E) Maintaining up-to-date knowledge of all codes and manuals, with all amendments, updates and corrections, as may be prescribed by DFCCIL or its constituent office. S/He

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shall be well acquainted with the rules, regulations and procedures concerning his work and duties as enjoined in these codes and manuals. S/He shall keep himself in touch with the orders and circulars issued by higher authorities from time to time and efficiently act upon them. He shall ensure that all staff working under him are well acquainted with the relevant rules and working methods and efficiently perform their duties.

- F) Co-ordination with civil & Bridge maintenance and other Staff at IMD/IMSD - The Sr Executive/Executive (Track) should keep close coordination with Civil & Bridge maintenance and other Staff at IMD/IMSD, when they are required to work jointly.
- G) Ensure proper field training of supervisors and officers sent to his depot for training.
- H) When accompanying higher officials on periodical or special inspections, he should have with him the following registers and documents pertaining to his section, other than the codes and manuals mentioned above:
 - I) Latest MIS reports generated from Maintenance management system including results of track recording and monitoring, all recent and relevant inspection reports and logs pertaining to track, curves, points and crossings, etc.
 - II) Latest applicable Time Table.
 - III) Permanent way diagrams of section and yards.
 - IV) Any other document as may be mandated by DFFCIL
 - V) List of Permanent and Temporary speed restrictions, if any
 - VI) List of works and other details.
- I) Assuring a fully equipped RCR with all the measuring devices and equipment as may be prescribed by DFFCIL or higher officials.
- J) The Sr Executive/Executive (Track) is directly responsible for the safety of the track. He shall be vigilant to locate faults in the permanent way and promptly remedy them. Track defects which are beyond his powers to remedy should be immediately brought to the concerned Depot Engineer's notice and mentioned in track inspection reports.
- K) Independent of detailed periodical inspections, the Sr Executive/Executive (Track), during his routine inspections, should watch for any signs of weakness in bridges and structures affecting track and promptly report any matter demanding the Depot Engineer's attention.
- L) Action in case of Emergency - On receipt of intimation of the occurrence of an accident (including breaches) affecting any part of track, restricting free passage of trains, the Sr

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Executive/Executive (Track) should coordinate with Depot Engineer and endeavour to proceed to site by the quickest available means. On the way he should collect information regarding the damage, the men and material requirement at site for restoration and arrange for movement of men and materials and thereafter the restoration

- M) Sr Executive/Executive (Track) shall keep himself updated with the safety and Incident management manual of DFCCIL and shall play all roles or tasks as may be assigned to him by DFCCIL, in accordance with such manuals.
- N) It shall be responsibility of the Sr Executive/Executive (Track) to ensure that all his tasks, duties and responsibilities are undertaken in accordance and absolute compliance with the DFCCIL's safety management system.
- O) Proper maintenance of all tools, plants, other equipment and small track machines under his charge.
- P) Personally, supervise various important track monitoring and maintenance activities like VERSE use for CWR Stress Free Temperature, Turnout maintenance, Ballast profiles, curve realignment, through tamping, distressing etc.
- Q) Maintaining the sectional register.

1.1.1.3 FOREMAN – LOCAL TEAMS (E0/E1)

- A) Foremen are responsible for the quality of the work performed by themselves and employees in their direct report. They are also responsible for ensuring the work is performed in a safe manner and in accordance to this Manual and the Indian Railways Permanent Way Manual. Where infrastructure is not in compliance with these Manuals, the foreman shall bring the infrastructure into compliance and if warranted, reduce the speed of trains or if deemed fit, stop movements altogether with immediate information to executive and depot engineer concerned. If needed, the foreman should also take measures to protect employees and the public at large.
- B) Every Foreman and his team members shall have the correct knowledge of hand and detonating signals and shall be conversant with the following rules:
 - I) Protecting the line in an emergency and during work affecting the track.
 - II) Method of fixing and safety range of detonators.
 - III) Action to be taken when a train is noticed to have parted.
 - IV) "Safety first" rules.
 - V) Action to be taken where sabotage is suspected, and patrolling in emergencies.

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- C) Every Foreman shall see that the signals, supplied to the team are kept in good order and ready for use and that every man in his team has a correct knowledge of all the signals.
- D) Every Foreman shall ensure that the following tools and equipment are with him at the site of work:
 - I) Level-cum-gauge, square, hemp cord, metre stick, keying and/or spiking hammer, fish-bolt spanner, Torque Wrench, 2 sets of H.S. flags, (2 HS lamps/Tricolour torches in the night), 2 nos. whistle thunderers, 10 detonators, marking chalk and Rail thermometer.
 - II) Sufficient number of shovels or phowrahs, beaters, crow bars, ballast forks or rakes, mortar pans or baskets and wooden mallet. 153.
- E) Observance of Sleeper Packing During Passage of Train– During the passage of the first and last trains in working hours, the Foreman and his men should stand on the cess, each about one rail length apart, and observe the effect on the sleepers. Loose sleepers should then be marked and adequately packed. On double line, the Gangs shall invariably stand on the cess side and not in between the tracks.
- F) Precautions when View is Obstructed–
 - I) On double and multiple lines on curves, the view is temporarily obstructed due to a train passing over a track other than that on which the Track Maintainers are working. It is worsened when trains are crossing each other. The noise of a train passing over one track prevents hearing the noise or whistle of another train approaching the work site.
 - II) When working at a place from which an approaching train cannot be seen, at least 600 metres away a MTS with hand signals should be sent out by the Foreman in the direction of approaching trains. It will be the duty of such Flagman to warn the Foreman by means of signals when a train is approaching. The Foreman will be responsible for warning the Team in good time to enable them to get clear off the track. It may be deemed expedient, as an additional precaution, to issue portable whistle boards of the type indicated in para 815(2) of IRPWM to the Foremen, who should fix them at least 600 metres from the work-site, in the direction the view obstructed to less than this distance.
- G) Patrolling During Abnormal Rainfall– During abnormal rainfall, the Foreman should organise patrolling on the gang-length, whether or not Patrolmen are on duty. In the event of damage being detected, action should be taken to safeguard traffic by protecting the line in accordance with para 161 of IRPWM.
- H) Preventing Trespass and Theft of P. way Fittings– Every Foreman and his men shall endeavour to prevent trespass in Railway limits by persons or cattle on his length of line

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and report any attempts at encroachment or unauthorised structures when noticed. He along with Team, should also attempt to prevent theft of P. Way fittings and report any attempt to steal, to his superiors.

- I) Foremen will be informed of defects (or non-compliance) by way of:
 - I) Inspections by Sr Executive/Executive (Track) and DFCCIL officials.
 - II) Results of Geometry Rail Car (GRC), Rail Flaw Detection (RFD) and other testing methodologies utilized by DFCCIL.
- J) In these cases, the foreman must undertake remedial action prescribed for the defect by the Sr Executive/Executive (Track) or the testing equipment. In addition, they must prepare a record of remedial action for each defect specifying:
 - I) The location of the defect
 - II) The date and time of repair, and
 - III) The remedial action.
- K) Defects (or non-compliance) observed by the foremen will be reported to the Sr Executive/Executive (Track) after appropriate immediate action is taken.

1.1.1.4 FOREMAN- MECHANIZED TRACK MAINTENANCE TEAM

- A) Foremen are responsible for the quality of the work performed by themselves and employees in their direct report. They are also responsible for assuring the work is performed in a safe manner and in accordance to this Manual, the IR LWR Manual and the Indian Railways Permanent Way Manual and IRTMM. Where infrastructure is not in compliance with these Manuals, the foreman shall bring the infrastructure into compliance and if warranted, reduce the speed of trains or if deemed fit, stop movements altogether and immediately inform Superiors. If needed, the foreman should also take measures to protect employees and the public at large.
- B) After the completion of track tamping work, foreman must inspect the track before permitting the passage of a train. As required, the foreman will place temporary speed restrictions before passage of trains; and promptly removed after the passage of requisite number of trains and/or passage of sufficient time. Defects (or non-compliance) observed by the foremen will be reported to the Sr Executive/Executive (Track) for the territory after appropriate immediate action is taken.
- C) Records of each inspection must be prepared on the day of the inspection and must specify:
 - I) The tracks and/or turnouts surfaced;
 - II) The date and time that work was undertaken and the inspection completed;

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- III) Location and nature of any deviation from maintenance standards (or defects); and the remedial action taken.

1.1.2 SYSTEM OF TRACK MAINTENANCE

The track should be maintained through optimum mix of Systematic Mechanized and 'Trouble-Shooting' local maintenance.

1.1.2.1 LOCAL TRACK MAINTENANCE TEAMS

- A) Four "local" Teams working from Integrated Maintenance Depots/ISMD will be responsible for:
 - I) Day-to-day maintenance;
 - II) Remedial action for defects and near-defects identified by inspection personnel and automated testing equipment;
 - III) Emergency response to derailments, system outages, and infrastructure damage.
- B) The four Teams, are as follows:
 - I) Track Maintenance Teams (TMT)
 - II) Rail Maintenance Teams (RMT)
 - III) Material Distribution Teams (MDT)
 - IV) Civil Maintenance Team (CMT)
- C) All Teams are led by a Track Maintenance Foreman except for Rail Maintenance Teams which are led by a Welding Foreman.
- D) Track Maintenance Teams are responsible for the maintenance of track (ballast and upwards) on their territory except for rail. The Rail Maintenance Teams are responsible for maintenance of rail on their territories. Civil Maintenance Teams will be responsible for the maintenance of infrastructure within the railway right-of-way excluding the track (ballast and upwards) on their territory. Material Distribution Teams are responsible for material distribution on their territory. Territory includes both mainline and non-mainline track.
- E) Day-to-day maintenance activities include:
 - I) Track Maintenance Teams (TMT)
 - a) Casual replacement of isolated sleepers when damaged or defective
 - b) Re-spacing and squaring of sleepers (if frequent occurrence or unusual movement, it shall be brought to Superiors notice for cause investigation)
 - c) Fish plates, bolts and other fittings and fastenings checked and tightened, where needed

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- d) Fish plates and bolts lubrication
 - e) Picking of slacks - Isolated cross-level and alignment correction
 - f) Ballast dressing at isolated locations (such as locations of trespassing)
 - g) Casual replacement of defective or broken rail, switch points or frogs
 - h) Lubrication and adjustment of switches and switch expansion joints
- II) Rail Maintenance Teams (RMT)
- a) Grinding and welding of switches and frogs;
 - b) Treatment and removal of rail defects;
 - c) Temporary and permanent closure of rail plugs;
 - d) Thermic welding of joints
 - e) Monitoring of temperature induced stress in rail and maintaining in a stress-free state; and
 - f) Removal and replacement of defect welds
 - g) Monitoring of Rail SFT through NDT methods like VERSE
- III) Material Distribution Teams (MDT)
- a) Distribution of material to work sites in advance of activities maintenance activities performed by Track and Rail Maintenance Teams.
 - b) Provision of lighting, tents etc.
- IV) Civil Maintenance Team (CMT)
- a) Clearing of long and cross drains, minor deweeding etc.
 - b) Making up of Cess and attending formation rain cuts, if any
 - c) Upkeep of approach roads in DFCC land boundary
- F) Teams and employees may be assigned to work outside of territories or areas of focus and Teams may be combined to work together as directed by DFCCIL officers. The foreman in-charge will be the foreman of the Track Maintenance Team unless otherwise indicated by a DFCCIL officer.

1.1.2.2 MECHANIZED TRACK MAINTENANCE TEAMS

- A) Four Mechanized Track Maintenance Teams will be responsible for the mechanized maintenance of track geometry and ballast profiling. The four Teams, are as follows:
- I) Conventional Track Tamping Team (CTTT)
 - II) Ballast Distribution Team (BDT)
 - III) Civil and Track Team

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IV) Switch Tamping Team

- B) All Teams are led by a Track Maintenance Foreman. Except for the Ballast Distribution Team, the Teams are responsible for the mechanical surfacing of track and profiling of ballast. Ballast Distribution Teams are responsible for the transport and distribution of ballast. All Teams and employees may be assigned to do other work as directed by DFCCIL officers.

1.1.2.3 MANAGEMENT OF TRACK MAINTENANCE

A) Maintenance Planning

- I) Every Depot Manager (DGM/Civil and above) must prepare a perspective yearly maintenance plan of the entire corridor, one month in advance of start of each new financial year. The plan shall be in complete harmony with this manual and should include,
- II) Schedule for normal inspection, inspection of point and crossings, curves and level crossings, realignment of curves, adjustment of creep, deep screening, casual renewal, renewal of points and crossings, welding of joints, destressing of long welded rails etc. so that optimum utilization of time and labour resources is possible.
- III) Schedule for deployment of various equipment for mechanized maintenance.
- IV) S/he should also ensure that arrangements are made for adequate materials, tools, labour, man power and necessary caution orders/books, as may be necessary.

B) Maintenance Management System

- I) All maintenance work shall be guided and managed through a maintenance management system. Such system shall cover entire lifecycle of maintenance including track inspection and defect remediation and include mobile application that enables track inspection and maintenance crews to document and record inspections in the field with easy-to-use electronic forms, work lists, records, register, team diaries, schedules, and precise GPS coordinates to record and assign tasks for defect remediation. All Depot Engineers and Sr Executive/Executive (Track) shall be provided access to hassle free record keeping, instant compliance review, and electronic territory management.
- II) The maintenance management system shall allow for “in the field” data entry of track inspections, defects, and repairs. Track inspections and defect records shall be transmitted to a central server via a satellite, radio frequency or modem connection. The central repository shall provide the means to store the Track Inspection records and make that data accessible for FRA review, management reporting, and track analysis.
- III) The system must have adequate functionalities including but not limited to:

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- a) Record Track Inspections
 - (i) Record inspection and defects while disconnected from the central server.
 - (ii) Identify inspector and recording date through secure log on.
 - (iii) Identify area inspected, inspection conditions, and beginning and ending mile poles.
 - (iv) Identify tracks inspected from a select list of existing tracks. Tracks are identified by subdivision, segment, and mile post, track class, description, and track type.
- b) Report Track Repairs including closure reports once the defect is repaired
- c) Report Track Exceptions or Defects
 - (i) Select track defect from a list based on the track monitoring data collected and estimated Track Quality Index
 - (ii) Record the defect including Description, defect measurements (if needed), corrective action, and date corrected
 - (iii) Identify the remedial action taken until the track is repaired
- d) Provide Central Repository and back-up inspection and defect records securely.
- e) Generate reports and MIS as may be required by DFCCIL and its track practitioners

C) Record of Team Work

- a) Each Foreman should be supplied with a team chart and a team diary. In the team chart, details of track maintenance work done over the team length, on a day-to-day basis shall be recorded by the Foreman, according to extant instructions. The work set to the team should also be indicated in the team chart by suitable notations. In the team diary supplied to each team, weekly programme of work should be entered by the Depot Engineer Track. At the end of the week, the Depot manager and Sr Executive/Executive (Track) should carry out a quantitative and qualitative assessment of the work done during the week after thorough inspection and make suitable observations in the team diary. Each team chart/diary should be digitized on a weekly basis by entering it into the maintenance management system. Temporary Teams employed in work allied to track maintenance, should also be supplied with team diary, wherein the details of the work set and the work carried out will be entered by the Depot Engineer-Track or Sr Executive/Executive (Track). Details of maintenance work carried out by these Teams should be entered in the team chart of respective permanent team and also into the maintenance management system on weekly basis.

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- b) Team charts/diaries should be checked and reviewed by the Depot Engineers and Depot Managers during their inspections. They should record their observation in the maintenance management system.
- c) Whenever the team equipments are checked by Depot Engineers or Sr Executives/Executives, the same should be recorded in team chart against the date on which such inspection is done as well as entered into the maintenance managements system.
- d) Special Reports on the condition of Permanent Way - Special Reports on the state of track in his charge shall be generated on monthly basis by the Depot Manager or its nominee for review of GM Infrastructure. In this report the Depot Manager in consultation with Depot Engineer Track and Sr Executive/Executive (Track) shall make candid statement of the defects in the track, reasons for defects and proposals for rectifying them. The GM Infrastructure should scrutinize such special reports with his comments and give his orders, to be also recorded in maintenance management system, thereon to the Depot Engineer Track through the Depot Manager. Orders issued by the GM Infrastructure should be promptly attended to by the Depot Engineer and Sr Executive/Executive (Track).
- e) Supervisory Inspections: The Depot Engineer-should check the track along with responsible Sr Executive/Executive (Track) at least monthly and verify the conditions mentioned by the Sr Executive/Executive (Track) and also study the proposed remedial actions. Remedial actions as necessary should be ordered within his power or referred to the Depot Manager for further orders. Already covered in duty list.

D) Sr Executive/Executive (Track)'s Section Register

- I) Each Sr Executive/Executive (Track)/Depot Engineer shall maintain a digital Section register containing all important information including a brief history of the section. Entries shall be brief and categorised under various sections as indicated below:
 - a) Administration -
 - (i) Change in Sr Executive/Executive (Track), Foremen and Clerks.
 - (ii) Change in jurisdiction.
 - b) Permanent Way -
 - (i) Formation - Sections giving frequent trouble with brief history and remedial measures adopted, if any.
 - (ii) Track structure, method of maintenance, details of particular locations giving frequent trouble and remedial measures adopted if any.

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- (iii) Details of kilometrages of track laid as short welded panels, long welded rails, continuous welded rails, etc. incidence of buckling, maximum and minimum rail temperatures observed, behaviour of Creep, S.E.J. and buffer rails.
- (iv) Grades - Regrading done, with brief details of lifting or lowering of track.
- (v) Curves - Realignment and/or transitioning of curves.
- (vi) Ballast - Kilometrages where there is deficiency of ballast and details of recouplement done. Particulars of deep screening carried out year wise.
- (vii) Creep Adjustment - Details of Creep adjustment done and action taken to reduce creep - Details of Gap survey carried out and adjustment done.
- (viii) Permanent Way renewals - Major renewal carried out as relaying, rerailing and resleepering; large scale renewal of track components at a section should also be shown.
- (ix) Depot yards and sidings - Extension or alteration to sidings, platforms and renewal of points and crossings.
- (x) Rail failures - Brief particulars of all types of rail failures, including weld failures should be noted in the section register, connecting references to the failure reports.
- (xi) Rail Testing and Renewals - Records of rail testing by various methods adopted - Brief details of all rails removed with reasons for removal. This will form the basis of justification for through rail renewals/ casual renewals.
- (xii) Brief particulars of fish-plate failures with details of fish plates and reasons for failure.
 - (a) Lubrication of rail joints: Particulars of work done with dates each year.
 - (b) Material under trial: Brief particulars -Connect reference to notes in the 'Materials-under-trial' register.
 - (c) Track recording: summary of the results of the various track recording runs

E) Bridges and Floods -

- I) Yearly record of rainfall showing month wise distribution
- II) Important repairs and renewal to bridges, details of extensive repairs to bridges, dismantling and rebuilding bridges, strengthening of girders, renewal of girders, extension of bridges and through renewal of sleepers, should be shown. Ordinary repairs need not be recorded.
- III) Damage due to floods: Extent of damage with particulars of rainfall, arrangements made for labour and material, time and labour spent for restoration and approximate cost. Cause of damage and notes of remedial measures

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- IV) List of Railway affecting Works with brief history.
- V) List of vulnerable locations, where stationary watchmen are to be posted.

F) Miscellaneous -

- I) Availability of Maintenance Teams on section for works.
- II) Encroachment and steps taken to remove them.
- III) Infringement particulars
- IV) Accidents attributable to Permanent Way with details.
- V) List of reference books available in the section.
- VI) Any other important information necessary or required by the maintenance management system.
- VII) The entries made in the section registers shall be brought up-to-date on quarterly basis and these shall be scrutinised and signed-off in the beginning of every year by each Depot manager for their respective sections.

G) Permanent Way Plans and Diagrams

- I) The maintenance management system shall have digitised complete sets of the following, with access to Depot Engineer and Sr Executive/Executive (Track):
 - a) The DFCCIL and I.R.S. Track manual or I.R.S. type plans, pertaining to track sections and turnouts extent over their jurisdictions.
 - b) Plans and longitudinal sections of the section showing the physical features, alignment, grades, location of bridges and level crossings. The longitudinal section of the line shall be updated by surveying the longitudinal profile of the line at least once in five years. The necessary action for elimination of humps, sags and unevenness or providing vertical curves as provided in Para 419 of IRPWM be taken if the survey reveals variations in grades. Such an action may also become necessary as a result of track works viz. renewal of rail/ sleepers, lifting/lowering of track, bridge works etc.
 - c) Drawing of bridges, level crossings and protective works and Depot yard layouts over their jurisdiction.
 - d) Working drawings or diagrams pertaining to track and components on their sections, issued from time-to-time.
 - e) The Permanent Way track diagram of the railway line showing the type of track and fittings when laid, type of ballast, type of formation with classification of soil (to be carried out as per RDSO's Circular No: GE-P1-May 2003 & GE-14), blanket thickness, type of formation trouble (if any) and indication of how the railway

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boundary is demarcated. Change points in the track diagram shall be indicated correct to the nearest meter

- f) The Permanent Way diagrams of Depot yards showing complete dimension of running lines, sidings, type of track and turnouts.
- II) The Depot Engineers and Sr Executive/Executive (Track) shall also have in their possession complete sets of drawings and diagrams, in printed hard copy, pertaining to their jurisdictions;
- III) Plans pertaining to their jurisdictions shall be maintained up-to-date by the Depot Engineer and Sr Executive/Executive (Track).

H) Checking work of Teams by Sr Executive/Executive (Track)

- I) Examination of Team's Work- The work done by a team either on the previous day or during the interval when the Sr Executive/Executive (Track)/Depot Engineer is next with the team should be examined for alignment, surfacing and boxing throughout. The Sr Executive/Executive (Track)/Depot Engineer should inspect rails and sleepers and their fastenings and check cross levels, gauge, squareness of sleepers, packing, joint maintenance, profile of ballast and depths of cess below rail level. The Foreman's Muster Sheet should be checked and signed-off, digitally, where possible. Instructions should be recorded in the team diaries through maintenance management system.
- II) Examination of tools and equipment - The Depot Engineer/ Foreman should examine every month and replace, when necessary, worn out tools and equipment. S/He should check the accuracy of the spirit level/gauge and straight edge every month, the result of this examination being entered in the Foreman's diary book.
- III) Each team should have the equipment as listed in ANNEXURE 7.
- IV) Instructions and Tuition -The Sr Executive/Executive (Track) and Foreman should ensure that every man in each team is aware of the following rules in which the men should be examined periodically and on appointment, promotion or transfer:
 - a) Protecting the lines in an emergency or during work affecting the running of trains.
 - b) Method of fixing and safety range of detonators.
 - c) Showing of signals with or without hand signal flags during day and with hand signal lamps during night.
 - d) Action to be taken when a train is noticed to have parted.
 - e) Safety first' rules as prescribed by DFCCIL Safety Management System.
- V) Patrolling of the line during heavy rains /storms and hot weather on CWR lengths. The Sr Executive/Executive (Track)/Depot Engineer should instruct the men in the proper

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use of tools and upkeep of the road. The instructions should not be of a casual nature; they should be demonstrative.

1.2 TRACK INSPECTION & TESTING

1.2.1 TRACK INSPECTIONS

Track inspection must be conducted to the frequencies and requirements of this section.

1.2.1.1 INSPECTION FREQUENCY

All track (including passing loops and cross-over tracks) must be inspected twice per week (Sunday to Saturday) with no more than 3 days between days of inspection.

1.2.1.2 SPECIAL TRACK INSPECTION

In the event of any occurrence that may have damaged the track (such as monsoon, derailment, temperature change, fire, or flood), a Special Track Inspection must be made of the track involved as soon as possible after the occurrence and, if possible, before the operation of any train and equipment movements.

1.2.1.3 WALKING INSPECTIONS

A walking track inspection must be completed on all track once per year. The inspection must allow the inspector a clear view of all track components including rail, ties, fasteners and ballast.

A walking inspection must be conducted on all mainline turnouts twice monthly. A walking inspection must be conducted on all reverse curves once during peak winter month and once during peak summer month.

1.2.1.4 DETAILED TURNOUT INSPECTIONS

Detailed inspections must be made of all mainline turnouts once in three months. The inspection is to be performed on foot so to assess the condition of turnout and must include the measuring and recording of the items mentioned in the proforma:

1.2.1.5 INSPECTION REQUIREMENTS

Unless otherwise specified, track inspections must be made on foot or by riding over the track in a rail-cum-road (RCR) or rail-bound vehicle at a speed that allows the person making the inspection to visually inspect and evaluate the track for compliance to maintenance standards.

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The speed of the vehicle must not be more than 10 kph when traversing grade crossings, turnouts.

An inspector may inspect up to two tracks at one time provided that:

- A) The inspector's visibility remains unobstructed by any cause and that the second track is not centred more than 9.1m (30 ft) from the track upon which the inspector is traversing.
- B) Each track must be traversed by the vehicle or otherwise at least once every week, and each siding and crossover must be traversed by the vehicle or inspected on foot at least once every month.

If the person making the inspection finds a deviation from the requirements of maintenance standards of the Indian Railways Permanent Way Manual, that individual must immediately initiate remedial action.

1.2.2 PORTABLE TRACK GEOMETRY MEASURING DEVICES

DFCCIL will use portable track geometry measurement devices to test track for geometry defects. They are to be strapped to the rear bumper of an inspection rail-cum-road vehicle so as to provide track geometry measurements to supplement visual observations.

Sr Executive/Executive (Track) are required to inspect all mainline track (including passing loops and crossovers) with Portable Track Geometry Measuring Devices once in a fortnight a week on main line. All detected defects are to be immediately inspected and treated as a visual defect with respect to remedial action and record keeping.

1.2.3 STANDARD DIMENSIONS

1.2.3.1 SOD AND INFRINGEMENT

The Track and Structures on DFCCIL should adhere to Standard Schedule of Dimension (SOD) for Eastern and Western Dedicated Freight Corridors (Jan. 2013)

The Sr Executive/Executive (Track) should refer any work resulting in infringement of SOD to the Depot Engineer for instructions. Work involving permanent infringement should be referred to the GM Infrastructure for sanction through the Commissioner of Railway Safety.

Permanent way staff shall be on the alert to prevent occurrence of:

- A) Errors in alignment causing the minimum distance to adjacent structures infringed, for example, overbridges, O.H.E. structures.

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B) Excessive lifting of the track, causing minimum height to overhead structure to be infringed, for example, underside of overbridges, roofs of tunnels, overhead contact wires.

1.2.3.2 VERIFICATION AND PREPARATION OF YEARLY STATEMENTS OF INFRINGEMENTS:

Once a year, the standard dimensions over their sections shall be verified personally by the Sr Executive/Executive (Track) according to the profiles shown in the schedule of dimensions and statements of infringements, if any, submitted to the Depot Engineer by the end of March. The Depot Engineer after scrutiny should forward these to the Depot Manager and GM Infrastructure or its nominee. The statement shall briefly indicate against each infringement the reasons for its continuance together with reference to the sanction of GM/Infrastructure /CTE DFCCIL . The GM Infrastructure after scrutinising the yearly returns will issue necessary instructions to the Depot Engineer through Depot Manager

1.2.4 TRACK TESTING

1.2.4.1 GEOMETRY RECORDING CARS (TRACK RECORDING CARS)- MAINLINE TRACK

DFCCIL will use Geometry Recording Cars (GRC) to measure, calculate and record geometric parameters of the track. The GRC will have a minimum vertical wheel load of 4536 kg (10000 lbs) and be capable of measuring:

- Surface / Longitudinal Profile
- Alignment / Curvature
- Superelevation / Cross level
- Gauge

The measurement chords shall be as per CS 149 and 150 to IRPWM. Track measurements obtained with these vehicles are considered dynamic geometry measurements representative of the track in a loaded condition.

All mainline track, passing loops and crossovers must be tested no less than 6 times per year. The time between successive tests must be no more than 3 months anywhere on mainline track.

A) Functionality

Geometry recording cars will be used for following purposes:

- I) Find urgent track defects and also feed databases with near urgent defects;

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- II) Produce Track Quality Indices (TQI) that direct tamping needs and evaluate rate of change of track quality;
- III) Incorporate rail head loss measurements and rail cant assessment;
- IV) Include rail head loss measurements that are processed through a program that projects wear vs. allowable limits and projects the timing for rail renewal.

B) Reporting

Records of tests by the Geometry Recording Car must specify:

- I) The track or tracks tested
- II) The date of the tests,
- III) Location and nature of any deviation from maintenance standards (or defect), and
- IV) The remedial action to be taken.

The Operator of the GRC is required to generate the report for the first three items and provide it in electronic format to the Track Maintenance Foreman in charge of the territory at the conclusion of testing on territory of a Track Maintenance Foreman. Within 12 hours of concluding the test; the track maintenance foreman is required to must submit the records electronically. In the event, the record is deemed to be incorrect or incomplete after submission, the record must be promptly re-submitted with an explanation as the reason for the correction or amendment. This amended or corrected electronic record will be electronically stored and retained apart from the original record

Sr Executive/Executive (Track) must inspect all defects and ensure appropriate remedial action on urgent items defined by GRC and electronically sign-off within reasonable time.

C) Missed Segments– Geometry Rail Car

If a portion of mainline track or passing loop cannot be tested within 3 months of the prior run, before the expiration of the 3-month period, the Sr Executive/Executive (Track) must:

- I) Inspect that segment of track with a Portable Track Geometry Measuring Device and be governed by the results of that inspection or
- II) Perform an additional visual inspection per week until the test with a Geometry Rail Car or Portable Track Geometry Measuring Device.

If a portion of a crossover cannot be inspected within 3 months of the prior test, the inspector must, before the expiration of the 3-month period, perform a detailed inspection of both turnouts and the track between the mainline tracks.

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1.2.4.2 RAIL FLAW DETECTION (RFD) VEHICLE – MAINLINE TRACK

DFCCIL will utilize a Rail Flaw Detection Vehicle to test rails for internal defects. All mainline track including sidings, turnouts and crossover tracks must be tested once in 6 months.

A) Functionality

Rail Flaw detection shall be undertaken on periodic basis through the use of **non-stop continuous testing**. Under this regime, rail flaw detector cars continue without stopping, at speeds at which they can accurately record the transmission and receipt of ultrasonic pulses into the rail. This enables the rail flaw detector car to test within the flow of traffic without disrupting train service and lowers the rail testing unit cost.

Information on detected defects is to be immediately communicated to track maintenance foreman for the track tested. Information to be communicated to include defect type and size; defect location (GPS and track kilometer); detection time and rail information at defect site such as rail weight and wear. In addition, each defect will be provided a defect number for tracking. Track maintenance foreman will promptly locate defects using GPS navigation and will verify defects using hand-held ultrasonic testers.

If warranted by the defect, the RFD vehicle operator must take all measures necessary to stop the passage of trains or reduce train speeds as required, until inspected and remedial action taken by the local track maintenance foreman.

B) Records

On a daily basis, RFD vehicle operators must submit records of daily tests including locations and times of testing; details of detected defects (type, size, location, number, time of detection); and track wear measurements.

Track maintenance foreman on a daily basis must report on defects detected in their territory. In addition to defect details (as above), details of remedial action taken must be recorded. Track maintenance foreman must continue to report further remedial actions thereafter.

Centralized data analyst will review testing and defect records on a daily basis. The data analyst shall overlay the ultrasonic signature for the same location 6-8 million gross tonnes of traffic in the past. He/she will then designate potential rail defects for field verification.

C) Missed Segments – Rail Flaw Detection

If a portion of mainline track or passing loop is not tested or not tested properly (due to rail surface condition or other reason) within 3 months of the prior run, before the expiration of

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the 3-month period, the Sr Executive/Executive (Track) must test the track with a hand or trolley rail flaw detection unit. Hand or trolley testing must be continued every 28 days until testing by the rail flaw detection vehicle.

1.2.4.3 RIDE QUALITY ACCELEROMETERS (RQA)

Ride quality measurements shall be taken on Fortnightly basis using wagon-equipped accelerometers aligned with mobile telemetry that would transmit the data to central servers for processing and recording as maintenance actions. Such wagons will be added to scheduled trains when they enter the DFCCIL network and removed before they exit.

1.2.4.4 TESTING OF NON-MAINLINE TRACK

A) Portable Track Geometry Measuring Device

DFCCIL will use Portable Track Geometry Measuring Device to measure, calculate and record geometric parameters of the track.

B) Rail Flaw Detection Vehicle

DFCCIL will utilize a Hand held /Trolley based Rail Flaw Detection Device to test rails for internal defects on other than main line Track.

1.3 TRACK DEFECTS AND REMEDIAL ACTIONS

1.3.1 SYSTEM OF TRACK MAINTENANCE

While maintenance of the Track based on Records through Geometry Records Cars or as the case may be shall be done based on MMIS, the basic principal of maintenance shall be guided by IRPWM and the latest guidelines of RDSO.

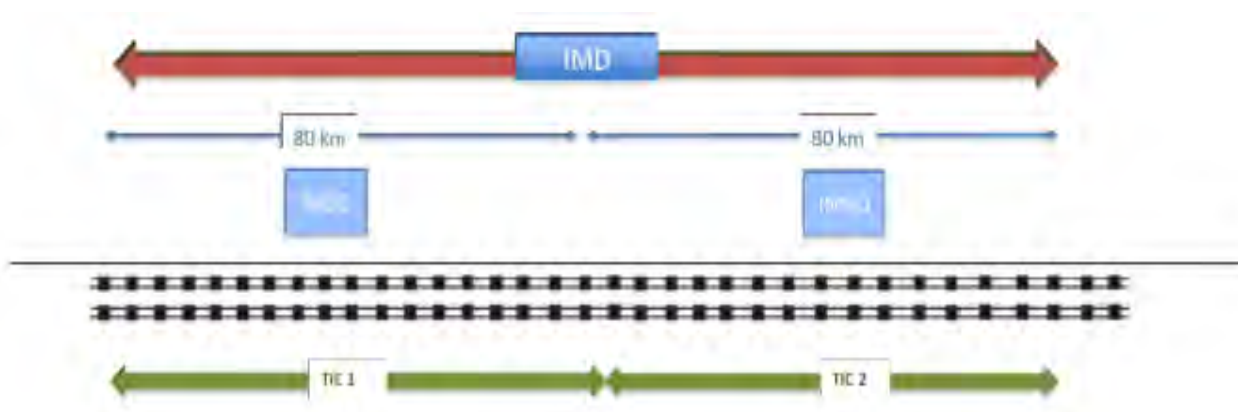
1.3.2 INTEGRATED MAINTENANCE DEPOTS

All track maintenance and engineering activities shall be guided and managed from Integrated maintenance depots of DFCCIL. The intention is that resources for most maintenance activities will be based at both IMD's and IMSD's. Such Integrated maintenance depots are set up at appx 160 kms distance, throughout the DFC, with sub depots appx 45 kms apart from the main depots. These depots and sub depots shall act as common hubs for Civil, Track, Electrical, Signal and Telecom engineering.

All Civil, Track, Electrical and S&T personnel will be located at IMD's and IMSD's with staff reporting to Depot Engineers- Civil, Electrical and S&T. Depot Engineers will report to Depot Managers who will be responsible for all maintenance activities under their jurisdiction.

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Figure 3 IMD Territory



1.3.3 ANNUAL TRACK MAINTENANCE PROGRAM OTHER THAN SYSTEMATIC THROUGH TAMPING

The annual program of regular track maintenance and works incidental thereto shall be based on the program given as shown below, with such variations to suit local conditions, as may be specified by the **GM/Infrastructure**.

Table 1 Annual Track Maintenance Program

S.No	Period	Work
1	Post monsoon attention: For about six months after end of monsoon	(a) Attention to run down lengths in the entire track under IMSDs to restore the section to good shape, (b) Systematic directed track maintenance from, one end of the IMSD jurisdiction area to the others. (c) Normally 4 to 5 days per week should be allotted for works under item (b) and the remaining days for picking up of slacks, attention to bridge approaches, points and crossings over the entire IMSD jurisdiction. Works such as lubrication of rail joints, joint gap adjustments as required and realignment of curves should be done during this period.
2	Pre-monsoon attention for about 2 months prior to break of monsoon.	Normally 2 to 4 days in a week should be devoted to clearing of side and catch water drains, earthwork repairs to cess, clearing water ways and picking up slacks. In the rest of the days normal systematic maintenance will be carried out.

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3	Attention during monsoon for about four months	Attention to track as required. This will consist primarily of picking up slacks and attention to side and catch water drains and water ways. During abnormally heavy rains, patrolling of the line by Teams should be carried out in addition to regular monsoon patrolling.
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1.4 TRACK MACHINES AND EQUIPMENT

1.4.1 EQUIPMENT

Every track maintenance staff must have adequate equipment prescribed in this manual to undertake assigned tasks. It shall be duty of Depot Manager and GM infrastructure or their nominee to ensure that adequate equipment is provided and maintained in proper working condition, in line with the requirements of Original Equipment Manufacturer (OEM).

1.4.2 MATERIAL TRAIN

Wherever a material train is required and retained by DFCCIL through any suitable agencies, detailed instructions as listed in Chapter 12 of IRPWM on working of such material trains shall be followed.

1.4.3 STORAGE AND STABLING OF TRACK EQUIPMENT

All IMD/IMSDs shall have adequate space for parking, stabling and maintenance of track machinery equipment assigned exclusively for their jurisdiction. For equipment expected to be shared by multiple IMD/IMSDs, General Manager Infrastructure or its nominee shall assign a base depot for such equipment's stabling, parking and maintenance. DFCCIL may leverage erstwhile construction depots at various locations for this purpose, by making suitable modifications.

The Depot Manager and GM Infrastructure or its nominee should maintain subdivision-wise a complete inventory in the form of a digital register in Maintenance management system of all equipment including but not limited to RCR, Track Geometry cars, Wagons etc. Such inventory must include following details:

- A) Equipment numbers;
- B) Type of equipment;
- C) Capacity;
- D) Condition of vehicle;

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- E) Locations and particulars of periodical overhaul, when carried out and due.
- F) Maintenance in-charge and schedule of maintenance with compliance reports
- G) Any other detail as may be required

The register should be kept current at all times.

A quarterly return of engineering vehicles and equipment on the IMD/IMSD should be submitted by the Depot manager to the GM Infrastructure with complete particulars of each Equipment for record in his office. It should be the Depot Manager's responsibility to keep track of all Engineering vehicles and equipment allotted to his IMD/IMSD and see that those that are sent to workshops for periodical overhaul are returned expeditiously.

1.4.4 WORKING OF TRACK MACHINE AND EQUIPMENT

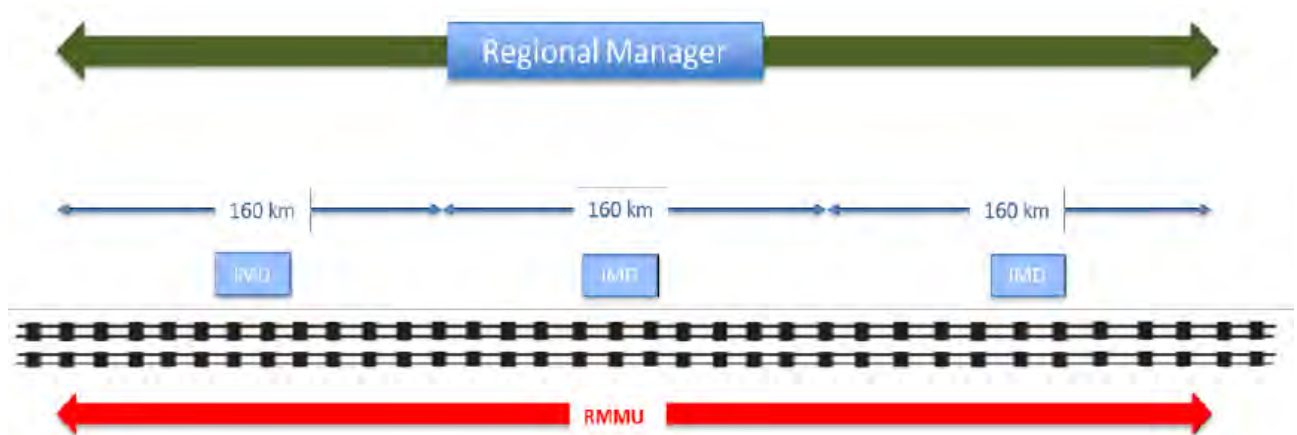
- A) All machines must be operated in harmony with the prescribed speeds, guidelines, limits and rules prescribed by Original Equipment Manufacturer. It shall be the duty of supervising engineering officials to ensure that machine is operated as per such OEM guidelines. Any breach and resultant damage must be reported immediately through maintenance management system. In all such cases, a damage assessment shall be undertaken by the supervisor and corrective maintenance action shall be taken immediately.
- B) All 'on-track' machines shall be worked only under traffic block with the permission of the concerned operations controller and in accordance with the special instructions issued by DFCCIL in this regard.
- C) Each machine shall be in direct charge of a nominated track machine operator. The operator shall be responsible for the working of the machine under his charge. He shall be fully conversant with the rules of working of equipment and of protection in case of emergency. He shall also ensure that the other staff deployed on the machine are fully conversant with the protection rules. He shall hold a valid certificate of competency for driving and working of the machine.
- D) The track machine (s) shall work under the direct supervision of an engineering official not below the rank of Depot Engineer or Sr Executive/Executive who will be responsible for taking the traffic block, for protection of the line while the work is in progress and clearing of the block after completion of the work when the last machine clears the block section and certifying that the track is fit for train movement.
- E) When the track machine is required to move from one block station to another block station, the operator shall run the machine with the proper authority to proceed as defined in IR GR 1.02/6.

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- F) While working on double/multiple lines or lines in adjacent to IR lines, the engineering officials supervising the work of track machine shall ensure that no part of the machine fouls the adjacent track. Any case of infringement to adjacent track is inherent to the machine working which can be cleared at a short notice, the work should be carried out after taking suitable clearance from appropriate authorities of Indian Railways and after protecting the infringed IR line by Engineering signals by Permanent Way Inspector as envisaged in Para 806 of IRPWM.

1.4.5 MAINTENANCE OF TRACK MACHINES AND EQUIPMENT

- A) The responsibility for acquisition and maintenance of all machinery, vehicles and testing equipment shall lie within each Corridor organization; specifically within the organization of GM Infrastructure. Deputy GM - Machine & Vehicles (who would report to GM Infrastructure) would be responsible for the acquisition and maintenance of machinery, vehicles and Portable Track Geometry Measurement Devices (PTGM). The responsibility for acquisition, operation maintenance of all testing equipment other than PTGM should lie with AGM Infrastructure who also would report to Deputy GM - Permanent Way.
- B) DFCCIL will be setting up maintenance yards and depots for parking, stabling and maintenance of large equipment for each corridor. Such maintenance depots must lie within the territory of each of the 8 Regional Managers (ideally in the confines of the central IMD). Adequate field mechanics shall be deployed at these depots to undertake and supervise equipment maintenance works.



- C) Inspection and maintenance requirements will adhere to schedules provided by Original Equipment Manufacturers (OEM's). The primary or initial inspection and maintenance shall be undertaken in the depots and in the field. Field mechanics would be assigned to Regional Mechanized Maintenance Units and to Corridor Maintenance Units and would undertake basic inspection and routine maintenance with the machine operator. In

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addition, more arduous scheduled inspection and maintenance (both scheduled and unscheduled) would be undertaken in maintenance depots.

- D) Specific to automotive vehicles, maintenance of all vehicles shall be undertaken by third parties as arranged by vehicle maintenance coordinators. As much as possible, maintenance agreements should be long-term with contractors near to where the vehicle is based.
- E) Testing Equipment: Responsibility for operation and maintenance of testing equipment (with the exception of Portable Track Geometry Measurement Devices) shall lie with Deputy-GM Infrastructure within each corridor. They will be responsible for developing and managing testing schedules, staffing testing equipment, and assuring equipment is maintained. It is recommended that maintenance for such equipment be completed by OEM's.

1.5 SCHEDULE OF TRACK INSPECTION

Sl. No	Item	Frequency	Remarks
1	Foot Inspection of vulnerable locations by Patrolmen	Once Daily	Beat-5 Km for Double Line, 8 Km for Single line. Vulnerable locations to be identified by CGM and approved by Director of the Section.
2	Track Measurement by measuring Trolley	Once a week	
3	Inspection by Executive/ Senior Executive track	Once a month	Beat-40 Km Double Line, 60 Km Single line.
4	Inspection by AM/ Manger Track	Once in 3 months	Beat-120 Km for Double Line, 120 Km for Single line.
5	Inspection by DY CPM/DGM Track	Once in 12 months	Beat-500 Km Double Line, 200 Km Single line.
6	Night Inspection	Executive/Sr. Executive- Once in Month	

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Sl. No	Item	Frequency	Remarks
		AM/Manager- Once in 3 months Dy. CPM/DGM – Once in 12 months	
7	Footplate/LV inspection by Executive / Sr. Executive Track	Once in month	In case of Double Line both lines should be inspected separately.
8	Footplate/LV inspection by AM/Manager Track	Once in 3 months	In case of Double Line both lines should be inspected separately.
9	Footplate/LV inspection by CPM/DGM Track	Once in 12 months	In case of Double Line both lines should be inspected separately.
10	Inspection by OMS	Once in Two months	Location of >0.3G to be inspected by AM/Manager and report submitted within 5 working days
11	Inspection by TRC	Once in 3 months	15% worst Km under each ISMD should be attended by temping machine.
12	USFD Testing of Rail and Weld	By Spurt Car-Once in a year By USFD machine-Once in 6 month or 8GMT whichever is earlier Verification of REM/IMR location reported by SPURT car	

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Sl. No	Item	Frequency	Remarks
13	Inspection of Curve	Once in 6 months (by Executive/Sr. Executive. Once in 12 month- by AM/Manager.	In case of reverse curve-Once in 3month) by Executive/Sr. Executive
14	Inspection of Points and crossings	Main line Once in 3 months by Executive/Sr. Executive. Once in 12 month- by AM/Manager. Loop line/sliding Once in 6 months by Executive/Sr. Executive. Once in 12 month- by AM/Manager for run through lines and 25% of sliding once in a year	
15	Inspection of LWR/CWR/SEJ	Once in 6 months by Executive/Sr. Executive (Summer/Winter). Once in 12 month- by AM/Manager	AM-Preferably a month ahead of onset of peak Summer/Winter month

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Sl. No	Item	Frequency	Remarks
16	Cold weather/ Hot weather Patrolling	Once in week by Executive/Sr. Executive (Summer/Winter). Once in month-by AM/Manager	Period and location to be approved by Manager/AM
17	Inspection of Side Drains, catch water Drains and water ways of Bridges including RUBs	Once in Year by Executive/Sr. Executive (Before Monsoon and every month during monsoon). Once in year during monsoon AM/Manager	
18	Inspection of Track on through Girder Bridges	Once in month by Executive/Sr. Executive Track. Once in 3 month-by AM/Manager	
19	Inspection on High Bank/tunnel and cutting	Once in Year by Executive/Sr. Executive (Before Monsoon and every month during monsoon). Once in year during monsoon AM/Manager	

CHAPTER II - THE MAINTENANCE OF PERMANENT WAY

PART A – GENERAL INSTRUCTIONS

Instructions are referred from **Chapters of IRPWM** and the additional provisions given in LWR manual and other IR manuals including the various instructions issued by IR/DFCCIL shall also be followed.

2.1 CLASSIFICATION OF LINES

DFC Lines have been classified as below:

- 1) Main lines with maximum train speed of 100Kmph
 - a) Up Line
 - b) Down Line
- 2) Connecting lines with maximum train speed of 100Kmph
 - Lines having the connection with IR
- 3) Loop Lines with maximum train speed of 50Kmph

2.2 RAILS

2.2.1 STANDARD SECTION OF RAILS

- Main lines and Points and Crossings- 60E1 1080 HH (as per IRS T-12)
- Other than Main lines & Points and Crossings- UIC 60kg 880grade
- For switch portion in turnout- 60 E1A1 1080 HH RAIL (as per IRS T-12)
- For stock rail, crossing portion, lead rail - 60E1 1080 HH RAIL (as per IRS T-12)
- For check rail- EN33C1
- For Guard rail- 880 grades rails UIC 60kg

2.2.2 HANDLING & STACKING OF RAILS

Shall be done as per Guidelines for Handling and Stacking of rails RDSO CT-35 Oct-2014

2.3 SLEEPERS

2.3.1 TYPES OF SLEEPERS:

- A) Line Sleepers as per drawing of RDSO-T/7008
- B) Turnout sleepers for 1 in 12 as per approved layout of VAEVKN09047-00
- C) Turnout sleepers for 1 in 8.5 as per approved layout of VAEVKN 05117-08
- D) Sleepers for derailing switches-RH as per approved layout of VAEVKN 07152-01

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- E) Sleepers for derailling switches-LH as per approved layout of VAEVKN 07153-01
- F) Sleepers for SEJ shall be followed as per VAEVKN08055-00
- G) Guard rail sleepers shall be followed as per RDSO/ T-8228
- H) Bridge Approach sleepers shall be followed as per RDSO/ T-8229 to 8237
- I) Level Crossing sleepers shall be followed as per RDSO/T-8225

2.3.2 STACKING & LOADING/UNLOADING OF SLEEPERS:

- A) The stacking of sleepers shall be done on levelled and consolidated ground, one over the other up to 25 layers.
- B) Each layer shall be separated by wooden/concrete battens of 50mmx50mm size of suitable lengths to avoid any damages.
- C) The loading of the sleepers which is properly marked and accepted by the designated official shall be loaded for dispatch. Loading in wagons shall be done by Gantries /Tower Cranes
- D) The sleepers shall be properly secured to avoid movement and displacement during transit.

2.3.3 SLEEPERS DENSITY

- A) Sleeper density on main line & Connecting lines shall be 1660 No's/km.
- B) Sleeper density on Loop lines shall be 1540 No's/km.
- C) Sleeper spacing on turnouts, DS and SEJ as per the approved layout drawing.

2.3.4 LAYING OF SLEEPERS:

- A) Sleepers shall be laid and maintained is square to the rails on straight and radially on curves.
- B) Sleeper spacing's should be marked on outer rails in case of curved track.

2.4 FASTENERS

2.4.1 PLAIN TRACK

- A) ERC-Mark V as per drawing no RDSO/ T-5919.
- B) Metal Liners as per drawing no RDSO/T-8254 & T-8255.
- C) GFN Liners as per drawing no RDSO/T-8222 & T-8223.
- D) CGRSP as per drawing no RDSO/T-7010.
- E) Guard rail sleepers as per drawing of RDSO/ T-8228

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F) Bridge Approach sleepers as per drawing of RDSO/T-8229 to 8237

G) Level Crossings as per drawing of RDSO/T-8225

2.5 TURNOUTS & DERAILING SWITCHES

Fasteners details for other than plain track is referred their respective approved drawings as below:

- A) Layout for 1 in 12 canted turnouts (B.G.) EN60E1A1 Curved Thick web tongue rail with 60kg Rail and weldable CMS crossing on PSC sleeper. (With piroll) as per drawing no- VAEVKN04617-08 (ANNEXURE 3)
- B) Layout for 1 in 12 canted turnouts (B.G.) EN60E1A1 Curved Thick web tongue rail with 60kg Rail and weldable CMS crossing on PSC sleepers (Without piroll) as per Drawing no- VAEVKN 09047-00 (ANNEXURE 3)
- C) Layout for 1 in 8.5 canted turnout (B.G.) EN60E1A1 Curved Thick web tongue rail with 60kg Rail and weldable CMS crossing on PSC sleeper as per Drawing no- VAEVKN05117-08 (ANNEXURE 3)
- D) 1 IN 8.5-218-60KG Thick web Canted “LH” Derailing switches with straight Thick Web Tongue Rail for 1676mm gauge and 25T axle load for Ballasted track as per Drawing no- VAEVKN 07153-01 (ANNEXURE 3)
- E) 1 IN 8.5-218-60KG Thick web Canted “RH” Derailing switches with straight Thick Web Tongue Rail for 1676mm gauge and 25T axle load for Ballasted track as per Drawing no- VAEVKN 07152-01 (ANNEXURE 3)

2.6 SWITCH EXPANSION JOINT

Normally the track is CWR without SEJs. In case there is need of SEJ, Improved SEJ as per drawing VAEVKN08055-00 with 80mm gap can be used.

2.7 BALLAST

Machine crushed 65mm (maximum size) stone ballast as per IRS-GE-1 (June-2004) is used in this project.

2.7.1 BALLAST CUSHION:

Ballast Cushion below the bottom of the sleeper at the rail seat for

- Main Line - 350mm
- Loop Line - 250mm
- Connecting lines - 350mm

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2.7.2 ASSESSMENT OF BALLAST REQUIREMENT:

Provision of IRPWM para 264 shall be followed.

2.7.3 COLLECTION & TRAINING OUT OF BALLAST:

Provision of IRPWM para 265,266 & 267 shall be followed.

2.7.4 UNLOADING OF BALLAST ALONG THE LINE:

Provision of IRPWM para 269 shall be followed.

2.7.5 SURPLUS BALLAST ALONG THE LINE:

Provision of IRPWM para 270 shall be followed.

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2.7.6 BALLAST PROFILE

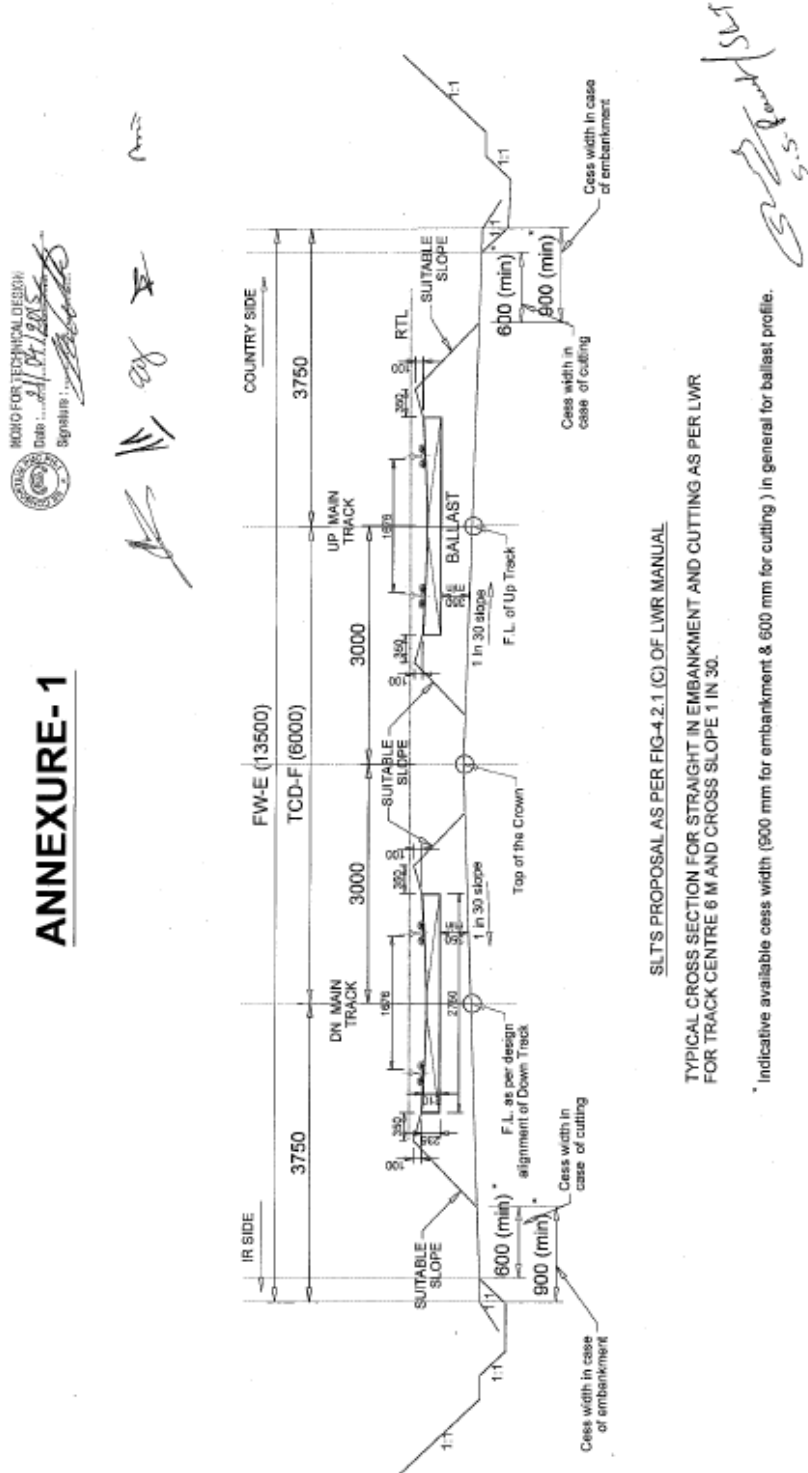


Figure 3 Approved Profile for lines in Straight

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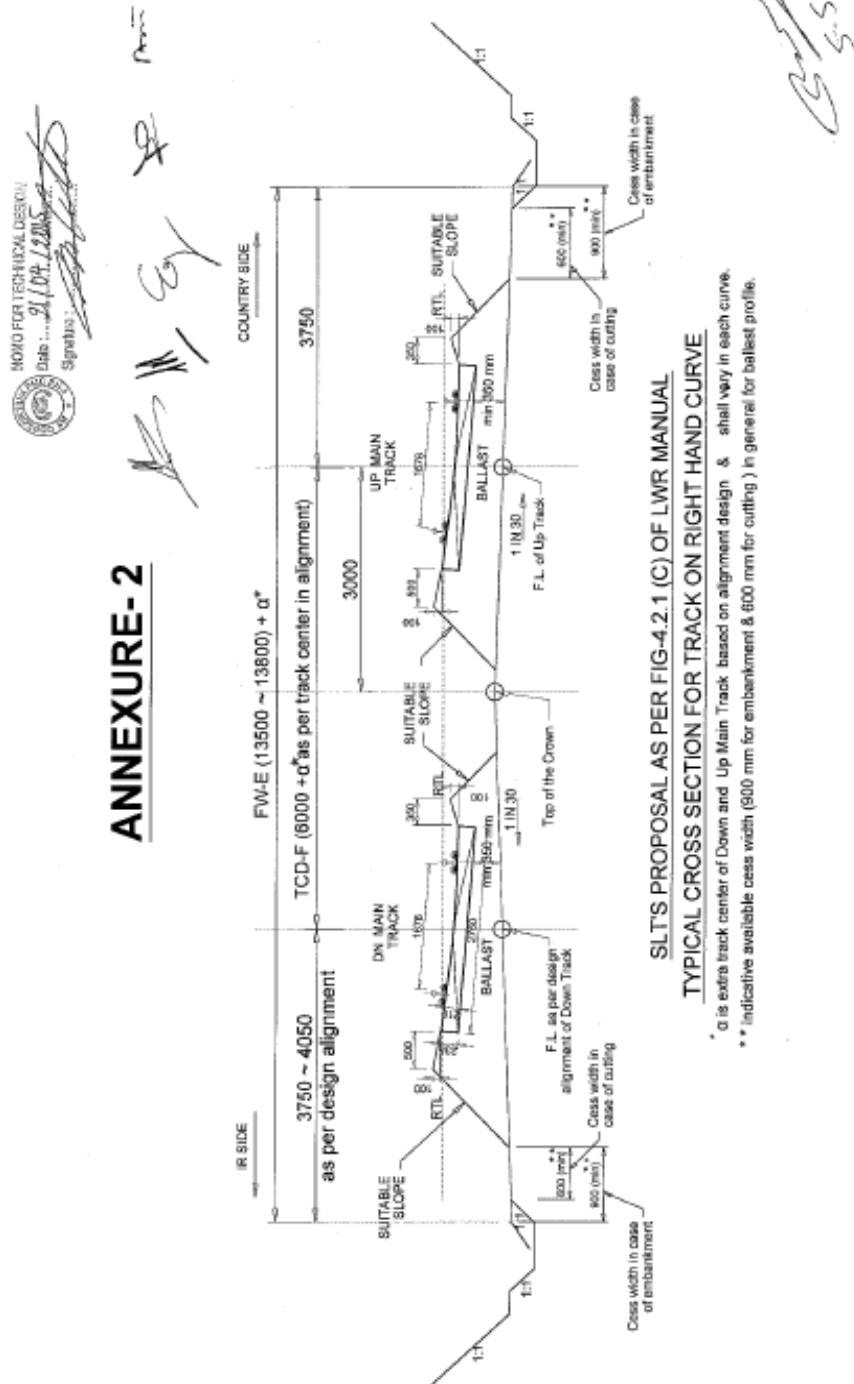


Figure 4 Approved profile for lines in Curves

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2.8 ATTENTION TO INSPECTION NOTES

Shall remain same as per the **IRPWM Clause 207**.

2.9 RECORD OF WORK OF WORKMEN EMPLOYED

The respective official described by DFCC shall scrutinise the work done by the workmen and make the required observation and shall be recorded in the workmen's diary.

2.10 SPECIAL REPORTS ON THE CONDITION OF PERMANENT WAY

Shall be followed as per **IRPWM Clause 209 (latest correction slip to be followed)**.

2.11 PERMANENT WAY SECTION REGISTER

Section register shall be maintained as per IRPWM para. 210 (1) (b), (c) & (d- iii & vi) and 210 (2).

2.12 PERMANENT WAY PLANS AND DIAGRAMS

To be followed as per **IRPWM Clause 211**.

2.13 VERIFICATION OF LAND BOUNDARIES

Respective IMD incharge shall be responsible for demarcation and maintenance of land boundary.

Rest of the contents are same as per **IRPWM Clause 219**.

2.14 TROLLEY REFUGES

Trolley refuges are not required, since no Push Trolley inspection is planned in the Section.

2.15 STANDARD DIMENSIONS

Standard Dimensions shall be followed as per WDFC SSOD.

2.16 FALLING OF TREES OBSTRUCTING VIEW

To be followed as per **IRPWM Clause 222**.

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PART B - TRACK MAINTENANCE

2.17 TRACK MAINTENANCE

An important prerequisite for proper functioning of CWR is its initial laying to a high standard and its subsequent maintenance by trained personnel possessing a valid competency certificate as per Cl. 2.19.1 of this manual and level of authorization shall be fixed by DFCCIL administration.

2.17.1 TYPE OF MAINTENANCE

Sl. No	Item	Frequency	Remarks
1	Casual Renewal of Track	As per Requirement	On inspection by DFCCIL official and approval by AM/Manager or above
2	Through Destressing of CWR	Once in 5 Year	20% track length should be covered in a year
3	USFD of Rail/Welds Manual testing	Once in 6 month or 8GMT whichever is earlier	
4	Deep Screening by BCM	Once in 10 Year	
5	Shoulder cleaning by FRM	Once in 5 Year	
6	Through temping of track by Temping express	Once in 2 Year	
7	Attention of run-down patches by temping machine based on TRC results	As per requirement	
8	Renewal of Fitting (Track)	Once in 7 Year	
9	Recoupmment of Ballast	Following FRM/BCM	
10	Rail Renewal	Based on GMT	
11	Renewal of Points and Crossings (Switch and crossings)	Main Line Once in 5 Year	

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Sl. No	Item	Frequency	Remarks
		Through Line -Once in 10 year Sidings –Once in 15 years	
12	Rail Grinding	Once before operation. Next run based on GMT	

2.18 SPECIAL EQUIPMENT FOR MAINTENANCE OF CWR

Staff responsible for maintenance of CWR shall be trained in using and equipped with additional equipment detailed below: -

Additional equipment with the Maintenance teams

- A) Super Puller and/or Rail Tensors
- B) A pair of joggled fishplates with bolted clamps.
- C) Rail thermometer with markings for temperature ranges for maintenance
- D) Special 1 metre long fishplates with screw clamps.
- E) Rail closure pieces.

Additional Equipment- Equipment as listed in Cl. 5.3 of IR LWR manual shall also be required.

2.19 CWR TRACK MAINTENANCE

2.19.1 GENERAL

- A) **Anchor (length)** is the length of the CWR track that is left clipped down during the Destressing operation to ensure that no movement occurs at the fixed ends of the length being destressed.
- B) **Certificate of Competence** as per Para 2.17 above.
- C) **Pull force** is the force applied to a rail in order to extend it.
- D) **Pulling point** is the location at which the rail is to be cut, tension applied and extension calculated.
- E) **Rail tensors** are hydraulic devices for extending CWR during Destressing. In this specification a set of tensors refers to sufficient equipment to tension both rails at the same time like Super Puller.

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- F) Reference points** are marks on each rail in line with a suitable reference mark on an adjacent unclipped sleeper at equal intervals, e.g. 100m, in order to ensure that the calculated extension is obtained uniformly throughout the length of free rail being destressed. They are numbered 1, 2, 3, etc. consecutively from the inner reference mark point towards the pulling point (see diagram in para. 2.20.7).
- G) Rollers** are devices used to support CWR clear of the sleepers so as to allow its free longitudinal movement during Destressing.
- H) Side rollers** (support arms) are devices used to prevent lateral movement of CWR during Destressing. Their design varies to suit the sleeper, fastening type and curvature of track.
- I) Stress-free Temperature (SFT)** is the rail temperature at which the rail is the same length as it would be in an unrestrained state and at which, therefore, there is no thermal force present.
- J) Destressing** is the process of stretching CWR so that the stress-free temperature of the rails is within the required range, i.e. $t_m + 5$ degree Celsius C to $t_m + 10$ degree Celsius C.
- K) Destressing naturally** is the clipping down of the rail without tensors being used, when the rail temperature at installation/repair equates to the desired SFT.
- L) Stress transition length** is the breathing length of track at each end of a length of CWR between the point of zero stress (the fracture point) and the point of full stress (i.e. stress free when the rail temperature equals the SFT). Note: a stress transition length may be used as an anchor for Destressing purposes but shall not contain any P&C even if that P&C is CWR-compatible.
- M) Tell-tales** are marks made on each rail in line with a suitable reference mark on an adjacent unclipped sleeper, in order to monitor the effectiveness of the anchor length. A reference mark is required at each end of each anchor length: that adjoining the free rail is the inner reference mark, the other is the outer reference mark (see diagram in para. 2.20.7).
- N) Watchman** is a person appointed to monitor the track when the rail temperature exceeds the CRT (W). He shall be equipped and competent to block the line in an emergency.

O) Person in Charge of CWR Track Maintenance

All Maintenance shall be supervised as per decision of DFCCIL.

The Person in Charge of Maintenance shall be in possession of a valid Certificate of Competence.

Important items of CWR Track maintenance are; besides regular gauge, line and level maintenance; replacement of rail & weld fractures if any, major lifting and lowering of track, adjustment of creep, destressing after repair works involving change in SFT or destressing when SFT lowers due to creep, Standard Deviation of SFT of sections along the CWR track

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length varies or planned scheduled SFT corrections to CWR track. CWR Track maintenance in the Plain line and the turnouts requires special knowledge and accordingly CWR Track Maintenance in Plain Line and Turnouts are dealt under different heads below.

2.20 CWR TRACK MAINTENANCE – PLAIN LINE

2.20.1 REPLACEMENT OF BROKEN, CRACKED OR DEFECTIVE RAILS or WELDS

2.20.1.1 EMERGENCY REPAIRS

The fractured rails shall be joined by using the arrangements shown below in Fig.4.4.3 (a), 4.4.3 (b1) and 4.4.3 (b2) of IR LWR manual. As soon as a rail failure is noticed by any Mate/Key man /Patrolman/Foreman, he should take first action to protect the track as per **IRPWM 1002** and inform the Executive in-charge / APM and nearest station master. He shall then take action to repair the track. If any fracture takes place, fractured rail shall be joint with 1m long fishplate & clamps. If gap at fracture does not exceed 30mm, no insertion of any closures is required. On the contrary rail closures of required length shall be inserted first in the gap and then tighten the fracture joint with 1m fish plate and clamps. If required PSC sleeper spacing may be reduced or wooden block supported. After emergency repair, traffic may then be restored at a speed of 20kmph. If in the meantime, some train has come and stopped, he shall pass the train only after emergency repair as mentioned above.

2.20.1.2 TEMPORARY REPAIRS

Ideally Permanent Repairs shall be carried out at the earliest after attending Emergency repair. If a rail fracture is skewed and rail fracture gap is high or welding party is not readily available, the fracture shall be repaired by using a cut rail (not less than 6.5 metre) and clamped as per arrangement shown in Fig. 4.4.3 (a), 4.4.3 (b1) and 4.4.3 (b2) of IR LWR manual.

- A)** A traffic block shall be taken as soon as possible preferably when the rail temperature is within the range specified for td in para 1.11 of IR LWR Manual.
- B)**
 - a) Two points on either side of the fracture shall be marked on the rail such that the length of closure rail (not less than 6.5 metre) to be inserted is equal to the total length of the rail pieces removed from the track minus allowances for two welds and saw cut (normally 51 mm).
 - b) Alternately two points on either side of the fracture shall be marked on the rail at a distance equal to the length of the available closure rail. The length of closure rail should not be less than 6.5 metre at the time of permanent repairs. See Fig. 7.2.4 of IR LWR Manual
- C)** The rails shall then be cut through at these points simultaneously, if possible. The closure rail shall then be inserted and joined. After joining, the traffic shall then be resumed at restricted speed in accordance with ANNEXURE-III of IR LWR Manual. In case closure rail

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as per para (B) (a) above is inserted, one of the joints may have to be provided with closure piece of adequate width and joined by one metre fishplate and clamps.

2.20.1.3 PERMANENT REPAIR

Permanent repair of rail / weld fracture shall depend on rail's temperature at that point of time of repair. If rail temperature is within desired destressing temperature range (t_m+5 degree Celsius to t_m+10 -degree Celsius degrees Celsius) then welding is done with gaps as just required for welding (i.e. without any adjustments for rail temperature difference). If it is below t_m+5 degree Celsius, gaps shall be made duly calculating extension required to bring rail SFT to $t_m+7.5$ degrees. In case rail temperature is above t_m+10 -degree Celsius degrees, need to wait and Permanent repair shall be undertaken once rail temperature reaches or falls below t_m+10 degree Celsius degrees. Only in exceptional cases, rail temperature may be brought down by applying cold water over estimated length and efforts made to keep rail temperature below t_m+10 degree Celsius degrees, till Permanent repair and Equalization of forces is completed.

- A) If the fracture is such that, wide gap AT welding can be adopted, then the total length of fractured ends to be cut shall be equal to the gap required for wide gap welding, duly making adjustments for rail temperature difference from SFT. The gap required for wide gap welding is 75mm. Once the two ends are cut, a gap required for wide gap welding will be created by using rail tensors and joint welded by wide gap AT welding technique.
- B) In case rail closure as per para. 2.20.1.2 B) a) of this manual has been provided for temporary repairs, one joint of the closure rail shall be welded without rail tensor after setting correct gap for welding. However, to ensure correct gap during welding of the other joint, tensor shall be used.
- C) In case rail closure as per para. 2.20.1.2 B) b) of this manual has been provided at the time of temporary repairs, the rail closure shall be suitably cut such that the length of the rail to be finally inserted in track is equal to length of rail removed from track after fracture minus allowances for two welds i.e. 50 mm. Once the closure rail is cut, the closure rail will be welded as given in para 2.20.1.3 B) of this manual.

2.20.1.4 EQUALIZATION OF FORCES

After welding of joints, a length of track equal to breathing length or about 125 meters on either side be unfastened and tapped to ensure equalization of stress and then refastened.

2.20.2 DESTRESSING

The following paragraphs set out the requirements for “destressing” when the above work is carried out. They apply principally to plain line. However, in P&C, where for example a switch half-set is being replaced, the stress shall be restored, and equalised as far as possible between the rails.

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The minimum length of rail which may be inserted into CWR is 6.5m (or 4.5m in cases of particular difficulty and subject to DFCC approval). In the event of a defect of a multiple type, the whole 18m length of rail shall be replaced.

Any movement of the rail relative to the sleepers which occurred either when the rail broke or when the rail was cut shall be noted and action taken as in Table 2.

Table 2 Action following movement at rail gaps

Rail Temperature	Movement of Rail		
	Gap closes (SFT is below rail temperature)	No movement (SFT approximately equal to rail temp.)	Gap opens (SFT is above rail temperature)
Below t_m+5 degree Celsius	Use tensors Re-stress fully before hot weather	No tensors required Re-stress fully before hot weather	Use tensors to restore original stress
Between t_m+5 degree Celsius and t_m+10 degree Celsius	No tensors required Re-stress fully before hot weather	No tensors required	Use tensors to restore original stress
Above t_m+10	No tensors required	No tensors required Re-stress fully before cold weather	No tensors required Re-stress fully before cold weather

If in any of the above cases the SFT appears to be below t_m+5 degree Celsius, or is in doubt and cannot be determined by investigative methods, destressing of the track shall be carried out.

2.20.3 REPLACEMENT OF A DEFECTIVE RAIL WHEN THE RAIL TEMPERATURE IS BELOW T_m+5 DEGREE CELSIUS

- Mark on the existing rail both cutting points for the replacement rail.
- Mark reference points on both the existing rail and corresponding unfastened sleepers/housings (with any sticking pads freed) clear of where the tensors will be located.
- Cut out the defective rail. Check any movement of the rail and take the appropriate action (see para.2.20.1).
- Weld in the replacement rail at one end, and when the weld has cooled, fit the tensors and pull the rails until the reference marks on both the rails and the sleepers correspond again.

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- E) Make the final cut for the welding gap and weld.
- F) Release and remove tensors. Replace pads and fastenings. Ensure all cribs are filled with ballast and sleepers packed. Remove weld debris from site.
- G) Necessary entry to be made in record as per ANNEXURE 8.

2.20.4 REPLACEMENT OF A BROKEN RAIL WHEN THE RAIL TEMPERATURE IS BELOW $T_M + 5^\circ \text{C}$

- A) Measure the gap between the rail ends and take the appropriate action (see para. 2.20.1).
- B) Mark on the existing rail both cutting points for the replacement rail.
- C) Mark reference points on both the existing rail and corresponding unfastened sleepers/housings (with any sticking pads freed) clear of where the tensors will be located.
- D) Cut out the defective rail. Weld in the replacement rail at one end.
- E) Position the tensors and, when the first weld has cooled, pull the rails until the reference marks on the rails have moved inward from the reference marks on the sleepers by a combined amount equal to the original rail gap.
- F) Make the final cut for the welding gap and weld.
- G) Release and remove tensors. Replace pads and fastenings. Ensure all cribs are filled with ballast and sleepers packed. Remove weld debris from site.
- H) Necessary entry to be made in record as per ANNEXURE 8.

2.20.5 NORMAL DESTRESSING PROCEDURE

Destressing of plain line CWR should normally be carried out as follows:

- A) Carry out preliminary site survey and select the positions and lengths of the anchors, lengths to be pulled, and pulling points (para. 2.20.6);
- B) Mark the outer and inner reference marks, any reference points and the pulling point (para. 2.20.7);
- C) Cut the CWR at the pulling point by about 1 m as per IR LWR manual.
- D) Unclip the rails from the pulling point towards the inner reference mark;
- E) Place unclipped rails on rollers and position side rollers as necessary (para. 2.20.9);
- F) Check any movement at the reference marks (para. 2.20.10);
- G) Measure rail temperature (para. 2.20.11);
- H) Calculate the extension(s) of the rails (para. 2.20.12);
- I) Mark the calculated extensions at the pulling point and at any reference points (para. 2.20.13);
- J) Cut the rails again at the pulling point to allow for the calculated extension, a 5mm working gap and any movement at the inner reference mark (para. 2.20.8);

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- K)** Fit tensors and apply tension to both rails until the required extension is achieved (para. 2.20.14);
- L)** Check any further movement of the inner reference mark to ensure that the anchors have held, and take the appropriate action (para. 2.20.10);
- M)** Fasten down 40m of rail on each side of the pulling point, outside the tensors. Cut the rail(s) again if necessary to produce the correct welding gap and complete the welds at the pulling point, with the tensors in position (para. 2.20.15);
- N)** Remove all rollers. Replace pads, insulators and clips;
- O)** Remove tensors (para.2.20.16). Remove welding debris. Fill cribs with ballast. Pack all sleepers.
- P)** Necessary entry to be made in record as per ANNEXURE 8.

2.20.6 PRELIMINARY SITE SURVEY

Before the planned date for Destressing the following shall be checked through the length to be destressed and the anchor lengths (in some cases the items may be attended to during the Destressing operation, rather than beforehand):

- A)** that the track is to the appropriate construction standard and fully ballasted;
- B)** rail fastening type(s) shall be identified, any seized fastenings freed and any missing or defective fastenings replaced;
- C)** worn pads and insulators shall be renewed;
- D)** any potential obstruction to the free movement of the unclipped rail and rollers shall be removed;
- E)** the correct high tensile steel bolts (to insulated joints, stress transfer blocks, etc.) shall be fitted and tightened to the correct torque; and
- F)** defective rails shall be replaced.

The position of anchor lengths, lengths of rails to be pulled and the location of closing welds shall be selected. Account should be taken of the influence which cuttings, embankments, the period of the year and the time of day, may have on rail temperatures during the Destressing operation.

If rail laid in an earlier shift is to be destressed, and the rail was laid in relatively cold weather, it is possible that the SFT of the adjacent existing rail will have fallen in the vicinity of the joint. Destressing should therefore include approximately an anchor length of the adjacent rail.

2.20.7 ANCHORS, REFERENCE MARK POINTS AND PULLING POINTS

The minimum length of CWR to be fastened down to form an anchor shall be as in Table 3:

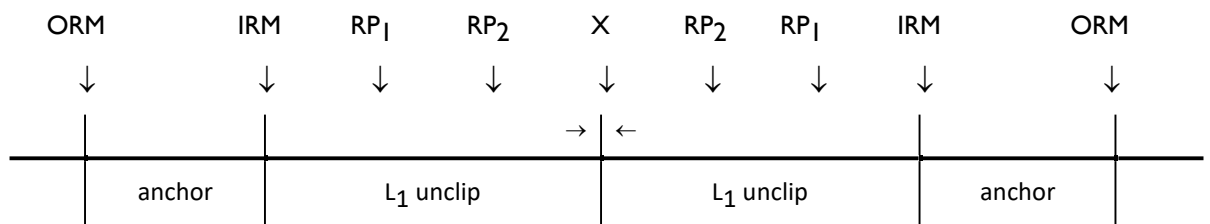
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Table 3 Anchor lengths

Track conditions	Minimum anchor length
where all fastenings are new Mark V and all present, installed with the correct tools and in good condition with all pads and insulators in place and good consolidated ballast conditions	20m or two sleepers per degC temperature difference from the proposed SFT, whichever is greater
ballast conditions poor (Note: para. 2.20.6 requires inadequacies in fastenings, rail pads or insulators to be made good prior to Destressing)	135m

Reference Mark points shall be marked (on both rail and sleeper) at each end of each anchor length in order to monitor the effectiveness of the anchor.

The relationship between the pulling point, the length of free rail to be pulled and the anchors is shown below.



ORM = outer reference mark IRM = inner reference mark RP = reference point
X = pulling point

L_1 = 100m minimum,

= 1000m maximum on straight track (but less on curves - see para. 2.20.14.4)

Anchor: see table above.

The pulling point shall be at least 4.5m from another weld or joint, mid-way between two sleepers and where possible should be the mid-point of the length of rail being destressed.

2.20.8 CUTTING THE RAILS

When cutting the rails for the extension, the required total extension shall be marked on the rail at the pulling point and allowance made for

- A) the calculated extension;
- B) a working gap between the extended rails of 5mm (the correct welding gap should be

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cut after the correct extension has been achieved; and

C) any movement of the rail at the inner reference mark away from the pulling point.

If the rail temperature is rising rapidly it may be necessary to make a number of cuts as the gap closes and new extension calculations will have to be made. If rail temperatures are within 2°C of their original reading re-calculation will not be required.

2.20.9 RAIL SUPPORTS DURING TENSIONING

Fastenings shall be released commencing from the cut or free end and the rails lifted clear of all obstructions and placed on solid rollers positioned at intervals not exceeding 12 sleepers.



Figure 4 Rollers placed under the rail head

Side rollers or support arms shall be used on the inside of curved track at the intervals in Table 4

Table 4 Side roller intervals (maxima)

Radius (m)	Interval	Radius (m)	Interval
> 2000	see below	800 - 601	6 sleepers
2000 - 1001	10 sleepers	600 - 400	4 sleepers
1000 - 801	8 sleepers	< 400	2 sleepers

They should be staggered so that there is only one roller per sleeper. Side rollers may also be used on the outside of the rail on curved track and on both sides of the rail on straight track to prevent tipping of the rail.

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2.20.10 MOVEMENT OF RAILS AT THE REFERENCE MARKS

Any movement of the rail at the reference marks, when the fastenings are released, shall be noted.

Movement at the outer reference mark indicates that the anchor is not satisfactory and a new anchor length shall be established.

Movement of the inner reference mark may be towards or away from the pulling point, depending upon the SFT of the anchor. Movement towards the pulling point should be ignored; movement away shall be added to the calculated extension.

2.20.11 MEASUREMENT OF RAIL TEMPERATURE

2.20.11.1 THERMOMETERS

Thermometers may be of mercury-in-glass, dial or digital display types. They shall be checked visually for condition on each occasion of use, and shall be verified annually for accuracy to within $\pm 1^\circ\text{C}$ against a master thermometer.

2.20.11.2 TAKING READINGS WITH THERMOMETERS

Rail temperature will vary according to location (cutting or embankment) and weather conditions. Readings should be taken at as many locations as necessary to allow for these variations.

Thermometers shall be placed on the rail foot or at the rail web on the shaded side of each rail. At least three thermometers should be used on each rail, located

- near to the pulling point;
- at the mid-point of the free rail;
- near to the anchor length.

Sufficient thermometers shall be used to achieve a maximum spacing of 200m between readings.

When the readings of all thermometers are reasonably uniform, and after not less than ten minutes, an average shall be taken. If any reading is obviously inconsistent with the remainder a check should be made by changing the thermometer(s) concerned. The readings used to calculate the extension shall be taken at the last possible moment before marking the reference points.

2.20.12 CALCULATION OF RAIL EXTENSION

The calculation of rail extension shall be based on the average rail temperature over the length of the rail to be destressed. The rail extension required, to produce a stress-free condition at a rail temperature of $t_m + 7.5$ degree, shall be calculated by the formula

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$e = 1000 L \alpha t$, where

e = extension in millimetres

L = the length of free rail in metres

α = coefficient of expansion of steel (1.152×10^{-5} per °C for rail)

t = difference between actual average rail temperature and required SFT.

2.20.13 RAIL EXTENSION REFERENCE POINTS

The full calculated extension shall be marked accurately at the free end (pulling point) and the proportional movement marked at regularly-spaced (e.g. 100m apart) intermediate reference points so that the rail movement may be monitored during tensioning. At the reference points any sticking pads shall be freed in order to prevent movement of the sleeper with the rail.

2.20.14 RAIL TENSORS

2.20.14.1 GENERAL

Destressing is normally carried out with hydraulic tensors or Super Puller. Tensors shall be used in accordance with the OEM operating instructions and operated only by persons suitably trained and certificated.

Tensor pressure gauges shall be checked and re-calibrated during regular servicing of tensors.

With the exception of an unpowered empty trolley or powered light empty trolley, no rail vehicle may pass over "obstruction less" tensors which are under pressure.

2.20.14.2 ALTERNATIVES TO TENSORS

Where it is not possible to obtain the recommended anchor lengths, e.g. in a CWR length of less than 180m with Turnouts at both ends, it will be necessary to obtain a satisfactory SFT without the use of tensors. This can be done when the rail temperature is between $t_m + 5$ degree Celsius and $t_m + 10$ degree Celsius. Fastening down shall be carried out from the anchor to the free end, care being taken to allow the free ends to pass each other, to permit free expansion.

Rail warmers may be used, subject to DFCC approval of the equipment and process.

2.20.14.3 TENSIONING - GENERAL

During tensioning, the rails shall be extended to the reference marks at the pulling point and the movement checked at any intermediate reference points. The force applied by the tensor shall be monitored during tensioning to ensure that it is reasonably related to the required temperature difference. If this is not so, it is likely that the calculated extensions have not been achieved and the rail shall be checked for possible obstructions to free movement. The tensors shall be checked for slippage of the clamps or for defects in the tensor dial pressure gauge.

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The maximum pull to be applied shall not exceed the lesser of

- 60 tonnes;
- 10 tonnes less than the maximum capacity of the tensors.

When this maximum pull is insufficient to achieve the full rail extension, an extension equivalent to the maximum pull shall be calculated and applied. In such cases the required SFT will not be obtained and it may be necessary for the CWR length to be destressed.

2.20.14.4 SINGLE PULL

In order to reduce the effect of frictional resistance and to ensure adequate stability during Destressing, the maximum lengths of CWR to be tensioned in one direction (i.e. measured between the pulling point and the inner reference mark) shall be as in Table 5.

Table 5 Tension length and track curvature

Radius of curvature (m)	Maximum length to be tensioned (m)	Radius of curvature (m)	Maximum length to be tensioned (m)
> 4000	900	1200 - 801	300
4000 - 2001	600	800 - 601	225
2000 - 1601	500	600 - 401	150
1600 – 1201	400	≤ 400	100

Where the radius changes through the pulling length, the maximum length to be pulled in one direction shall be determined by the **minimum** radius.

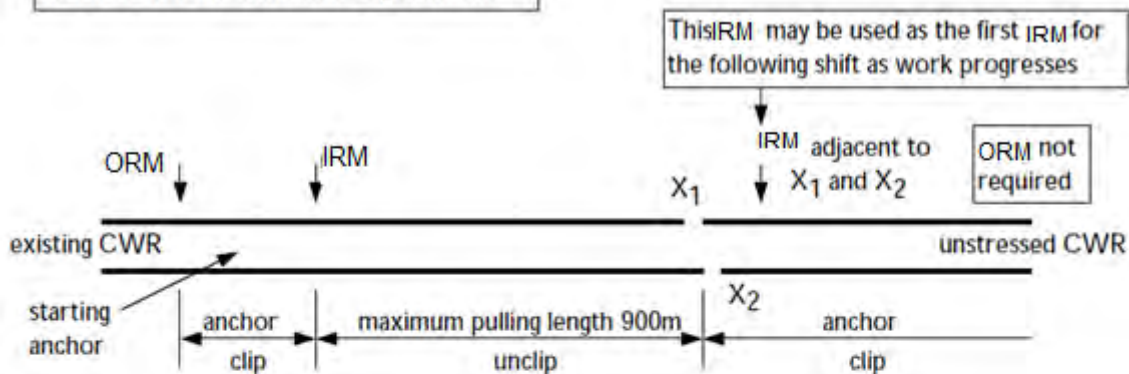
Provided that the overall length to be destressed does not exceed double the lengths in Table 4 and the tensor capability (load and extension) is adequate, the required extension should be obtained over the whole length by “pulling both ways” from a central pulling point: see diagram in para. 2.20.8. When pulling both ways, the lengths to be pulled should be as nearly equal as possible.

2.20.14.5 MULTIPLE PULLS

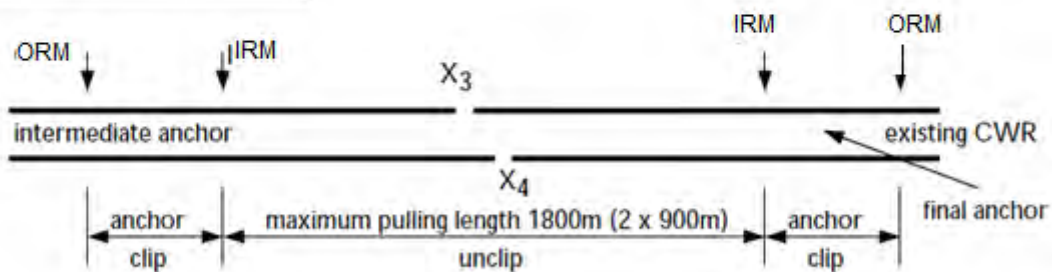
The recommended sequence for tensioning rails upto 1800m is shown below.

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1. First and intermediate operations



2. Final operation



IRM - Inner Reference Mark

ORM - Outer Reference Mark

X - Pulling Points

2.20.15 DESTRESSING OF CWR ON CURVES

Destressing of rail on curves may induce a lateral shift of the track and for this reason the maximum pull applied shall be limited as follows:

2.20.15.1 SIMPLE CURVES

Maximum pull shall be corresponding to the radius of curvature's the maximum tension length as specified in Table. 5 above.

2.20.15.2 REVERSE CURVES - ANY RADIUS

Destressing shall be carried out in stages, with the length of rail being pulled at any time contained fully within one curve or the other, not straddling the point of reversal (i.e. when "pulling both ways" from a central pulling point, the pulling point shall be located at the point of reversal).

2.20.16 WELDING THE FINAL JOINT AND FASTENING DOWN

When the required total extension in each rail has been obtained, at least 40m of rail on each side of the tensors shall be fastened down. The welder shall check that the welding gap is

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correct for the type of weld to be made, and re-cut the gap if necessary. Once a weld is started there shall be no disturbance to that rail or the opposite rail until the weld is completed and the cooling period has elapsed.

On sharp curves fastening down shall follow within 10 sleepers of the removal of each side roller, in order to minimise the tendency of the distressed rail to leave the rail seating.

2.20.17 RELEASE AND REMOVAL OF TENSORS

The tensors shall remain in position for the minimum cooling time specified in AT Welding manual, after the weld has been made, in order to avoid hot tears.

2.20.18 RECORDS

A register of CWR shall be kept, in a form approved by DFCC, containing complete current records for all CWR track, including SFT values. The register shall be held by, or be available to, the person with current responsibility for maintenance of the track. It shall also be available to the DFCC Track Engineer. Certificates and all relevant information concerning Destressing (including details of achieved SFT and site conditions) shall be passed to the holder of the register within seven days of any disturbance to, or any destressing of the track.

Where there are reasons to expect that the original stress condition has changed, e.g. where excessive rail creep has occurred, the SFT should be measured with NDT equipment like VERSE and satisfied.

2.20.19 EQUALITY OF SFT BETWEEN RAILS

The SFTs of the two rails of the track should not differ by more than 6C. If only one rail is being replaced the other rail shall therefore be destressed at the same time, unless the SFT of the other rail can be measured and shown to be not more than 6C different from the relevant SFT values.

For CWR lengths which have an SFT of less than $t_m + 5$ degree Celsius a prioritised programme of Destressing shall be produced. The prioritisation shall be carried out in accordance with the methodology, which shall be used also for the situation where Destressing records do not currently exist.

Records of SFT shall also be available to DFCC at all times.

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2.21 CWR MAINTENANCE PROFORMA

1. Maintenance Details

Year/Year/Year/Year

- i) Method of packing at the time of laying, manual/by machines
- ii) If by machine, type of machine/ tamper used
- iii) Packing during maintenance - manual beater/manual MSP/by machines
- iv) If by machines, type of machine/ tamper used
- v) Whether directed maintenance/ systematic through packing is done and if the latter, the period when it is done
- vi) Location requiring repeated maintenance, if any
- vii) Quantity of ballast recouped

2. Climatic Details

(Measurements shall be taken on representative CWR only)

- i) Maximum daily variation in temperature, vide para 5.2 of LWR manual
- ii) Max. rail temperature °C
- iii) Max. ambient temperature °C
- iv) Minimum rail temperature °C
- v) Minimum ambient temperature °C

3. Details of Installation, Destressing etc.

Items (i) to (iv) to be entered soon after CWR is laid and destressed for the first time.

- i) Installation temperature
- ii) Mean rail temperature for the locality

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- iii) Temperature at the time of destressing t_d
- iv) Reasons for carrying out the destressing
- v) Subsequent destressing done - temperature and date

4. Unusual Occurrences

- i) Rail fractures
- ii) Buckling (Location and reasons for buckling)
- iii) Replacement of components in SEJ assembly
 - a) Fastenings
 - b) Bolts
 - c) Sleepers
 - d) Longitudinal ties
 - e) MS Bracket
 - f) Chairs
 - g) Rubber pads
 - h) Tongue rails
 - i) Stock Rails
- iv) Derailments and accidents on the CWR portion
(give km of the affected portion)
- v) Replacement of sleepers (give km and the number of sleepers replaced and reasons)
- vi) Replacement of rails
(indicate length, km and reasons)

5. Formation

- i) Any trouble subsequent to laying and the treatment given

Note: Above chart has been reproduced from IR LWR Manual Para 8.2.1

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CHART FOR MEASUREMENT OF CREEP OF CWR

CREEP POST NO.....		LOCATION.....				Rectification carried out		Remarks
Distance (mm) between Ref. mark and mark on rail at various Creep posts of CWR		Measured By.	On date	By				
Date of Measurement	Time of measurement	Rail temp	Right rail/Left Rail	Measurement at CWR creep post				

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2.22 CWR TRACK MAINTENANCE – TURNOUTS

CWR Turnout maintenance includes adjustments of gaps & torques, replacement of defective switch-stock rail, crossing, anti-creep device, rail / weld fracture etc. Defective rails / parts replacement requires cutting of CWR track and whereas broken stock rail or lead rail automatically creates stress free rail ends. Turnouts/ Turnout components replacement or adjustments involving change to CWR SFT, shall be destressed as per 2.23.

Emergency, Temporary and Permanent Repairs are brought out as below:

2.22.1 EMERGENCY REPAIRS

2.22.1.1 REPLACEMENT OF STOCK & SWITCH RAIL

When a broken or defective stock or switch rail warrants replacement, the following procedure shall be followed:

- A) No traffic is allowed till its replacement.
- B) Gap at fracture shall be measured and noted.
- C) All fixtures like S&T gears & SSD shall be disconnected
- D) A cut mark near AT Weld (including the HAZ portion of weld) shall be done
- E) For replacement of Stock rail, two cuts on Stock Rail, (one at SRJ & another at junction of lead rail, including HAZ portion of weld), shall be done.
- F) For replacement of Switch rail, one cut at Junctions of Switch Rail & lead rail covering heat affected zone portion of A.T. Weld shall be done.
- G) All fixtures and fasteners are then removed from Stock and Switch Rail
- H) Replace the fractured / defective Tongue and Switch Rail unit
- I) Adequate size of Rail closures is then inserted at all three joint locations and then tightened with one meter long fish plate and clamps
- J) All fixtures and fasteners are again put back at their proper place and tightened.
- K) All activities of replacement of Switch Rail/Tongue rail shall be done in presence of S&T staff.

Note:

1. Whenever any replacement of tongue rail or stock rail, on account of fracture, is required same shall never be replaced individually. Stock Rail & Tongue Rail will always be replaced together, to achieve the proper housing.
2. If Stock Rail is defective or broken, keeping emergency traffic conditions in mind, emergency repair of Stock rail alone may be taken up in rare cases and trains at 20 km/h can be permitted.
3. If one side tongue rail is fractured, the traffic on other side tongue rail route, which is intact, can be permitted, if it is main line, at a speed of 50km/h and if it is diverging line at 20 km/h. Traffic on route where tongue rail is broken shall be permitted only

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after its replacement. The speed can be relaxed to normal after permanent repair only.

2.22.1.2 REPLACEMENT OF WELDABLE CMS CROSSING

If a CMS Crossing warrants Emergency repair by replacement,

- A) No Traffic shall be allowed till its replacement.
- B) Isolate the CMS Crossing by cutting all four sides of it including heat affected zone.
- C) Remove the fittings from crossing.
- D) Before placing new crossing over sleepers all the rubber pads shall be replaced, if required.
- E) Replace the fractured crossing with new crossing.
- F) Insert the closure rail piece of suitable size at all the four joints.
- G) During replacement of CMS Crossing it must be ensured that TNC positions should not be disturbed & all the one meter fish plated & clamped joints shall be supported with wooden blocks.
- H) After tightening the all four joints of CMS Crossing, traffic shall be allowed at speed of 20 km/h.

2.22.2 TEMPORARY REPAIRS

Temporary repairs shall be resorted to only when permanent repairs carrying out is extremely difficult and traffic loss in the meanwhile will be immense.

2.22.2.1 REPLACEMENT OF STOCK & SWITCH RAIL

- A) Arrange three number rail closure (not less than 6.5m length) at site and keep them near cut locations - one at each cut.
- B) Following activities shall be done under traffic block of suitable time.
- C) Create one cut ahead of SRJ & towards CWR side at a distance of length of closure Rail +50 mm (2 A.T. welding gap).
- D) Create another cut ahead of junction of Lead and Stock rail, towards lead rail side at a distance of (length of closure – gap) +50mm 2 A.T. weld minor fractured gap.
- E) Create a cut on lead rail of Tongue Rail side at distance of Length of Closure Rail +50mm (2 A.T. welding gap)
- F) If fractured location on Switch Rail is between ATS and ACD, Gap at fractured location shall not be considered during repair work. On the contrary, fracture is between ACD & Junction of lead rail, gap at fractured location must be considered while repair is going on.
- G) All the three Rail Closures are inserted into the Track, after removal of their respective rail pieces and joints are tightened with 1-meter long fish plate & clamp. Before tightening of free joints, it must be ensured that rail pieces of suitable size are put into the gap of each joint.
- H) All the one meter fish plated joints shall be clamped and supported with wooden blocks.

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- I) Traffic block is then cancelled and traffic is allowed to pass over from the site at a speed of 30kmph.

2.22.2.2 REPLACEMENT OF WELDABLE CMS CROSSING

- A) A traffic block of suitable duration shall be obtained.
- B) Create rail closure of 4.050m/4.100m in length from first cut point & tightened with one meter long fish-plate & clamp.
- C) All the four created joints must be supported with the wooden blocks & passed the traffic at a speed of 30kmph after cancellation of traffic block (only in case welding party is not readily available)
- D) If welding party is available & traffic block is also available then the activities mention in permanent repair shall be followed.

2.22.2.3 REPLACEMENT OF LEAD RAIL

- Action shall be taken as per para. 7.2.4 of LWR Manual.

2.22.3 PERMANENT REPAIRS

2.22.3.1 REPLACEMENT OF STOCK & SWITCH RAIL

- A) Arrange three welding parties with all tools & equipment.
- B) Study Rail Temperature record of last seven days and check the probability of achieving td aril temperature. Else arrange for Tensor Destressing.
- C) One of the gaps at all three closures is adjusted to 25mm by longitudinal slewing of closure & welding is performed. The other gap is cut to 25mm and welded.
- D) Before carrying out welding of another end of lead rail which is welded the tongue rail the position of anti-creep device must be ensured for matching with initial laying position.
- E) After fixing the anti-creep device at correct position, welding of free joint shall be performed at td. If gap is more than 25mm then lead rail is make free from fastenings upto the toe of crossing & rail is tamped with wooden mallet. Thus, the gap of 25 mm is adjusted and welding is performed.
- F) Similarly welding on either end of stock rail closures shall be welded.
- G) All the fittings & fastening are tightened.
- H) All the new A.T. Welded joints shall be protected with jogged fishplate and clamp and remain supported with the wooden blocks.
- I) Destressing to be done as applicable as per para 2.30.5 and equalization of forces is carried out over 125m on either side.
- J) Check for proper Switch housing, ACD gaps,
- K) Traffic block is cancelled and train is allowed at normal speed.

Note:

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1. AT Welding of 60Kg 90 UTS rails shall be preferably be done with using three piece pre-fabricated mould and single shot crucible with automatic tapping. For 1080 HH rails two piece prefabricated mould technique will be used. For the combination of 1080 HH rail and 90 UTS rails single shot crucible shall be used.
2. New AT welded joints will remain protected with joggled plates till they are declared good after USFD Testing.

2.22.3.2 REPLACEMENT OF WELDABLE CMS CROSSING

- A) Study Rail Temp. Record of last seven days/or one week and knows the probable time of td.
- B) Arrange two welding parties with all tools & equipment.
- C) Arrange traffic block at the time when td is expected to be achieved.
- D) One of the gap at all three closures is adjusted to 25mm by longitudinal slewing of closure & welding (SKV) performed. If the other gap is also 25mm the other joint could also be welded. However, if other gap is not 25mm then the other gap is fish-plated & tightened with clamps and traffic block is cancelled & Traffic is allowed with respective speed of 30kmph.
- E) If gap at other end of Rail Closure is more than 25mm, Lead Rail of suitable length shall be opened by removing sleeper fastenings and tapped with wooden. Thus the Gap is adjusted and welding is performed
- F) All the fittings & fastening is tightened.
- G) Equalization of forces shall be done up to length of anti-creep device from toe of crossing and upto a length of 125m from back of crossing.
- H) Traffic is allowed with normal speed.

Note:

1. AT Welding of 60Kg 90 UTS rails shall be preferably be done with using three piece pre-fabricated mould and single shot crucible with automatic tapping. For 1080 HH rails two piece prefabricated mould technique will be used. For the combination of 1080 HH rail and 90 UTS rails single shot crucible shall be used.
2. New AT welded joints will remain protected with joggled plates till they are declared good after USFD Testing.
3. Sequence of Welding: -
 - Straight lead and back of crossing.
 - Turnout side lead & back of crossing.

2.22.3.3 REPLACEMENT OF LEAD RAIL

- A) Action shall be taken as per para. 7.2.5 (i), 7.2.5 (ii) & 7.2.5 (iii) of LWR Manual.

Note:

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1. Tensor shall not be used to ensure the correct gap of 25 mm. Ensure the correct gap of 25 mm following activities shall be done.
2. Fittings shall be removed up to the junction of tongue rail under traffic block. When temperature is within the range of T_d , thus adjusted the gap & welding is performed.

Rail temperature at the time of tightening the fittings shall be recorded. If it is found within the range of T_d ($T_m + 5$ to $T_m + 10$), no further destressing work on turn out shall be required. If Rail temperature is found beyond the prescribed range, then this will be treated as temporary destressing of the particular portion of turnout. On all three sides approach of turnout, equalisation of forces/stresses must be done in a length of 125 meters at temporary destressing temperature and then traffic shall be allowed as per their prescribed normal speed. Since in turnout portion, tensor could not be used being crossing is in fixed portion so strict vigil shall be maintained on the prevailing rail temperature.

But as soon as the T_d is found before the onset of summer, this whole turnout will be destressed at T_d by isolating the turn out from all three-side approach, subject to the condition that three repairs have been taken place in the affected turnout.

Under unavoidable circumstances where destressing could not be done soon after and not likely to be done within a reasonable period, a strict vigil shall be maintained on the prevailing rail temperatures, and if the rail temperature rises more than 20 degree Centigrade above the rail temperature at which welding of rails/laying of welded panels was done, temporary destressing shall be undertaken at a rail temperature of 10 degree Centigrade below the maximum rail temperature likely to be attained until final destressing. If the rail temperature comes down appreciably, cold weather patrolling as per para 9.1.2 (ii) of IR LWR Manual should be introduced. Final destressing shall be done after consolidation of track as per para 1.18 of IR LWR Manual has been achieved.

On the Contrary no destressing is required in the turnout portion. Only strict vigil to be kept over the said turnout and if temperature rises more than 20degree Centigrade above at which welding of rails was done, temporary destressing shall be undertaken at a temperature 10 degree Centigrade below the temperature which is likely to be attain.

Note:

1. AT Welding of 60Kg 90 UTS rails shall be preferably be done with using three-piece pre-fabricated mould and single shot crucible with automatic tapping. For 1080 HH rails two-piece prefabricated mould technique will be used. For the combination of 1080 HH rail and 90 UTS rails single shot crucible shall be used.
2. New AT welded joints will remain protected with joggled plates till they are declared good after USFD Testing.

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2.22.3.4 REPLACEMENT OF ANY RAIL IN CWR TERRITORY

Procedure shall be followed as per 2.20.3, 2.20.4, & F) and equalization of stresses shall be done on either side of replaced rail in a length of 125m.

2.22.4 INSPECTION/MAINTENANCE/SPECIAL REPAIR IN TURNOUT PORTION

2.22.4.1 MAINTENANCE - GENERAL

- A) Points and crossings are laid with 1 in 20 canted rail seats.
- B) Turnouts of 1 in 12 thick web switches on main lines and loop lines.
- C) Turnouts of 1 in 8.5 thick web switches on sidings.
- D) Taking CWR through Points & Crossings with weldable CMS crossings
- E) 1 in 12 turnouts have a permissible speed of 50 kmph and 1 in 8.5 turnouts have a permissible speed of 25 kmph on diverging track.
- F) No fishplates Joint and approach SEJs in Turnout zone.

In turnout designs one anti-creep devices are fitted between each switchblade and its matching stock rail near the heel end of the switch rail (between sleepers 23 and 24 in 1 in 12 and between 18 & 19 in 1 in 8.5 turnouts) This is designed to prevent differential longitudinal movement of the switchblades relative to their stock rails caused through temperature variation.

The derailing switches are fitted with two anti-creep devices between the switch blade and its stock rail (between sleepers 17, 18 and 19).

2.22.4.2 MAINTENANCE OF ROLLERS AND SLIDE CHAIRS

The installation of the PIROLL system cannot compensate the condition poor maintained turnout. The used roller material and the lubrication-free bushing material is resistant to environmental influences like heat, cold or dirt. To avoid material damage, the PIROLL must not be heated above 120 ° C.

When installing the turnout into the track, i.e. during tamping works or maintenance tamping work, the PIROLL can remain mounted on the base plates.

The Inspection and Maintenance of PIROLLS shall be followed as per the instructions given in ANNEXURE 2.

2.22.4.3 MAINTENANCE OF ANTI-CREEP DEVICE

A) General

Anti-creep device contains two parts viz pin & fork. The pin and fork are fixed to the rails using bolt and nuts. The ideal gap between the pin and fork is 7 mm on either side between the pin and fork.

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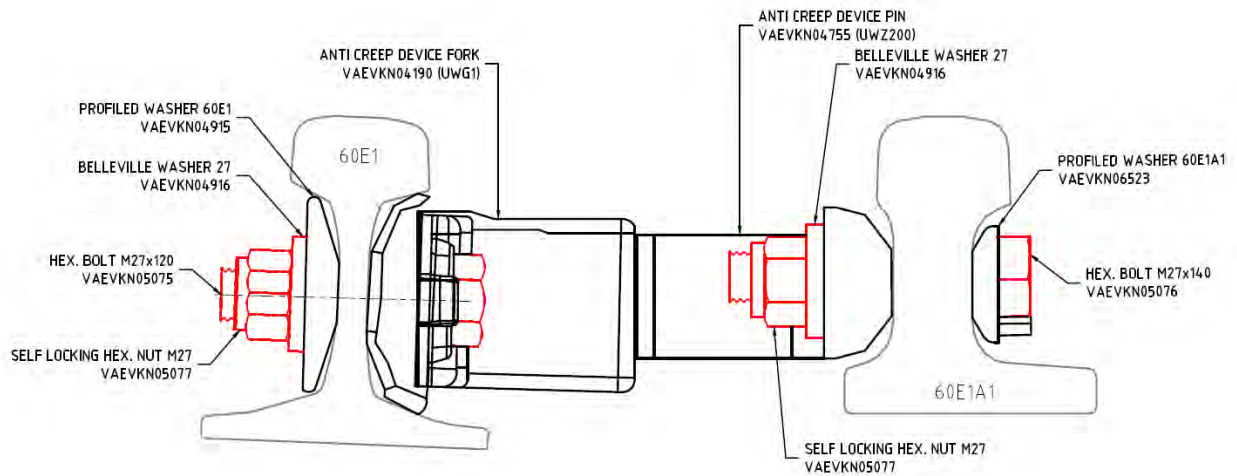


Figure 5 Various components of ACD

In turnout designs one anti-creep devices is fitted between each switchblade and its matching stock rail near the heel end of the switch rail and in derailing switch, two anti-creep devices are fitted. This is designed to prevent differential longitudinal movement of the switchblades relative to their stock rails caused through rail creep. The location of anti-creep devices shall be as per respective layouts attached as ANNEXURE 3.

There are gaps between these two blocks, usually from 4 mm to 7 mm. When creep occurs between the switch and stock rail under thermal force, the two blocks will move closer in one side and away on the other side. With the increasing of the thermal force, these blocks will contact each other on one side, and then thermal stresses are transferred from the switch to the stock rail via the anti-creep blocks. This design does not maintain the switch in its proper location when the creep occurs and it can also release part of the thermal stress applied on the stock rail.

The design of the turnout is in such a way that there is minimum creep in the tongue rail, as it is acting as a breathing length. However, it has to be ensured that the crib ballast is properly maintained to allow the turnout to act as a proper CWR.

B) In 1 in 8.5

Design considerations of ACD in 1:8.5 turnout is summarized as below:

The maximum expansion of lead rail length from starting of crossing to the ACD is 6.1 mm, which is less than the 7.0 mm designed gap between the pin and fork. If in any case the ACD gap closes up or the ACD breaks in extremes, all the rubber pads and the rail clips shall be inspected, toe load of fasteners and ballast resistance- cushion and profile - Creep shall be checked, cause to be investigated and necessary required action of renewal of fasteners or

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destressing or other shall be taken. Regular tightening of bolts of ACD with torque wrench shall be done at a maximum torque of 900 Nm.

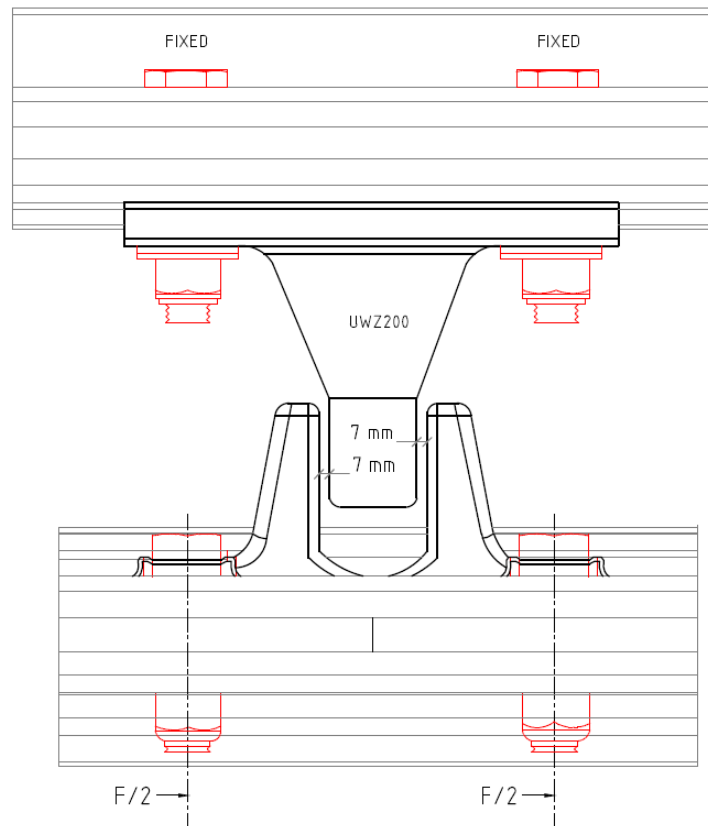
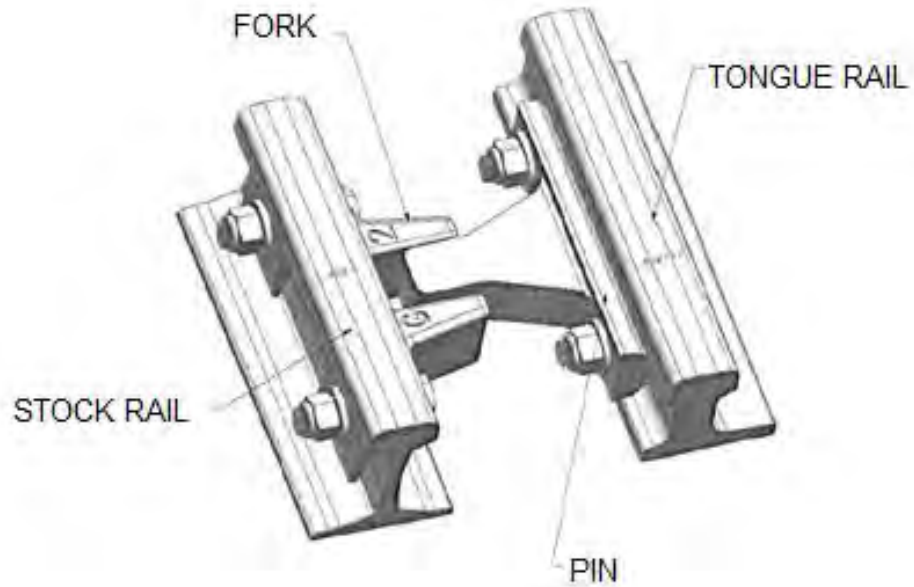
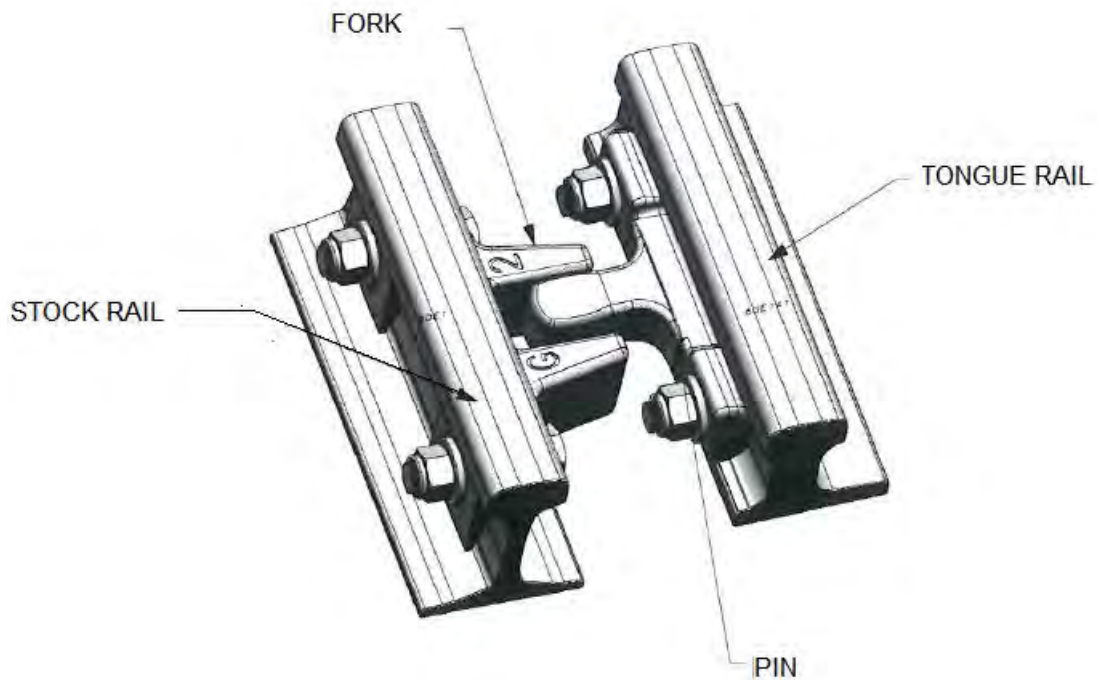


Figure 6 Ideal Position of pin and fork

C) In 1 in 12

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Design considerations are that the maximum expansion of rails from starting of crossing to the ACD is 9.9 mm. In this case, the expansion exceeds from the maximum designed gap of 7mm by 2.9mm, which is equivalent to 204 kN force. The longitudinal restrain force offered by the rail fasteners and the shear capacity of ACD design is much greater than the thermal force generated. Thus, any breakage of ACD of the turnout in service indicates either the deviation of rail fasteners quality from the designed quality or excessive creep. Proper investigation shall be done and corrective action shall be taken based on the findings of the investigation. Regular tightening of bolts of ACD with torque wrench shall be done at a maximum torque of 900 Nm.



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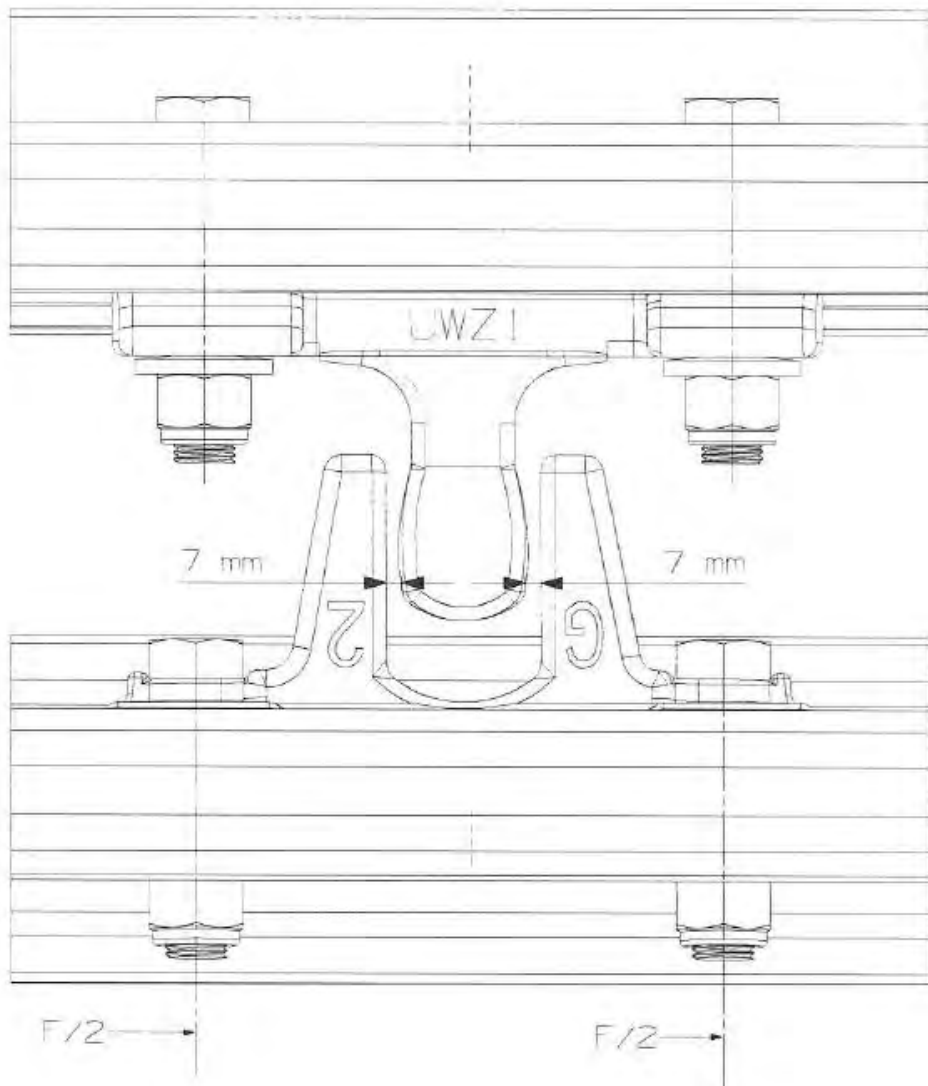


Figure 7 Ideal Position of pin and fork

D) In Derailing Switch

Design considerations are that the maximum movement of rail in the worst-case scenario (at lowest 7 kN per rail seat longitudinal resistance) is 15.9 mm i.e. equivalent of 306 kN. During operation if the worst-case scenario crosses, the ACD may break but the second ACD acts as a redundant system and gets into action (note additional 7mm gap available to take forces). When first ACD breaks, proper investigation of the ballast cushion and profile and the rail to sleeper fasteners effectiveness shall be checked. After rectification of potential cause of breakage, the broken ACD shall be replaced duly keeping designed first & second ACDs at designed gaps of 7mm. The bolts of the ACD shall be tightened with torque wrench at a maximum torque of 900 Nm at regular intervals or if found loose. The ACD that is away from the toe of switch (UWZ200 & UWG1) of the Derailing switch can fail first and the other ACD acts as redundant system.

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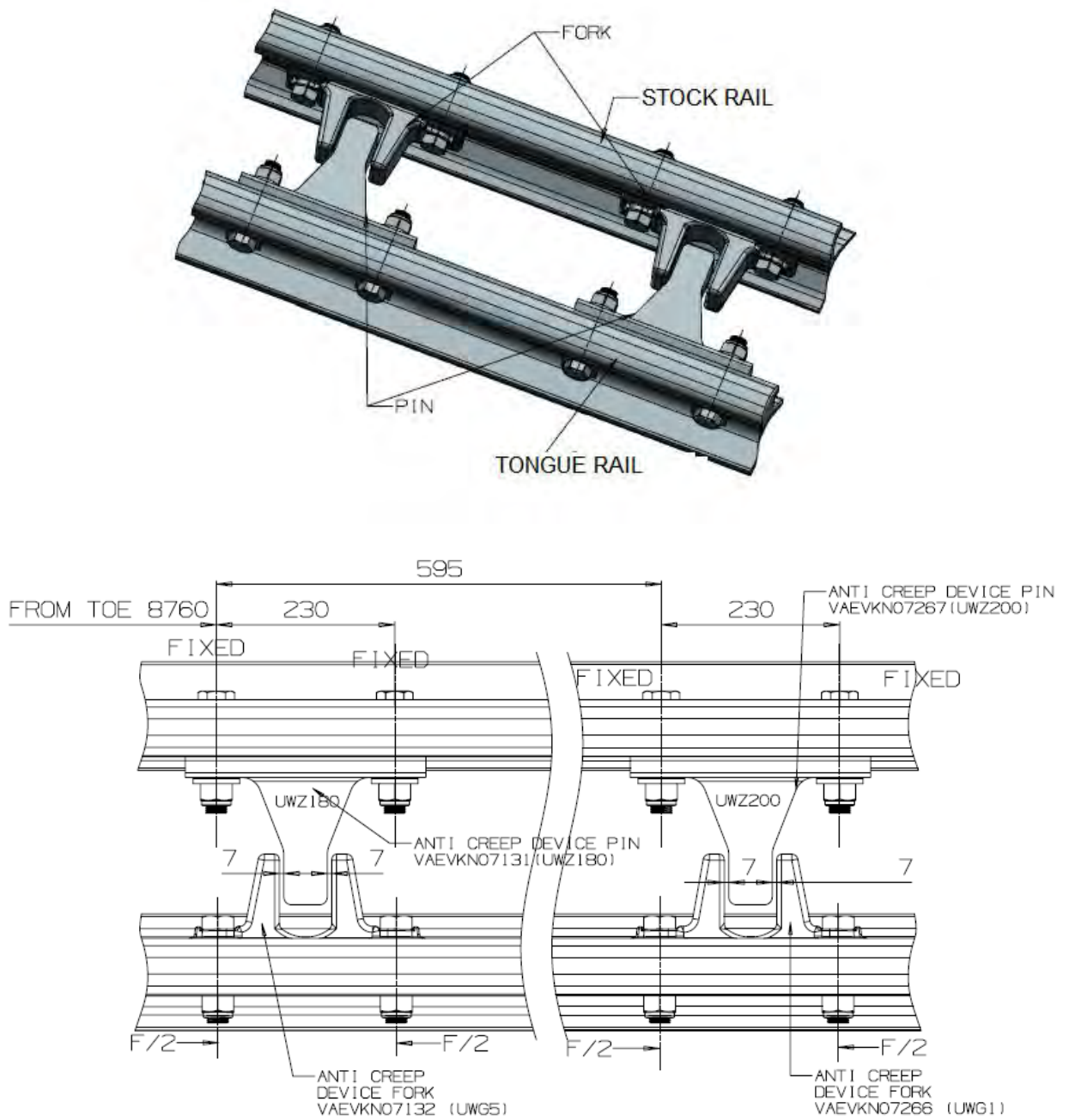


Figure 8 Ideal position of ACD in Derailing Switch

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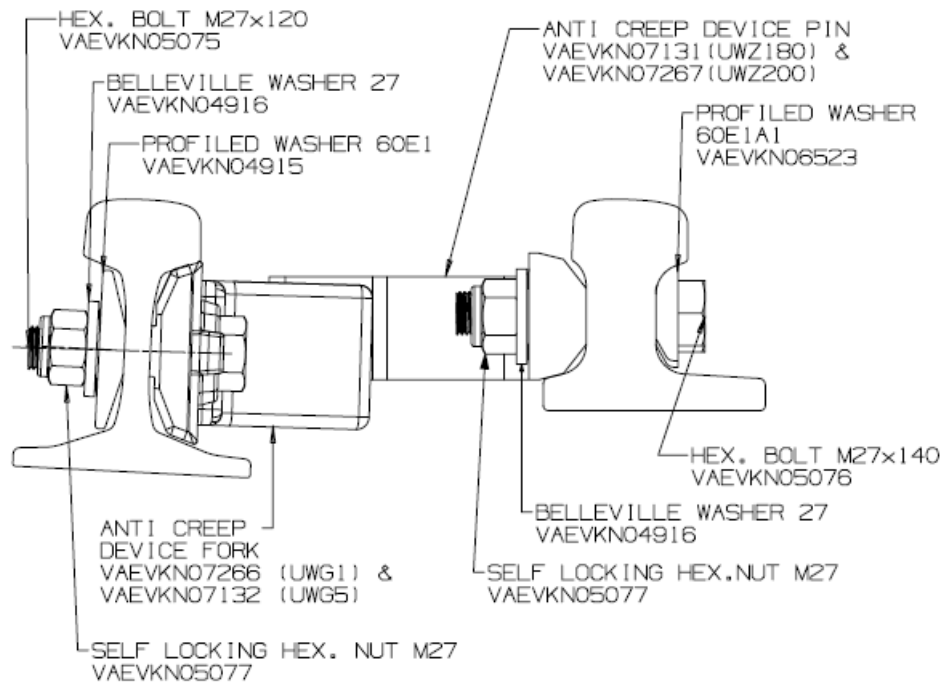


Figure 9 Various components of ACD in Derailing Switch

2.22.4.4 MAINTENANCE OF SWITCHES

- A) If tongue rails are chipped or cracked over 200mm length within 1000 mm from ATS, switch shall be replaced.
- B) Tongue rail is twisted or bent or causing 5mm or more gap at toe, switch rail shall be replaced.
- C) If tongue rail is knife-edged, switch rail shall be replaced.
- D) If vertical wear exceeds more than the prescribed limit as mentioned in Cl. 2.33.2 of this manual, switch shall be replaced.
- E) If any burr/metal flow is noticed, same should be grinded. Attention shall be given during grinding to protect the original profile of tongue rail/stock rail.

2.22.4.5 MAINTENANCE OF BASEPLATES/SPECIAL BEARING PLATE AND FASTENINGS

A) Switch baseplates

- I) Broken baseplates/Special Bearing plates indicate either that the sleepers are becoming indented or that the baseplate was incorrectly seated at the time it was installed.
- II) Remove the baseplate/Special Bearing plates to assess its seating on the sleeper. If the sleeper is severely or unevenly indented then change the affected sleeper and baseplates/Special Bearing plates using the procedure in Renewing turnout sleepers.

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- III) If sleeper seating is suitable, clear away debris from the baseplate seating and fit a new baseplate/Special Bearing plates using the procedure in Replacing a Base plate.
- IV) Make sure that the new baseplates are correctly seated on the sleepers and held to the rails with new fastenings as required.
- V) When fitting fastenings hold the baseplate tightly to the rail until the fastening is fully home. If you don't do this the new casting will fail immediately.

NEVER use the elastic rail clip to pull the baseplate or baseplate and sleeper up to the rail as the additional loading imposed on the clip housing will cause the casting to shatter.

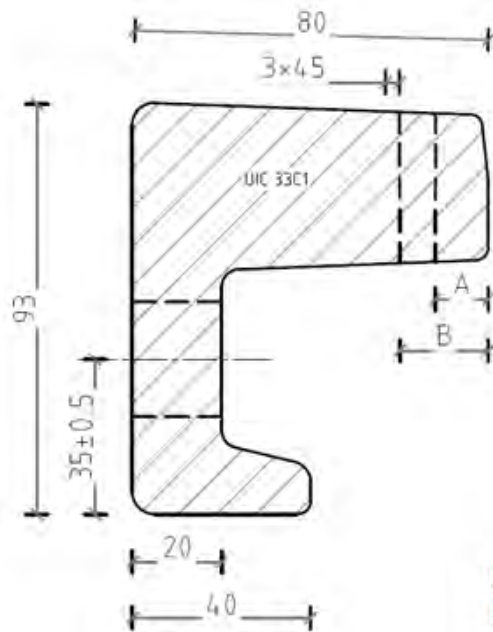
2.22.4.6 MAINTENANCE OF WELDABLE CMS CROSSINGS

- A) Crossings nose and wing rails shall be reconditioned on reaching the wear limit as mentioned in Cl. 2.33.2 of this manual. Reconditioning of CMS crossing may be carried out either in depot or in-situ as per IR provisions.
- B) Burr/metal flow over weldable CMS crossing shall be grinded to the extent so that original profile is not damaged.
- C) Pack sleepers, particularly under the nose of the crossing.
- D) Replace broken baseplates using the procedure in **Replacing a Base plate**.
- E) This can usually be done without the need to remove the crossing from the track.
- F) When the broken baseplate is removed, carefully check the condition of the sleeper supporting the baseplate to make sure that the sleeper condition is not causing the problem.
- G) Replace the crossing, using the procedure in **Replacing a Crossing** if it cannot be repaired.

2.22.4.7 MAINTENANCE OF CHECKRAILS

Checkrail clearance should be within 44-48 mm opposite nose of crossing and at the flare end it should be 66 mm at entry and 83 mm at exit. The maximum checkrail clearance permitted is 48 mm. The length of checkrail in 1 in 8.5 on the turning side is 4600mm and on the straight track is 5800mm. The length of checkrail in 1 in 12 on the turning side is 4700mm and on the straight track is 6100mm. The cross section of the check rail and check rail assembly for easy reckoning is as under.

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Note:
For dimension of A&B please refer
Annexure 4

Figure 10 Check rail cross section

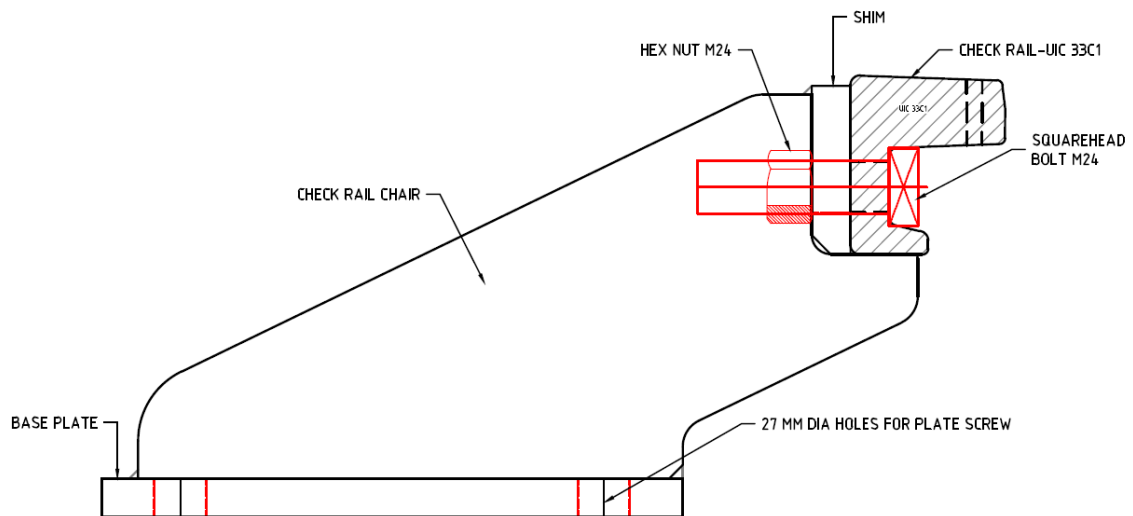


Figure 11 Check rail assembly

If checkrail clearance could not be maintained within the prescribed limit following activities should be undertaken:

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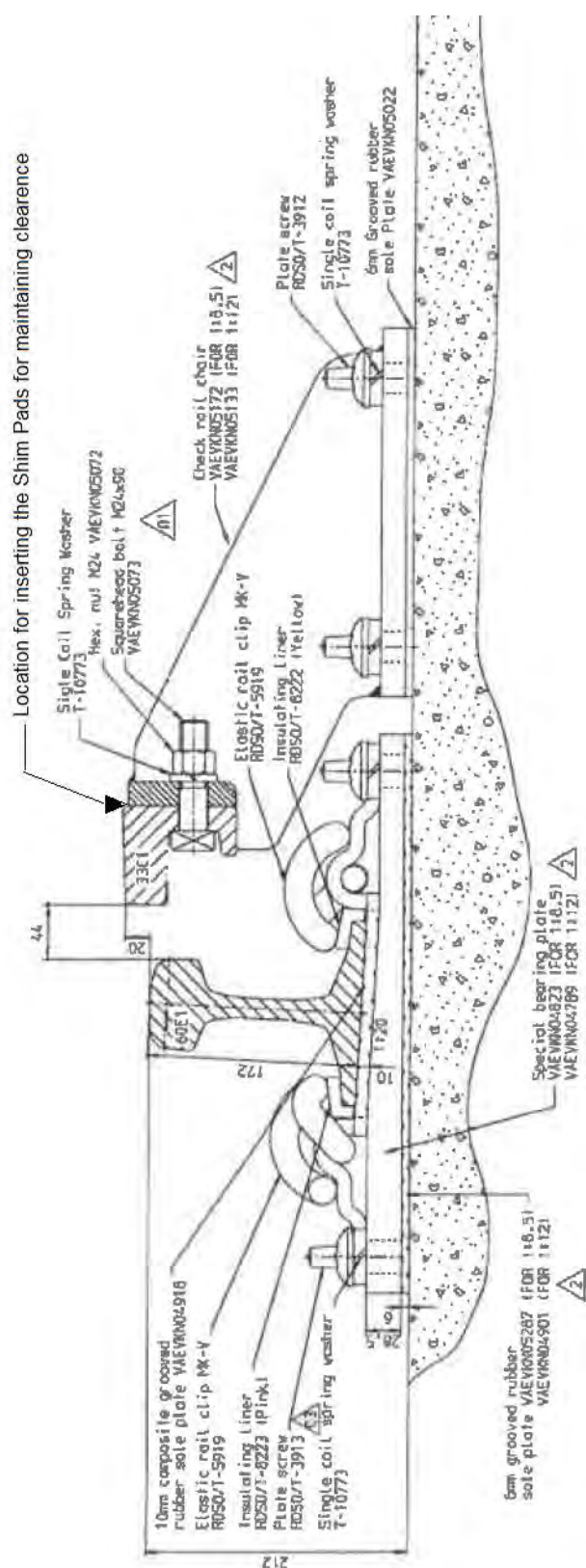


Figure 12 Location for inserting Shims to maintain clearance

- A) Remove or add spacer shims in the checkrail flange way until the correct flange way

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clearance and checkrail effectiveness measurement is obtained.

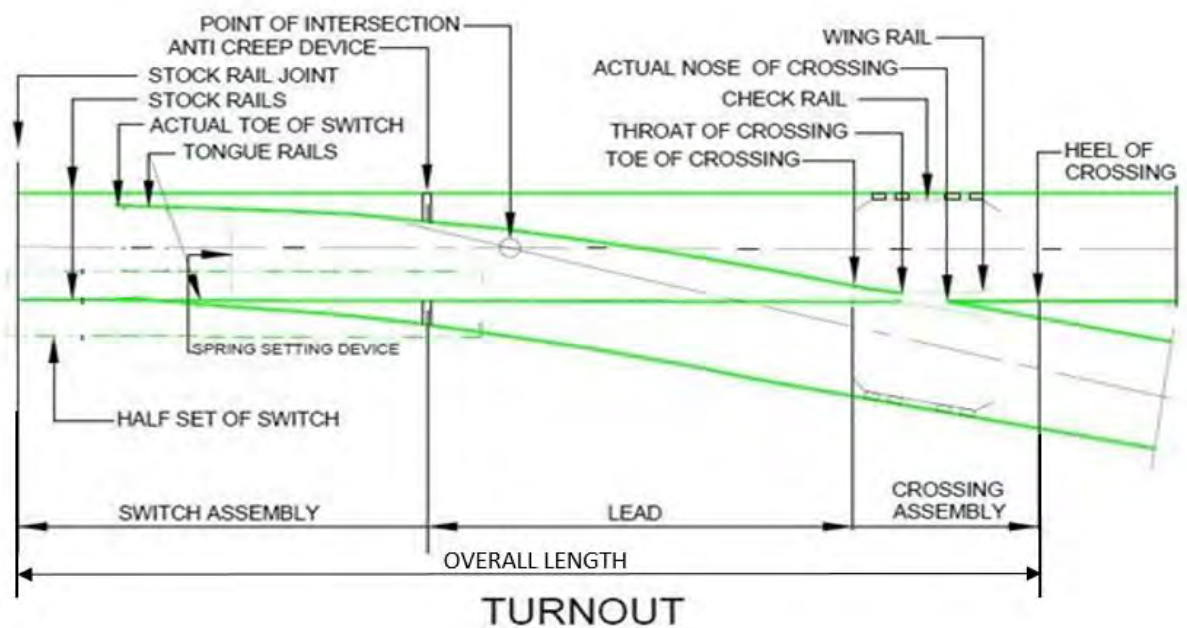
- B)** If the flange way is WIDE make sure that the correct check rail bearing plates are fitted. If not, replace them with the correct bearing plates at time of replacement of checkrail. Otherwise replace the checkrail.
- C)** If the flange way is TIGHT replace the worn check rail bearing plates or grind checkrail carrier rail overflow.
- D)** Correct the flare at the end of the check rail.
- E)** Replace the checkrail, using the procedure in Replacing of Check rail, if it cannot be repaired.

2.22.4.8 PROFORMA FOR INSPECTION OF TURNOUTS AND INSTALLATION/MAINTENANCE CHECKS

STATIC DATA FOR TURNOUTS USED IN WDFC

TYPE OF TURNOUTS:

	OVERALL LENGTH (IN m)	RADIUS (IN m)	CROSSING ANGLE	SWITCH ANGLE
1 IN 12	42.667	460	4°45'49"	0°20'00"
1 IN 8.5	30.768	218	6°42' 35"	0°36'08"
D/S	12.204			0°36'08"



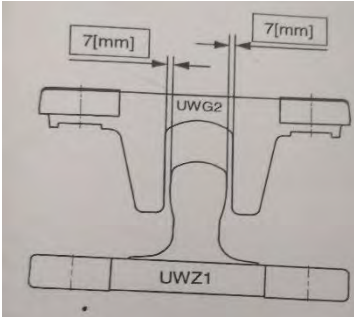
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PROFORMA FOR INSPECTION OF POINTS AND CROSSINGS

Station:		Point No.:	
Location:		Rail Section:	
Type of Sleeper/assembly:		Angle of Crossing:	
Nominal Gauge of Turnout:		Left Hand or Right Hand:	
Laid on Straight or on Curve of Radius		Similar/Contrary flexure:	
Date of Laying Sleeper (mm/yyyy):		Type of Crossing:	
Date of laying new crossing (mm/yyyy):		1st	
Crossing Unique Number			
Manufacturer			
Date of laying new Switch (mm/yyyy):		1st	
LH:			
RH:			
Particulars	Tolerance w.r.t design	Details of Inspection	
1	2	3	
I General		Date of Inspection:	
1 Condition of ballast and drainage in turnout. (clean cushion)			
II SWITCH ASSEMBLY AND LEAD 2 Condition of sleeper, slide chairs, plate screws, other fittings of switch, pirolls (if provided) & their tightness.			
		LH	RH
3 Condition of tongue rails: a) Whether chipped or cracked over 200mm length within 1000 mm from ATS.			

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(b)Whether twisted or bent (causing gap of 5 mm or more at toe)			
(c)Remarks over condition of tongue rail, whether it requires replacement.			
4 . Condition of stock rail, burr formation to be mentioned specifically.			
5 Creep and squareness of tongue rail at toe of switch.	5mm		
6 Straightness of straight stock rail if laid on straight:(Measured on 7.5m chord)	+/-5mm		
7 Packing conditions under the switch assembly (preferably to be observed under traffic.)			
8. Throw of Switch	160mm (+)1mm to (-) 0 mm	LH:	RH:
9 Housing of stock and tongue rails:	Proper/ Not Proper		
10 Working of SSD	Proper/ Not Proper		
11 Whether anti-creep device pin and fork are in correct position (i)Gap between pin and fork to be measured and recorded on each side:			
a) Towards ATS		LH	RH
b) Towards Lead			
(ii)Standard gap for each side should be 7mm 			

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iii) Tightness of pin and fork to be checked			
12 Gauge and cross level in switch and lead.		G	XL
a) At 450mm ahead of toe of switch	Gauge 1676 (Tolerance=0 to +3 mm)		
b) At ATS between two stock rail.	Designed Gauge 1682 and tolerance upto +6mm		

Particulars	Tolerance w.r.t design	Details of Inspection			
1	2	3			
c) Gauge and cross-level for ML and Turnout side.	Gauge 1676 (Tolerance=0 to +3 mm)	ML		T/O	
STATION		G	XL	G	XL
Station 0 -	do				
1	do				
2	do				
3	do				
4	do				
5	do				
6	do				

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7	do							
8	do							
9	do							
10	do							
Particulars	Designed Versine				Details of Inspection			
1	2				3			
d) Versine of stock rail for turnout side upto end of lead.	1 in 12		1 in 8.5		1 IN 12 ML	1 IN 12 T/O	1 IN 8.5 ML	1 IN 8.5 T/O
	Curved Stock	Tongue Rail	Curved Stock	Tongue Rail				
STATION					Ve	Ve	Ve	Ve
Station 0 - Away from SRJ/ATS (mm)	0	0	0	0				
1	9.8	9.8	20.7	20.7				
2	9.8	9.8	20.8	20.7				
3	9.8	9.8	20.9	20.7				
4	9.8	9.8	20.8	20.8				
5	9.8	9.8	21.0	20.8				
6	9.8	9.8	21.1	20.9				
7	9.9	9.8						
8	9.9	9.8						
9	9.9	9.8						
10	9.9	9.9						

Note:

1) Stn. No. 0 to be marked at heel of switch for straight switch & at ATS for curved switch. Subsequent stations shall be marked at every 3m. Versines to be recorded on 6m chord length commencing from station no. 1

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2) In case of gap between T/R and S/R, that should be added to gauge measurement.

III. Crossing Assembly

Particulars	Details of inspection		
1	2		
13. Condition of crossing a) Sign of propagation of crack (if any) in crossing assembly:			
b) Burring on top surface at nose:			
14. Wear of crossing (to be measured with straight edge at 100 mm from ANC)*	LH wing wall	Nose	RH wing wall

*For CMS crossing, Actual wear for 60 kg section = Measured wear - 2.5mm

15. Gauge and cross level at crossing	ML		T/O	
	G	XL	G	XL
a) 1m ahead of ANC				
b) 150 mm behind ANC				
c) 1m behind ANC				
16. Condition of check rail fitting e.g. Bearing plates and elastic fastening.				

Particulars	Tolerance w.r.t design	Details of inspection	
1	2	3	
17 Clearance of check rails.		LH	RH
a) Opposite ANC:	44mm - 48mm		
b. At 1st block towards toe of crossing and 1st block towards heel of crossing	44mm - 48mm		

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c. At the flared end towards heel and at the flared end towards toe	Entry-66mm Exit-83mm			
IV Turn in curve:				
18 Turn in curve: Stations to be marked at 3m interval. Versines to be measured on 6m chord station no 0 to be marked at the center of the last long sleeper in case of PSC sleeper.				
STATION NO.		Ve	G	XL
0				
1				
2				
3				
4				
5				
6				
7				
19. Availability of 150mm additional ballast shoulder with on outside of turn in curve.				
V. General 20. Any other special feature /defects.				
21. Signature of the inspecting official with date				

NOTE:

1. Locations where the gauge, cross levels & offsets are to be measured will be marked and painted on gauge face side

2. Locations of Versines will be marked & painted on non-gauge face side.

2.23 DESTRESSING OF P&C WITHIN CWR

2.23.1 ANCHOR LENGTHS AND REFERENCE MARKS

P&C layouts may be used to form part of an anchor length provided that all P&C units are of

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an approved design. However, no P&C element is sufficient on its own to provide an anchor. Movement of switch tips and crossings shall be minimised by selecting appropriate pulling points. Subsequent testing and adjustment of any switch detection shall be carried out.

2.23.2 PRELIMINARY SURVEY AND CONSIDERATIONS

The layout shall be to the design line and level and shall be packed and correctly ballasted. Whenever possible, pulling points should be in plain track and approximately 18m away from any P&C element.

The positions of existing welds and insulated joints shall be checked to ensure that at least 4.5m is maintained between welds/joints.

2.23.3 TENSOR EQUIPMENT

The distance between switches installed toe-to-toe may cause problems in the fitting of tensors. For example, switches with short fronts of 1650 mm, and with signalling equipment in the bed immediately in front of the switches towards the stock rail end, provide a clearance for the tensors of just 2600mm. The site shall be examined prior to Destressing to ensure that appropriate tie bars are supplied with the tensor equipment.

2.23.4 GENERAL SITE PROCEDURE FOR DESTRESSING IN P&C

CWR Points and Crossings are though rigid track-structure and resistant to buckling, they are sensitive to temperature forces and movements and thus need to monitor closely SFT for smooth performance.

If CWR turnout and its approaches tracks are maintained properly and creep is controlled duly maintaining required ballast cushion & profile & healthy rail fasteners, isolated destressing of parts like Switch Rail or Lead rail or Crossing is not required. De-stressing of parts of CWR turnout may be resorted to only when required and where possible all lengths to be destressed are should be done at the same time as in RDSO CT-48 report. Care shall be taken not to shift Stock Rail-Actual Toe of Switch point by destressing either at both sides or on one side, keeping ATS as no movement Outer reference mark sign. While doing complete turnout destressing, to avoid relative movement of pin and fork of ACD, steel wedges of designed gap i.e 7mm shall be temporarily placed and same is taken out after destressing and equalization of forces. Check for ACD gaps on either side and satisfy that it is within ± 1 mm of designed gap. Once ACD gaps are temporarily filled with the steel wedges, Destressing of 1:12 and 1:8.5 CWR turnouts is similar except accounting for variation in their lengths and required extensions at welding joints.

Where above is not possible the procedure below to be taken into account.

- A)** Cut both rails approximately 18m from the unit to be destressed on all three sides.
- B)** To stress stock rails, release fastenings from the plain track up to the switch toes. Do not

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disturb the ballast, sleepers or fastenings of the stock rail.

- C) To stress CWR to the heel of a crossing, release fastenings up to the third sleeper from the crossing joint/weld. Do not disturb the ballast, sleepers or fastenings of the crossing.
- D) If it is not possible to destress all lengths of track at the same time, the toes of switches should be destressed first and short lengths (less than 55m) should be destressed last, e.g. heel to heel.
- E) Mark the rails of the plain line clear of where the tensors will be located and provide any necessary reference points on unfastened sleepers/housings corresponding to the marks. Any sticking pads shall be freed, to prevent the reference sleepers moving with the rail.
- F) Apply tension until the required extension of the plain track is achieved (but if this is less than 5mm, pressure shall be applied to the tensors until either the required extension is achieved or the calculated pressure is shown on the pressure gauge, whichever is reached first).
- G) Check for any movement of the reference marks.
- H) Fasten down and complete the closing weld in the normal way. After switches have been destressed, arrange for any switch detection to be checked and, if necessary, adjusted.

Care shall be taken to preserve the correct crossing nose to toe (ANC to ATS) dimensions.

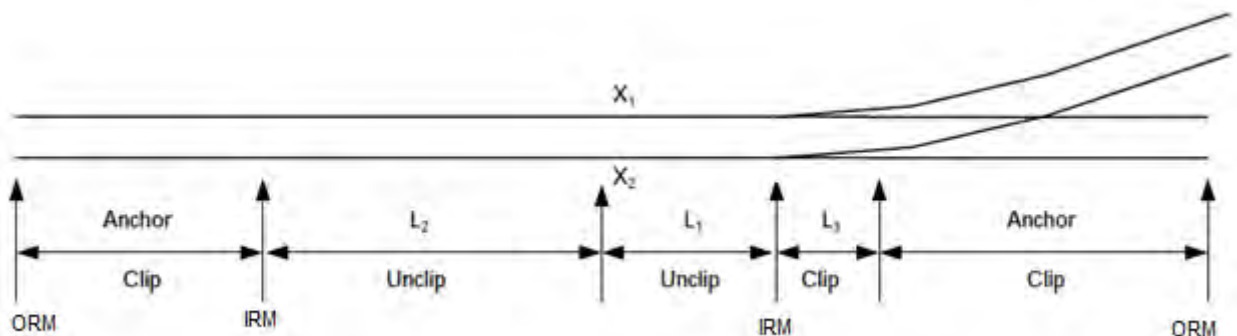
2.23.5 PROCEDURE FOR DESTRESSING COMMON P&C UNITS

The following instructions are additional to the general procedures for destressing given above.

- A) Para. 2.23.5.1: Destressing at switch toes
- B) Para. 2.23.5.2: Destressing at a crossing vee
- C) Para. 2.23.5.3: Destressing between switch heel and crossing vee

In all the examples the adjoining plain line CWR is assumed to be correctly destressed. If it is not known to be correctly destressed then destressing of the plain line should be carried out at the same time as the destressing of the P&C or, failing that, before the onset of hot weather.

2.23.5.1 DESTRESSING AT SWITCH TOES



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X_1X_2 = pulling points IRM = inner reference mark (see Note 2)

L_1 = 18m approx. ORM = outer reference mark

L_2 = 90m minimum

L_3 = distance from the switch tips to the ACD

Anchor: see para. 2.20.7

Notes: (1) Reference points will be required within L_2 if $L_2 > 90m$

(2) The IRM is next to the switch toes on the first plain sleeper, which shall be unclipped.

In order to maintain squareness across the switch toes, and to maintain the SFTs of the rails within $6^\circ C$ of each other, it may be necessary to stress both rails even if only one half-set of switches has been disturbed (e.g. through maintenance replacement).

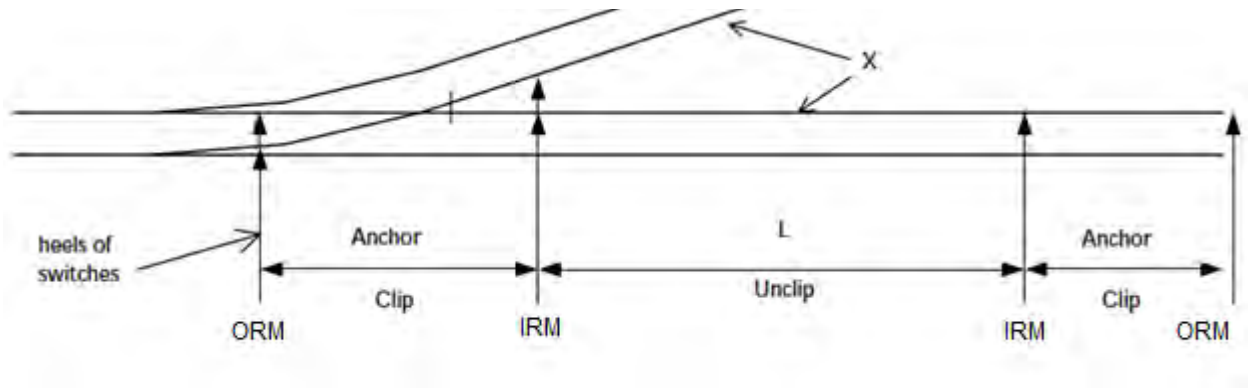
Method:

1. Choose the position for the tensors, X_1X_2 , approximately 18m (L_1) from the switch toes and mark the reference mark points for each anchor length.
2. Cut rails at X_1X_2 . First cut shall be gas cut. Cut length of rail shall be 1m or as per instructions of the Tensor OEM Manual.
3. Unclip both rails and place on rollers from the switch toes for a distance of L_1+L_2 .
4. Mark both rails clear of the tensor positions and, when necessary, provide reference points on unfastened sleepers/housings in the plain line corresponding to the marks.
5. Calculate the extension required for L_1+L_2 and the pull force.
6. Cut again, allowing for the extension and the welding gap.
7. Fit the tensors at X_1X_2 and apply tension to both rails until the required extension or calculated pressure is achieved.
8. Check reference mark points to ensure that the anchors have held and check the switch toe dimensions.
9. Remove all rollers and replace pads, insulators and clips, remove tensors, ensure that all cribs are filled with ballast, sleepers packed and do necessary welding including rail changing with minimum rail length and welding debris removed.
10. Arrange for switch detection to be checked and, if necessary, adjusted.

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2.23.5.2 DESTRESSING AT A CROSSING VEE

The rails attached to the back of the crossing shall be destressed as follows:



X = pulling points, ideally at mid-point of L IRM = inner reference mark (see Note 2)
 ORM = outer reference mark L = 18m minimum Anchor: see para. 2.20.7

Notes: (1) Reference points will be required within L if L > 90m

(2) The IRM shall be behind the crossing vee even if the anchor (measured from anti-creep device at the switch heel) is thereby longer than required by para. 2.20.7

Both rails shall be destressed to the same SFT in order to prevent any tendency to rotate the crossing.

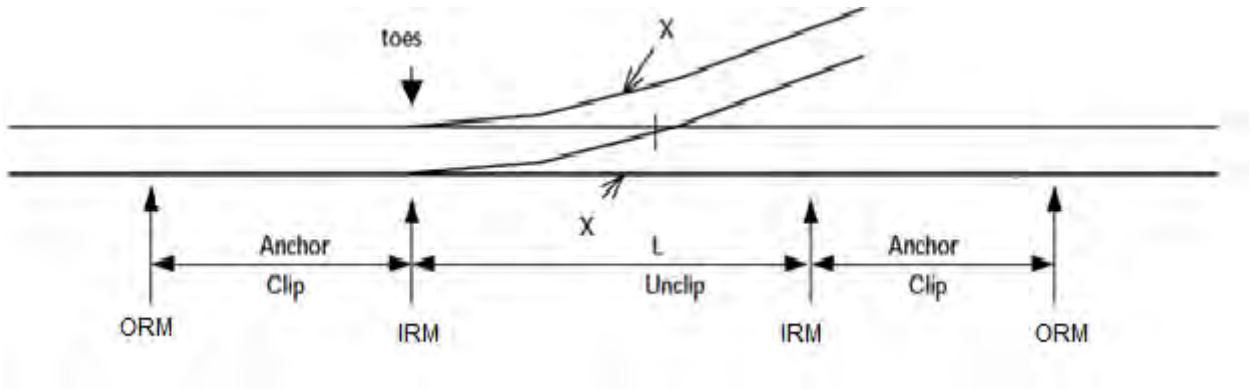
Method:

1. Choose the position for the tensors, X, and mark reference mark points for each anchor length.
2. Cut rails at X. First cut shall be gas cut. Cut length of rail shall be 1m or as per instructions of the Tensor OEM Manual.
3. Unclip both rails and place on rollers for a distance of L.
4. Mark both rails clear of the tensors and, when necessary, provide reference points on unfastened sleepers/housings corresponding to the marks. Provide reference mark marks for each anchor length.
5. Calculate the extension required.
6. Cut again, allowing for the extension and the welding gap.
7. Fit the tensors at X and apply tension to both rails until the required extension or calculated pressure is achieved.
8. Check the reference mark marks to ensure that the anchors have held.
9. Remove all rollers and replace pads, insulators and clips, remove tensors, ensure that

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all cribs are filled with ballast, sleepers packed and do necessary welding including rail changing with minimum rail length and welding debris removed.

The rails alongside the crossing, i.e. the stock rail continuations, shall be destressed as follows:



X = pulling points, ideally at mid-point of L IRM = inner reference marks ORM = outer reference marks Anchor see para. 2.20.7

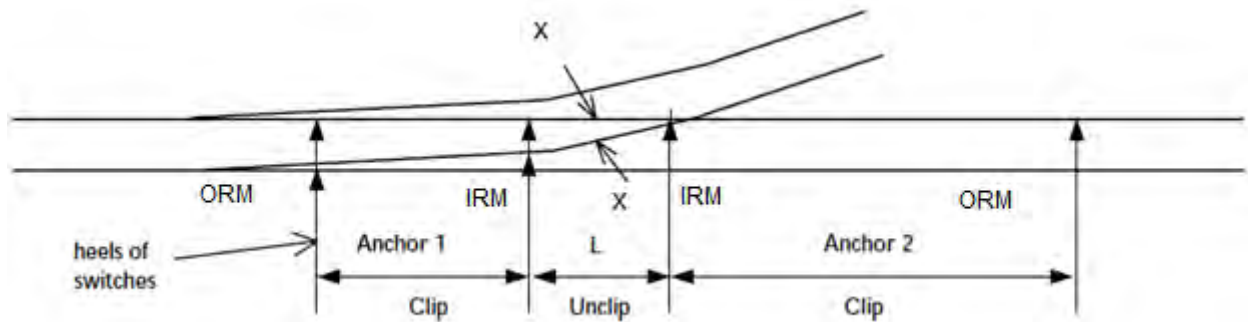
Any movement of the stock rail at the switch toes shall be minimised: the effect of destressing the stock fronts is likely to compensate. Having established the anchors, the method is the same as above.

2.23.5.3 DESTRESSING BETWEEN SWITCH HEEL AND CROSSING VEE

After the stock rails have been destressed, the switch rails shall be set to the correct position relative to the stock rails. With a welding gap at a suitable position in each closure rail, and all fastenings on the toe side of that gap released, the switch rails shall be adjusted longitudinally so that the pin (assumed to be fitted to the switch rail) of the anti-creep device is central within the fork at a rail temperature of $t_m + 10$ degree Celsius. Interpolation shall be made for intermediate temperatures. After the correct adjustment of switch and the stock rail, insert two steel wedges into the 7mm gaps on either side of the pin. Remove the wedges only after the neutralization of forces on either side of the weld.

The maximum tolerable force occurs after approximately 40m, so closures longer than this should be destressed. If distance "L" in the diagram precludes the use of tensors, rollers, and suitable weld-to-weld distances, natural destressing shall be used. Movement of ACD and crossings shall be minimised by selecting appropriate pulling points.

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X = pulling points IRM = inner reference marks ORM = outer reference marks

Anchor 1, in each closure rail, is from the first fastening at the heel, and at least 20m long, or two sleepers per degree C difference between rail temperature and SFT, whichever is longer.

Anchor 2, incorporating the crossing and each rail behind, is to commence at the crossing weld and be of a length according to para. 2.20.7.

Both rails shall be destressed during the same shift.

Method:

1. Choose the position for the tensors, X, approximately midway between the two IRMs.
2. Cut rails at X. First cut shall be gas cut. Cut length of rail shall be 1m or as per instructions of the Tensor OEM Manual.
3. Check setting of anti-creep device: if necessary, adjust by placing temporary steel wedges to correct gap (7mm). Reclip anchor 1, leaving length L unclipped, and place L on rollers.
4. Mark both rails clear of the tensors and provide reference points on unfastened sleepers/housings corresponding to the marks. Provide reference mark marks on each anchor length.
5. Calculate the extension required for L. Note it differs for 1:8.5 and 1:12.
6. Cut again, allowing for the extension and the welding gap.
7. Fit the tensors at X and apply tension to both rails until the required extension or calculated pressure is achieved.
8. Check the reference mark marks to ensure that the anchor has held and check the switch toe dimensions.
9. Remove all rollers and replace pads, insulators and clips, remove tensors, ensure that all cribs are filled with ballast, sleepers packed and do necessary welding including rail changing with minimum rail length and welding debris removed

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10. Arrange for switch detection to be checked and, if necessary, adjusted.

2.24 WORK WHICH REDUCES THE STABILITY OF CWR

2.24.1 GENERAL

The stability of CWR will be reduced if the SFT is reduced (para. 2.24.2) or the lateral resistance is reduced (paras 2.24.3).

2.24.2 REDUCTION IN SFT

2.24.2.1 CAUSES OF REDUCTION IN SFT

Reduction in SFT shall be assumed to occur if

- A) the rail is broken, or
- B) the rail is cut, e.g. for the purpose of renewing or replacing insulated joints, catch points or P&C components (e.g. switches), or
- C) the track is slued inwards on a horizontal curve, lifted on a concave vertical curve (trough) or lowered on a convex vertical curve (summit), or
- D) the track is re-ballasted, re-sleepered or re-laid, or
- E) an underline bridge or level crossing is reconstructed or repaired, if this requires the track to be removed or its alignment disturbed.

(**Note:** slewing outward on a horizontal curve, lowering a concave vertical curve or lifting a convex vertical curve will increase SFT.)

For a particular value of slue/lift/lower the effect on SFT is most pronounced over small wavelengths on small radius curves and least pronounced over long wavelengths on large radius curves.

Normal tamping, packing/straightening of dipped welds, fine lining, the treatment of wet spots, the installation or replacement of insulated joints carried out with the use of tensors should not cause significant change in the SFT.

2.24.2.2 WHEN TO DESTRESS AFTER LIFTING, LOWERING, RE-CANTING AND RE-ALIGNING CWR

Destressing is not necessary when:

- A) the line and level were previously determined by fixed references and the original line and level are restored to within acceptable limits (see a) to e) below), and
- B) the CWR was previously correctly destressed and has not been cut, and
- C) adjustment switches (SEJs) are absent or confirmed not to have been affected.

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Where the design line and level is not established by fixed references, or an established design line and level is to be altered, destressing is required as follows (slues, lifts etc are from the position at previous Destressing):

A) Alteration to line: sluing inward on horizontal curves

- I) Curve radius 750m or less: destress when maximum slue is 30mm or more.
- II) Curve radius over 750m: destress when maximum slue in mm exceeds 0.04 times radius in m.

B) Alteration to line: sluing outward on horizontal curves

- I) Re-stress when maximum slue in mm exceeds 0.08 times curve radius in m.

C) Alteration to level: lifting and/or increasing cant through concave vertical curves or lowering and/or reducing cant over convex vertical curves

- I) Vertical curve radius 1875m or less: destress when maximum lift/lower/re-cant is 75mm or more.
- II) Vertical curve radius over 1875m: destress when maximum lift/lower/re-cant in mm exceeds 0.04 times vertical curve radius in m.

D) Alteration to level: lifting and/or increasing cant over convex vertical curves, or lowering and/or reducing cant through concave vertical curves

- I) Destress when maximum lift/lower/re-cant in mm exceeds 0.08 times vertical curve radius in m.

E) Alteration to level and line (in combination)

- I) Add together the individual values of alteration to line and level at each design station and determine the maximum combined value for each radius. Compare these combined values with the limits set for alterations to line or level appropriate to the curve radii. If in any single case the limit is exceeded, restressing shall be carried out.

2.24.2.3 DESTRESSING AFTER THE SFT HAS BEEN ALTERED

Whenever CWR has been disturbed such that the SFT has been altered, and/or any lifting/lining/recanting is outside the above limits, either

A) destressing shall be carried out according to 2.20.2, or

B) destressing shall be carried out to cover at least 100m on each side of the disturbance.

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2.24.3 REDUCTION IN LATERAL RESISTANCE

2.24.3.1 GENERAL

The lateral resistance of the track will be reduced by any work which moves or removes sleepers, or removes or loosens the ballast, including tamping/lining of undisturbed track.

2.24.3.2 TEMPERATURE LIMITS – LINES OPEN TO TRAFFIC

Work which may reduce the stability of track which is open to traffic

- A) shall not be started when the rail temperature exceeds $t_d + 20$ degree Celsius;
- B) should not take place when the rail temperature is likely to exceed $t_d + 20$ degree Celsius during the work;
- C) should not take place when the rail temperature is likely to exceed $t_d + 20$ degree Celsius within the next three days.

If the rail temperature rises above $t_d + 20$ degree Celsius after work has commenced the work shall be stopped as soon as possible, leaving the track in the most stable condition possible.

Persons in charge of work shall carry rail thermometers and take rail temperatures frequently when these are approaching the specified limit. The organisation carrying out the work shall check rail temperatures for the three days after the work is done.

2.24.3.3 TEMPERATURE LIMITS – LINES UNDER POSSESSION

Tamping/lining of undisturbed track (i.e. not associated with renewals work or with any other work which has disturbed the track) shall not be carried out when the rail temperature is above $t_m + 20$ degree Celsius. Otherwise, there is no restriction but on re-opening of the line to traffic and the precautions in paras 2.27.1 and 2.27.2 shall apply.

2.24.3.4 OPENING OUT

If sleepers are to be packed manually, or require to be squared, not more than six alternate half beds or two consecutive full beds may be opened out at one time. Care shall be taken to retain the top and line, and to restore the ballast profile before opening out adjacent sleepers.

2.24.3.5 CHANGING DEFECTIVE SLEEPERS

Defective sleepers shall be changed singly and on completion the new and adjoining sleepers shall be packed, the ballast between the sleepers and on the ballast, shoulders shall be replaced to the correct profile, and the ballast consolidated or other appropriate precautions taken.

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2.24.3.6 SLURRIED BALLAST

Where there are wet and slurried ballast conditions the foul ballast shall be removed from around the sleepers and a graded slope created for drainage purposes. Fresh ballast shall then be placed, packed and compacted. Not more than six alternate half beds or two consecutive full beds should be opened out at one time.

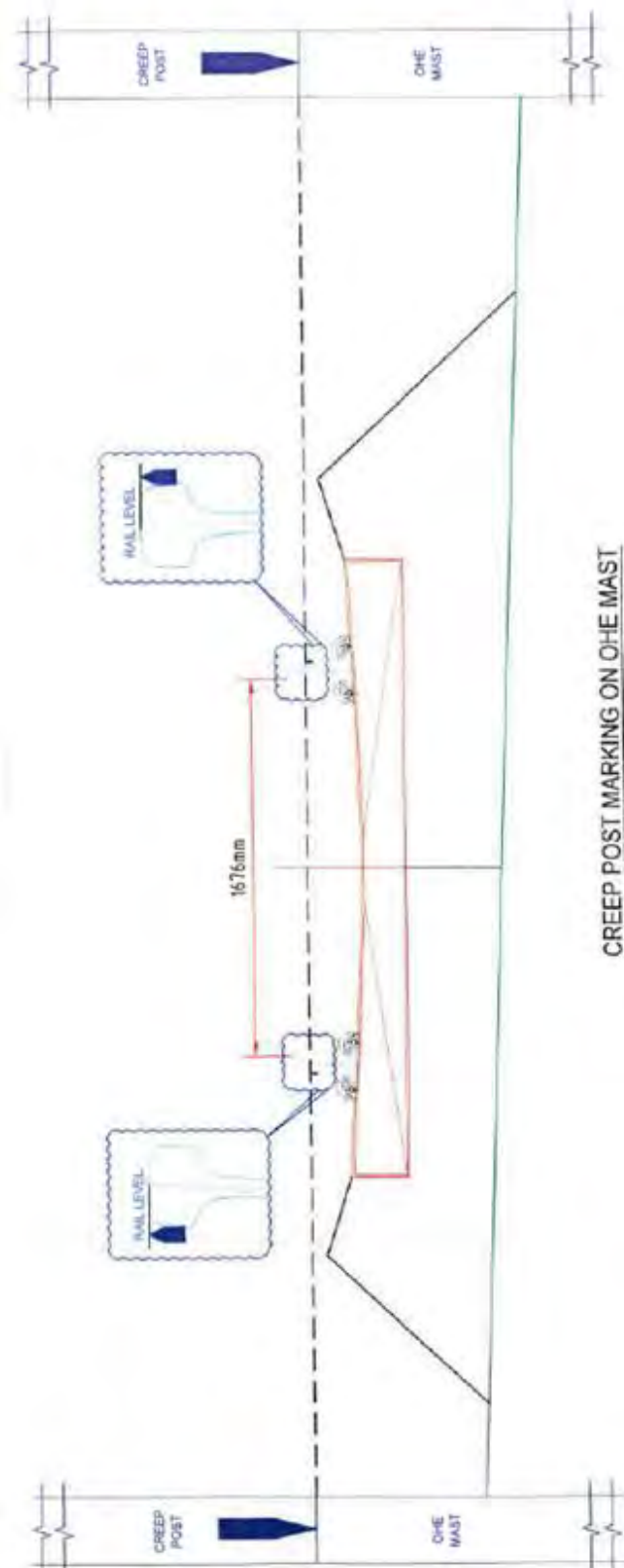
2.25 CREEP

Creep is defined as longitudinal movement of Rails. Rails tend to move gradually the direction of dominant traffic or Down gradients in the direction of traffic with regular braking applications or Up gradients in the direction of traffic with traction loads. It is believed to be caused by ironing out of yielding track by the moving load, augmented by moving load, augmented by breaking load & by impact of wheels on the running-on ends the rail, particularly at times when they are in state of expansion and contraction. Creep monitoring shall be done by frequent measurements of Creep and SFT.

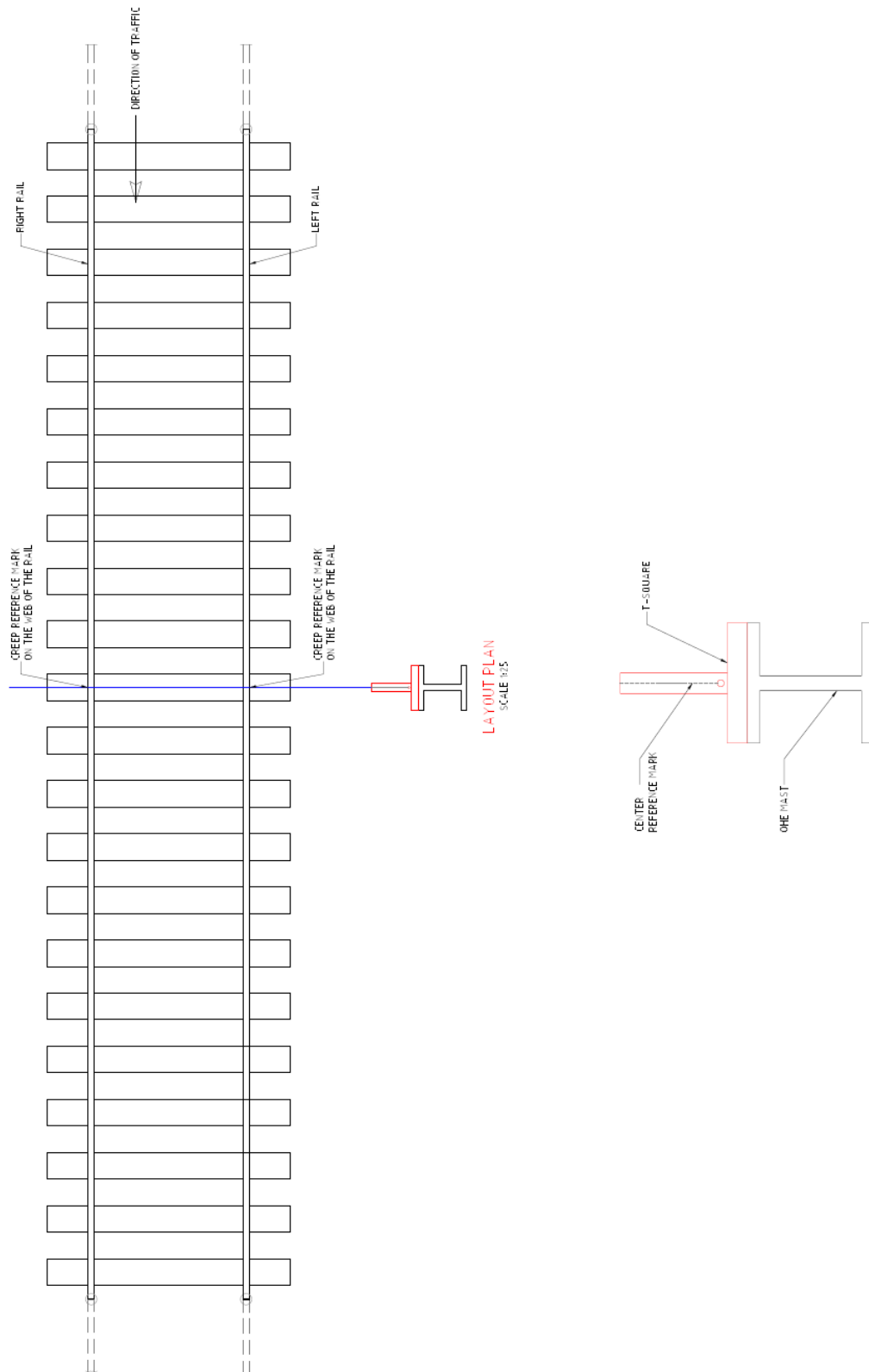
Creep Markings & Measurement by T-Square:

Creep markings are provided on OHE masts at every one km along the block section and on a T-section at every mainline / loop line in turnout approach.

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2.25.1 METHOD OF MONITORING AND MEASURING OF CREEP

2.25.1.1 CREEP POSTS FOR BLOCK SECTION

OHE Posts will be used as Creep post at every Km. Arrow mark on OHE posts will be marked and

- A) Point of arrow should be 25 mm above rail level and “CREEP MARK” will be written.
- B) Corresponding creep marks on web of the rail on non- gauge face.
- C) OHE masts (with creep marks) on both sides of track will be indicated on drawing for clarity of field staff.

2.25.1.2 CREEP POSTS FOR OTHER THAN BLOCK SECTION

Creep posts will provide posts at every Km as below: -

- A) OHE Mast / OHE Portal exists in yard shall be used as creep post.
- B) Corresponding creep marks on non-gauge face side of rail heads will be mentioned.
- C) T- Section Creep posts shall be erected at Mainline / Loop line Turnout at
 - I) Actual Toe Switch (ATS);
 - II) Actual Nose of Crossing (ANC);
 - III) One in lead portion;
 - IV) One at 20 m from ATS; and
 - V) One at 50 m beyond ANC.

2.25.2 GENERAL

2.25.2.1 CAUSES OF CREEP

- A) Rails not properly secured to the sleepers due to inadequate toe load.
- B) Less Ballast resulting in adequate ballast resistance to the movement of sleepers.
- C) Yielding formation.
- D) Cracking of Sleepers, uneven spacing of sleepers.
- E) Lack of proper drainage.
- F) Loose Packing.
- G) Inefficient or badly maintained Rail joints.

Additional cause points are:

- A) poor fastenings or pads (where lack of longitudinal restraint is suspected);
- B) situations at the bottom of an incline or valley curve where traction or braking may cause bunching;
- C) Locations where repeated braking or acceleration occurs (stations, signals, junctions etc.)

2.25.2.2 EFFECTS OF CREEP

Rail creep is a significant contributor to changes in SFT. Effects include:

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- A) Sleepers get out of Square.
- B) Distortion of Gauge.
- C) Loosening of soils.
- D) Shearing & Breaking of Spikes, Bolts & fish plates.
- E) Buckling in extreme cases.

2.25.2.3 PERMISSIBLE AMOUNT OF CREEP

In LWR portion Rail creep in excess of 20 mm (± 20 mm) shall not be permitted. Left and Right Rails absolute creep difference shall also not to be more than 30mm.

2.25.2.4 MEASUREMENT OF CREEP & FREQUENCY

Creeps should be measured at every Km by stretching a nylon chord from T-Square and readings to be recorded in prescribed register. Frequency is given below: -

Every fortnight during the two coldest and two hottest months of the year at about minimum and maximum temperature.

Alternately once in two months during the other months of the year.

Performa for CWR Section creep register is given in para 2.21.

2.25.2.5 MONITORING OF CREEP

Monitoring is to be carried out by periodic inspection as part of the Welded Track. The difference occurring in the position of the rail paint marks and the original recovered position is to be checked. When Creep/difference in movement of rail with respect to original exceeds 20mm. Full investigation shall be carried out & remedial measures undertaken.

2.25.3 RAIL CREEP MEASUREMENT SCHEDULE

2.25.3.1 SEJ LOCATION (IF ANY)

Creep post shall be provided at the centre of the SEJ and on both the sides at the end of breathing length portion. The creep shall be measured at all these three locations as per the frequency decided in the Indian Railway practice.

2.25.3.2 OTHER THAN SEJ LOCATIONS

Marking for the purpose of measurement of creep, OHE mast will be used as creep post and at every kilometre proper marking on the OHE post and in yards in addition to OHE masts a T shape creep post erected at locations mentioned as per proforma at 2.25.4 shall be done and reference mark shall also be made on the non-gauge face side rail head on both the rails on all the tracks. The marking on the OHE mast shall be done at a level of 25mm above the rail level.

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The frequency of measurement shall be as below (CI 8.1.5 of IR LWR Manual):

By Sr. Executive / Executive (Track)

- A) Every fortnight during the two coldest and two hottest months of the year at about minimum and maximum temperatures.
- B) Alternately once in two months during the other months of the year.

By Depot Manager

- A) At least once in 6 months, preferably during coldest and hottest months.

2.25.3.3 PRECAUTIONS TO REDUCE CREEP

- A) In case excessive creep is observed on PSC Sleeper load, the condition of elastic fastenings, sleepers and adequacy of ballast resistance should be examined.
- B) Action for replacement/renewal of fittings, sleepers and providing adequate ballast resistance etc. should be taken as necessary.
- C) Creep anchors to be provided, if required.

2.25.3.4 RECTIFICATION OF CREEP

Refer para.8.2.4 of IR LWR manual.

2.25.4 PROFORMA FOR MEASUREMENT OF CREEP IN YARDS

PROFORMA FOR MEASUREMENT OF CREEP OF CWR IN YARDS				
1	LOCATION			
1.1	Station Name			
1.2	Main Line			
1.3	Loop Line and line number			
1.4	Turnout Number			
1.5	Chainage			
2	INSPECTION DETAILS			
2.1	Date of measurement			
2.2	Time of measurement			
2.3	Rail Temperature			
2.4	Creep Measured on Main Line	Left Rail:	Right Rail:	
2.5	Creep Measured on turnout			
	I) at Actual Toe Switch (ATS)	Left Rail:	Right Rail:	
	II) at Actual Nose of Crossing (ANC)	Left Rail:	At ANC:	Right Rail:
	III) at lead portion	Straight Track		Diverging Track
		Left Rail:	Right Rail:	Left Rail: Right Rail:
	IV) at 20 m from ATS	Left Rail:	Right Rail:	

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	V) at 50 m beyond ANC	Left Rail:	Right Rail:
2.6	Condition of Anti-Creep Device and its fittings		
2.7	Condition of welded joints of CMS crossing		
2.8	Measured By		
3	RECTIFICATION (if any)		
3.1	Rectification Carried out on		
3.2	Rectification Carried out by		
4	Remarks		

2.26 PRECAUTIONS TO BE TAKEN BEFORE THE ONSET OF HOT WEATHER

CWR Track parameters like Creep, SFT shall be monitored closely to avoid buckling during summer. SFT measurement is an important activity to determine precautions to be taken before the onset of hot weather.

2.26.1 STRESS FREE TEMPERATURE

Stress Free Temperature or SFT of any block section by non-destructive method shall be done by using verse equipment.

2.26.2 VERSE (Vertical Rail Stressing Equipment)

VERSE is a non-destructive method of measuring the Stress Free Temperature (SFT) of Continuous Welded Rail (CWR). VERSE uses the long established Hooke's Law (extension is directly proportional to the load) to calculate SFT.

VERSE has been designed for ease of use. VERSE is a combined mechanical/software system that will produce initial results trackside from a hand held computer (supplied). Uniquely VERSE measures SFT directly VERSE is supplied complete with both handheld and PC software, the later incorporating a PC data base that can hold comprehensive data results for a whole rail network. A VERSE measurement entails lifting the rail to 10 kN (monitored by load transducer) and logging the load and displacement (monitored by a displacement transducer) at regular increments throughout the lifting cycle. The outputs from both the transducers are routed via a dedicated signal condition system to the hand held computer. Along with some other data such as ambient rail temperature, rail profile (a simple selection from the software menu) and height of the rail, the handheld software produces the SFT result. The height of the rail is included to take account of rail wear/grinding which will naturally affect the stiffness of the rail.

2.26.3 ADVANTAGES OF VERSE

Non-destructive – Many unnecessary destressing operations have been avoided. Eliminates an unnecessary weld. Additionally, VERSE is ideal for checking that newly installed/commissioned track has been set to the specified SFT, due to the fact that it is non-destructive.

Accurate – Demonstrated to be within ± 0.2 °C with a standard deviation of 1.3 °C when compared to strain gauges.

Portable – The framework folds to aid storage when not in use. Supplied in purpose made roller case that provides safe secure storage for all VERSE equipment. The roller case can fit in a small estate car or medium hatchback, so it can be easily deployed on site.

Cost effective – due to the time saving of obtaining SFT values and the elimination of unnecessary destressing operations when using VERSE, the cost is estimated to be less than 5% of traditional methods, all of which rely on cutting and re-welding the rail. The payback for the unit would be realised once 3 to 4 destressing operations have been avoided with VERSE.

Speed of operation – Typical procedure time 20 to 30 minutes (using 4 men) for both rails, including the unclipping and reinstating of rail. Significantly quicker than traditional methods.

Improved safety – VERSE provides fast, accurate, non-destructive SFT measurements, which enable rigorous, pro-active neutral temperature management. This strategy can bring real benefits of reduced dangerous incidents, increased track availability, improved understanding of the behaviour of the track, better maintenance planning, reduced risk of litigation and better maintenance expenditure. These are all powerful motivators in making the railway more competitive and cost effective and above all, safer.

The OEM's standard operating procedure and the operating procedure by the experienced International Railways has been given as ANNEXURE 4.

Low installation temperature and/or low SFT

If any portion of CWR track has an SFT below t_m+5 degree Celsius (whether because of failure to stress, low temperature, or track curvature (see below)) it shall be destressed before the onset of hot weather.

2.26.4 CWR WITH INCORRECT OR UNKNOWN STRESS-FREE TEMPERATURE

2.26.4.1 WHEN STRESS-FREE TEMPERATURE IS KNOWN TO BE BELOW t_m+5 DEGREE CELSIUS

All sections of CWR shall be destressed where the SFT is known to be, or can be expected to be, less than t_m+5 degree Celsius. This may be as a result of disturbance of the track, including where defects have been cut out without destressing.

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2.26.4.2 *WHEN STRESS-FREE TEMPERATURE IS UNKNOWN*

A resourced plan shall be in place to destress or measure all CWR where the SFT is unknown. This must meet with the approval of the DFCC Head of Track Engineering.

2.26.5 **SPECIFIC ITEMS TO BE CHECKED AND RECTIFIED**

2.26.5.1 *ADJUSTMENT OF SWITCH EXPANSION JOINT (SEJ) GAPS AND OVERLAPS, IF ANY AVAILABLE*

Switch Expansion Joint (SEJ) gaps and overlaps shall be checked against the values in LWR Manual and reset if required. Any gap of 60mm or less shall be recorded with the rail temperature, and the record retained until the gap is reset. The end of each switch rail shall be within the clamp plate by not less than 60mm. Accessible sliding contact surfaces shall be lubricated once a year.

2.26.5.2 *P&C OR JOINTED PLAIN LINE ABUTTING CWR AT AN ADJUSTMENT SWITCH*

Where jointed plain line or P&C layouts abut CWR at an adjustment switch, fastenings shall be maintained in good order, expansion gaps adjusted, fishplates oiled, and ballast shoulders and cribs fully made up for at least 90m into the jointed track. Where rail creep in jointed track adjoining CWR is known to occur, rail anchors shall be provided.

2.26.5.3 *VOIDS AND MINOR MISALIGNMENTS*

Voids and minor misalignments shall be eliminated particularly in or adjacent to welded joints, insulated joints, catch points, P&C, underline bridges (especially those with longitudinal timbers) and level crossings.

2.26.5.4 *BALLAST SHOULDERS*

The ballast shoulders especially in curves and reverse curves shall be checked and corrected if necessary.

2.26.5.5 *FASTENINGS, INSULATORS AND RAIL PADS*

Fastenings shall be examined periodically for rail pad movement and for distortion or excessive wear on insulators or other components. If loss of toe load (which can cause track instability if over several consecutive sleepers) is suspected, the defective components shall be changed. Badly-worn insulators and pads shall also be changed before excessive wear results in track circuit failure. Worn or damaged rail pads shall be replaced and displaced pads repositioned.

Individual pads may be replaced during traffic provided that not more than eight consecutive pairs of clips on one rail are released at once.

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2.27 PRECAUTIONS DURING HOT WEATHER

2.27.1 WATCHMEN

To be followed as per para 9.1.2 of IR LWR manual.

2.27.2 SPEED RESTRICTIONS

To be followed as per Annexure-III of IR LWR manual.

2.27.3 ADDITIONAL PRECAUTIONS DURING EXCEPTIONALLY HOT WEATHER

Even if the above restrictions have not already been applied, differential speed restrictions (as defined in the Rule Book) shall be imposed as follows:

- A) when the forecast air temperature is over 36°C (equivalent to possible rail temperatures over 54°C): 45/90mph between the hours of 12.00 and 20.00;
- B) additionally, when the forecast air temperature is 41°C or more (equivalent to possible rail temperatures in excess of 58°C): 30/60mph between the hours of 14.00 and 18.00.

These shall be imposed by Zonal/Regional Control offices on the basis of weather forecasts provided on the previous day.

2.27.4 LIMITS ON WORK DURING HOT WEATHER

See paras 2.24.3.2 and 2.24.3.3.

2.28 PREPARATION FOR AND PRECAUTIONS DURING COLD WEATHER

2.28.1 MONITORING ALIGNMENT

Ballast provision shall be such as to resist the tendency for the track to “pull in” on curves.

Alignment shall be monitored, against fixed references, where there is known to be a possibility of lateral movement (**Note:** this is likely to be when the rail temperature is below 0°C, or where the track radius is below 600m or where the cant is 100mm or more).

When inward movement of a curve occurs, there will be a reduction in SFT. The alignment should be restored and if necessary, see paras 2.24.3.3 and 2.24.3.4) (the CWR destressed before the onset of hot weather. When structural, OLE and/or passing clearances are unacceptable immediate action shall be taken to restore and hold the original alignment.

2.28.2 LIMITS ON WORK DURING COLD WEATHER

See para. 2.24.3.3

2.29 TRACK BUCKLING

2.29.1 TRACK CONDITION AND BUCKLING

A buckle in CWR track is not created solely by very hot weather but requires the presence of at least one other factor, the most common being:

- A) ballast disturbance, e.g. tamping, opening out;
- B) ballast shortage, e.g. voided sleepers, insufficient ballast shoulder;
- C) poor top and/or line;
- D) low SFT.

A low SFT may be associated with a build-up of non-thermal compressive stress in the rail. This will occur where rail creep encounters a resistant feature such as P&C or a change in fastening type.

Attention shall be paid to any rail creep and ballast deficiency before the onset of hot weather.

2.29.2 TRACK SAFETY AT BUCKLE SITES

If a track buckle occurs the first requirement is to safeguard traffic and advise the controlling Station or signalling centre. Before any traffic is allowed to pass over the buckled length the track may first need to be slued, preferably outwards on a curve or to an easy curvature on the straight, and a speed restriction imposed. Structural clearances and the track position relative to any OLE equipment shall be checked. If required, an adjacent track may need to be temporarily slued to provide satisfactory clearances and a speed restriction imposed. Adjustment of the track to treat the cause of the buckle and to restore the original alignment shall not be made until the rail temperature is less than $t_m + 20$ degree Celsius.

The most common causes of buckles are:

- A) Track not correctly adjusted to be stress free at t_d .
- B) Prevalent of excessive misalignments which trigger buckling under wheel dynamic loads.
- C) Lateral resistance gets impaired due to inadequacy of ballast and due to carrying out of operations such as deep screening, lifting of track and slewing of track without adequate precautions.
- D) Lateral resistance of curve sections is not adequate to resist combined CWR forces and long train drawbar pull forces.

2.29.3 PRECAUTIONS AGAINST BUCKLING

- A) Measurement of Rail SFT by NDT before onset of peak Summer month and adjustments / remedial measures, if any required.

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- B) Operations which impair the lateral resistance of track are not carried out when rail temperatures are high.
- C) All precautions to be taken to reduce creep in CWR sections.
- D) Particular attention is also paid to stretches of track one liable to creep and other held against creep.

2.29.4 BUCKLING AND ITS INVESTIGATION

Instruction contained in CI 7.4.2 of IR LWR manual shall be followed.

2.29.5 ACTIONS TO BE TAKEN ON BUCKLING OF TRACK

Instructions contained in para 243(4) of IRPWM should be adhered to.

2.29.6 REPAIRS TO BUCKLED TRACK

When the track actually buckles, the traffic shall be suspended and the cause of buckling ascertained. The rectification shall normally be carried out in the following stages under the supervision of Sectional Engineer/competent authority.

- Emergency repairs
- Permanent repairs
- Destressing

2.29.6.1 EMERGENCY REPAIRS

The buckled rails shall preferably be gas cut adequately apart not less than 6.5 metre. The track shall then be slewed to the correct alignment and cut rails of the required lengths shall be inserted to close the gaps making due provision for welding of joints on both sides. The cut rails shall then be connected by use of special fishplates and screw clamps and the line opened to traffic with speed restriction as indicated in ANNEXURE-III of Manual of instructions on LWR.

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ANNEXURE-III

SCHEDULE OF SPEED RESTRICTIONS

Sl. No.	Conditions of track	Restriction imposed in km/h
1.	When 1 metre long fishplates (Fig. 4.4.3(a)) with special clamps (Fig. 4.4.3(b)) or jogged fishplates with bolted clamps (Fig. 4.4.3(c)) are used at a temporary rail joint and there is 24 hrs. watch Fig. 4.4.3(a), 4.4.3(b) and 4.4.3(c) are to be referred from IRLWR manual	30
2.	When other clamps are used at a temporary rail joint	20
3.	When sleeper fastenings on alternate sleepers are loosened before destressing	30
4.	At fracture after emergency repairs are completed:- i) First train ii) Subsequent trains	STOP DEAD & 10 20
5.	After emergency repairs to track after buckling:- i) First train ii) Subsequent trains	STOP DEAD & 10 20
6.	Speed restriction during consolidation period of track, after regular track maintenance operations when rail temperature exceeds $t_b + 20^{\circ}\text{C}$: i) When shoulder and crib compaction has been done:- For BG ii) When shoulder and crib compaction has not been done:- For BG	50 30

2.29.6.2 PERMANENT REPAIRS

- A) Creep is measured on both side of the affected portion.
- B) Instructions contained in CI 7.4.5 of IR Manual of instructions on Long Welded Rails shall be adhered.
- C) The track length found under creep should be marked.
- D) In case of directional traffic, one point is marked at the location where creep is not found

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and another point to be marked in the direction of traffic at a length of about 1 km.

- E) Length from where the creep is found not more than 5mm should be isolated from one end by cutting the rail and on another end second cut is made after a length of 1 km.
- F) If affected length is greater than 1 km, then that length should be made stress free as per para 5.7 of IR LWR manual.
- G) After destressing and welding the joints at both ends, the track equal to breathing length (100 m) on either side be unfastened and tapped to ensure equalisation of forces and then refastened.
- H) Then traffic shall be restored at normal speed.

2.29.6.3 DESTRESSING

This shall be referred as per Cl. 2.30.1 of this manual.

2.29.6.4 DO'S FOR P.WAY OFFICIALS

- A) Confine essential maintenance to period when the temperature is below $t_d + 10$ degree centigrade.
- B) Know the t_d of the C.W.R. of the section and check the stress-free temperature before ONSET of summer with a NON-DESTRUCTIVE EQUIPMENT like VERSE.
- C) Start patrolling of track as soon as rail temperature rises to $t_d + 20$ degree centigrade. Give attention to frequent misalignment zones, sharp curves and reverse curves.
- D) Impose suitable speed restriction if temperature exceeds $t_d + 20$ degree centigrade during track stabilisation period.
- E) Keep the ballast section full, including extra ballast widths on curves, and recoup the deficiency of ballast well before the onset of summer.
- F) Keep the Anchors, where ever provided, always butting against the sleepers.
- G) Renew fittings only on one sleeper at a time, if requires.

2.30 DESTRESSING IN PLAIN LINE (OTHER THAN TURNOUTS)

Abnormal behaviour of CWR whenever gets manifested in one or more of the following, destressing shall be undertaken as per procedure laid down in Cl 5.7 of IR LWR manual.

Destressing of CWR shall be done by cutting it into LWR'S of about 1 km or shorter in length as per Table.5 which shall be joined after destressing in accordance with below procedure.

Destressing shall be done:

- A) When stock/tongue rail crosses the mean position.
- B) After special maintenance operations mentioned in para 6.3 of IR LWR Manual.
- C) After restoration of track following an unusual occurrence mentioned in para 7.1 of IR LWR Manual.

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D) If number of locations where temporary repairs have been done exceed three per km.

2.30.1 PROCEDURE

General: The following procedure shall be read in conjunction with Para 2.19.1

The work of destressing shall be done during a traffic block under the personal supervision of a Foreman/Executive (Track) as laid down in ANNEXURE-VI of Manual of instructions on LWR.

It is preferable to impose a speed restriction of 30 km/h before actually obtaining the traffic block and to loosen/remove fastenings on alternate sleepers to reduce total duration of the traffic block.

2.30.2 SEQUENCE OF OPERATIONS

The procedure to be adopted for destressing consists of the following steps:-

- A) Remove impediments to free movement of rail such as rail anchors, guard rails, check rails etc.
- B) Create gap of about 6.5 metre at the centre of LWR during a traffic block and insert a closure rail there at a restricted speed.
- C) Mark anchor lengths at either end of the proposed LWR in accordance with CI 2.20.7 of this manual.

Note:

- a) Anchor length shall be determined on the basis of the lowest value of t_p at which destressing is likely to be carried out.
- b) Anchor length shall be increased suitably if the fastenings, rubber pads, liners or ballast conditions are poor.
- D) Erect marker pillars at the ends of anchor lengths on either side and at 100 metre intervals thereafter.
- E) Obtain a traffic block when t_p is less than the desired installation temperature (t_o), remove the closure rail at the centre and unfasten the full length of rails leaving only the anchor lengths at either end. Mount the rails on the rollers.

Note: Side rollers shall also be used while undertaking destressing on curved track. Side supports on the inside of curve should be spaced at every n th sleeper; Outside supports shall be used in addition at the rate of one for every three inside supports.

- F) Fix rail tensor across the gap at the centre and apply tension so as to get the required amount of extension as provided in ANNEXURE –VII of Manual of instructions on LWR.
- G) Re-fasten the rails starting from the anchor lengths at either end after removing the rollers progressively and adjusting tension at the rail tensor to make sure that the required extension has been achieved at each marker pillar.
- H) 16 where, $n = \text{Radius of curve (R)} \times \text{No. of sleepers per rail length } 50 \times (t_o - t_p)$

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- a. Put paint marks on either side of the gap at the centre, spanning over the gap at a distance of 6.5 metre. Insert a closure rail of a length equal to (6.5 metre - 2 gaps for welding + 1 mm for saw cut ends) and clamp as per arrangements shown in Fig.4.4.3(a),4.4.3(b1) and 4.4.3(b2) of IR LWR manual or 4.4.3(c) of IR LWR manual.
- b. Release and remove the rail tensor.
- c. During another traffic block, weld both the joints of the closure rail at the centre.
- d. Equalise stress at the centre and at the anchor lengths.

2.30.3 DESTRESSING OPERATION OF CWR WITH THE USE OF RAIL TENSORS

For destressing of CWR with the use of rail tensor, the following procedure shall be adopted:

-

- A)** During the first traffic block, create a gap of 1 metre at location 'B' i.e. centre of CWR (Fig. 5.7.3.(b) of IR LWR Manual). Introduce rail closure as required and fasten with special fishplates and clamps. Allow traffic at restricted speed.
- B)** Mark the anchor length A1 A2 and C1 C2 each equal to la at either end of the length A2 C2 to be destressed (Fig. 5.7.3.(b) of IR LWR Manual).

Note:

The anchor length ' la ' should be determined on the basis of the lowest value of tp at which the destressing is likely to be carried out.

- C)** Erect marker pillars W0 W1 etc., on each of the length A2 B and C2B. Transfer the marks W0 onto the rail foot (Fig: 5.7.3(b) of IR LWR Manual)

Note:

The distances W0 W1, W1 W2 etc. shall be marked at about 100 metre intervals, the distance from the previous pillars and the last pillar WB may be less than 100 metre.

- D)** During the second traffic block, when tp is less than the desired to (Fig. 5.7.3 (b) of IR LWR Manual), destressing operation shall be carried out for the lengths A2 B and C2 B as described below: -

- I) Remove the closure rail from location 'B.' Unfasten and mount on rollers the portion from A2 C2.
- II) Fix the rail tensor across the gap at 'B' and apply tension so as to obtain some movement at W0 to remove any kinks or misalignment and to minimise the friction in the rollers etc. Release the tension and note the movement Y0 at W0.
- III) Transfer marks W1, W2, onto the rail foot and note temperature tp .

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IV) Calculate the required movement at W1 as under: -

Movement at W1 = $Y_0 + \text{elongation of length } W_0 W_1 (L) \text{ due to temperature difference } (t_o - t_p) = Y_0 + L(t_o - t_p)$

V) Calculate the required movement at W2 as under: -

Movement W2 = Movement at W1 + elongation of length W1W2 (L) due to temperature difference (to - tp).

Similarly, calculate the required movements successively at each of the remaining points.

Mark the above calculated extensions with respect to the transferred marks referred at (c) above on the rail foot on the side away from the tensor. Apply the tension by means of rail tensor till the mark of required extension comes opposite to the mark on the marker pillar W1. Fasten down the segment W0W1. Then check at W2, bring the mark of required extension at this location opposite to the mark on the marker pillar W2, by adjusting the tensor either by reducing or increasing tension and fasten down the segment W1W2. Similarly, check the remaining marks, adjust the tension as required and fasten down each segment before proceeding to the next.

Note:

- I) ANNEXURE-VII of Manual of instructions on LWR gives the value of $L(t_o - t_p)$ for different values of L and (to - tp).
- II) Only one value of tp has to be taken at the time of marking W1, W2 etc. on the rail foot. The values of tp is not required to be taken thereafter. The variation of temperature, if any during the destressing operation shall automatically be taken care of by reducing or increasing the tensile force from the tensor, while coinciding the reference mark on rail with the corresponding mark on pillars.
- III) If for any reason, both the lengths A2B and C2B cannot be fastened down simultaneously, the final adjustment in pull and fastening down of the individual segments may be done in series, first from A2 to B and then, from C2 to B.
- IV) After the fastening down of the last length A2B and C2B is completed, make a paint mark near free end of one rail at a distance of (6.5 metre + 2 x 25 mm -1 mm), measured from the end of the other rail across the gap spanned by the rail tensor.
- V) Remove the tensor, close the 1 metre gap temporarily by using rail closure and clamping with fishplates and allow traffic at restricted speed.

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VI) During another traffic block, cut the rail at the paint mark, insert a rail closure of length exactly equal to 6.5 metre and weld one end thereof. If the gap at the other end is also 25 mm, it can be welded in the same block. Otherwise, fasten with special fishplates and clamps and allow traffic at restricted speed. In the latter case, during a subsequent block, when t_p is not greater than t_o , release rail fastenings on either side to the required extent and pull the rails with rail tensor to get the desired gap of 25 mm; refasten the rail and weld the joint. Release the tensor after a lapse of a minimum of 20 minutes after pouring of the weld metal.

VII) During a subsequent traffic block, when t_p is less than t_d , equalise the forces in the rail by releasing the fastenings over a length of 100 metre on either side of location 'B' and tapping with wooden mallets etc. Fasten down the rail and allow traffic.

2.30.4 DESTRESSING OPERATION DURING TEMPERATURE RANGE SPECIFIED FOR DESTRESSING

In case rail temperature at the time of destressing is within the range specified in para 2.30, detailed procedure, as given in ANNEXURE –VIII Manual of Instructions on LWR, without using rail tensor may be adopted.

2.30.5 DESTRESSING WITHIN TURNOUT PORTION

For the desired repair/replacement of rail/sleepers/fittings in case of any fracture/buckling/special maintenance, the entire turnout portion shall be cut and isolated from all the three sides of the approach tracks, the necessary repair/maintenance work shall be done as per the requirement and after that destressing of turnout and joining with the approach tracks shall be done as per the approved method statement.

Destressing of turnout shall be done in following steps: -

- A) Speed restriction of 30Kmph shall be imposed in the turnout portion, which is to be destressed.
- B) Traffic block of sufficient time shall be taken when the prevailing temperature is within the range of T_d .
- C) The turnout shall be cut from all the three and isolated from all the three sides of approach track by removing a closure of 6m or more length from all the three approach tracks.
- D) All the fittings and fastening to be removed.
- E) The entire turnout shall be placed on rollers and wooden malleting shall be done to permit free movement.
- F) The rollers shall be removed and turnout shall be placed in correct position.
- G) Anti - creep device shall be adjusted in the central position.
- H) The gap shall be measured on all three sides of turnout and rail closures of suitable length

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considering the gap for AT welding shall be provided.

- I) After ensuring that the entire turnout is free from all the stresses, the welding and fastening of turnout shall be provided and tightened starting from SRJ and proceeding to back of crossing.
- J) AT welding of all the joints shall be done within the range of Td.
- K) Point interlocking to be checked and adjusted after destressing, if required.
- L) The fittings of approach track of all the three sides shall be opened in a length of 125m (as per Cl. 7.2.5(4) of LWR/CWR manual) for doing equalization of forces and again after that fastening shall be provided and tightened properly.

2.30.6 DESTRESSING ADJACENT TO THE ACTUAL TOE OF SWITCH (ATS)

When destressing adjacent to a turnout is to be undertaken the following points must be checked.

- A) Turnout must be fully ballasted and lifted to line and level.
- B) Whenever a new turnout is to be installed or replaced, all three sides of the turnout shall be isolated by inserting the closure rail of more than 4.0 m in length. All the joints shall be welded including closure rails as free welds.
- C) Through road (facing & trailing roads) shall be destressed first and separate element. The turnout road is to be completed in the last.
- D) It must be ensured that rails through the turnout are not to be destressed.
- E) Anchor length shall be marked from Sleeper no. 03 to last Sleeper of heel of Crossing.
- F) Sleeper No. 03 over which the tongue rail toe rest, shall remain fastened as the anchor point sleeper and is the interface location between the Destressing length and the anchor length.
- G) Switch toes must be maintained square and therefore both the rail shall be destressed preferably at the same time of destressed consecutively during the same work shift during a period of stable rail temperature.
- H) Create a cut at distance of 500m from ATS.
- I) Keep the road ahead of cut mark fully encored leaving a distance of 500m.
- J) Remove the fittings on either side of cut mark in the length of 500m.
- K) Destressing shall be carried out as per para 2.30.1 of this manual.

2.30.7 DESTRESSING ADJACENT TO THE HEEL OF CROSSING

- A) Establish the anchor length from second long turnout sleeper and extending through the turnout for full anchor length.
- B) Second last long Sleeper shall remain fastened at the anchor point sleeper & is the interface location between the destressing length and anchor length.
- C) ATS and anti-creep device must be maintained square and therefore both the rail adjacent

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to the turnout shall be destressed preferably at the same time or consecutively on the same work shift during a period of stable rail temperature.

D) Destressing shall be carried out as per para 2.30.1 of this manual.

2.30.8 DESTRESSING OF CROSSING

It will be same as procedure given as per Cl. 2.30.5 of this manual.

2.30.9 WHEN RAIL TEMPERATURE APPROACHES DESIGNED STRESS FREE TEMPERATURE

At Rail temperature within a few degrees of designed stress-free temperature, when extension gap may be too small to achieve movement at the reference marks when the rail is pulled up. In such situations one or more of the following techniques may be implemented.

- A) Increase the design stress free temperature upto 40 degree centigrade in the extension calculation. This will increase the required gap.
- B) Overlap the rail ends to achieve the required extension with the assistance of the rail lifting plate.
- C) Use a shorter Destressing length (for shorter length gap may be sufficient to allow the rail to be Destressed).
- D) Cut the weld gap prior to the pulling the rails instead of after words, optionally in the combination of wide gap weld.

Cut a very wide extension gap & after destressing and while the rail is relaxed, cut and weld in a closure rail.

2.30.10 DESTRESSING OF TURNOUTS & JOINING OF TURNOUT PORTION WITH APPROACH TRACKS

For the desired repair/replacement of rail/slippers/fittings in case of any fracture/buckling/special maintenance, the entire turnout portion shall be cut and isolated from all the three sides of the approach tracks, the necessary repair/maintenance work shall be done as per the requirement and after that destressing of turnout and joining with the approach tracks shall be done as per the approved method statement.

Destressing of turnout shall be done in following steps: -

- A) Speed restriction of 30Kmph shall be imposed in the turnout portion, which is to be destressed.
- B) Traffic block of sufficient time shall be taken when the prevailing temperature is within the range of Td.
- C) The turnout shall be cut from all the three and isolated from all the three sides of approach track by removing a closure of 6m or more length from all the three approach tracks.
- D) All the fittings and fastening to be removed.
- E) The entire turnout shall be placed on rollers and wooden malleting shall be done to permit

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free movement.

- F) The rollers shall be removed and turnout shall be placed in correct position.
- G) Anti - creep device shall be adjusted in the central position.
- H) The gap shall be measured on all three sides of turnout and rail closures of suitable length considering the gap for AT welding shall be provided.
- I) After ensuring that the entire turnout is free from all the stresses, the welding and fastening of turnout shall be provided and tightened starting from SRJ and proceeding to back of crossing.
- J) AT welding of all the joints shall be done within the range of Td.
- K) The fittings of approach track of all the three sides shall be opened in a length equal to breathing length (as per Cl. 7.2.5(4) of LWR/CWR manual) for doing equalization of forces and again after that fastening shall be provided and tightened properly.

2.30.11 INSTALLATION PROFORMA

DETAILS OF STRUCTURE OF CWR AS LAID:

1. GENERAL

- i) Kilometres to
- ii) Between stations and
- iii) Up/Down/Single line
- iv) Date of laying

2. TRACK STRUCTURE

2.1 Rails

- i) Sectional weight
- ii) Rolling mark
- iii) Year of laying
- iv) Length of rails as rolled
- v) Plant where rails welded
- vi) Types of welding - Flash Butt/ATW
- vii) Length welded into panels
- viii) Thermic welding done in-situ/on cess by ordinary alumino-thermic process/SKV/ FB by Mobile plant

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2.2 Sleepers

Type

- a) Density or No. per km.
- b) Type of fastenings
- c) Rail anchors if in use
- d) Details of sleepers anchored
- e) Lengths of track box-anchored, location, reasons

2.3 Ballast and Sub-Ballast

- i) Size (mm)
- ii) Depth of cushion (mm)

2.4 SEJ (if any provided in future)

- i) Location (km)
- ii) Date of laying
- iii) Maximum gap possible
- iv) Drawing no.
- v) Manufactured by
- vi) Whether joined to CWR by welded joint

2.5 Girder Bridges

- i) Location and No.
- ii) Lengths and spans of bridges with CWR
- iii) Type of fastenings used
- iv) Any other remarks

2.6 Level Crossings

Location and No.

2.7 GRADES, ALIGNMENT AND FORMATION

- i) Steepest gradient
- ii) Maximum degree of curvature
- iii) Formation (indicate type of soil)
- iv) Particulars of trouble with formation and treatment given, if any, at the time of laying. (Sketch showing grades and curves with locations to be

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attached).

2.31 MEASUREMENT OF BREATHING LENGTHS

The end portion of all the dead ends, sidings i.e., overrun lines, Hot Axle sidings and other sidings where Buffer stops have been provided at the end will behave as breathing length, the length of which will be approximately 80 m. If there is any movement in the CWR portion beyond the breathing length, then the necessary action shall be taken as per this manual, cited for creep adjustments.

The necessary records of the measurement of breathing lengths shall be as per the Indian Railway Practice.

2.32 SYSTEMATIC OVERHAULING

Procedure and precautions shall be followed as per para 227 of IRPWM.

2.33 RAILS

2.33.1 CAUSES OF RAIL DETERIORATION

The possible reasons of rail deterioration are listed in ANNEXURE-1B

2.33.2 PERMISSIBLE WEAR LIMITS

Permissible wear limits		
Rails	Vertical wear (in mm)	Lateral wear (in mm)
60 E1 1080HH	13	10 (Curve portion) 8 (straight portion)
UIC 60 kg 880 gr	13	10 (Curve portion) 8 (straight portion)
Tongue rail (60E1A1)	8	8
Stock rail	13	10 (Curve portion) 8 (straight portion)
Weldable CMS crossing	(8mm-2.5mm) = 5.5mm	-

2.34 RAIL MAINTENANCE TO REDUCE RAIL DETERIORATION

Provision of IRPWM para 250 shall be followed.

2.35 INSPECTION OF RAILS IN SERVICE

Provision of IRPWM para 252 shall be followed.

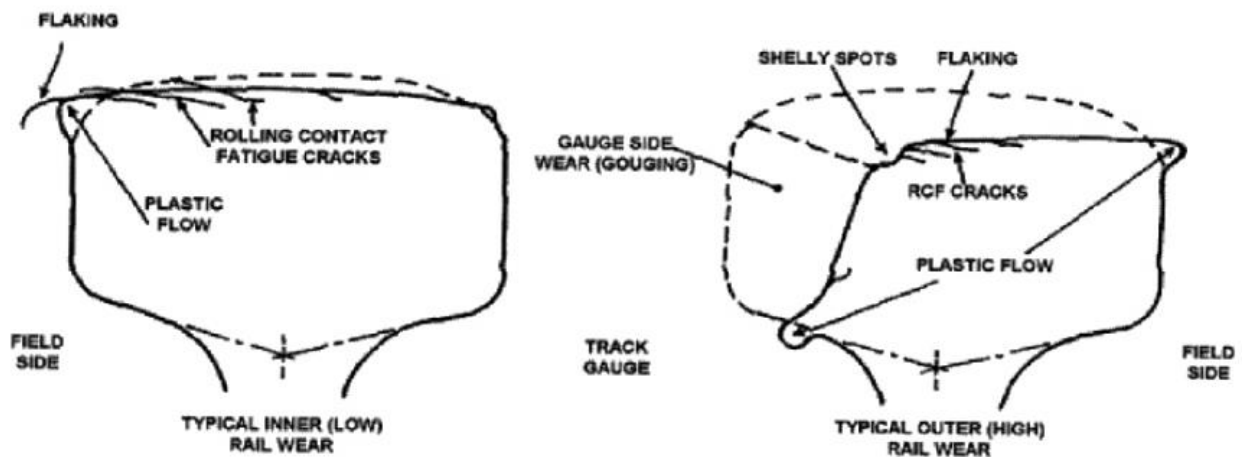
2.36 RAIL GRINDING

2.36.1 INTRODUCTION

Rail grinding maintains optimal contact conditions and controls the propagation of rail surface fatigue defects. Guidelines are provided in this chapter for best practice preventive grinding for; rail grinding metal removal, rail grinding cycles (in gross million tonnes, GMT), acceptable surface finish after grinding and grinding of turnouts.

2.36.2 REASONS FOR RAIL GRINDING

As the rail wears with accumulating tonnage, changes to the wheel/rail contact geometry can promote excessive wheel/rail contact stress that causes rail-surface plastic flow and surface fatigue (flaking, spalling, shelling and head checks).



Excessive rail wear, plastic deformation, and surface fatigue will result if rail is not maintained by rail grinding.

Rail grinding controls rail (and wheel) surface plastic deformation, rolling contact fatigue (RCF) cracks on the surface and sub-surface, improves bogie steering and dynamic stability and increases rail and wheel life.

2.36.3 RAIL GRINDING DEFINITIONS

Corrective grinding removes deep rail surface defects. Preventive grinding is the removal of small quantities of rail metal to maintain rail profiles and control the development and

propagation of initiating RCF cracks and corrugations before they enter the stage of rapid growth.

2.36.4 PREVENTIVE GRINDING METAL REMOVAL RATES

Instructions shall be followed as content in chapter V of Indian railway track machine manual.

2.37 RAIL CLOSURE

Rail closure of not less than 6.5m shall always be used of the same grade and section of rail at the location where it is to be inserted.

Provision of para 256 of IRPWM shall be followed, for non-CWR track. However, in CWR track for any reasons if a Rail Closure is inserted with one metre fish plates, speed restriction of 30 km/h shall be imposed and stationary watchman shall be posted till track is welded to CWR and equalization of forces is done within 125 m on either side of the affected portion.

Use of rail closure should be limited and used to the minimum possible. Closure should not be located near each other in the same portion of track nor in proximity to bridges, level crossings etc.

2.38 RAIL FAILURES

A rail is said to have failed if it has fractured in track or it is considered necessary to remove it from the track on accounts of defects other than those due to accidental damages, due to buckling, kinking, derailments, abnormal wheel burns etc.

2.38.1 REPORT OF RAIL FAILURES

Modified Rail Failure Report shall be prepared as the proforma given below:

CORRIDOR

Pro forma for Reporting Rail Failures

Broken/Cracked /Defective rail removed on day of 20.....

1. General Information
 - 1.1. IMD/IMSD/RMMU
 - 1.2. Section (name of line or branch)
 - 1.3. Between stations.....and.....
 - 1.4. Kilometrage
 - 1.5. Line

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- 1.5.1. Up/Down/Single.
- 1.6. Alignment
 - 1.6.1. Straight/Curve (indicate degree of Curvatures).
 - 1.6.2. Inner/Outer in case of curve.
2. Characteristic of Traffic and Traction
 - 2.1. Traffic Density in G.T.K.m./annum.
 - 2.2. Total traffic carried in G.M.T before failure (for released rails, also add previous traffic carried)
 - 2.3. Maximum axle load with type of vehicle on section.
 - 2.4. Maximum permissible speed.
3. Characteristic of Rail
 - 3.1. Rail Sectionkg/m
 - 3.2. Rolled Marking.....
 - 3.3. Total number of years in service....
 - 3.4. Cast No.
4. Characteristic of Track
 - 4.1. If fracture occurred at or within 100mm of weld. Indicate date & type of welding.
 - 4.2. Position of fracture.
 - 4.3. Types of sleepers and density.
 - 4.4. Depth of clean ballast cushion.
5. Particulars of defects and Fractures (to be filled by Depot Manager)
 - 5.1. Detected-Visually/By flaw detector.
 - 5.2. Remarks if any.....
 - 5.3. Classifications of failure in code.....

Signature of Executive Track	Signature of Depot Manger	Signature of GM/AGM Infra
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METALLURGICAL EXAMINATION OF FRACTURED PIECE: -

If any rail failure is detected visually, a rail piece of approximately 2m length shall be sent for metallurgical test. The test pieces for metallurgical examination shall be sent if rail failure occurs-

- a. Within 10 years of laying
- b. Within guarantee period if rails are imported from other countries – In all cases of failure of imported rail, occurring within the guarantee period irrespective of the type of failure/ flaw a 2m piece containing the fracture/ flaw detected visually or by ultrasonic flaw detector, should be sent to the Chemist and Metallurgical together with a rail failure report for metallurgical investigation.

2.38.2 TYPE OF RAIL FAILURE

- a) 100/200 – Transverse breakage with apparent origin (sudden breakage)
- b) 1212/2212 – Head, surface, shallow surface defect (line)
- c) 1321/2321 – Web horizontal crack at fillet radius
- d) 1322/2322 – Web horizontal crack at bottom fillet radius
- e) 1323/2323 – Web horizontal crack not at fillet radius
- f) 238 – Web diagonal crack not at a hole
- g) 253 – Foot, vertical, longitudinal crack in foot half-moon break
- h) 1511/2511 – Foot transverse break at rail seat
- i) 1512/2512 – Foot transverse break not at rail seat
- j) 111/211 – Internal flaw in head, transverse breakage
- k) 112/212 – Internal flaw in head, horizontal crack
- l) 113/213 – Internal flaw in head, vertical longitudinal split
- m) 133/233 – Web, vertical longitudinal splitting
- n) 139/239 – Web, lap
- o) 153/253 – Foot, vertical longitudinal split

2.38.3 PROCEDURE FOR SENDING SAMPLES FOR METALLURGICAL INVESTIGATION

Instructions contained in para 257(7) of IRPWM shall be followed.

2.38.4 DO'S of CWR FOR P.WAY OFFICIALS

The below listed points are in addition to IR LWR manual ANNEXURE-XIV

- a. Confine essential maintenance to period when the temperature is below $t_d + 10$ degree centigrade.
- b. Know the t_d of the C.W.R. of the section and check the stress-free temperature before ONSET of summer with a NON-DESTRUCTIVE EQUIPMENT like VERSE.

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- c. Start patrolling of track as soon as rail temperature rises to $td+20$ degree centigrade
- d. Impose suitable speed restriction if temperature exceeds $td+20$ degree centigrade during stabilisation period.
- e. Keep the ballast section full and recoup the deficiency of ballast before onset of summer.
- f. Keep the Anchors, where ever provided, always butting against the sleepers.
- g. Renew fittings only on one sleeper at a time, if requires.

2.38.5 DONT'S of CWR FOR P.WAY OFFICIALS

To be followed as per IR LWR manual ANNEXURE-XIV.

2.39 REPLACEMENT/RENEWAL OF TURNOUT/TURNOUT COMPONENTS

1. All activities shall be done under direct supervision of competent supervisor.
2. All renewal activities shall be done after isolating the turnout from CWR. The isolation shall be done on all three sides of the turnout by providing rail closures of not less than 4m.
3. All activities for isolation shall be done under traffic block of suitable time.
4. If switch portion is to be renewed, pre-assembled switch should be kept outside on cess and get tested jointly with signalling department with respect to setting than only be replaced with existing one under traffic block.
5. After removal of existing switches, all rubber pads and base plates/slide chairs (if any broken or cracked) be replaced in advance before placing the new switch.
6. All other activities are listed below.

2.39.1 REPLACING SWITCHES

Replace a switch by undertaking the following steps:

1. Check that the replacement switches matches the existing (rail size, length, hand and style (angle, undercut etc.) Check that the new switch matches the specification.
2. Disconnect the switch.
Arrange disconnection of the signalling equipment from the switch.
 - a) Measure and mark on rail location for a cut.
 - b) Cut the rail.

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3. Remove the switch.
4. Clean the base plates.
5. Lift new switch into position. Make sure the switch is supported at multiple lifting points to prevent hogging or crippling of the switch.
6. Locate correct position for switch tip and adjust position. In newer turnouts the design position will be marked with a punch mark on the outside head of the stockrail.
7. Weld the switch to the closure rail using the ATW procedure Install the heel block or creep block.
8. Check the match between head of the switch and the stockrail and grind the stockrail if required.
9. Check and correct the bearing of the switch on the bed plates.
10. Check fit of switch to stockrail between point and full head (crow if necessary).
11. Arrange reconnection of the signalling equipment to the switch.
12. Arrange check of the switch operation.
13. Check for uniform alignment along the running edge of the blade and for the integrity of fastenings. Adjust the switch if required.
14. Check line and level through switch.
15. Make sure all keys and pins are in slide chair plates.

2.39.2 REPLACING A SWITCH/STOCKRAIL ASSEMBLY

Replace stockrail/switch assembly by undertaking the following steps:

1. Check the new switch/stockrail assembly for match with existing (rail size, length, type).
2. Check the new switch/stockrail for match with specification.
3. Disconnect the switch.
4. Measure and mark on rail the location to cut the stockrail and switch
5. Cut the rails

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6. Remove stockrail fastenings.
7. Remove switch and stockrail.
8. Clean base plates.
9. Lift new switch and stockrail into position
10. Check that the stockrail foot rests fully on chair plate prior to operation and prior to the addition of signalling equipment such as signal motors.
11. Take particular care with the support of the stockrail on the chair plate at newly constructed turnouts and the positioning of turnout ties in the switch area.
12. Check bolt tightness after a day or so and re-tighten bolts as required.
13. Locate the correct position for the switch tip and adjust the position of the switch and stockrail to suit.
14. Install closures (if necessary) and welds. Welds and closures should be minimized as alignment and weld fatigue reduce the asset life.
15. Install stockrail fastenings.
16. Check match between rail ends of the stockrail and grind the stockrail if required.
17. Check fit of switch to stockrail.
18. Adjust the switch if required
19. Check line and level through switch.
20. Make sure all keys and pins are in slide chair plates.

2.39.3 REPLACING A CROSSING

Defects within 10mm from the surface and defects connected with the outside and presence of cracks/hot tears shall be cause for rejection of Weldable CMS Crossing in accordance to clause 6.2.2 of IRS T 29-2016

Replace crossing assemblies by undertaking the following steps:

1. Check the quality of the new crossing.
2. If the old crossing is worn on the gauge face of the nose, check and adjust the checkrail effectiveness to the correct dimension before installing the new crossing.
3. Measure new to old crossing (length, crossing rate and leg openings) to establish that the replacement is correct.
4. Mark recovery points for location of the theoretical point of the crossing.
5. If there is evidence to suggest that the crossing is in the wrong place, check the location of the existing crossing by measuring the distance from the theoretical point to the switch tip and comparing it to the design dimensions.

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6. Measure and mark locations on crossing legs for cutting
7. Remove rail fastenings from the crossing.
8. Cut rails to remove the crossing.
9. Remove the crossing from the turnout using a mobile crane or off-track plant and stack it clear of the work site.
10. Check the condition of the sleepers associated with the crossing and replaces any that have failed or have poor fastening strength.
11. Check condition of base plates and replace if necessary.
12. Place the new crossing into position using a mobile crane, on-track or off-track plant.
13. Check the location of the theoretical point using the recovery points marked in Step 5. Adjust the crossing location if required.
14. Check the crossing alignment (including gauge and checkrail effectiveness) and correct if necessary.
15. Re-install fastenings.
16. Lift and pack track.
17. Carry Aluminothermic welding for welding the crossing to the rails.

2.39.4 REPLACING A CHECKRAIL

Replace raised checkrail assemblies by undertaking the following steps:

1. Check new checkrail for match with existing (height, length, bolt hole number and location).
2. Check the new checkrail for match with specification.
3. Remove the checkrail fastenings.
4. Lift out the checkrail.
5. Install the new checkrail.
6. Install fastenings.
7. Check height, flangeway clearance and checkrail effectiveness and adjust if required.

2.39.5 REPLACING A PLATE

Replace track plates by undertaking the following steps:

1. Remove fastenings.

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2. Remove the ballast on either side of the plate.
3. Reduce rail bearing on plate and knock plate sideways out into the hole.
4. Pull out the old plate.
5. Place the new plate in the ballast crib.
6. Lift up the rail using a jack or bar. Do not raise adjacent sleepers.
7. Position the new plate, replace fastenings and restore ballast profile.

There should always be a minimum clearance of at least 15mm between plates on turnouts.

2.39.6 RENEWING TURNOUT SLEEPERS

Establish the size of replacement timber sleepers using the following information:

1. Select the sleeper to replace.
2. Check that the sleeper is the correct one for the job (right gauge) and all components are available.
3. Arrange disconnection of signalling equipment if required.
4. Remove rail fastenings.
5. Remove shoulder ballast.
6. Clear ballast from the crib on either side of the sleeper to the underside of the timber.
7. Remove sleepers by sliding out manually, using mobile crane or off-track plant.
8. Break up the ballast bed under the old sleeper to a depth of at least 50mm.
9. Insert the new sleeper from the side using a mobile crane or off-track plant. DO NOT place sleepers underneath insulated joints.
10. Insert plates under the rails and adjust the longitudinal position to correspond with the requirements of the turnout sleeper set out diagram. When renewing plates near an insulated joint check the design requirements. Special plating arrangements generally apply.
11. Line up the end of the new sleeper with the ends of the adjoining sleepers on the straight stockrail.
12. Gauge the curved stockrail, bore and fasten.
13. Restore ballast profile.
14. Check alignment and line track if required.
15. Lift and pack sleepers.

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2.40 TRACK MAINTENANCE BY MACHINES

Track maintenance by machines shall be followed as per **IRPWM Clause 226.**

2.41 3-TIER SYSTEM OF TRACK MAINTENANCE

To be followed as per **IRPWM Clause 228.**

2.42 PICKING UP SLACKS

Shall be followed as same as **IRPWM Clause 229.**

2.43 OBSERVANCE OF SLEEPERS UNDER PASSAGE TRAFFIC

Shall be followed as same as **IRPWM Clause 230.**

2.44 SAMPLE OF STANDARD SECTION OF TRACK

A sample track shall be maintained on approach of every beat of section.

2.45 LIFTING OF TRACK

Shall be followed as per **IRPWM Clause 233.**

2.46 LOWERING OF TRACK

Shall be performed as per **IRPWM Clause 234.**

2.47 FOULING MARKS

To be followed as per **IRPWM Clause 236.**

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PART C - WORKS INCIDENTAL TO REGULAR TRACK MAINTENANCE

2.48 DEEP SCREENING OF BALLAST

Shall be followed as per **IRPWM Clause 238** and **LWR/CWR manual Clause 6.3.2**

2.49 BUFFER STOPS

The Installation and maintenance shall be as per the ANNEXURE 5 & 6.

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PART D - MAINTENANCE OF TRACK IN TRACK CIRCUITED AREAS

2.50 PROVISION AND MAINTENANCE OF SIGNALLING FIXTURES IN TRACK

To be followed as per **IRPWM Clause 279.**

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PART E – SIGNAGES AND SIGN BOARDS

2.51 WAYSIDE SIGNAGES

2.51.1 KILOMETER POST:

Kilometre Posts are located at every km point along the alignment on the left-hand side of the track towards the termination of DFC line. The distance from the Zero Chainage is shown in numerals increasing from Zero to the termination engraved 6 mm deep on both sides of the post.

2.51.2 HECTOMETER POST:

Hectometre posts are located at every 100m between the Kilometre Posts. Letters in numerals are engraved 6 mm deep on both sides of the post.

2.51.3 GRADIENT POSTS:

Gradient posts are located at every change of gradient along the vertical alignment on the left-hand side of the track toward the termination of DFC line. Letters showing forthcoming gradients are engraved 6 mm deep on both sides.

2.51.4 CURVE BOARD:

Curve boards are located at the middle point of a horizontal curve. They include Curve Number, Radius of Curvature, Cant, Circular Curve Length and Tangent Curve Length, which are embossed onto a metal plate.

2.51.5 TRANSITION CURVE POST:

Transition posts are located at the beginning and end of each transition curve. Letters painted in white indicating "BTC" or "ETC" are engraved in front of a black- painted wooden post with the Chainage written on the back side in the same way.

2.51.6 FOULING MARKS:

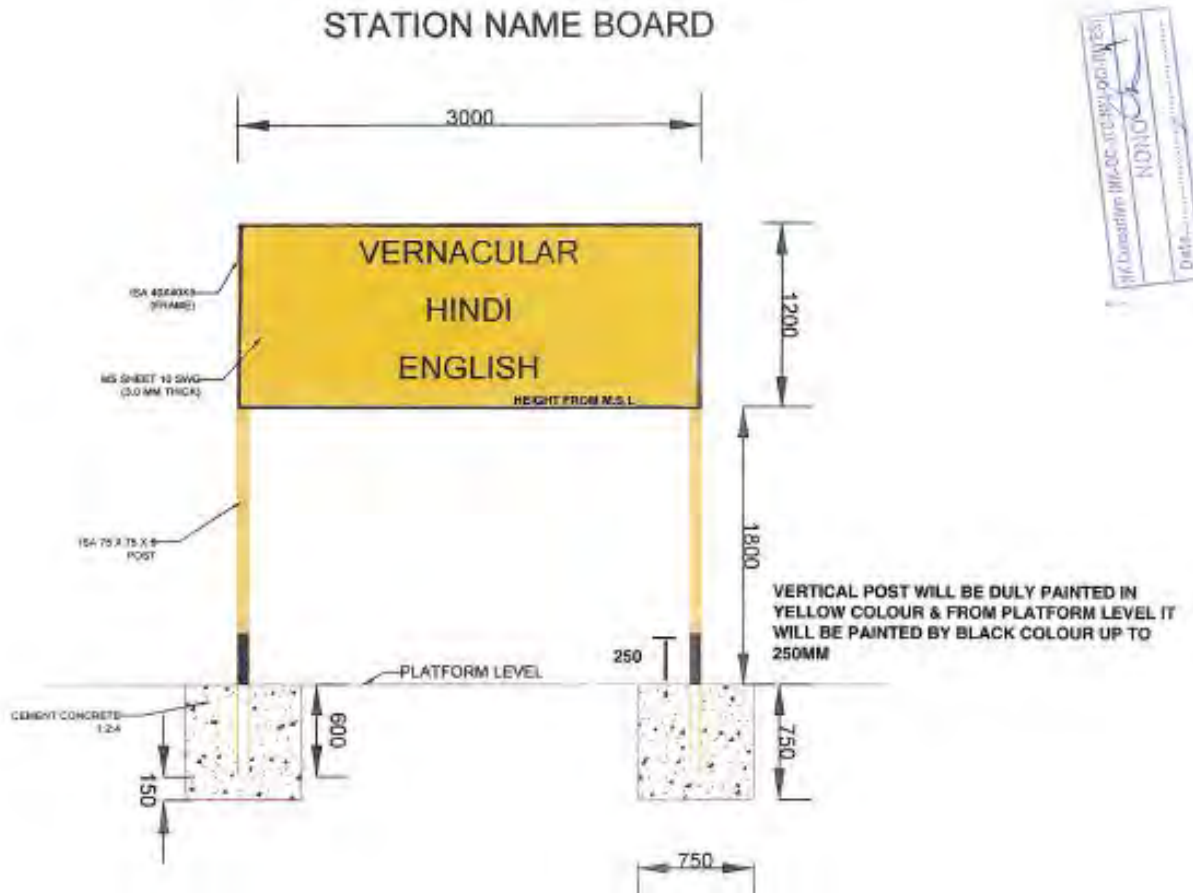
Fouling marks are located between two converging tracks within the ballast to indicate the safe stabling limit for trains approaching the convergence point. This point ensure that no contact is made with a train passing on the adjacent track. Its location shall be based on the width of the MMD as described in the Specifications between the two track centre lines.

2.51.7 STATION NAME BOARDS:

Station Name Boards (which are provided at both ends of the Station and at the Station Building) on which the name of the station shall be embossed onto a metal plate, or similar

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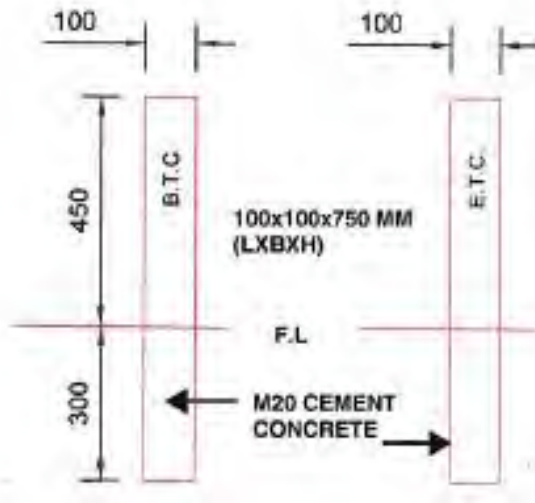
are placed at both ends of the Station at the main Station Building in the station area facing to the Main Line and shall comprise a metal plate, a metal frame supported on steel posts.



2.51.8 RAIL POSTS INDICATING TANGENT POINTS:

On the inside of the curve, rail posts are erected on each approach of the curve, to indicate the positions of the beginning and end of transition curves. These rail posts are painted in red and white colours respectively.

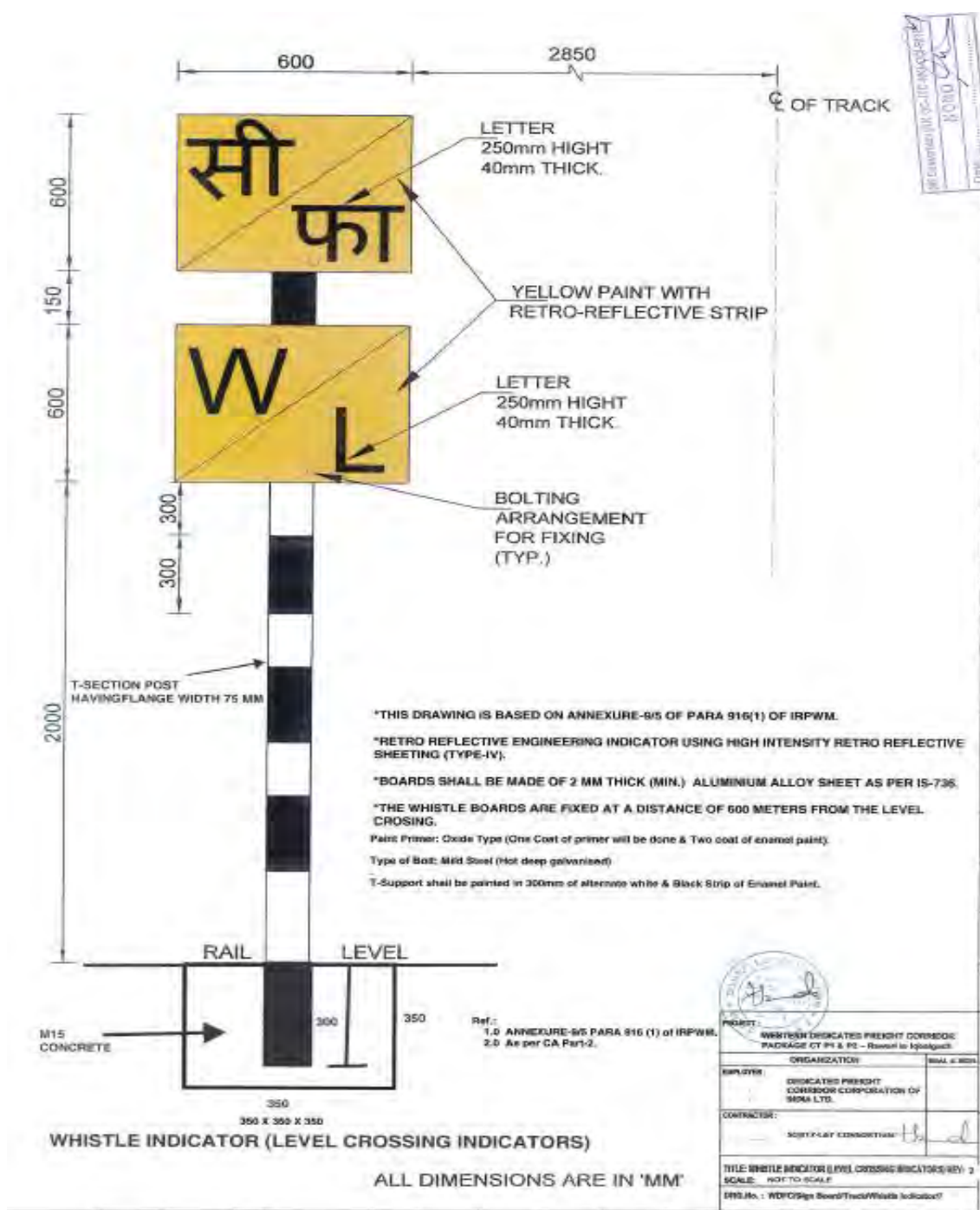
MANUAL FOR MAINTENANCE OF TRACK



2.51.9 WHISTLE INDICATOR:

Whistle boards shall be provided at the rear of all places where the view of the Drivers is obstructed by cuttings or tunnels or curves and where it is necessary to give audible warnings of the approach of a train to those working on the track. The whistle boards are fixed at a distance of 600 meters. The whistle indicator shall consist of two 600 mm square boards painted yellow and bearing letter W/L in black. Its height shall be 2100 mm from the rail level to the underside and the post on which it is fixed, painted with 300 mm high bands in white and black.

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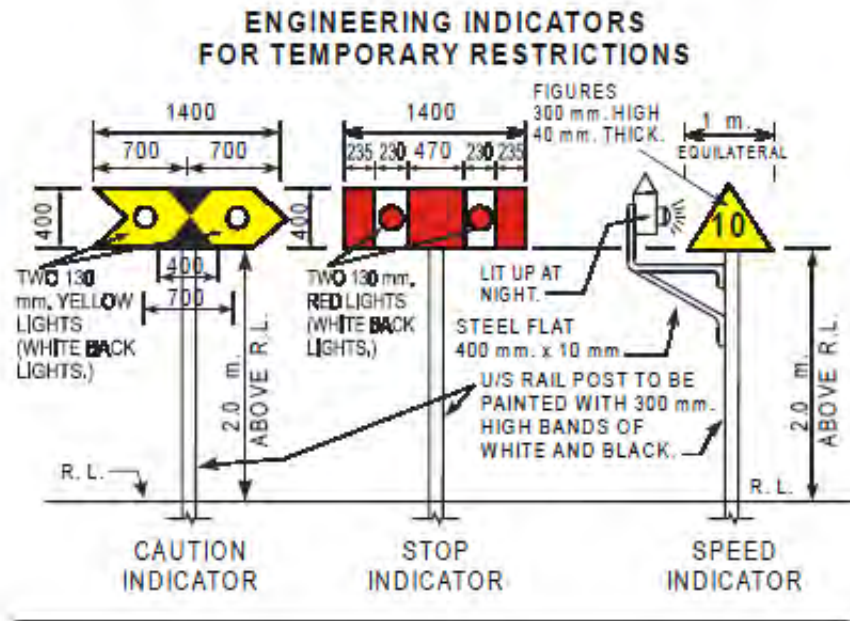
2.51.10 STOP BOARDS

Stop Boards shall be provided on the DFC side of the road approaches to all unmanned 'C' class Level Crossing. This shall consist of a board 675mm X 525mm on a suitable post bearing the indication of an engine and the legend '**Stop, look out for trains before crossing**'.

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2.51.11 ENGINEERING RESTRICTIONS BOARDS



2.51.12 SECTION LIMIT BOARDS

1. Boards at jurisdictional limits should be provided thus:

a) DFCC :- From Ch. to Ch.

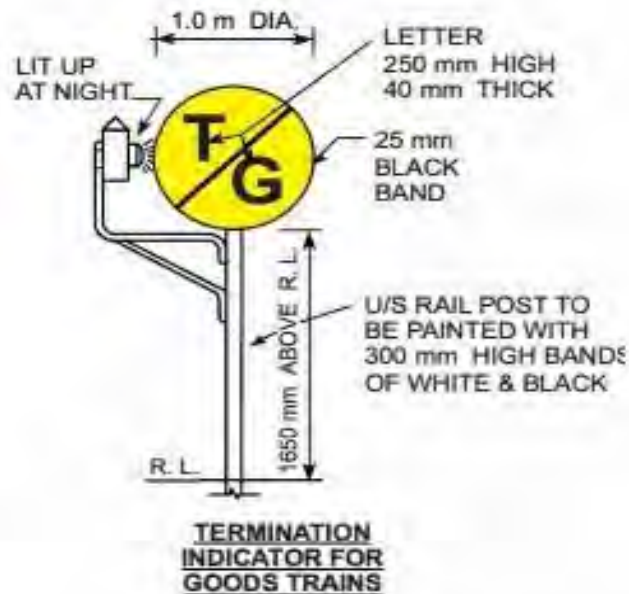
RMMU....	RMMU....
REGIONAL MANAGER.....	REGIONAL MANAGER.....
DEPOT MANAGER.....	DEPOT MANAGER.....
Sr. EXECUTIVE / EXECUTIVE TRACK....	Sr. EXECUTIVE / EXECUTIVE TRACK....

b) GANG LENGTH

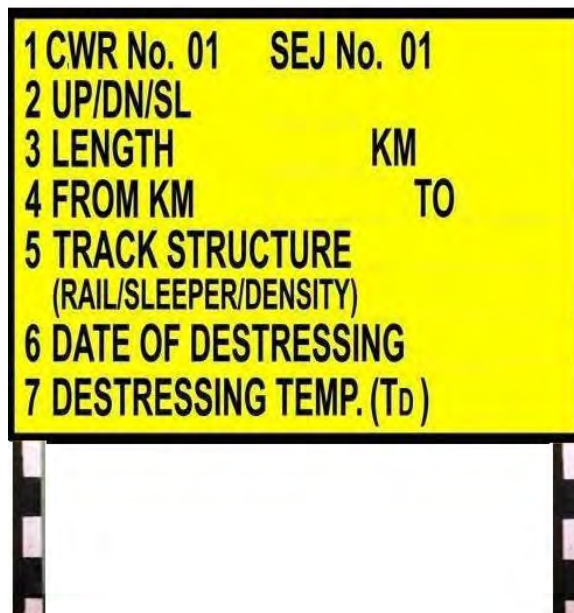
1. If the gang beat ends in a curve the beat should be so adjusted that the entire curve lies in one of the beats. Similarly, in case of yards gang beats should be so adjusted that the yard is maintained as far as possible by one gang, exception being in the case of big yards where the yard may have to be maintained by more than one gang.
2. Suitable boards should also be provided indicating the state and district boundaries.
3. When a board has to be located at an exact kilometre, it should be fixed by the side of the kilometre post.
4. The boards which may be of scrap iron or RCC should throughout a division be fixed on the cess on the same side of the line. The letters and figures should be painted black on white background.

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2.51.13 TERMINATION BOARDS



2.51.14 CWR BOARDS



2.51.15 SHUNTING LIMIT BOARDS

Shunting Limit Boards shall be provided at an adequate distance in advance of the trailing points. This shall consist of 600 mm. x 1000 mm. rectangular board painted yellow with a black cross on the top and words "shunting limit" written in black below it. Its height should

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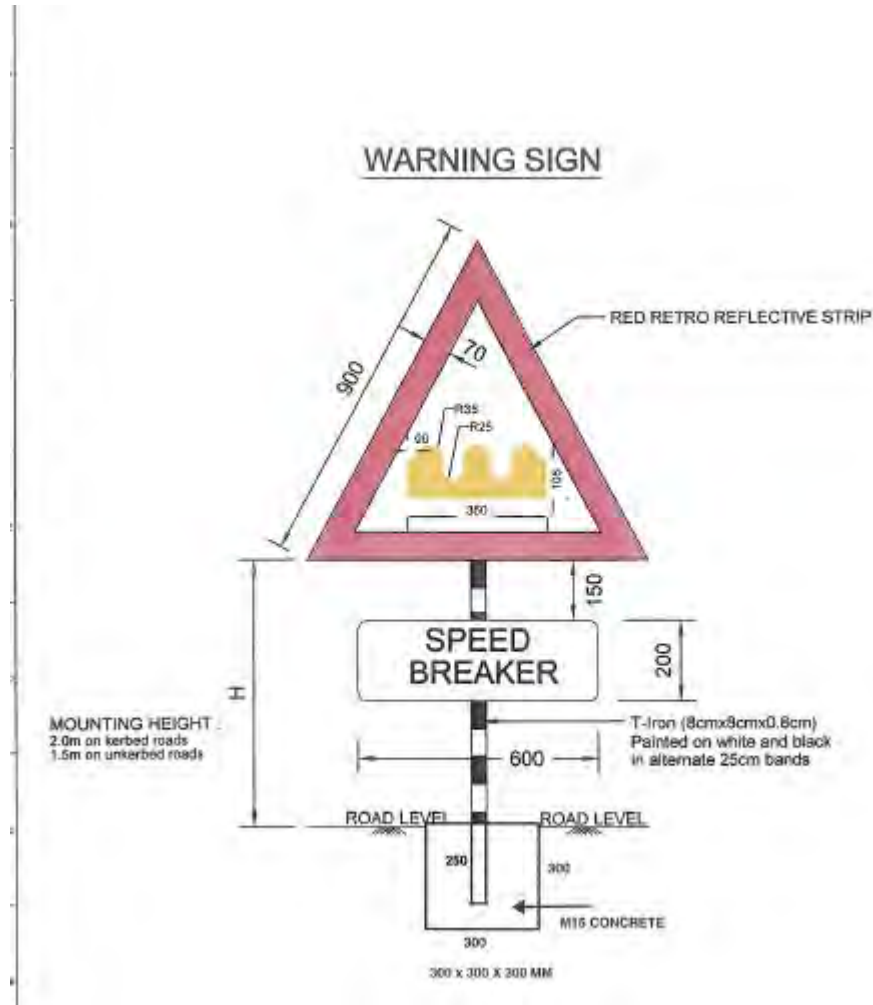
be 2 metres from the rail level to the underside of the portion containing the cross and the post on which it is fixed, painted with 300 mm. high bands in white and black. It shall be fitted with a lamp showing white light in both directions.

SHUNTING LIMIT BOARD



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2.51.16 ROAD SIGN BOARDS



HEIGHT GAUGE BOARD
50m FROM HEIGHT GAUGE



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AS PER MOTOR VEHICLE ACT-1988 (C-37)
60 m AWAY FROM PROP.L-XING BOARD



AS PER MOTOR VEHICLE ACT-1988(C-37)
200 m AWAY FROM PROP.L-XING BOARD



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PART F – TREATMENT OF FORMATION

To be followed as per Formation maintenance manual for CTP 1&2.

CHAPTER III - PERMANENT WAY RENEWALS

3.1 CLASSIFICATION OF RENEWALS

All the track renewals can be classified into one of the following categories:

- Complete Track Renewal (CTR)
- Through Track Renewal (TRR)
- Through Sleeper Renewal (TSR)
- Casual Renewals

3.2 FACTORS GOVERNING PERMANENT WAY RENEWAL

Shall be followed as per **IRPWM Clause 302.**

3.3 PLANNING OF RENEWALS

Shall be followed as per **IRPWM Clause 303.**

3.4 TRACK RENEWAL PROGRAMME

Shall be as per **IRPWM Clause 304.**

3.5 TRACK STANDARDS FOR RENEWALS

Shall be as per **IRPWM Clause 305.**

3.6 PLANNING FOR POSTING OF STAFF AND OTHER FACILITIES

To be described by DFCC

3.7 TRAFFIC FACILITIES FOR RENEWALS

To be described by DFCC

3.8 SPEED RESTRICTIONS

Speed Restrictions may be followed as per the **IRPWM Clause 308 Table II.**

3.9 PROJECT REPORT FOR TRACK RENEWAL WORKS & PRELIMINARY WORKS

To be followed as per **IRPWM Clause 309 A & 309 B.**

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3.10 UNLOADING OF RAILS, SLEEPERS AND FASTENINGS

To be followed as per **IRPWM Clause 310.**

3.11 METHODS OF CARRYING OUT RENEWAL

Relaying of track by mechanical method is to be adopted while carrying out renewal works.

3.12 RELAYING WITH MECHANICAL EQUIPMENT

To be followed as per **IPWM Clause 312.**

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3.13 TRACK LAYING STANDARDS

The track laying standards shall be as per the table given below:

Table 6 Track laying standards as per Employer's Requirements

Sl. No.	Description	Value
1	Maximum difference of any point in relation to the designed layout	a) Vertical : ± 10 mm b) Horizontal : ± 10 mm
2	Gauge (with reference to 1676 mm)	a) Maximum variation over the prescribed track gauge : +3mm to 0mm b) Maximum variation in track gauge from sleeper to sleeper : 2mm c) Average track gauge (mean over 100 m length); +1.8mm to 0mm
3	Misalignment :	± 5 mm
4	Vertical Un-evenness (left & right hand rails)	+2mm to (-) 1 mm
5	Maximum deviation of measured versine over its designed value on a 20 m chord (half overlapping)	± 5 mm
6	Cant/Cross Level (to be measured at every 4 th sleeper)	a) Straight track and curved track: ± 3 mm b) Sleeper to sleeper variation of cant/x-level : ± 1 mm
7	Twist (maximum value on base of 3 m)	a) Straight and circular portion of curve = ± 1 mm/m b) On transition portion of curve (over designed value) = +0.5mm/m
8	Turnouts	a) Stock rail joint (longitudinal location) : ± 15 mm b) Nose to nose of Xing in crossovers : ± 10 mm c) Flangeway clearance at the end of the switch planning : +5mm to (-) 0mm d) Switch toe opening : +1mm to (-) 0mm e) Switch toe squareness : 5 mm f) Deviation of measured versine over its designed value for switches, lead track and

3.14 MARKING OF PERMANENT WAY MATERIAL

To be done as per IRPWM Clause 321.

3.15 IDENTIFICATION OF DIFFERENT QUALITIES OF RAILS IN FIELD

To be done as per IRPWM Clause 322.

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3.16 WORKS TO BE ATTENDED AFTER COMPLETION OF RELAYING

To be done as per **IRPWM Clause 323.**

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CHAPTER IV – CURVED TRACK AND REALIGNMENT OF CURVES

4.1 INSTALLATION PROFORMA FOR CURVES

Table 7 Proforma for Recording Curve Installation

Section _____ Line (UP/DOWN/LOOP) _____ From _____ to _____

Sl. No. (1)	Curve No. (2)	Between Stations (3)	Kilometres		R.H.or L.H. (6)	Degree of curve (7)
			From (4)	To (5)		

Radius of curve (8)	S.E. (9)	Length of		Total Length of curve (12)	Whether joints are square or staggered (13)	Whether reference pillars have been provided (14)
		Transition (10)	Circular curve (11)			

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4.2 INSPECTION AND MAINTENANCE PROFOMA FOR CURVES

Table 8 Proforma for inspection and maintenance of curves

Line (UP/DOWN/LOOP) _____

Curve No. Degree of curve.....
From Km. To Km. Section.

Station No. (1)	Prescribed (Ideal)			Date of check (5)	Measurements recorded		
	V (2)	SE (3)	G (4)		V (6)	SE (7)	G (8)

Action to be taken (9)	Date of realignment or local adjustment (10)	Measurements recorded after adjustment with date of check		
		V (11)	SE (12)	G (13)

- Frequency of inspection- (ref. Para 420(2) of IRPWM)
- Instructions contained in Chapter IV for maintenance of curve track in IRPWM shall be followed.

CHAPTER V– WELDING OF RAILS: CONTINUOUS WELDED RAILS & TESTING OF RAILS

5.1 GENERAL

Welding of rails to achieve Continuous Welded Rails on DFCC track is done by:

1. Alumino – Thermic Welding
2. Flash – Butt Welding

5.2 ALUMINO THERMIC WELDING OF RAILS

AT shall be done with use of 3-piece mould (Zincon washed) compressed air petrol pre-heat single shot crucible with auto tapping for 60Kg 90 UTS rail and compressed air petrol pre heating two-piece mould for 60Kg HH rails. For combination AT welding of 60 Kg HH rails and 90 UTS rails 3 piece mould and single shot crucible with portion for 60 Kg 90 UTS rails (kept in green cover bags as per IRS T-19) and compressed air petrol pre-heating to be used for welding 60 Kg 90 UTS with 60 Kg HH rails.

5.2.1 PORTIONS TO BE USED

In order to have ease in identification of portions of various rail sections & grade as well as various techniques in field by welder, the packing of portions shall be done as under:

(i) The portion shall be packed in bags of different colour as per Rail Grade. The colour of bags containing portions shall be as per table below:

S. No.	Rail Grade	Colour
1.	90 UTS	Green
2.	Head Hardened	Yellow
3	Combination of 1080 HH and 880 grade	Green

(ii) The insignia of the firm and rail Section (60 Kg) should be printed in contrast colour on the exterior of the cloth bag of above mentioned colour so that it is clearly visible.

(iii) Till the portions with manual tapping system are being supplied, such portions shall be marked with Bold Letter “M” on the bag in same colour as used for marking the Rail Section. The letter size indicating Rail section & Tapping System shall be of minimum 150 mm high.

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(iv) Thermic portions confirming to 90 UTS rails, packed in green colour bags shall be used while joining the 90UTS rail with the 110 UTS rail. The welding shall be done at 25mm gap with 3 piece mould with compressed air petrol pre-heating system with single shot crucible fitted with Automatic tapping system.

5.2.2 TOLERANCES ON FINISHED WELDED JOINTS

Provisions of IRS T-19 and IR Manual on AT welding shall be followed.

5.2.3 WIDE GAP WELDING TECHNIQUE

Provisions of IRS T-19 and IR Manual on AT welding shall be followed.

5.2.4 AT WELDING OF DIFFERENT RAILS

Provisions of IRS T-19 and IR Manual on AT welding shall be followed.

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5.2.4.1 CHECKLIST FOR INSPECTION OF AT WELDS

CHECK LIST FOR INSPECTION OF A.T. WELD

SECTION..... RMMU KM/TP..... LINE:UP/DN/LOOP

DATE..... BLOCK TIME.....

1. Equipment are complete as per Annexure – 2 of A.T. welding Manual–2012 Yes/No
If no then deficiencies are

2. Composition of welding team:-

- a) Welder having competency certificate Yes/No
b) Welding Supervisor having competency certificate Yes/No
c) Welding being done departmental or Departmental/Contractual
through contractual agency
d) Name of Welder

3. Details of Rails being welded:

- a) Section & UTS
b) whether rail is free from corrosion or excess wear Yes/No
c) Whether rails has been USFD tested before welding. Yes/No
If yes, result of USFD testing. O.K./Not O.K.
d) Condition of rails Good/Satisfactory/Not satisfactory

4. Condition of Rail ends:

- a) Battering in mm
b) Hogging in mm
c) End cuts are truly vertical Yes/No
d) Ends are cleaned by K-oil & wire brush Yes/No
e) Whether fish bolt holes have been eliminated or not Yes/No

5. Portions details:

- a) Manufacturer
b) Date of manufacture
c) Condition of polythene packing Sealed/Broken
d) Condition of portion Wet/Dry
e) Portion no. & UTS

6. Gap in mm(25±1):mm

7. Alignment of rail with 1m straight edge:

- a) Lateral ± 0.5mm
b) For 90UTS/HH rails Higher (by 2-2.5mm)

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- | | |
|---|--------------------------|
| 8. Whether fastening of 5 adjacent sleepers removed: | Yes/No |
| 9. Mould shoes are fixed correctly
(should be perfectly vertical shall not be skew) | Yes/No |
| 10. Pressure in compressor tank ($7 \pm 0.7 \text{ kg/sq.cm}$): | |
| 11. Preheating time (in minutes): | |
| 12. Tapping time in seconds (20 ± 3): | |
| 13. Mould waiting time (4-6 minutes): | |
| 14. Trimming is done by: | Weld trimmer/Manual |
| 15. Wedge was removed after (20 minutes): | Minutes |
| 16. Grinding is done with the help of : | Mech. Grinder/Manual |
| 17. Tolerances of finished weld: | |
| a) Vertical alignment with 1m st.edge ($+1.0, - 0.0 \text{ mm}$) | |
| b) Lateral alignment with 1m st. edge ($\pm 0.5 \text{ mm}$) | |
| c) Finishing of top surface at the
end of 10cm st.edge ($+0.4, -0.0 \text{ mm}$) | |
| d) Finishing on sides of Rail head at the
centre of 10cm st.edge ($\pm 0.3 \text{ mm}$) | |
| 18. Weld is serially numbered as per
(Para 5.6 of A.T.Welding Manual) If
yes, then give the no. | Yes/No |
| 19. Whether USFD testing of weld done or not
If yes, result of USFD testing | Yes /No
O.K./Not O.K. |
| 20. Whether painting of thermit weld done or not | Yes/No |
| 21. Whether sample test joint made and
sent for testing or not (one out of 100 joints) | Yes/No |
| 22. Weld register is maintained as per manual (Annexure-1) | Yes/No |

Signature

Name of inspecting authority:

Designation:

Date:

5.3 FLASH BUTT WELDING OF RAILS

5.3.1 PRE-STRAIGHTENING OF RAILS

Provisions of IR Manual for Flash Butt Welding of Rails shall be followed.

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5.3.2 END RAIL CLEANING

Provisions of IR Manual for Flash Butt Welding of Rails shall be followed.

5.3.3 ELECTRICAL CONTACT

Provisions of IR Manual for Flash Butt Welding of Rails shall be followed.

5.3.4 RAIL ALIGNMENT

Provisions of IR Manual for Flash Butt Welding of Rails shall be followed.

5.3.5 WELDING SEQUENCES

Provisions of IR Manual for Flash Butt Welding of Rails shall be followed.

5.3.6 POST WELD HEAT TREATMENT

Provisions of IR Manual for Flash Butt Welding of Rails shall be followed.

5.3.7 FINISHING OF WELDS

Provisions of IR Manual for Flash Butt Welding of Rails shall be followed.

5.4 MARKING OF WELDS AT SITE

5.4.1 MARKING OF AT WELD AT SITE

Provisions of IRS T-19 and IR Manual on AT welding shall be followed.

5.4.2 MARKING OF FBW JOINT AT SITE

Provisions of IR Manual for Flash Butt Welding of Rails shall be followed.

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CHAPTER VI – TRACK RECORDING ANALYSIS AND MONITORING

To be carried out as per latest **Correction Slip 149** to IRPWM.

CHAPTER VII – ACTION DURING ACCIDENTS INCLUDING BREACHES AND PRE-MONSOON PRECAUTIONARY MEASURES

7.1 ACTIONS DURING ACCIDENTS INCLUDING BREACHES

Instructions contained in Part A & Part B of IRPWM shall be followed.

7.2 PATROLLING OF TRACK IN EMERGENCIES

Types of Patrolling:

- i) Keyman's daily patrol
- ii) Gang patrol during monsoon
- iii) Night patrolling during monsoon
- iv) Security patrolling during civil disturbances and for movement of VIP specials.
- v) Hot weather patrolling for long welded rails/continuous welded rails
- vi) Watchmen at vulnerable locations.

Details of the above mentioned patrollings can be referred from Chapter 10 of IRPWM.

ANNEXURE 10A, 10B, 10C of Manual of Instructions on LWR shall be followed which are as reproduced below:

7.2.1 HOT WEATHER PATROLLING:

Period for hot weather patrolling shall be laid down by the GM Infrastructure/ equivalent for each section and patrol charts prepared where necessary. Patrolling shall be organized by Sr. Executive / Executive (Track) accordingly. In addition, the Sr. Executive / Executive (Track) and the Gangmate shall be vigilant during summer and on hot days. Patrolling will also be introduced, when the rail temperature rises above $td+25^{\circ}\text{C}$. Following general guidelines may be followed for issuing detailed guidelines by the GM Infrastructure/ equivalent: -

1. Hot weather patrolling will be carried out as follows: -

- i) On single line or where only one road in a double line section is having LWR/CWR - One patrolman for 2 km.
- ii) On double line section when LWR/CWR exist on both roads - One patrolman for 1 km length of UP and DN road. The beats of each hot weather patrolman will thus be restricted to 2 km.

2. The hot weather patrolman should always carry the following equipment: -

- HS Flags - Red 2
- Staff for Flags 1

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- Detonators 10
- Canne-a-boule 1
- Patrolman number plate (for showing his presence in field for the officials carrying out foot plat inspections during hot weather)

3. Duties of hot weather patrolman are as follows: -

He will walk over his beat slowly over one rail in one direction and on the other rail in the return direction. On double lines, he will repeat this procedure alternately on UP and DN tracks. He will be vigilant and look out for kinks in the rail especially during the hottest part of the days. When a kink is observed, he shall immediately examine at least 100 sleepers ahead and in the rear of the kink for any floating condition of track. He should meticulously sound each and every sleeper, 100 sleepers on either side of the kink, to determine any floating condition. The sounding will be done by dropping a canne-a-boule on each end of the sleeper to determine the extent of void under the sleeper. Should the sounding reveal a floating condition, under which a buckle may be anticipated or the patrolman has detected actual buckling of track, he will take immediate steps to protect the affected portion by display of hand signals as per rules in force. After protecting the track, the patrolman will arrange to advise the Gangmate, Sr. Executive / Executive (Track) of his apprehension of a buckle/actual buckle.

4. The Gangmate on receipt of advice of a danger of buckle, will proceed to the site quickly with all available men. On arrival at site, he will first ensure protection of affected portion. He should then inspect the condition of track 100 m on either side of this suspected zone and commence heaping of surplus ballast, if available, on the shoulders and up to the rail head and keep on compacting the ballast with available tool. **No attempt should be made to slew or align the track or disturb the existing ballast section.** The mate should continue to remain at site till the arrival of Foreman / Sr. Executive / Executive (Track). The rail temperature will also be noted by one of these officials at the place of apprehended/ actual buckle. The rail facing the sun will be covered up to the level of rail head on the outside by ballast or leaves etc. to bring down the temperature of the rail.

7.2.2 COLD WEATHER PATROLLING:

Cold weather patrolling shall be introduced when rail temperature is less than $t_d - 30^{\circ}\text{C}$. Period and section where cold weather patrolling is to be done shall be laid down by the GM Infrastructure and patrol charts prepared where necessary. Patrolling shall be organised by Sr. Executive / Executive (Track) accordingly. Following guidelines may be followed for issuing detailed guidelines by the GM Infrastructure.

1. Cold weather patrolling shall be carried out as follows: -

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- i) On single line or where only one line in a double section is having CWR – One patrolman for five kilometres.
- ii) On double line section when CWR rest on both rail line - One patrolman for three kilometres length of UP and DN road.
- iii) Changes in beat length and man power deployment as given above if found necessary, may be decided by the GM Infrastructure depending on prevailing local conditions, frequency of train service, weather conditions etc.

2. Cold weather patrolman should carry the following equipment: -

- a) 10 fog signals in a tin case
- b) Two tri-colour hand signal lamps
- c) One match box
- d) Two red flags and one green flag
- e) One three-cell electric torch
- f) One staff
- g) Number plate

3. Duties of cold weather patrolman are as follows: -

He will walk over his beat slowly along one rail in one direction and on the other rail in the return direction. On double line, he will repeat this procedure alternately on UP and DN tracks. He will be vigilant and look out for rail/weld failure. He will also notice the gaps at SEJs if they fall in his beat. In case he notices a rail/weld failure or gap at SEJ becomes more than the designed maximum gap, he will take immediate action to suspend the traffic and protect the line. After protecting the track, the patrolman will arrange to report to Keyman/ Gangmate/ Foreman / Sr. Executive / Executive (Track), who shall arrange for making emergency repairs to pass the traffic immediately.

7.2.3 MOBILE WATCHMAN:

- 1. Mobile watchmen shall be posted, till period of consolidation is over, on stretches where after maintenance operation, rail temperature has exceeded $t_d + 20^{\circ}\text{C}$ during the period of consolidation as given in para 1.18 of IR LWR manual. They would be withdrawn when the period of consolidation is over and the track behaves satisfactorily.
- 2. Number of Mobile watchman required to be posted would depend upon the length of track attended by manual or by tampers. Normally one Mobile watchman would be sufficient for one kilometer of track.

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3. Mobile watchman will be provided with all equipment as per para 2 of Cl. 7.2.1 and 7.2.2 of this manual as the case may be.
4. Mobile watchman posted on these sites would patrol the section, which has been freshly packed and for which period of consolidation is not over. He shall be vigilant and will look out for incipient buckles and kinks in the rails. He will also observe for rail/weld failures and gaps at SEJs.
5. When a kink is observed, he will take action as specified in Para 3 of 7.2.1 of this manual.
6. If a rail/weld failure is observed or gaps at SEJs become more than the designed maximum gap of the SEJs, action as specified in Para 3 of Cl. 7.2.2 of this manual will be taken.

CHAPTER VIII - LEVEL CROSSING AND GATEMAN

To be followed as per approved working rule.

CHAPTER IX – PROCEDURE FOR OBTAINING SANCTION AND CARRYING OUT WORKS AFFECTING SAFETY OF THE RUNNING LINE AND FOR OPENING NEW WORKS

To be followed as per **IRPWM Chapter 13.**

CHAPTER X – TRAINING FOR PERMANENT WAY STAFF

To be followed as per **IRPWM Chapter 15.**

SECTION 3 – ANNEXURES

MANUAL FOR MAINTENANCE OF TRACK

C. ANNEXURES

ANNEXURE 1 LIST OF DRAWING NUMBERS OF VARIOUS COMPONENTS AND THEIR SOURCE

Sl.No.	Description	Drawing Number	Source
1	Check Rail for Points and Crossings for 1 in 8.5 T/outs	Drawing number VAEVKN05126-02 to VAEVKN05128-02	VAEVKN
2	Check Rail for Points and Crossings for 1 in 12 T/outs.	Drawing number VAEVKN05019-02 to VAEVKN05021-02	VAEVKN
3	Guard Rail	UIC 60	SAIL, JSPL
4	Normal Sleepers	RDSO-T/7008	RDSO approved vendor
5	Turnout sleepers for 1 in 12	VAEVKN 04617-08	VAEVKN
6	Turnout sleepers for 1 in 8.5	VAEVKN 05117-08	VAEVKN
7	Sleepers for derailing switches- RH	VAEVKN 07152-01	VAEVKN
8	Sleepers for derailing switches- LH	VAEVKN 07153-01	VAEVKN
9	Sleepers for SEJ	RDSO/T-8224	RDSO approved vendor
10	Guard rail sleepers	RDSO/ T-8228	RDSO approved vendor
11	Bridge Approach sleepers	RDSO/ T-8229 to 8237	RDSO approved vendor
12	Level Crossing sleepers	RDSO/T-8225	RDSO approved vendor
13	ERC-Mark V	RDSO/ T-5919.	RDSO approved vendor
14	Metal Liners	RDSO/T-8254 & T-8255.	RDSO approved vendor
15	GFN Liners	RDSO/T-8222 & T-8223.	RDSO approved vendor
16	CGRSP	RDSO/T-7010.	RDSO approved vendor

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19	Turnout for 1 in 12 with rollers	VAEVKN 04617-08	VAEVKN
20	Turnout for 1 in 12 without rollers	VAEVKN 09047-00	VAEVKN
21	Turnout for 1 in 8.5	VAEVKN 05117-08	VAEVKN
22	Derailing switches-RH	VAEVKN 07152-01	VAEVKN
23	Derailing switches-LH	VAEVKN 07153-01	VAEVKN
24	Improved Switch Expansion Joint	VAEVKN08055-00 with 80mm gap	VAEVKN
25	Improved Switch Expansion Joint	VAEVKN 05689-00	VAEVKN
26	Rail (60 kg, 1080 HH 60E1)		NSSMC
27	Rail (60 kg, 880 Gr, 90 UTS)		SAIL, JSPL

MANUAL FOR MAINTENANCE OF TRACK

ANNEXURE 2 INSTALLATION AND MAINTENANCE OF PIROLLS (PARA. 2.22.4.2)

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Installation, Operation and Maintenance Manual PIROLL

Doc. No.: - TD-519

voestalpine VAE VKN India Pvt. Ltd.

<http://www.voestalpine.com/vaevkn/en/>

voestalpine
ONE STEP AHEAD.

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Caption	Name	Department	Date	Signature
Created	Shubham Tyagi	QA	22/11/2019	Shubham Tyagi
Checked	Govinda Raj	QA & Technical	23/11/2019	Govinda Raj
Approved	Govinda Raj	QA & Technical	23/11/2019	Govinda Raj

VERSION HISTORY:

Version No.	Modified	Date	Approved	Comment

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Version No.	Company	Copies	Date	Comment

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1. Safety Instructions

The area of tracks and especially the area of switch devices and crossings must be considered as a special dangerous area to perform measurements and maintenance work. By entering such areas, the operation- and safety instructions of the railway organization and the instructions of the railway personnel in charge must be strictly observed. Inspections and maintenance work can be done during track traffic, but the workers must be well informed about the time schedule and the time left to the next train by safety personnel at the working site. If maintenance activities or measurements will be performed in switch devices / moveable crossings it has to be ensured by preventive measures that an unexpected movement of points will not occur. Hands, fingers and feet shall, if possible, never be put between moveable parts of turnouts, an unexpected movement will cause grave bodily harm.

Continuously welded rails may be under high tension - or pressure forces, depending on the actual rail temperature. During cutting of such rails for exchange of parts or installation of insulated joints, unexpected sudden movements of the separated rail ends may occur, if equalising measures have not been taken. At separating, drilling, chamfering of rails, eyes and skin may be injured by cut chips, this can be avoided by wearing personal protective equipment as listed below.

Admittance to track and turnout areas may only be given if the safety of maintenance personnel is provided by adequate measures, such as safety guards and if necessary temporary closing of road traffic.

Please consider:

- Risk and safety precautions
- Hazards associated with work activities
- Miscellaneous body injuries resulting from being struck by moving railway vehicles
- Getting hands or feet caught during installation of track work material
- Observe local safety instructions for working near or on the track
- Do not get in contact of skin and eyes with resin and hardener
- Do not breathe in exhalation from resin and hardener (Fabrication of insulated rail joints)
- Personal protective equipment
 - Wear high visibility clothing and safety shoes/-boots
 - Wear helmets when using lifting gear
 - Wear safety gloves
 - Wear safety glasses and ear plugs in case of grinding works
 - Personal protective equipment for welding tasks

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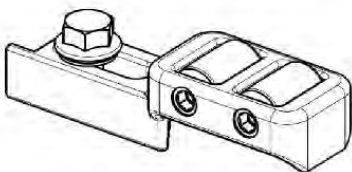
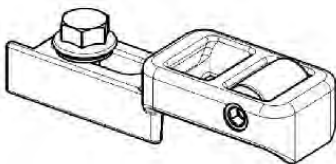
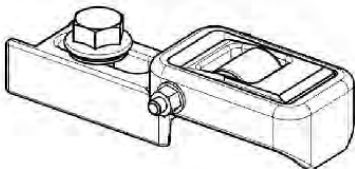
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2. Part Description

2.1 PIROLL

PIROLL Types

TYPE	Function and Characteristics
TYPE 1R (L): with 2 rollers 	Lifts the switchblade during setting movement min. 1mm to max 2,5mm from the slide chair respectively the slide plate. For switch opening from 0mm to 170mm.
TYPE 2R (L) 	Lifts the switchblade during setting movement min. 1mm to max 2,5mm from the slide chair respectively the slide plate. For switch opening from 0mm to 70mm.
TYPE S (Spring) R(L) 	For installation in the area of the switch heel. Lifts the switch blade max. 1mm from the slide chair respectively slide plate – spring supported.

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All PIROLL types (1R (L), 2R (L) & S (R/L)) are mounted directly on the base plates in case of turnouts with flexible switches and spring-rail switches, if the slide chair height is min. 30mm. Base plates with mounted PIROLL do not project over the outer edge of the sleepers. On all turnouts with rail switches the PIROLL types (1R (L), 2R (L) & S (R/L)) are mounted vertically displaced on the base plate directly beside the sleeper.

For mounting of the PIROLL threaded holes have to be provided on the base plates.

The switch point roller PIROLL ensures that the slide chairs and slide plates in the area of the switch device do not have to be lubricated anymore.

The switches roll on the rollers mounted in the PIROLL holder during the setting operation. Therefore, the sliding friction is replaced by the minor rolling friction.

The PIROLL type S (spring) lies under the foot of the switch when the switch is closed and opened and supports the switch in the area of the very short horizontal setting distances in a spring-loaded way, i.e. the full wheel load is carried via the slide chairs and slide plates and not via the rollers.

It is not obligatory that all of above mentioned types are used in a switch device. The exact positions and type can be found on the layout drawings.

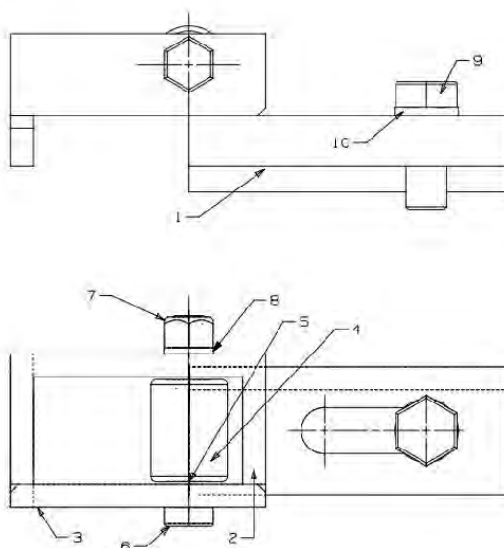
Typical arrangement drawing: VAEVKN04757-07.

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PIROLL with one Roller



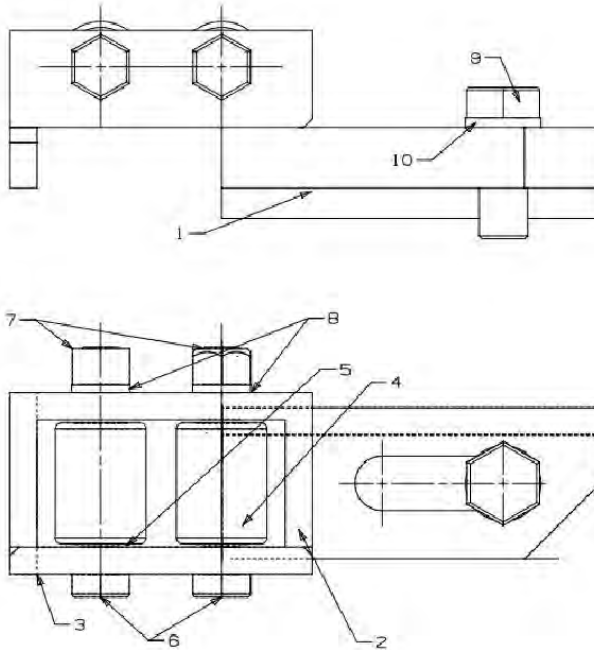
1. PIROLL Frame 01
2. PIROLL Frame 02
3. PIROLL Frame 03
4. PIROLL Rollers
5. PIROLL Bush
6. Screw M 12X75-10.9
7. Nut M12-10
8. Washer NL12
9. Hex Bolt with flange M 16x40-8.8 zinc coated SW24
10. Key Locking washer bonded NL 16F

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PIROLL with Two Roller



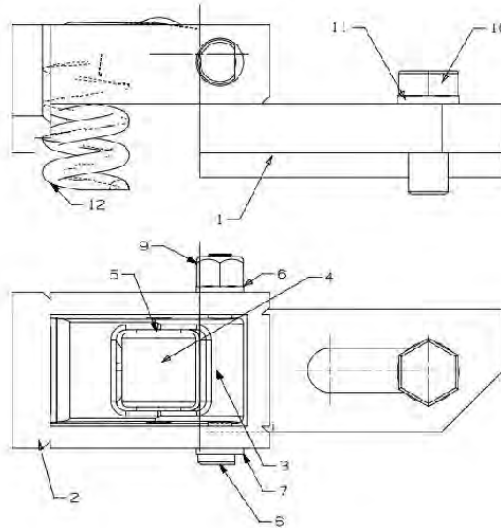
1. PIROLL Frame 01
2. PIROLL Frame 02
3. PIROLL Frame 03
4. PIROLL Rollers
5. PIROLL Bush
6. Screw M 12X75-10.9
7. Nut M12-10
8. Washer NL12
9. Hex Bolt with flange M 16x40-8.8 zinc coated SW24
10. Key Locking washer bonded NL 16F

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PIROLL S (Spring)



1. PIROLL Frame 01
2. PIROLL Frame 02
3. Rollentraeger PIROLL Spring AS
4. ROLL loose
5. ROLLER Pin
6. Thread Bolt with head D10/M8x78
7. Washer ISO 7090-10-200HV zinc coated
8. Washer ISO 7090-8-200HV zinc coated
9. Hex Nut ISO 10511 M8-10A2P
10. Hex Bolt with flange M16x40-8.8 zinc coated SW24
11. Key Locking washer bonded NL 16-F
12. Helical Spring

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3. Installation/Removal Piroll

Preparation:

- Clean the contact area on baseplates for the roller units.
- Check the slide chair support of the switch rails. Correct, if the switch rail is not supported continuously.
- Completeness check of items according to packing list / part list
- Remove the protective plugs from the threaded holes
- Clean and lubricate the threaded holes

Tools needed for installation/maintenance:

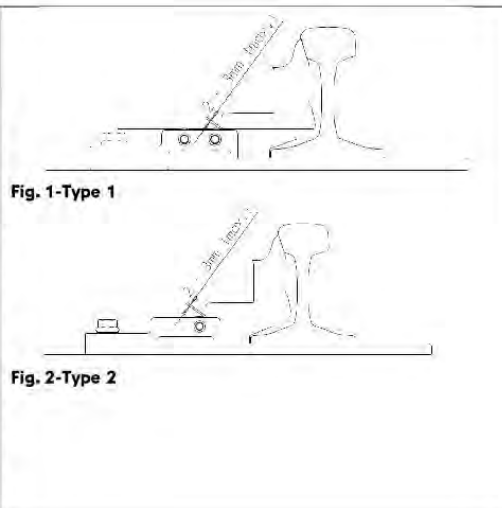
- Torque wrench
- Socket, width across flats 24mm
- Spanner, width across flats 13 and 24mm
- Hammer
- Feeler gauge, 1 to 2mm
- Screwdriver flat, for bearing bolts with slot (supplies up to 2010)
- Hexagon socket wrench, for bearing bolts with hexagon socket, width across flats 10mm (supplies since 2010)
- Mechanical blocking of the open switch (e.g. by wooden post, turnout locking device)

3.1.1 Installation

For the arrangement (position) of the PIROLL the VAEVKN turnout drawing plan, respectively the site specific layout plan has to be respected.

PIROLL TYPE 1 to 3

1. Lock and secure the turnout.
2. Place the PIROLL according layout plan on the inner side of the track close to their assembly position.
3. Fasten the PIROLL (Type 1 to 2) on the side of the closed switch blade with the hexagon bolt in way that the roller has a gap of 2 to 3mm to the rail foot of the blade.
4. If applicable continue with PIROLL type S (Spring).
5. Tighten all hexagon bolts M16 with a fastening torque of 200Nm.
6. Set the turnout to the other side and continue installation as per point 1 to 5.
7. To assure the exact position of the PIROLL, the torque of the M16 bolt has to be checked and, if required, adjusted once after opening after approximately 1.000 train passages respectively latest 14 days, depending on the traffic density.



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PIROLL TYPE S (Spring)

1. Lock and secure the turnout.
2. Place the PIROLL according layout plan on the inner side of the track close to their assembly position.
3. Place PIROLL type S (Spring) on the side of the closed switch blade on the base plate and shift the roller unit towards the stock rail in horizontal direction until the roller is positioned in the center of the rail foot, then fix with hexagon bolt M16.
4. Tighten all hexagon bolts M16 with a fastening torque of 200Nm.
5. Set the turnout to the other side and continue installation as per point 1 to 4.
6. To assure the exact position of the PIROLL, the torque of the M16 bolt has to be checked and, if required, adjusted once after opening after approximately 1.000 train passages respectively latest 14 days, depending on the traffic density.

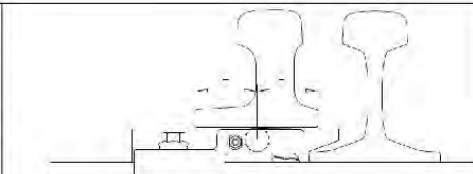


Fig. 4-Type S (Spring)

3.1.2 Removal:

1. Loosen and remove the M16 hexagon bolt fixing the roller unit on the base plate.
2. Remove roller unit.

4. Maintenance PIROLL

The installation of the PIROLL system cannot compensate the condition poor maintained turnout.

The used roller material and the lubrication-free bushing material is resistant to environmental influences like heat, cold or dirt. To avoid material damage, the PIROLL must not be heated above 120°C.

When installing the turnout into the track, i.e. during tamping works or maintenance tamping work, the PIROLL can remain mounted on the base plates!

Tools and consumables needed for maintenance:

- Torque wrench
- Socket, width across flats 24mm
- Spanner, width across flats 13 and 24mm
- Hammer
- Feeler gauge, 1mm to 2mm
- Screwdriver flat, for bearing bolts with slot (supplies up to 2010)
- Hexagon socket wrench, for bearing bolts with hexagon socket, width across flats 10mm (supplies since 2010)
- Mechanical blocking of the open switch (e.g. by wooden post, turnout locking device)
- Metal glue, in case exchanging rollers

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4.1.1 Adjustment of PIROLL, Type 1 to 2

1. Loosen the hexagon bolt M16.
2. Shift the roller unit horizontally to the proper position.
3. The correct position of the roller is 2 to 3mm away from the rail foot of the switch blade.
4. Fix the hexagon bolt with a torque of 200Nm.

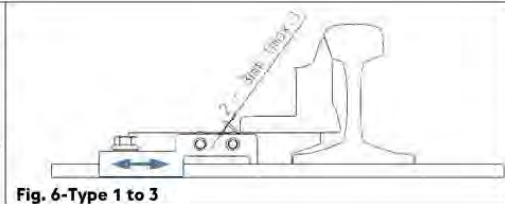


Fig. 6-Type 1 to 3

4.1.2 Adjustment of PIROLL, Type S (Spring)

1. Loosen the hexagon bolt M16.
2. Shift the roller unit horizontally to the proper position.
3. The correct position of the roller is centered under the rail foot.
4. Fix the hexagon bolt with a torque of 200Nm.

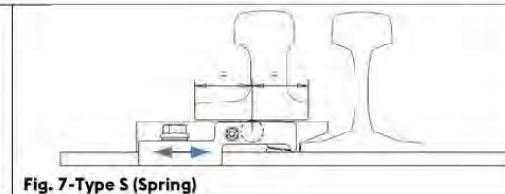


Fig. 7-Type S (Spring)

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4.1.3 Adjustment of height

1. Mark the position of the roller unit if OK. If the position is not OK, re-adjusted it after installation of shims (see chapter installation PIROLL).
2. Unscrew the hexagon bolt M16 and remove the roller unit.
3. Insert a height adjustment shim.
4. Position the roller unit properly and fix with hexagon bolt M16, fastening torque 200Nm.

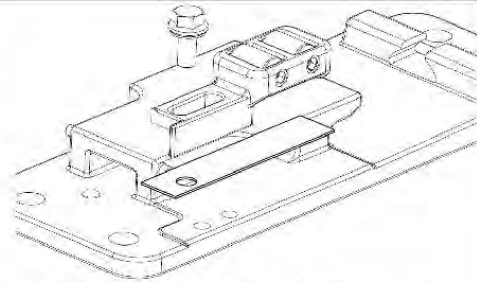


Fig. 11- Height adjustment (Assembly shown without rails and fastening material)

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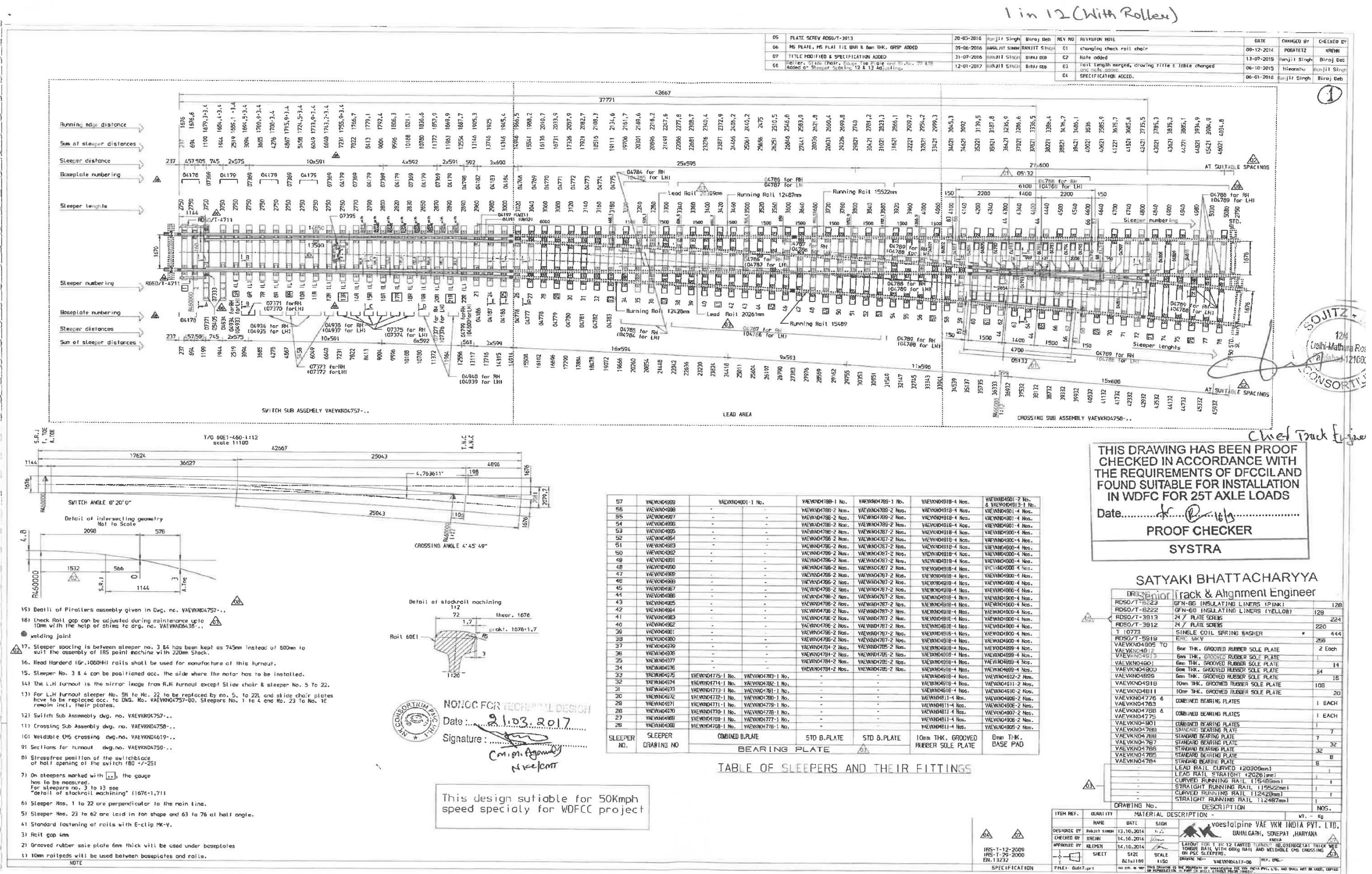
5. Maintenance Schedule

The given inspection and maintenance intervals are averaged values and have to be adapted by the railway operator in case of excessive load or severe environmental conditions, respectively could eventually be extended when turnouts are not used very often.

PIROLL		
Item serviced	Procedure	Recommended interval
Visual inspection	- Visual inspection of the roller system regarding existence of parts, especially fastening material. - Missing bolts to be replaced immediately to avoid increased setting forces. - Broken or damaged components to be replaced.	6 months
Position of rollers	Gap between rail foot and roller approx. 2 to 3mm PIROLL Types 1 to 2, centered position of roller under switch rail foot in case of PIROLL Type S (Spring).	Max. 12 months or ca. 100,000 turnout settings
Flat spots	- The rollers have to be checked for flat spots. In case of signs of wear on the slide chairs in the moveable part of the switch the rolls need to be exchanged. - Trace of contact on the slide chair below the closed switch is normal (the switch is carried by the slide chair in the closed position) and is not a reason for changing the adjustment of the roller or replacing the roller.	Max. 12 months or ca. 100,000 turnout settings
Lift of switch	- The open switch blade shall continuously be lifted min. 0,1mm from the slide chairs. (Adjustment normally during installation). - If this is not the case use height adjustment shims to lift the rollers. - In case of Type S (Spring) the gap between rail foot and roller has to be approx. 1mm. - The height of worn Rollers can NOT be compensated with height adjustment shims. In this case the Rollers have to be replaced.	6 to 12 months or in case of increased setting force

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ANNEXURE 3 DRAWINGS OF TURNOUT LAYOUT AND DS (PARA. 2.22.4.3)

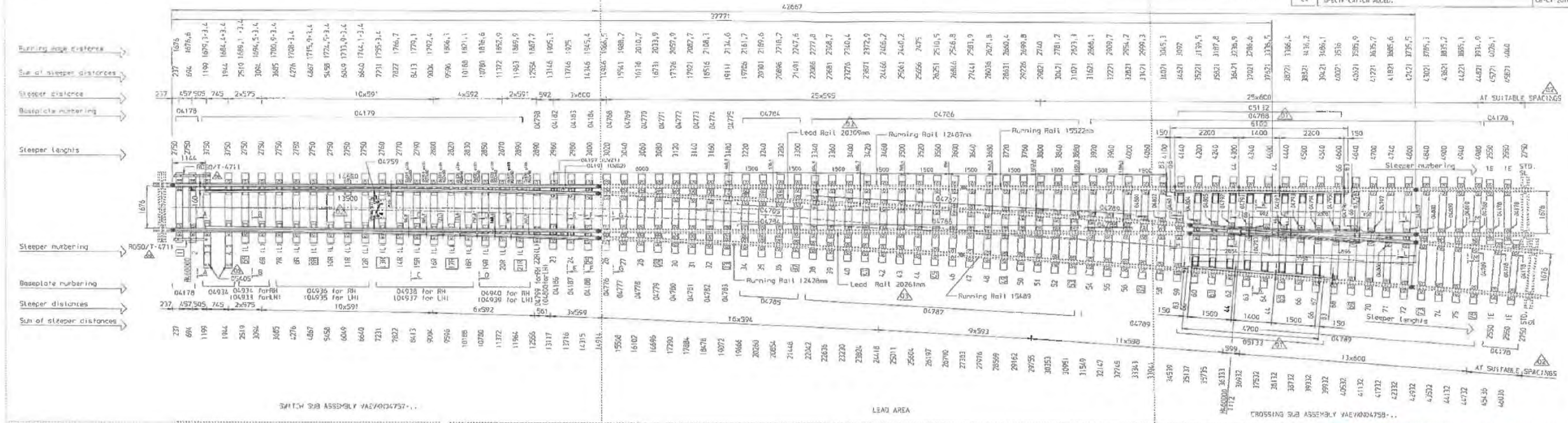


1 in 12 (Without Roller)

Annexure-3

1

05	PLATE SCREW R50/T-3913	26-05-2016	Ranjit Singh	Biraj Das	REV 1.0	REVISION NOTE	DATE	CHANGED BY	CHECKED BY
06	MS PLATE, MS PLAT TIE BAR & 6mm R-X, GRSP ACCED	26-06-2016	RANJIT SINGH	BIRAJ DAS	01	changing check rail chair	09-12-2014	PODARTEZ	IRENN
07	FILE MODIFIED & SPECIFICATION ADDED	11-07-2016	RANJIT SINGH	BIRAJ DAS	02	Note added	11-07-2015	Ranjit Singh	Biraj Das
					03	Rail Length changed, drawing title & table changed	06-10-2015	Ranjit Singh	Biraj Das
					04	SPR. SOLE PLATE	06-07-2016	Ranjit Singh	Biraj Das
					05	SPECIFICATION ADDED			



THIS DRAWING HAS BEEN PROOF CHECKED IN ACCORDANCE WITH THE REQUIREMENTS OF DFCCIL AND FOUND SUITABLE FOR INSTALLATION IN WDFC FOR 25T AXLE LOADS

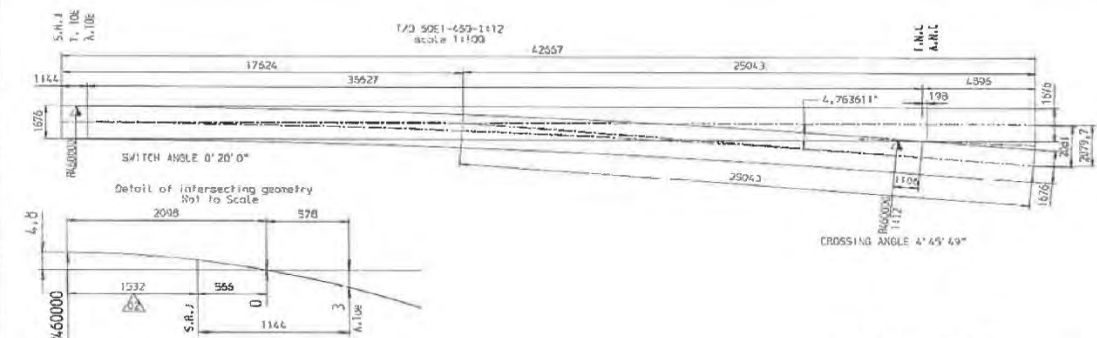
Date:.....
PROOF CHECKER
SYSTRA
SATYAKI BHATTACHARYA
 Senior Track & Alignment Engineer
SYSTRA

ORG. NO.	DESCRIPTION	QTY
R50/T-8223	GFN-68 INSULATING LINERS (PINK)	128
R50/T-8222	GFN-65 INSULATING LINERS (YELLOW)	128
R50/T-3913	24 / PLATE SCREWS	224
R50/T-3912	24 / PLATE SCREWS	224
T-10723	SINGLE COIL SPRING WASHER	224
R50/T-3913	24 / PLATE SCREWS	224
VAEVND4905	8mm THK. GROOVED RUBBER SOLE PLATE	2 Each
VAEVND4906	8mm THK. GROOVED RUBBER SOLE PLATE	14
VAEVND4907	8mm THK. GROOVED RUBBER SOLE PLATE	64
VAEVND4908	8mm THK. GROOVED RUBBER SOLE PLATE	18
VAEVND4909	10mm THK. GROOVED RUBBER SOLE PLATE	108
VAEVND4910	10mm THK. GROOVED RUBBER SOLE PLATE	20
VAEVND4911	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4912	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4913	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4914	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4915	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4916	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
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VAEVND4921	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
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VAEVND4988	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4989	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4990	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4991	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4992	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4993	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4994	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4995	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4996	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4997	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4998	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND4999	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH
VAEVND5000	10mm THK. GROOVED RUBBER SOLE PLATE	1 EACH

SLEEPER NO.	DRAWING NO.	COMBINED B.P. PLATE	STD B.P. PLATE	STD B.P. PLATE	10mm THK. GROOVED RUBBER SOLE PLATE	8mm THK. BASE PAD
01	VAEVND4901	-	-	-	-	-
02	VAEVND4902	-	-	-	-	-
03	VAEVND4903	-	-	-	-	-
04	VAEVND4904	-	-	-	-	-
05	VAEVND4905	-	-	-	-	-
06	VAEVND4906	-	-	-	-	-
07	VAEVND4907	-	-	-	-	-
08	VAEVND4908	-	-	-	-	-
09	VAEVND4909	-	-	-	-	-
10	VAEVND4910	-	-	-	-	-
11	VAEVND4911	-	-	-	-	-
12	VAEVND4912	-	-	-	-	-
13	VAEVND4913	-	-	-	-	-
14	VAEVND4914	-	-	-	-	-
15	VAEVND4915	-	-	-	-	-
16	VAEVND4916	-	-	-	-	-
17	VAEVND4917	-	-	-	-	-
18	VAEVND4918	-	-	-	-	-
19	VAEVND4919	-	-	-	-	-
20	VAEVND4920	-	-	-	-	-
21	VAEVND4921	-	-	-	-	-
22	VAEVND4922	-	-	-	-	-
23	VAEVND4923	-	-	-	-	-
24	VAEVND4924	-	-	-	-	-
25	VAEVND4925	-	-	-	-	-
26	VAEVND4926	-	-	-	-	-
27	VAEVND4927	-	-	-	-	-
28	VAEVND4928	-	-	-	-	-
29	VAEVND4929	-	-	-	-	-
30	VAEVND4930	-	-	-	-	-
31	VAEVND4931	-	-	-	-	-
32	VAEVND4932	-	-	-	-	-
33	VAEVND4933	-	-	-	-	-
34	VAEVND4934	-	-	-	-	-
35	VAEVND4935	-	-	-	-	-
36	VAEVND4936	-	-	-	-	-
37	VAEVND4937	-	-	-	-	-
38	VAEVND4938	-	-	-	-	-
39	VAEVND4939	-	-	-	-	-
40	VAEVND4940	-	-	-	-	-
41	VAEVND4941	-	-	-	-	-
42	VAEVND4942	-	-	-	-	-
43	VAEVND4943	-	-	-	-	-
44	VAEVND4944	-	-	-	-	-
45	VAEVND4945	-	-	-	-	-
46	VAEVND4946	-	-	-	-	-
47	VAEVND4947	-	-	-	-	-
48	VAEVND4948	-	-	-	-	-
49	VAEVND4949	-	-	-	-	-
50	VAEVND4950	-	-	-	-	-
51	VAEVND4951	-	-	-	-	-
52	VAEVND4952	-	-	-	-	-
53	VAEVND4953	-	-	-	-	-
54	VAEVND4954	-	-	-	-	-
55	VAEVND4955	-	-	-	-	-
56	VAEVND4956	-	-	-	-	-
57	VAEVND4957	-	-	-	-	-
58	VAEVND4958	-	-	-	-	-
59	VAEVND4959	-	-	-	-	-
60	VAEVND4960	-	-	-	-	-
61	VAEVND4961	-	-	-	-	-
62	VAEVND4962	-	-	-	-	-
63	VAEVND4963	-	-	-	-	-
64	VAEVND4964	-	-	-	-	-
65	VAEVND4965	-	-	-	-	-
66	VAEVND4966	-	-	-	-	-
67	VAEVND4967	-	-	-	-	-
68	VAEVND4968	-	-	-	-	-
69	VAEVND4969	-	-	-	-	-
70	VAEVND4970	-	-	-	-	-
71	VAEVND4971	-	-	-	-	-
72	VAEVND4972	-	-	-	-	-
73	VAEVND4973	-	-	-	-	-
74	VAEVND4974	-	-	-	-	-
75	VAEVND4975	-	-	-	-	-
76	VAEVND4976	-	-	-	-	-
77	VAEVND4977	-	-	-	-	-
78	VAEVND4978	-	-	-	-	-
79	VAEVND4979	-	-	-	-	-
80	VAEVND4980	-	-	-	-	-
81	VAEVND4981	-	-	-	-	-
82	VAEVND4982	-	-	-	-	-
83	VAEVND4983	-	-	-	-	-
84	VAEVND4984	-	-	-	-	-
85	VAEVND4985	-	-	-	-	-
86	VAEVND4986	-	-	-	-	-
87	VAEVND4987	-	-	-	-	-
88	VAEVND4988	-	-	-	-	-
89	VAEVND4989	-	-	-	-	-
90	VAEVND4990	-	-	-	-	-
91	VAEVND4991	-	-	-	-	-
92	VAEVND4992	-	-	-	-	-
93	VAEVND4993	-	-	-	-	-
94	VAEVND4994	-	-	-	-	-
95	VAEVND4995	-	-	-	-	-
96	VAEVND4996	-	-	-	-	-
97	VAEVND4997	-	-	-	-	-
98	VAEVND4998	-	-	-	-	-
99	VAEVND4999	-	-	-	-	-
100	VAEVND5000	-	-	-	-	-

TABLE OF SLEEPERS AND THEIR FITTINGS

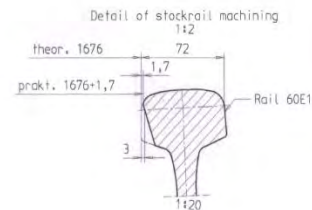
NONOC FOR TECHNICAL DESIGN
 Date: 19.8.2016
 Signature: [Signature]



- Check Rail gap can be adjusted during maintenance upto 10mm with the help of shims to org. no. VAEVND4938-...
- Welding joint
- Sleeper spacing is between sleeper no. 3 & 4 has been kept as 742mm instead of 600mm to suit the assembly of IRS point machine with 220mm Stock.
- Head Mordard (R-1000) rails shall be used for manufacture of this turnout.
- Sleeper No. 3 & 4 can be positioned acc. the side where the motor box to be installed.
- The L.H turnout is the mirror image from R.H turnout except Slide chair & sleeper No. 5 to 22.
- For L.H turnout sleeper No. 3R to No. 22 to be replaced by no. 3L to 22L and slide chair plates have to be replaced acc. to 3R's. No. VAEVND4757-01. Sleepers No. 1 to 4 and No. 21 to No. 12 remain fact. their places.
- Switch Sub Assembly org. no. VAEVND4757-...
- Crossing Sub Assembly org. no. VAEVND4753-...
- Weldable DMS crossing org. no. VAEVND4619-...
- Sections for turnout org. no. VAEVND4750-...
- Stress-free position of the switchblade of half opening of the switch (R/L-25)
- On sleepers marked with [Symbol], the gauge has to be measured.
- For sleepers no. 1 to 13 see "Detail of stockrail machining" (1676-1.7)
- Sleeper nos. 1 to 22 are perpendicular to the main line.
- Sleeper nos. 23 to 52 are laid in fan shape and 53 to 75 at rail angle.
- Standard fastening of rails with 5-clip R-X.
- Rail 230 mm
- Grooved rubber sole plate from track will be used under baseplates
- 10mm railpads will be used between baseplates and rails.

JOSEPH PASCAL
 CHIEF TRACK ENGINEER (SLT)
 17.08.16
 This design suitable for 50Kmph speed specially for WDFCC project

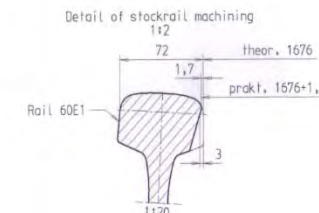
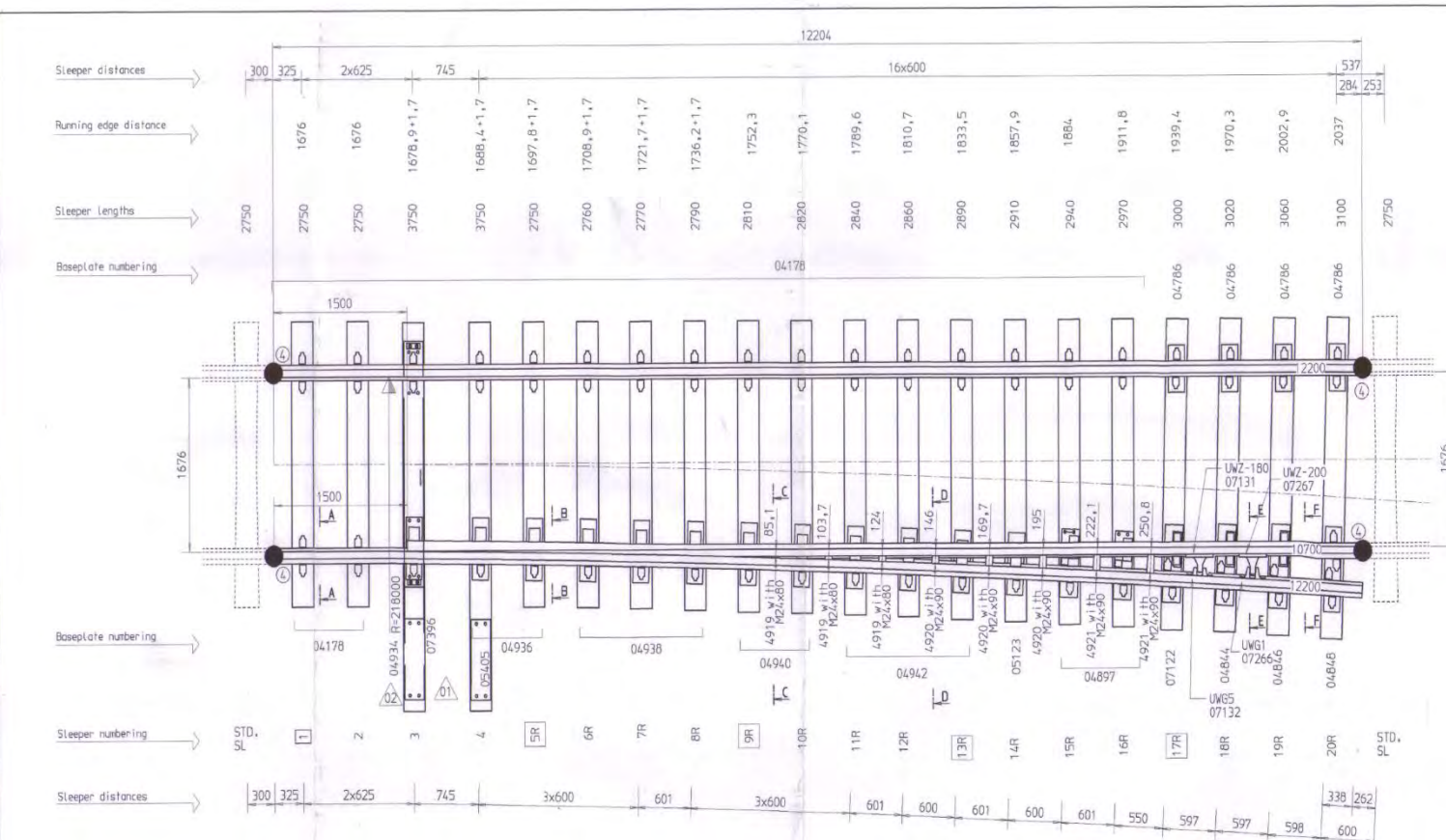
Figure 15 Layout without rollers of 1 in 12

[illegible]


NOVOC FOR TECHNICAL DESIGN
 VUD NO. 2174/2019
 Date: 6-MAY-2020 Pmc-NO2-11
 (F-02) TDDO
 Signature: 
 (Official Stamp)
 THIS DRAWING HAS BEEN PROOF
 CHECKED IN ACCORDANCE WITH
 THE REQUIREMENTS OF DFCCIL AND
 FOUND SUITABLE FOR INSTALLATION
 IN WDFC FOR 25T AXLE LOADS
 Date: 30.1.2020
 PROOF CHECKER 
 SYSTRA

IRS: T 10-2000
SPECIFICATION

185



IMPORTANT NOTE:-
MINOR CORRECTIONS IF ANY, MAY HAVE TO BE
INCORPORATED IN THE DRAWING AT THE TIME
OF SUBMISSION OF FINAL LAYOUT DRAWING.

- Welding joint
- The hardness of rail head will remain same as per requirement of RH rails after head machining of Switch Blade
- Head Harder (Gr.1000H) Tongue & Stock rails shall be used for manufacture of Switch.
- Sleeper spacing is between sleeper no. 3 & 4 has been kept as 745mm instead of 600mm to suit the assembly of IRS point machine with 220mm Stock
- Sections for switch device dwg.no. VAEKND6772-00
- Sleeper no.3 and no.4 can be positioned acc. the side where the switch machine has to be installed
- The RH switch device is shown.
- Stressfree position of the switchblade at half opening of the switch 180 +/-25°
- On sleepers marked with [X], the gauge has to be measured.
- For sleepers no.3 to no.8 see "Detail of stockrail machining" 1676+1.7mm
- Detail of stockrail machining" 1676+1.7mm
- Rail gap 4mm
- Grooved rubber sole plate 6mm thick will be used under baseplates
- 10mm railpads will be used between baseplates and rails.

FOR RH SWITCH DEVICE									
20R	VAEKND7173	-	-	-	VAEKND4786-1 No.	-	VAEKND4848-1 No.	VAEKND4199-2 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
19R	VAEKND7172	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
18R	VAEKND7171	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
17R	VAEKND7170	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
16R	VAEKND7169	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
15R	VAEKND7168	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
14R	VAEKND7167	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
13R	VAEKND7166	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
12R	VAEKND7165	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
11R	VAEKND7164	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
10R	VAEKND7163	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
9R	VAEKND7162	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
8R	VAEKND7161	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
7R	VAEKND7160	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
6R	VAEKND7159	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
5R	VAEKND7158	-	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
4	VAEKND7157	VAEKND5405-1 No.	-	-	VAEKND4786-1 No.	VAEKND4848-1 No.	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
3	VAEKND7156	-	VAEKND7396-1 No.	VAEKND4936-1 No.	VAEKND4786-1 No.	-	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
2	VAEKND7155	-	-	-	VAEKND4786-2 Nos.	-	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
1	VAEKND7154	-	-	-	VAEKND4786-2 Nos.	-	-	VAEKND4199-1 No. & VAEKND4918-1 No.	VAEKND4900-1 No. & VAEKND5282-1 No.
SLEEPER NO.	SLEEPER DRAWING NO.	M.S. plate	Gauge Tie Plate	Base plate with S. PLATE NO.	SPL Plate with S. Chair	Standard Plate	Machined Plate	Combined Plate	10mm THK. GROOVED RUBBER SOLE PLATE
									6mm THK. BASE PAD
									24 Dia Plate Screw T3912
									24 Dia Plate Screw T3913
									GFN Liners (Yellow) RT-8222
									GFN Liners (Pink) RT-8223
									ERC MK V RT-5919
									SCSW T10773
									KEY VAEKND 04196
									LEAF SPRING VAEKND 04195

REV NO	REVISION NOTE	DATE	CHANGED BY	CHECKED BY
01	Gauge Tie Plate & M.S. Plate Added and BOM & Table revised	13-02-2017	Ranjit Singh	Biraj Deb
02	Gauge Tie Plate extended & M.S. Plate removed	13-04-2017	Ranjit Singh	Biraj Deb



THIS DRAWING HAS BEEN PROOF CHECKED IN ACCORDANCE WITH THE REQUIREMENTS OF DFCOIL AND FOUND SUITABLE FOR INSTALLATION IN WDFC FOR 25T AXLE LOADS
Date: 30.1.2020
PROOF CHECKER: [Signature]
SYSTRA

DRAWING No.	DESCRIPTION	RH
VAEKND7173	Head Harder	8
VAEKND7172	Head Harder	8
VAEKND7171	Head Harder	8
VAEKND7170	Head Harder	8
VAEKND7169	Head Harder	8
VAEKND7168	Head Harder	8
VAEKND7167	Head Harder	8
VAEKND7166	Head Harder	8
VAEKND7165	Head Harder	8
VAEKND7164	Head Harder	8
VAEKND7163	Head Harder	8
VAEKND7162	Head Harder	8
VAEKND7161	Head Harder	8
VAEKND7160	Head Harder	8
VAEKND7159	Head Harder	8
VAEKND7158	Head Harder	8
VAEKND7157	Head Harder	8
VAEKND7156	Head Harder	8
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VAEKND7153	Head Harder	8
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VAEKND7150	Head Harder	8
VAEKND7149	Head Harder	8
VAEKND7148	Head Harder	8
VAEKND7147	Head Harder	8
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VAEKND7144	Head Harder	8
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VAEKND7100	Head Harder	8
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VAEKND7085	Head Harder	8
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VAEKND7074	Head Harder	8
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VAEKND7068	Head Harder	8
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VAEKND7051	Head Harder	8
VAEKND7050	Head Harder	8
VAEKND7049	Head Harder	8
VAEKND7048	Head Harder	8
VAEKND7047	Head Harder	8
VAEKND7046	Head Harder	8
VAEKND7045	Head Harder	8
VAEKND7044	Head Harder	8
VAEKND7043	Head Harder	8
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VAEKND7041	Head Harder	8
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VAEKND7039	Head Harder	8
VAEKND7038	Head Harder	8
VAEKND7037	Head Harder	8
VAEKND7036	Head Harder	8
VAEKND7035	Head Harder	8
VAEKND7034	Head Harder	8
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VAEKND7030	Head Harder	8
VAEKND7029	Head Harder	8
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VAEKND7027	Head Harder	8
VAEKND7026	Head Harder	8
VAEKND7025	Head Harder	8
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VAEKND7007	Head Harder	8
VAEKND7006	Head Harder	8
VAEKND7005	Head Harder	8
VAEKND7004	Head Harder	8
VAEKND7003	Head Harder	8
VAEKND7002	Head Harder	8
VAEKND7001	Head Harder	8
VAEKND7000	Head Harder	8

ITEM REF.	QUANTITY	MATERIAL DESCRIPTION	WT. - Kg
DESIGNED BY	NAME	DATE	SIGN
CHECKED BY	NAME	DATE	SIGN
APPROVED BY	NAME	DATE	SIGN
SHEET	SIZE	SCALE	SCALE
FILE	DATE	DATE	DATE

Figure 17 Right Hand Derailing Switch



ANNEXURE 4 VERSE (PARA. 2.26.2)



Some current VERSE® users:

VERSE - Non Destructive Stress Free Temperature Measurement System



Proven and Accepted Worldwide with over 150, 000 accurate readings

Approved by DB Netz AG



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**AT LAST -
cost effective and practical
checking of temperature stress
in rail**

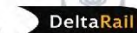
VERSE®

**Non-Destructive Stress Free
Temperature Measurement**

**A giant leap
for track safety
management,
risk reduction
and new track
quality acceptance**



a joint development with



10°C rise in rail temperature creates a compressive force of 18 tons

Every railway engineer knows that rail should be at the correct **Stress Free Temperature** (SFT) (also referred to as **Neutral Temperature**). Whilst care is normally taken to ensure that new track is laid in this condition, there are many factors which can cause local changes over time.

Rail outside of its SFT limits is at risk of:

Buckling in hot weather

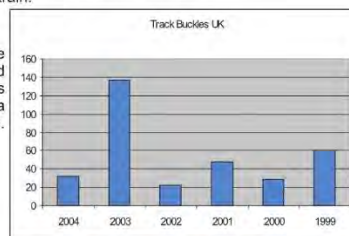
Fracturing in cold conditions

Pull-in on curves when cold

In addition it is possible for the left and right rails to develop a different SFT which can produce uneven loads with unpredictable results, as has been found in recent measurements.

Anyone of these track failures will, at best, cause considerable disruption to the network and, in the worst case, cause catastrophic de-railment of a high speed passenger train.

Most railways endure both rail buckles and breaks. The statistics for the UK reveal a typical experience.



The fact that buckles and breaks are prevalent, is in many respects due to the absence of a cost effective and practical method of checking SFT.

Until now the only choice has been to cut the rail, measure the gap, manually calculate the SFT and finally re-weld the rail. This traditional method is high cost, labour intensive, frequently inaccurate, and introduces an additional and unnecessary weld. Not surprisingly most railways have tended to limit such measurements of SFT.

A non-destructive, cost effective and practical method for checking SFT will provide today's railway engineer with the means to measure the network, determine the change and rate of change of SFT and its causes, identify high risk sites and plan pre-emptive measurement and re-stressing programmes.

Through such management, the modern railway can demonstrate a determined approach to risk reduction, and add another level of safety to the rail transport network.

Accuracy means risk reduction



Since the prevention of rail buckling is often the prime consideration (the greatest safety hazard) many railway engineers have preferred to err on the side of caution and have set the SFT above the average. This then risks a higher number of rail breaks. Now that **VERSE**® has made SFT measurement so much easier and more accurate, the SFT can be set much more sensitively to mitigate against both rail breaks and rail buckles. A step forward in track safety management and risk reduction.

New track quality acceptance

Much new track is now laid by contractors on behalf of the infrastructure authority who are increasingly seeking assurances related to track quality at the time of hand over. SFT can now effectively be added to a number of track parameters that constitute quality assurance which in turn improves track safety.



Predictive maintenance



Although it is well understood that SFT in rails can, and does, change over time it has not previously been economical to study the cause and effect relationship for specific sites in a way that will aid predictive maintenance. The **VERSE**® method now provides that option.

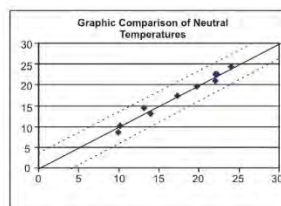
Most railway engineers are able to predict which parts of their network are likely to be vulnerable to SFT change. SFT measurement can therefore be well targeted in the first instance to bring maximum benefits. It is now possible to plan cost effective re-stressing programmes which will help to bring significant improvements in confidence and safety across the rail network.

Development and Validation

Validation of the **VERSE** technique was carried out by AEA Technology-Rail during the research phase of perfecting the methodology. These tests showed an accuracy of 0.2° C and a standard deviation of 1.3° C. Tensors were used to pull the rail to known tensions, as read by strain gauges, and the rail measured using **VERSE**.

In addition, tests were conducted under service conditions during which the rail was first checked with **VERSE**, and then cut and the SFT measured using strain gauges.

Validation has since been carried out by other technical rail organisations. The Transportation Technology Centre Inc. (TTCI), Pueblo, Colorado, conducted tests on large section American rails. Results indicated a mean error of 2.50°F and a standard deviation of 1.170°F.



The Railway Research Institute in Prague, Czech Republic, conducted validation of **VERSE** using similar techniques to TTCI and produced the results shown in the graph.

Bibliography of Tests and Demonstrations

A selection of some of the many tests carried out:

Canada	April 1999	USA	July 1999
Finland	September 2000	Denmark	April 2001
Italy	August 2001	France	January 2000
Czech Republic	March 2000	Germany	April 2002
China	October 2002		

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VERSE

**Non-Destructive Neutral
Temperature Measurement**



VERSE

**Non-Destructive Neutral
Temperature Measurement**

Proven with many thousands of measurements taken in service conditions by different railways across the world.



non-destructive no need to cut the rail unless SFT check proves re-stressing necessary.

quick & easy only 20 to 30 minutes per measurement depending on type of rail clip.

low cost less than 5% of the cost of performing the traditional cut, measure, and re-weld method.

accurate excellent reproducibility with a mean error of 0.2°C and standard deviation of 1.3°C so you can be confident in the result.

versatile suitable for all rail profiles.

instant result follow on-screen instructions on hand held computer for easy measurement and instant result.

database files from the hand-held can be downloaded into a PC database to produce a record of your whole rail network.

portable supplied in a convenient carry case which will fit into small van or estate car.



VERSE delivers a direct SFT result without needing to know any residual stress history of the rail.

VERSE measurement is made whilst the rail is in tension, at an ambient temperature lower than the existing SFT of the rail.

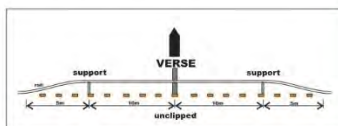
Only one measurement at one ambient rail temperature is required.

Making a Measurement

Measurement entails unclipping approximately 30 metres of rail.



At points 10 metres each side of the centre point, the rail is lifted onto a rail support spacer. This leaves a 20 metre length of freely suspended rail.



The **VERSE** equipment is positioned over the central point, and the rail is jacked steadily to a load of 10 kN and monitored by a load transducer. Displacement is simultaneously monitored for each increment of load by another transducer attached to the rail.

The output from both transducers is sent to a hand-held computer.



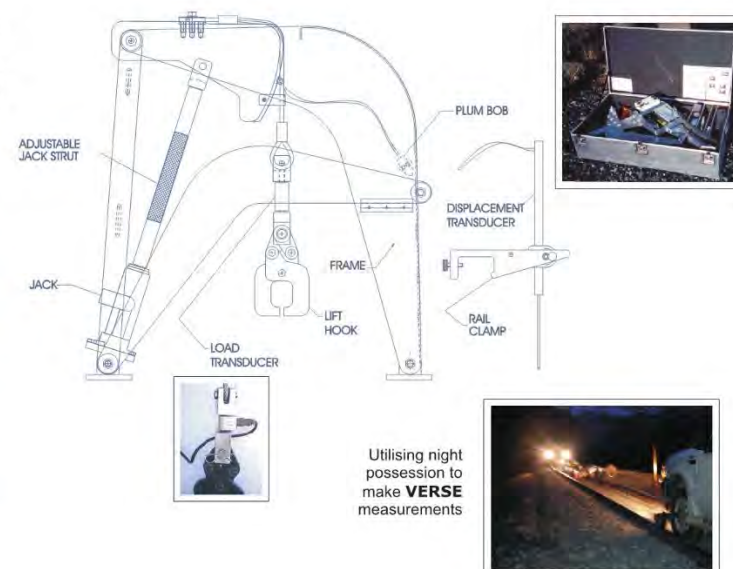
The software menu prompts the user to enter certain site specific data such as ambient rail temperature, rail profile and actual rail height.

The software then calculates the SFT directly. **VERSE** equipment is self calibrating for each lift and the software is able to check the validity of each measurement. Time on site is between 20 and 30 minutes per rail depending on the type of rail fastening.

VERSE Non-Destructive Neutral Temperature Measurement

The Equipment

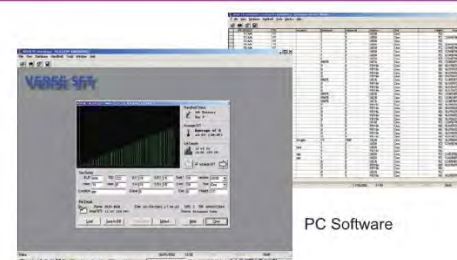
The **VERSE** equipment comes ready assembled for positioning over the rail. Once the electrical connections have been made between the transducers, signal processor, battery pack and hand-held computer, measurement can commence.



The Software

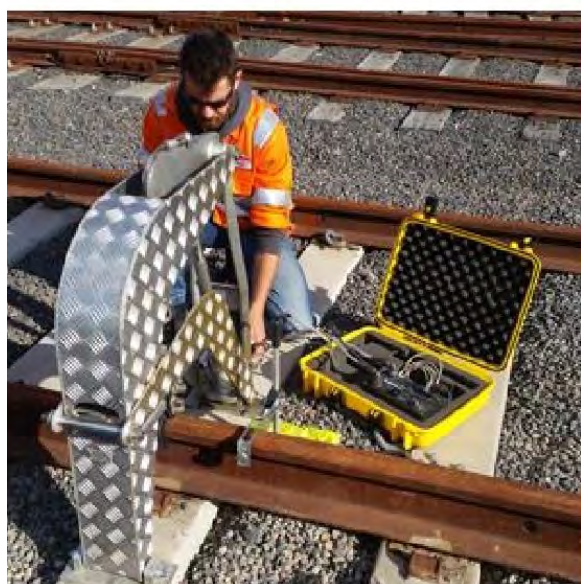
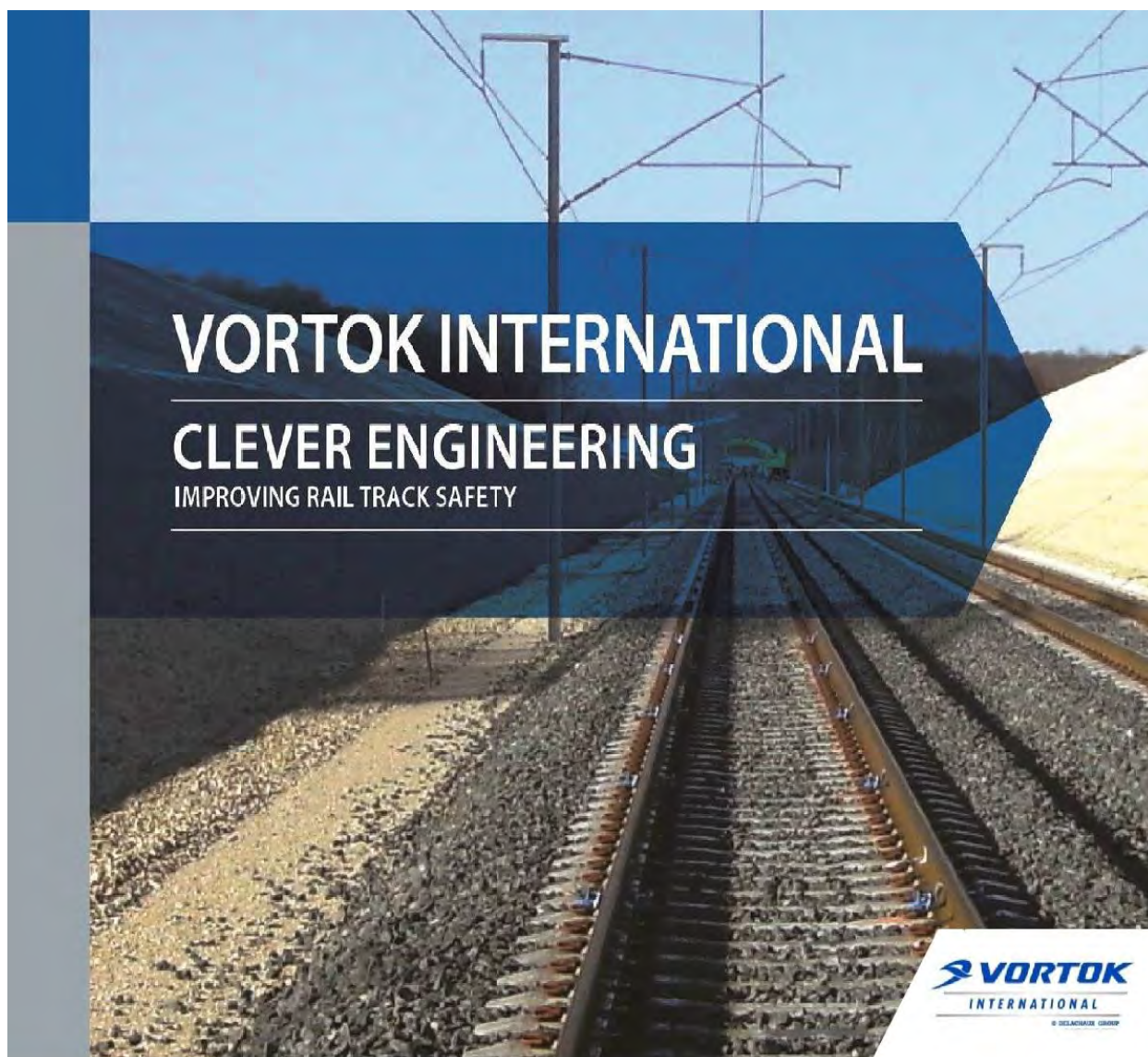
The hand-held computer software guides the user through the measurement procedure, logs certain site specific data and produces a track side SFT result. The rail profile is easily selected from a drop down menu.

Data sets for each measurement can subsequently be uploaded to the **VERSE** PC software and stored in a database as a permanent record of rail network measurements and can be interrogated at any time.



PC Software

Data can also be exported and analysed with other databases such as MS Excel or MS Access types where it can be integrated with other track data for maintenance management decision making.



VERSE[®]

USER GUIDE

Document number: VE200911 Revision: 2 Page 1 of 64

Approval	Name	Position	Signature	Date
Author	Marcin Stokowski	Design Engineer	PP	26/05/17
Reviewer	Neil Algar	Head of Sales and Marketing	Neil	26/05/17
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WARNING! This symbol indicates that failure to follow directions could result in serious injury.



CAUTION: This symbol indicates that failure to follow directions could result in damage to equipment or loss of information.

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1 Getting Started

This guide covers the steps of setting up a **VERSE**® system, how to use the handheld computer Allegro²™, the SFT measurement and how to perform the post measurement data analysis.

1.1 System Description

VERSE® (Vertical Rail Stiffness Equipment) is a system developed by **VORTOK INTERNATIONAL** for measuring the Stress Free Temperature (SFT) of Continuous Welded Rails (CWR). The system is based on the principle that the vertical force required to lift a rail varies with the axial force contained within. Conventional SFT measurement techniques involve cutting the rail. Such destructive measurements are expensive, disruptive and slow. **VERSE**® fulfils the need for a simple non-destructive SFT measurement technique.

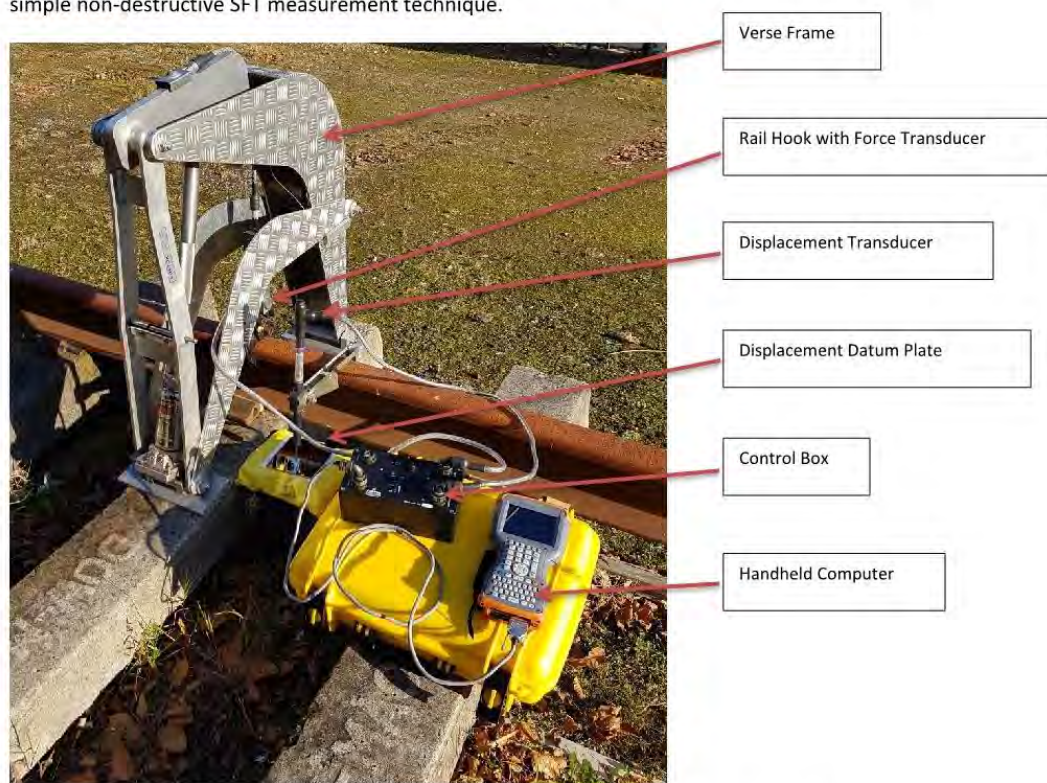


Figure 1 Verse kit set up over rail - main components displacement transducer to be shifted for clarity. Note, displacement transducer positioned for clarity.

The equipment consists of a portable loading frame, which is positioned over the rail to be measured. An upward vertical force is applied to the rail via the system hydraulics. Transducers are used to measure the applied force with respect to the vertical displacement of the rail. This data is stored in a small robust handheld computer, which, along with a signal conditioning system, powers the transducers, captures their output signals and calculates the SFT. **VERSE**® SFT results are based on calculating the axial force in the rail from the measured load and displacement data.



The **VERSE**® system is supplied complete in its own container. The main frame is folded into its storage position, and the remaining items of equipment are located in compartments set into the container. The instrumentation is secured in a separate case within the main container

1.2 Safety Requirements

A track possession with the appropriate safety management system is required prior to attempting to make **VERSE**® measurements.

- The lifted 20m length of rail shall not contain any insulated rail joints (IRJ's).
- **VERSE**® measurement shall not be taken within 30m of insecure S&C
- Take care whilst measuring SFT at tight curves, especially during low temperatures, as rail tension will act to straighten the rail curve.
- Ensure any bonded cable connections do not impede the process of lifting the rail.
- Use approved methods of lifting rails when installing support blocks – Do not use **VERSE**® frame to lift rail onto support blocks.
- On completion of the **VERSE**® measurement ensure that the track is correctly reinstated and all equipment is removed from site.

1.3 Application

- **VERSE**® lift shall not be carried out when sudden changes in rail temperature are occurring, e.g., when the sun shines on the rail which has previously been in the shade during a measuring run.
- **VERSE**® measurements shall not to be carried out at rail temperatures above the minimum SFT tolerance. For Example, 21°C for the UK , or if there is a suspicion that the rail might be in compression.
- **VERSE**® measurements shall not be carried out at rail temperatures below -5°C. These values will be different in other climates as declared by the local rail authority.
- The rail within the lifted length should be the same section type i.e. not rails of different sections welded together.
- The ballast adjacent to the rail to be measured should be at such a level that it does not fall beneath the foot of the rail upon lifting, thus preventing the rail from settling back to its original position once the lift load is removed. If higher than the rail foot, the ballast will need to be scraped away over the lifted length.
- If possible avoid rail welds within site length, move the measurement site away from welds. In sites where it is not possible to avoid welds, organise the site that the welds are away from lifting point. The lifting point shall not be directly above the weld. The SFT result is affected by presence and quality of welds. As each site will be unique the operator shall be responsible to interpret measurement results.
- Move the site away from closure rail (a rail height difference will affect the SFT result)



2 Site and Equipment Preparation and Methodology

2.1 Site Preparation

- Before commencing any work, ensure conditions outlined in section 1.2 and 1.3 are satisfied. This will require an initial approximate temperature reading of the rail to be taken. If satisfied select approx. 30m site to unclip the rail. When measuring rail greater than 60kg/m the site length can be extended to approx. 33m to ensure correct SFT results.
- Providing all safety precautions have been set in place select the approximate location of the site, and mark the position of any welds that could not be avoided within the 30m length to be unclipped.

2.2 Marking Out

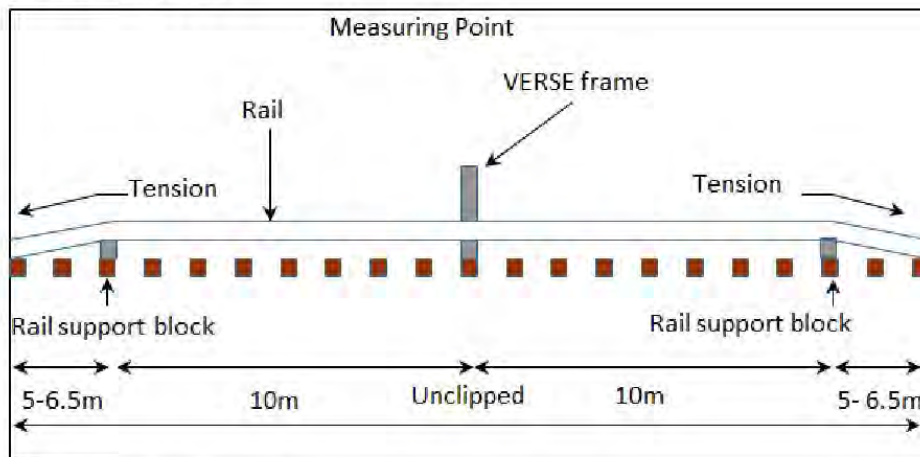


Figure 2 Positioning of the spacer blocks

- Mark lifting point on rail head.
- Mark 10m either side of the lift point to where support blocks can be located (Inner Span). Extend length if sleeper spacing denotes. Length adjustments must be made with the position of each spacer block to ensure they are as evenly spaced as possible.
- Mark 15m either side lift point to indicate first anchorage position (Total Span). Extend length to suit sleeper spacing.



CAUTION! When measuring heavier gauge rails (60E1 etc), the unclipped length is extended to ensure that during the jacking, that the rail does not cause the first fastened sleeper at each end of the unclipped length to rise out of the ballast.



CAUTION! Note down the measurements, these will be needed to enter to the handheld.



CAUTION! Take accurate measurements (to the nearest cm) of each side of the unsupported length of rail and the two outer end spans and record these details. A 1cm error in the measurement of the support block distances will produce a nominal error of 0.1°C in the SFT result. 1cm error in

the measurement of the support end span distances will produce a nominal error of $<0.05^{\circ}\text{C}$ in the SFT result. The distance measured is that between the lift point and the position at which the last fastening restrains the rail.



Figure 3 Measuring of the position of spacer blocks.

2.3 Unclipping

Unclip the length of rail between marked anchor points.



WARNING! Use appropriate tools to unclip the rail.



CAUTION! Ensure rail fastenings are fully disengaged to ensure free lift.

2.4 Rail Lifting

Jack the rail up and insert the rail support blocks between the rail and the sleepers at the 10m positions.



Figure 4 Spacer blocks - placing examples

⚠ CAUTION! Remove any pads or insulators which might be in direct rail contact over measured rail length.

⚠ WARNING! Use approved methods of lifting the rails.

⚠ WARNING! Do NOT use **VERSE®** frame as method of lifting rail onto the support blocks.

2.5 Equipment Setup

- Setup the **VERSE®** equipment in accordance with Section 4.
- Position VERSE frame over lift point.

⚠ CAUTION! Position **VERSE®** frame with a lifting hook vertically aligned above marked lift point. Utilise plumb bob to assist with alignment.

2.6 Rail Temperature Measurements

Take accurate rail temperatures readings by fitting the thermometer to the rail foot. Usually only one reading is required, however if temperature variation is suspected, up to three readings should be taken along the unclipped length of the rail. Note that the kit is supplied with a single thermometer, if additional units are required, contact Vortok.



Figure 5 Rail thermometer, left sunny side of rail, right shaded side of rail

⚠ CAUTION! Ensure that the rail temperature is measured on the shaded side of the rail and not in direct sunlight.

⚠ CAUTION! Ensure temperature output is stable before taking a reading

⚠ CAUTION! Note down the measurements, these will be needed to enter to the software.



CAUTION! Note that a 1°C error in the temperature measurements will lead to a 1°C error in the SFT result

2.7 Rail Section Measurements

Determine the rail section by reading the brand marks on the rail web, then measure the rail height to an accuracy of $\pm 0.1\text{mm}$ using supplied extended jaw Vernier calliper. Record and note the results



Figure 6 Rail height measurements



CAUTION! 0.5mm error in the measurement of the rail height will produce a nominal error of $<0.3^\circ\text{C}$ in the SFT result.

2.8 Input Data

Input site and track data into **VERSE**® handheld. See section 0 for guidance on handheld usage.

2.9 Perform Measurement



WARNING! When the rail is being measured, always check that personnel are clear before operating the hydraulic jack.

Conduct **VERSE**® measurement runs using handheld computer (see section 0 and 6) for guidance. Carry out at least three runs per section of measured rail.



Figure 7 VERSE kit fully set up



3 Special Conditions

3.1 Lifting on Curves

It is universally recognised, that as the radius of a track curve reduces, the accuracy of the **VERSE®** measuring technique is reduced. The following procedure is applied in the handheld computer when taking measurements in tight curves to compensate for the error in the resultant SFT.

When a curved rail is below its neutral temperature and is released from its fasteners, the tension in the rail pulls it towards the centre of the curve. This process reduces the length of the rail as it cords across the curve and so reduces the tension in it, which reduces its effective neutral temperature.



WARNING! Take extra care whilst unclipping a rail on curve. The accumulated tension in rail might cause injury

The **VERSE®** software has the ability to compensate for this change; the simple table below can be used to guide the operator on how that compensation is applied to the result measured by **VERSE®** to achieve an accurate measurement under these conditions.

Before the rail is removed from its fastenings, a 20m chording line is stretched from one rail support block position to the other. The distance of the centre of the curve (the point at which the **VERSE®** measurement is to be taken) to the cording line is measured in millimetres – this is known as the versine. See Figure 8.

The rail is then released from its fasteners, lifted onto its spacer blocks and the versine is measured again these two measurements are extrapolated into the table below to calculate the temperature error to be added to the **VERSE®** measurement.

	Unclipped versine (mm)																
Clipped versine (mm)	10	20	30	40	50	60	70	80	90	100	110	120	130	140	150	160	167
	SFT correction (°C)																
70	3	3	2	2	1	1	0										
80	4	4	3	3	2	2	1	0									
90	5	5	4	4	3	3	2	1	0								
100	6	6	6	5	5	4	3	2	1	0							
110	7	7	7	6	6	5	4	3	2	1	0						
120	9	8	8	8	7	7	6	5	4	3	2	0					
130	10 <td>10<td>10</td><td>9</td><td>9</td><td>8</td><td>7</td><td>6</td><td>5</td><td>4</td><td>3</td><td>2</td><td>0</td><td></td><td></td><td></td><td></td></td>	10 <td>10</td> <td>9</td> <td>9</td> <td>8</td> <td>7</td> <td>6</td> <td>5</td> <td>4</td> <td>3</td> <td>2</td> <td>0</td> <td></td> <td></td> <td></td> <td></td>	10	9	9	8	7	6	5	4	3	2	0				
140	12	12	11	11	10	10	9	8	7	6	5	3	2	0			
150	14	13	13	13	12	11	11	10	9	8	6	5	3	2	0		
160	15	15	15	15	14	13	13	12	11	9	8	7	5	4	2	0	
167	17	17	16	16	15	15	14	13	12	11	10	8	7	5	3	1	0

Table 1 Versine matrix

The above table is based on clipped and unclipped versine measured on 20m chord.

The values in green area are in °C (Celsius)

If value of clipped versine is higher than value of unclipped versine then add the SFT correction factor to the measured SFT to get true SFT. This should be done only when curve compensation is added after the measurement (on PC) and was **not** used during the lift.

If value of clipped versine is lower than value of unclipped versine then the rail is in compression at current temperature.

⚠ CAUTION! Should the rail be in compression no rail SFT measurements shall be taken

The versine is shown on a diagram as below.

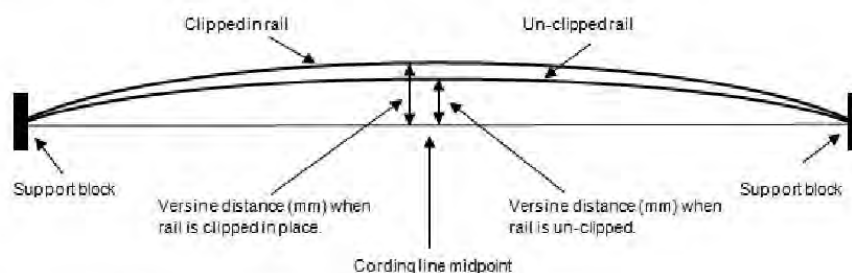


Figure 8 Clipped and unclipped versine explanation diagram

Curve compensation feature is available on handheld, see section 5.3.7..

3.2 Delayed Start

Sometimes, when a rail is in a suspended condition, it can still be in contact with a fastening or housing. This can occur when measuring in a curve. The rail tension causes the rail to straighten over rail fastenings, and the rail makes contact with a fastening or a screw spike. If the rail is in contact with an object before the lift, it is not freely suspended, some of its mass is supported at the contact point, this causes an error in the calculation. It is essential for the rail to be freely suspended at the commencement of the lift to make an accurate **VERSE®** measurement. The load transducer must start measuring the force from a zero.

Where possible the obstruction should be removed to allow the rail to be freely suspended before the lift. If this is not possible the Delayed Start feature should be selected from the handheld menu. The data collection from the lift to be delayed until the fixture is clear. See section 5.3.8.1.

3.3 Partial Lift

At any time during the lift process a part lift option is available. Selection of the part lift will stop the data collection by the handheld computer. As long as over 30 data points have been collected this data will be included into the final SFT calculation, see section 5.3.8.2 for how to select a partial lift feature.

Partial lift may be required on small rail sections where a significant deflection occurs during the lift.



4 Equipment Preparation

The main components of the frame are as follows

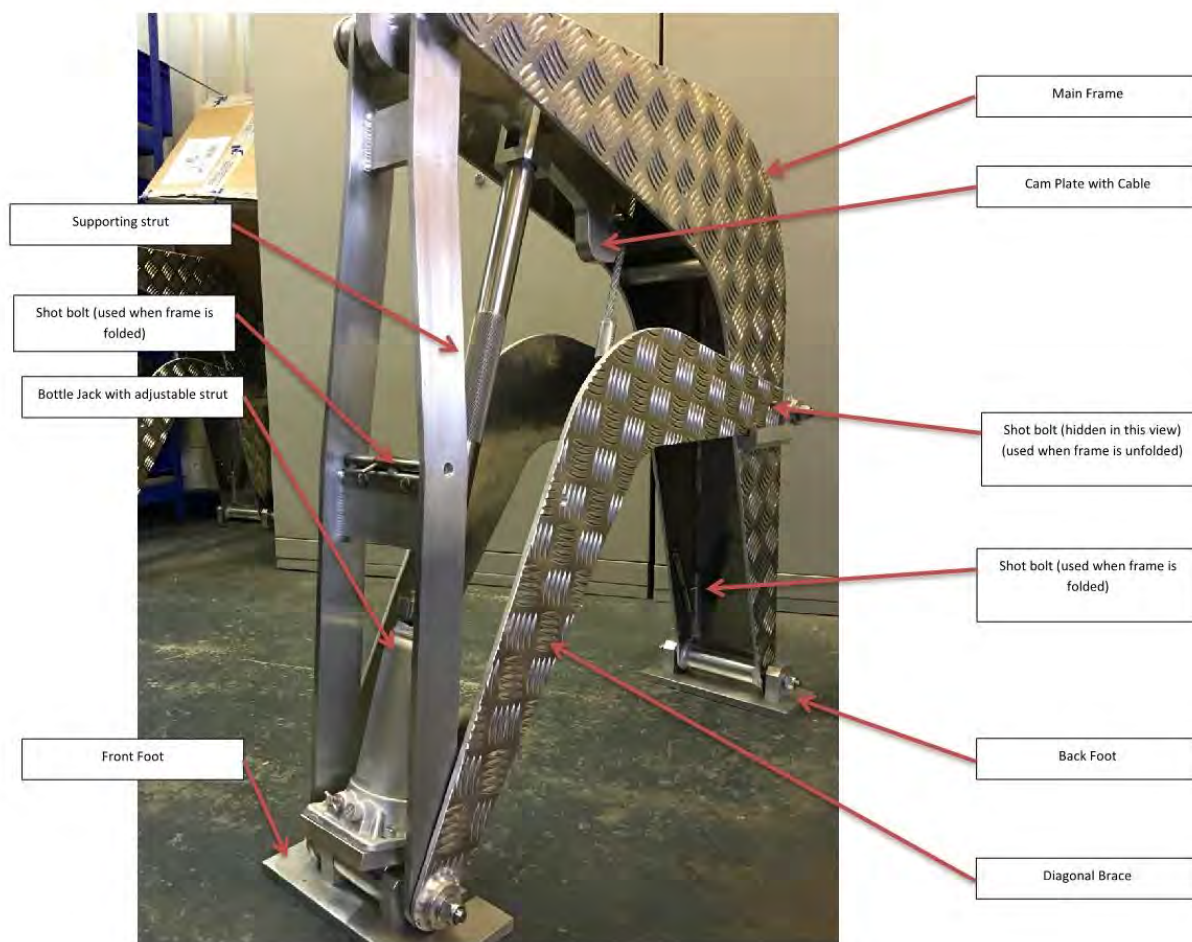


Figure 9 Frame main components



WARNING! The frame weighs approximately 32Kgs. Please refer to and be aware of your local health and safety recommendations/rules for the safe handling of heavy equipment. Care must always be taken when unfolding and folding the lift frame due to loose components moving in a scissor action. Always ensure that hands and fingers are clear of any freely pivoted parts when handling the frame. The safest method to unlock and unfold the lifting frame is shown below.

- Ensure shot bolts are engaged and frame is lifted in a folded condition
- Lift the frame as shown below and place it on the ground.

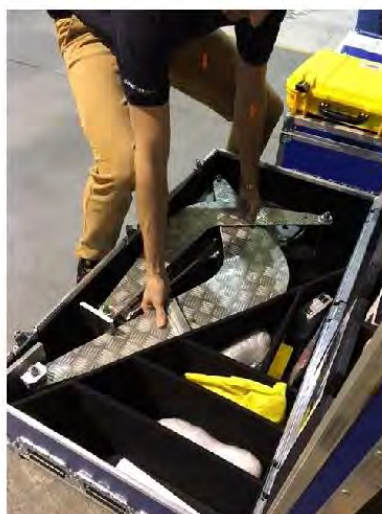


Figure 10 Safe handling of the frame

- Unfasten the shot bolt securing the foot to the frame. The cam plate should be grasped in one hand and the pivot straps' coupling bar grasped in the other



Figure 11 Location of safety catches that need to be released to unfold the frame

- The pivot straps should be raised and the coupling bar lifted over the main side-plates, whilst steadying the cam plate.
- Unfasten the shot bolt securing the pivot strap.

- The process should continue until the pivot straps locate in their stops



Figure 12 Sequence of unfolding the frame

- Finally, the feet should be rotated, if necessary, to sit flat on the supporting surface



Figure 13 Unfolded frame with security catch bolt close up

- Lift the frame to position spanning the rail above the lift point.



Figure 14 Correct handling of frame to lift point

4.1 Equipment Connections

- Connect the force transducer (load cell) and rail hook assembly to the lift cable via the clevis pin connector. The yokes must not be done up tight as there must be some allowance for rotation during both positioning and lifting operations.



Figure 15 Load cell attached to a rail hook

- Adjust jack strut so that the cable is just slack.



WARNING! Do not apply any preload to the jack strut.

- Ensure lifting cable is just slack by rotating the bottle jack strut



Figure 16 Jack strut adjustments



WARNING! Do not bend the cables. Do not carry the transducers by holding by the cables to avoid pulling out internal wiring.



CAUTION! Be sure the cables are going over the diagonal brace not under it. If cables go under the brace the risk of cables being bent during the lift is increased.

- Place the displacement datum plate (sandbag) firmly in the ballast at the side of the sleeper as close to the lifting point as possible, see below:



Figure 17 Mounting displacement to the rail

- Next fit the displacement transducer with the clamp on to the head of the rail over the datum plate (taking care not to over-tighten the transducer clamp) and adjust the carrier

until the transducer is centred and vertical over the datum plate. Check the moving arm has approximately 25mm of free travel when installed. It should not be down hard against the plate as it can fatally damage the transducer.



CAUTION! Be sure to fold the transducer back into the clamp assembly to protect the unit when not in use.



Figure 18 Verse frame and transducers prepared to do a lift

- Once all transducers are attached to the frame, connect the leads to the control box as per image below.



Figure 19 Connections to Control Box



5 Handheld Computer Operation, **VERSE[®]** Software Operation

5.1 Basic Information

The **VERSE**® system is supplied with an Allegro 2™ Standard Rugged Handheld computer. This is a small, robust, lightweight, battery-powered PC.



Figure 20 Allegro 2 handheld computer and sockets designations for **VERSE**®

5.2 First Use

Upon receiving the **VERSE**® kit the handheld computer is switched off for battery conservation. To switch it on press Power button  located on right bottom of the keyboard. The **VERSE**® software can be operated using the full QWERTY and numerical keyboard. The Tab key is to switch/highlight the on screen buttons. Enter key to accept the choice of the menu/button

5.2.1 Charging the Unit

The handheld computer can be charged via AC wall adapter that was shipped with unit or optional Ethernet single dock

To charge it via AC adaptor follow as below

- Plug the handheld computer into the AC wall adapter that was shipped with your unit.

5.2.2 Keyboard Features

The Allegro²™ has a numeric keyboard, function keys, and a QWERTY keyboard. The keys are sealed and have backlight illumination.

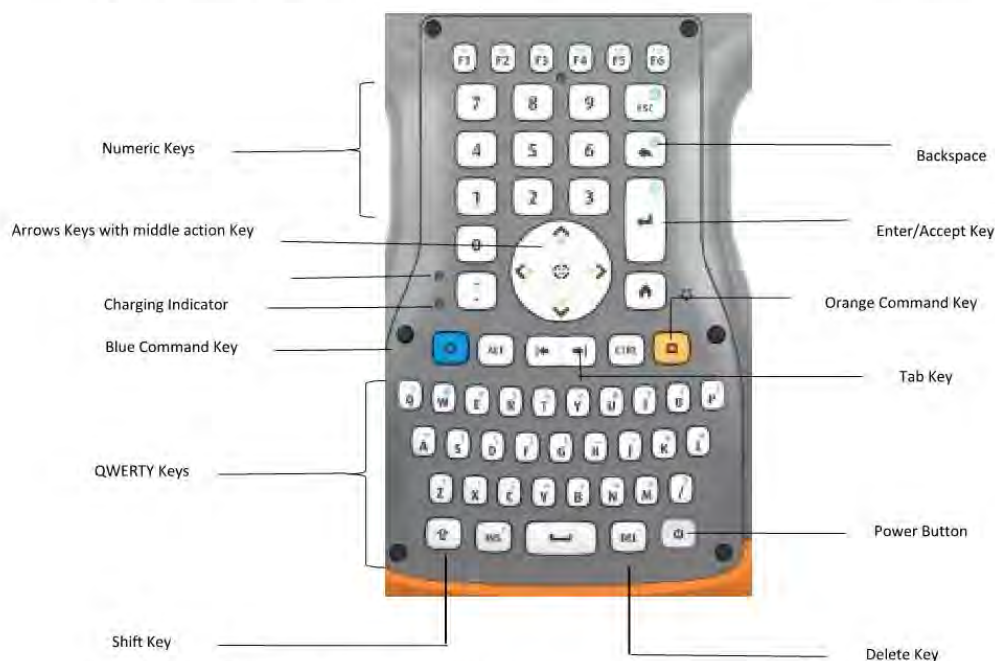


Figure 21 Allegro 2 keyboard, keys active on VERSE

When entering information in the **VERSE**® software, lower case is a default case. To change it to upper case press  Shift key once or press twice to be a sticky key.

All buttons of the QWERTY keys have a second function.

By pressing  Blue Command Key the second key function of keyboard can be chosen.

By pressing  Orange hot key and one of the arrow keys   the sound volume can be controlled.

By pressing  Orange hot key and   the brightness can be controlled

By pressing  Orange hot key and   the keyboard brightness can be controlled.

Shift, Blue and Orange Command keys are sticky on second press.

Tab Key is used to tab between windows when entering data

5.2.3 Maintenance

Follow the instructions in this chapter to properly maintain and recycle the Allegro²™.

Storing the Allegro²™ and Battery Pack

When the handheld computer is not being charged and is suspended, it draws a small amount of power. This power draw is used to maintain the memory (RAM) of the handheld computer in the

same state it was when it was suspended. We recommend charging the handheld computer each night or weekend when it is in suspend mode. See charging unit section.

If the Allegro 2 is not charged while in suspend mode and the battery reaches a low charge, it automatically powers off to prevent further drain on the battery.

Note: Data and programs are secure as long as they have been saved, even if the battery pack becomes discharged. The handheld computer does not depend on the battery to store the data for extended periods.

5.2.3.1 Cleaning

Screen

- ✓ Suspend or Power off the unit.
- ✓ Apply warm water or a mild cleaning solution to a microfiber cloth and gently wipe off the touchscreen.



CAUTION! Do not use tissues, paper towels, soft bristle brushes, or harsh cleaning solutions on the screen.

- ✓ Rinse the screen with water and dry it with a microfiber cloth.
- ✓ Apply a screen protector if you want to (not included). To clean a screen protector, follow the instructions provided with the package.

Case, Bumpers, and Connector Module

Make sure the battery door is securely installed. Use warm water, a mild cleaning solution, and a soft bristle brush to clean the case, bumpers, and connector module.



CAUTION! Do not direct a high-pressure stream of water at the device to clean it. This action could break the seal, causing water to get inside the device and voiding the warranty.



CAUTION! Exposure to some cleaning solutions may damage your device, including automotive brake cleaner, isopropyl alcohol, carburettor cleaner and similar solutions. If you are uncertain about the strength or effect of a cleaner, apply a small amount to a less visible location as a test. If any visual change becomes apparent, promptly rinse and wash with a known mild cleaning solution.

5.2.3.2 Recycling the Allegro²_{TM} and Batteries

When the Allegro 2 reaches the end of its life, it must not be disposed of with communal waste. It is your responsibility to dispose of your waste equipment by handing it over to a designated collection point for the recycling of waste electrical and electronic equipment. If you cannot find a location, contact local authorities, Vortok International or Allegro 2 manufacturer Juniper Systems for information about disposal. The Li-Ion battery packs for your Allegro²_{TM} are recyclable. Avoid placing them in the trash or communal waste system.

5.3 Track Data Input and **VERSE**® Software Operation

5.3.1 Startup

Power on or wake up the Allegro2 by pressing the Power Button  briefly.

You can choose to restart the **VERSE**® software by holding the Power key .

The power button options window will appear. Using   choose Reset button as shown below.



Figure 22 Power key options menu

VERSE® - Splash Screen

Once Allegro2™ rebooted the **VERSE**® software first screen appears.

Press OK using  Enter Key.



Figure 23 VERSE - First screen

5.3.2 Language

VERSE® - Language

Using   choose the operating language from the menu.

09 Mar 2017  100% 15:40:51 09 Mar 2017  100% 15:41:33



Figure 24 VERSE - Language option menu

Tab through the windows using Tab keys  .

5.3.3 Demo Mode

VERSE® - Demo Mode

The software has a demonstration mode which will allow the program to simulate a **VERSE®** measurement for demonstration purposes. The Demo Mode check box is shown below the language option. If the check box is selected, the handheld computer will operate independently from the instrumentation and will give a simulated result.

09 Mar 2017  100% 15:42:11



Figure 25 VERSE - Demo Mode

 **CAUTION!** In order to prevent false data being collected, ensure that Demo Mode is deselected before carrying out a true **VERSE®** measurement.

5.3.4 Main Menu

VERSE® - Main Menu

The **Main Menu** consists of 8 buttons which can be chosen and accessed using Tab   keys and Enter key .

09 Mar 2017  100% 15:42:28

Measurement Menu	Edit Date and Time
History	Suspend
System Information	Return to first Screen
System Settings	Restart VERSE

Figure 26 VERSE - Main Menu

Measurement Menu – is to start the **VERSE®** measurement

09 Mar 2017  100% 15:42:42

Enter Site Details

Enter City/Country

Enter Roll Number

Enter Date/Time

View Result when T1 is completed

On Measurement Run

Notes

Exit Menu

Figure 27 VERSE - Measurement Menu

5.3.5 History

History - is to review previous measurements

09 Mar 2017 100% 15:43:58

Filename 13 / 21

Demo13

Date 08 Mar 2017, 11:15:05

Location 0

Run	SFI	Error	Points
1	25.3	0	80
2	25.3	0	80
3	25.3	0	80

Cancel

Figure 28 VERSE – History

5.3.6 Handheld Information

System Information – is to access licence holder information handheld computer ID and displacement and load transducers serial numbers and last calibration date.

09 Mar 2017 100% 15:44:38 09 Mar 2017 100% 15:44:59

This software is registered to:-
marcin test
marcin test

UID/Ref No: 3138323437390000

Memory usage: 56057 of 475648 total Kbytes.

Load Cell Serial No: 20254
Load Cell Calibration Date: 15.35 14/02/2017
Disp Transducer Serial No: 03A02929
Disp Cell Calibration Date: 15.30 14/02/2017

Next **OK**

Figure 29 VERSE – System Information

System Settings - allows adjustments to be made to the program via password

Edit Date and Time – allows adjustments to local date and time via password (sysdate)

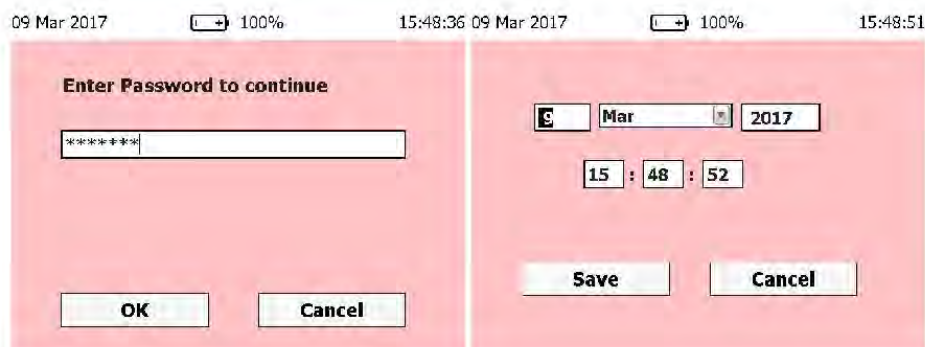


Figure 30 VERSE - Edit Date and Time Menu

Suspend – takes the handheld computer in standby mode, same as pressing power button

Restart VERSE – Takes the software back to first screen

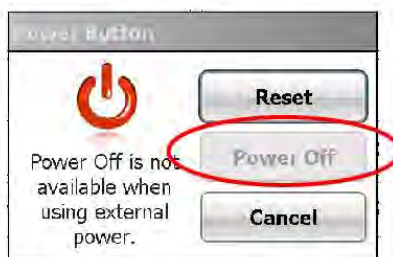
Reboot Device – reboots the handheld computer.

Reboot can also be done using power button.  located on right bottom of the keyboard.

Hold briefly the power button and using arrows choose Power Off.



CAUTION! Power off is inactive when unit is on charge



5.3.7 Taking Measurements

VERSE® - Measurement Menu, Enter Site Details

To carry out a **VERSE®** measurement, enter **Measurement Menu**.

Initially the only options available are:

Enter Site Details, Notes or Exit Menu.

Choose to **Enter Site Details** using Tab keys   and Enter key .

09 Mar 2017 100% 15:49:20 09 Mar 2017 100% 15:50:12

Enter Site Details
Enter Site Details
Enter Rail Details
Curve Compensation
View Measurement Information
Do Measurement Run
Notes
Exit Menu

ELR SWM	Track ID 114F	Miles 12
Yards 150	Curve Radius 0	Cant 20
Location Plymouth		

Save
Cancel

Figure 31 Measurement Menu - Enter Site Details

The details entered onto the Site Detail Editor are to provide information relating to the location of the **VERSE**® measurement. These boxes will be specific to each individual user, and the box titles can be customised and set in the software menu file according to the users' choice.

The boxes identify the route of the track, its number, the distance from the terminus, in either miles/yards or Km/m, and the actual location of the measurement. None of these inputs are used in the **VERSE**® calculation of the SFT. They are however recorded into the data file generated by the **VERSE**® measurement. Enter the details using the keyboard, and move onto the next box by pressing using Tab keys   and Enter key .

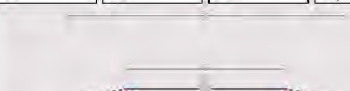
VERSE® - Measurement Menu, Enter Rail Details

Choose to **Enter Rail Details** using Tab keys   and Enter key .

09 Mar 2017 100% 15:51:47

Rail
Left

Span Total 1 15	Span Inner 1 10	Span Inner 2 10	Span Total 2 15
---------------------------	---------------------------	---------------------------	---------------------------



Save
Cancel

Rail Section
UIC50

Rail Height (mm) 1
152
Rail Height (mm) 2
152



Save
Cancel

Left

Figure 32 VERSE - Enter Rail Details Menu

The details entered into the **Rail Info Editor** are to record information relating to the actual rail being measured. This information should have been gathered during the site preparation, in section 2.

The **Rail Section** box is a pull down list. Using arrows   scroll down the list of rails and choose the correct rail type

Span Inner 1 and 2 are the measurements taken from the point of the **VERSE®** rail lift, to the support block upon which the rail is suspended.

Span Total 1 and 2 are the measurements taken from the **VERSE®** rail lift point to the first location where the rail is clipped, as per section 2.

Enter the details of these measurement into each box using the key pad, and scroll through each using the Tab keys  .

Enter Rail Depth as recorded previously during the site preparation. Work down through all of the boxes, and if satisfied that the inputs are correct, use Tab   keys to **Save** button and select Enter key .

VERSE® - Measurement Menu, Curve Compensation

In the Site Detail Editor If curve radius has been previously selected as zero, do not use the Curve Compensation feature. If however a curve radius has been included, the user may use the Curve Compensation to improve the accuracy of the **VERSE®** measurement.

To include curve compensation in the calculation, check the **Use Curve Compensation** box. The versine measurements should have been taken across the 20m span from one spacer block to the other as described in section 2. The values must then be inserted into the boxes and the save key selected.

When applying the compensation to a **VERSE®** measurement the operator has two choices

1. Add the curve compensation after the measurement has been taken and carry out versine calculation as per Table 1.
2. Include the curve compensation before the measurement by selecting Use Curve Compensation.

Enter the versine into the curve compensation window as below,

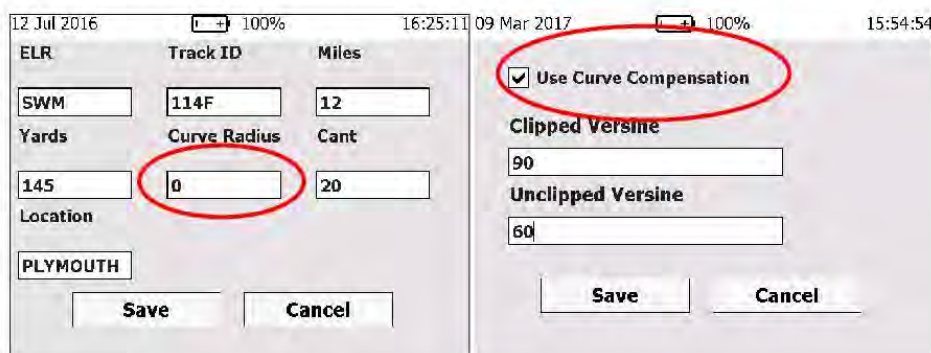


Figure 33 VERSE - Curve Compensation

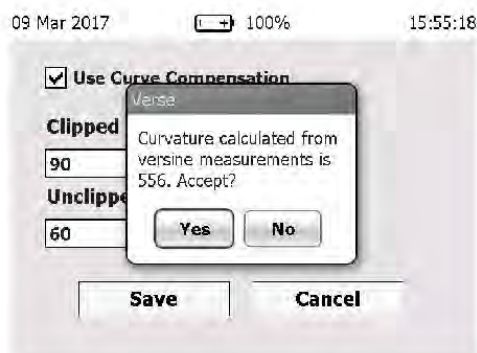


Figure 34 VERSE - Curve Compensation - Curvature calculation

The software will then calculate the radius of the curve and ask if the value is to be accepted. If "Yes" is chosen, the calculated radius of curve will be back written over the original curvature inserted in the Site Detail Editor. The curve compensation process will determine how much rail tension has been lost during the measurement process. The calculated relaxation of the rail in the curve will then be added to the calculated SFT and this will be used to improve the accuracy of the final result.

VERSE® - Measurement Menu, View Measurement Information

Select **View Measurement Information** to check that the data relating to the VERSE measurement has been entered correctly. A window will show a graphical representation of the measurement site. Use this window to check that all the data has been entered correctly.

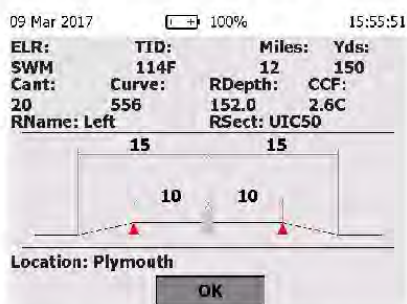


Figure 35 VERSE - Measurement Information

VERSE® - Measurement Menu, Do Measurement Run

Prior to carrying out the initial lift of the rail, the user will be asked to input the rail temperatures. If the readings from three thermometers located along the unclipped length are different, (e.g. some of the rail is in sunlight, and some is in the shade) **Temp Varies** must be selected and temperatures must be taken from three locations along the unclipped length and entered into the 3 boxes provided.

09 Mar 2017
 100%
15:56:28

Temp 1

10

☒ Temp Varies

Temp 2

10.5

Temp 3

9.6

Save

Abort

Figure 36 VERSE - Measurement Run - rail temperature before lift

09 Mar 2017
 100%
15:56:56 09 Mar 2017
 100%
15:57:15

Warning! About to start run.
Please ensure equipment is setup and safe before starting to record data.

Pressing the Abort button at anytime during a run will abort the run.

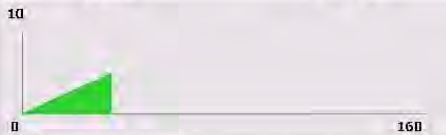
☐ Delay start of run

OK

Abort

Measurement Run: 1

Load (kN)	Disp (mm)
4.530	36.48



Part Lift

Abort

Figure 37 VERSE - Warning Screen and lift in progress

09 Mar 2017
 100%
15:57:37 09 Mar 2017
 100%
15:58:08

Start New Run

Save Data and Exit

Edit Notes



Graph for run 1

Temp 1

10

☒ Temp Varies

Temp 2

10.1

Temp 3

10.3

Save

Figure 38 VERSE - Complete graph run and end of lift rail temperatures

5.3.8 Special Lift Conditions

5.3.8.1 Start Delay

If the conditions require a part lift (see section 3.2) Follow these steps.

At the commencement of the measurement run, the warning window is displayed. In the warning window there is a tick box entitled **Delay start of run**. Checking the delay start of run box will set the software to enable the rail to be lifted clear of any obstruction. Once clear of the obstruction the measurement can begin.



Figure 39 VERSE - Delay Run option and graph

5.3.8.2 Part Lift

If the conditions require a part lift (see section 3.3) Follow these steps.



Figure 40 VERSE - Part Lift



Figure 41 VERSE - SFT results, example with curve compensation

VERSE® - Measurement Menu, Notes

At any stage through the measurement process the Notes button may be selected which will open up a notes window. The Notes window is used to record extra detail relating to the measurement or track related issues. It is good way of communication between track workers and other departments. This information will be saved into the data file and may be viewed at a later stage on the PC software.

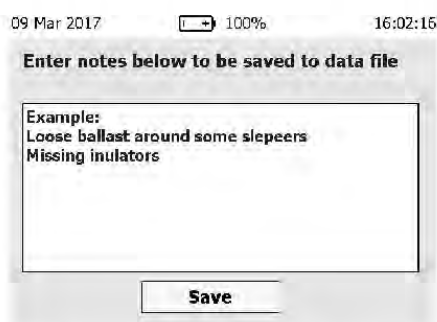


Figure 42 VERSE - Notes menu, example



6 Rail Measurement

6.1 Method of SFT measurement

Once the site and equipment are prepared as per section 2 the rail measurement may commence.



WARNING! When the rail is being measured, always check that everyone is clear before undoing the release valve on the hydraulic jack or releasing the rail jack when inserting or removing the spacer blocks.

1. Ensure the site and rail information are entered into the handheld computer as described in section 5.3.
2. Check all the connections to the control box has been made and switch the control box on.
3. Close the valve on the bottle jack and start lifting the rail through operating the bottle jack.



Figure 43 Bottle jack release valve

4. Operate the handle smoothly and steadily. Graphical output on the handheld computer should be linear. Once load reaches 10kN the data collection is suspended for that lift.



WARNING! Do not apply load to jack handle once the jack is at the end of its stroke. This is misuses of equipment.

09 Mar 2017 100% 15:56:28 09 Mar 2017 100% 15:56:56

Temp 1
10

☒ Temp Varies

Temp 2
10.5

Temp 3
9.6

Save Abort

Warning! About to start run.
Please ensure equipment is setup and safe
before starting to record data.

Pressing the Abort button at anytime during a
run will abort the run.

☐ Delay start of run

OK Abort

Figure 44 VERSE - Measurement Run - rail temperature before lift

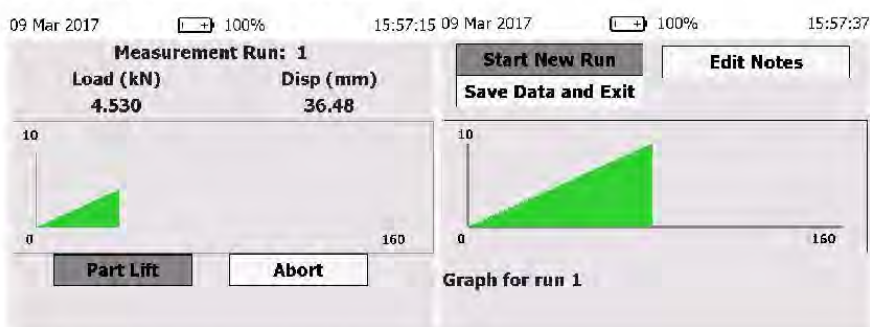


Figure 45 VERSE - Warning Screen and lift in progress

5. Lower the rail by releasing the pressure in the bottle jack, next select Start New Run and lift the rail as per guidance above.
6. Repeat for at least 3 lifts per measurement site
7. At the end of the third lift select Save Data and Exit and record end lift rail temperature.

09 Mar 2017 100% 15:58:08

Temp 1
10

☒ Temp Varies

Temp 2
10.1

Temp 3
10.3

Save

Figure 46 End lift temperature record

8. Reinstall the rail to a previous state
9. For measuring in curves or special conditions see section 3
10. Copy the measurement lifts to a PC as per section 7



7 Data Transfer

7.1 Allegro²_{TM} /PC - PC Communications

Before connecting the handheld computer to a desktop computer, verify that you have the needed connection software (with drivers) installed:

- Microsoft Windows Mobile Device Center (WMDC, used for connecting Windows Mobile/CE devices to Windows Vista/7/10, older version comes pre-install in Vista)
- Microsoft ActiveSync (used for connecting Windows Mobile/CE devices to Windows XP/2000/98)

Once communication has been established the icon of Allegro 2 should appear and green tick saying **Connected**.



Figure 47 WMDC with connected Allegro²_{TM}

7.2 Connection Issues and Suggested Solutions

Below is a list of the common issues encountered when connecting to Microsoft ActiveSync / Windows Mobile Device Centre (WMDC):

- Installing Microsoft ActiveSync (Version 4.5 or later) / WMDC
- Firewall and virus scan programs blocking connection
- Port availability
- Cables and hardware

Please contact IT department or Vortok for support with connection issues.

7.3 Accessing Allegro²_{TM} Directory in Windows Mobile Device Centre

- Click on **Connect without setting up your device**
- Next hover your mouse over **File Management** and click on **Browse the contents of your device**



Figure 48 Windows Mobile Device Center - Main Window with status connected

- ✓ Click on the Root Directory
- ✓ Then Navigate to Verse→Data. The **VERSE**® measurement data files recorded from the **VERSE**® measurements are held in the Data folder

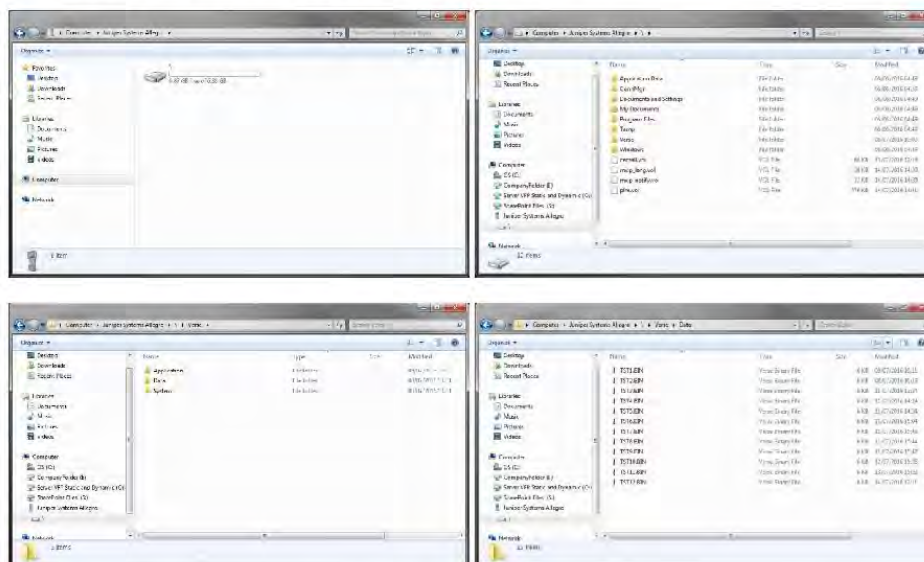


Figure 49 Path to data folder

The files are recorded in a numerical order and will remain on the handheld computer until they are deleted.



8 Lift Data Analysis

8.1 VERSE® PC Software Analysis

In order to read **VERSE®** data files PC **VERSE®** software must be installed on PC. The PC **VERSE®** software is included in the kit.

The data files must be copied out from the handheld computer and onto the PC. This can be achieved by selecting the files , then coping and pasting in the chosen location on PC, or by selecting the data files and dragging and dropping them in the selected folder on PC using one of two methods.

8.2 VERSE® PC Software Installation



Figure 50 Splash Screen

Insert the USB drive with software **VERSE®** into the USB socket on the PC. From the directory click on open Menu.exe. The USB Menu will appear.

Next click button called Install and follow prompts on the screen. The **VERSE®** software will be installed. The software shortcut will be created on the desktop. Click the shortcut to open the **VERSE®** PC software. You will initially see the splash screen with handheld computer serial number and Owner/User names.

Go to **File/Open** or click on the **Open** icon  to open the data file.

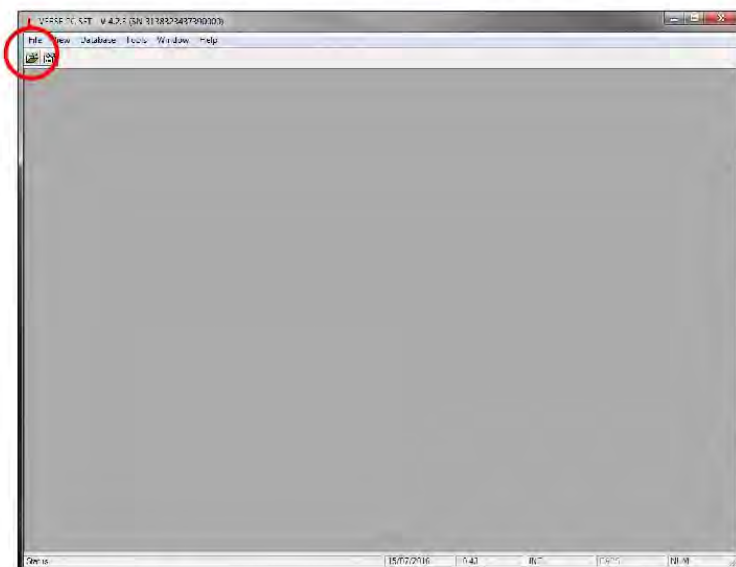


Figure 51 VERSE PC software main window

Navigate and choose a file you want to analyse and then click Open. Multiple files can be selected if required.

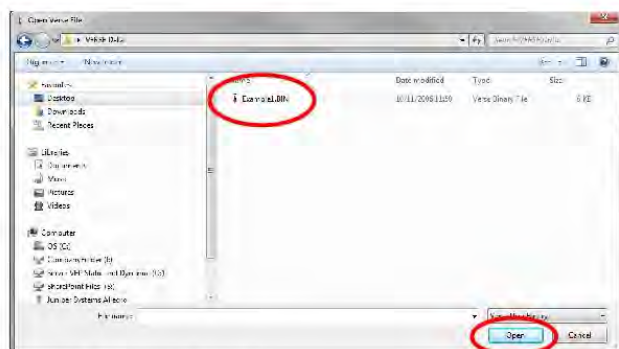


Figure 52 Open VERSE file window

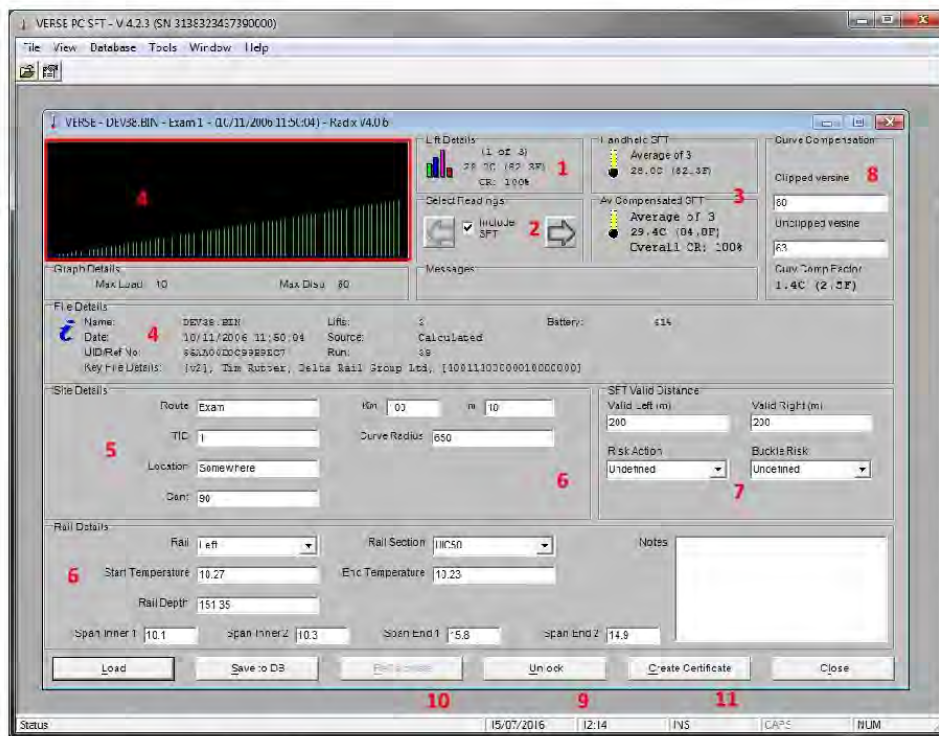


Figure 53 VERSE – Lift results, example

1. The **Lift Details** are shown in a box, and the lower panel gives the details of the .BIN file, the time and data of the measurement, the serial number of the Radix, the number of lifts recorded and the run number
2. To scroll through the readings taken in one full measurement use the arrow keys in the **Select Readings** box. If a lift is to be excluded from the calculations remove the “tick” from the Include SFT box by clicking on it.
3. The average of the results calculated by the handheld computer is shown in the upper window. The colour of the thermometer icon indicates if the result is in tolerance. (Note: Setting the tolerances and colours is detailed in section 8.3). The lower window displays the average of the results calculated by the PC. The user has the option to recalculate the result from the handheld automatically upon opening the file by checking the box in the tools options menu
4. The upper left display is a bar graph of the load/displacement. Each of the green lines represents the acquisition of data taken during the lifting process. The Load is displayed on the vertical axis. The max Load and Displacement for the lift are displayed below the bar graph
5. The **Site Details** entered into the handheld computer for the measurement run are shown in this area of the window and can be checked for accuracy.
6. Rail Details entered into the handheld computer for the measurement run are shown in this area of the window and can be checked for accuracy. Any notes, which

- may have been recorded on the handheld computer at site, will be displayed in the Notes window.
7. The **SFT Valid Distance** can be inserted into the area shown for recording on the database. The valid distance is that which is considered by the engineer to be consistent with the SFT result. Pull down options also give the user choices with regard to **Buckle Risk and Risk Action**.
 8. Curve Compensation information as per input in the handheld computer.
 9. If errors in the input data are noted, they can be corrected by unlocking the data file with the **Unlock** button at the bottom of the window.
 10. The details in the each box can now be edited. Those boxes with the colour changed to red, will affect resultant SFT if modified. The **Recalculate** button must be selected if one or more of the red boxes are altered. The file modified will then be displayed in the **File Details** area as **User Altered**. This provides a full auditable trail of modifications and corrections. The original files cannot be modified and the recalculate file cannot be saved. Modified files can however be exported to the database for future reference.
 11. Once the operator is satisfied with the details the file can be saved to the database by clicking **Save to DB** button. Repeat this for each new file (measurement).

8.3 Temperature, Tolerances and Colours Set Up.

The thermometer icon in the file window displays a colour dependent on the chosen settings in **Colour Options**. The purpose of this feature is to assist the operator in the easy identification of sites, which may need urgent attention. If the colour is red it is because the result is much too low, or much too high, it indicates urgent corrective action. If the colour were yellow, it would indicate the result is out of tolerance and corrective action is required but is not urgent. If the colour is green the result is in tolerance

The operator can select the colours and the temperatures at which the colours should change. This is done in the Colour Options window, which is opened by clicking the **Colour Options** icon or selecting **Tools/Colour Options** from the **Menu** bar.

In the **Min** box, enter the temperature below which any result would require urgent corrective action to the stressing of the rail. In the "A" box, enter the temperature, which is the lower limit of the acceptable tolerance. In the "B" box, enter the temperature, which is the upper limit of the acceptable tolerance. In the **Max** box, enter the temperature above which any result would require urgent corrective action to the stressing of the rail. Clicking on either of the colour boxes opens a colour panel, which allows the choice of colour options according to user requirements. For Example: green, yellow and red. Choose to work in **Degree Celsius** or **Degrees Fahrenheit** by clicking the appropriate button in the top right corner of the window. When all the settings have been chosen click the **Apply** button.

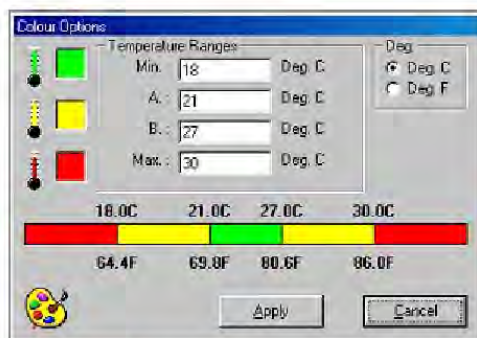
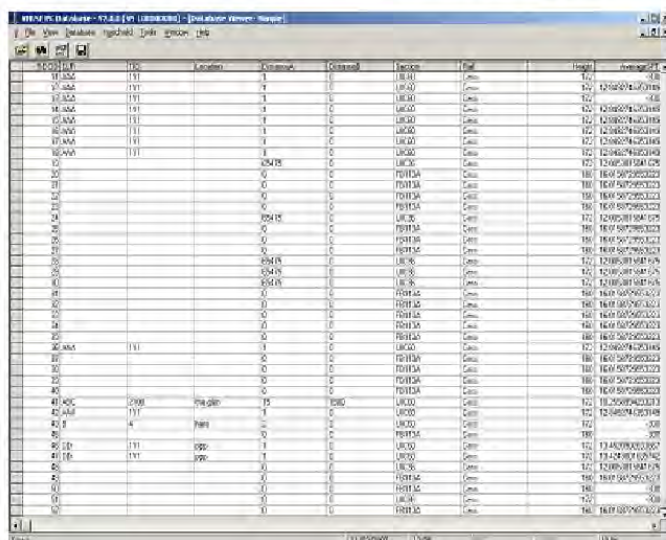


Figure 54 VERSE - Colour option menu

8.4 The Database

The database can be viewed in two modes: the **Simple** form or the **Full** form. To view the **Simple** database click the Database menu and select Display (Simple) and the key items for each measurement processed is shown. (Note: if -300 or -1000 is shown as the result, it indicates that there was no SFT result. A zero could be confused with an actual result of 0°C or 0°F)



ID	Location	Equipment	Component	Section	Test	Result	Average (°C)
11	111	1	1	1	1	1	1
12	111	1	1	1	1	1	1
13	111	1	1	1	1	1	1
14	111	1	1	1	1	1	1
15	111	1	1	1	1	1	1
16	111	1	1	1	1	1	1
17	111	1	1	1	1	1	1
18	111	1	1	1	1	1	1
19	111	1	1	1	1	1	1
20	111	1	1	1	1	1	1
21	111	1	1	1	1	1	1
22	111	1	1	1	1	1	1
23	111	1	1	1	1	1	1
24	111	1	1	1	1	1	1
25	111	1	1	1	1	1	1
26	111	1	1	1	1	1	1
27	111	1	1	1	1	1	1
28	111	1	1	1	1	1	1
29	111	1	1	1	1	1	1
30	111	1	1	1	1	1	1
31	111	1	1	1	1	1	1
32	111	1	1	1	1	1	1
33	111	1	1	1	1	1	1
34	111	1	1	1	1	1	1
35	111	1	1	1	1	1	1
36	111	1	1	1	1	1	1
37	111	1	1	1	1	1	1
38	111	1	1	1	1	1	1
39	111	1	1	1	1	1	1
40	111	1	1	1	1	1	1
41	111	1	1	1	1	1	1
42	111	1	1	1	1	1	1
43	111	1	1	1	1	1	1
44	111	1	1	1	1	1	1
45	111	1	1	1	1	1	1
46	111	1	1	1	1	1	1
47	111	1	1	1	1	1	1
48	111	1	1	1	1	1	1
49	111	1	1	1	1	1	1
50	111	1	1	1	1	1	1
51	111	1	1	1	1	1	1
52	111	1	1	1	1	1	1
53	111	1	1	1	1	1	1
54	111	1	1	1	1	1	1
55	111	1	1	1	1	1	1
56	111	1	1	1	1	1	1
57	111	1	1	1	1	1	1
58	111	1	1	1	1	1	1
59	111	1	1	1	1	1	1
60	111	1	1	1	1	1	1
61	111	1	1	1	1	1	1
62	111	1	1	1	1	1	1
63	111	1	1	1	1	1	1
64	111	1	1	1	1	1	1
65	111	1	1	1	1	1	1
66	111	1	1	1	1	1	1
67	111	1	1	1	1	1	1
68	111	1	1	1	1	1	1
69	111	1	1	1	1	1	1
70	111	1	1	1	1	1	1
71	111	1	1	1	1	1	1
72	111	1	1	1	1	1	1
73	111	1	1	1	1	1	1
74	111	1	1	1	1	1	1
75	111	1	1	1	1	1	1
76	111	1	1	1	1	1	1
77	111	1	1	1	1	1	1
78	111	1	1	1	1	1	1
79	111	1	1	1	1	1	1
80	111	1	1	1	1	1	1
81	111	1	1	1	1	1	1
82	111	1	1	1	1	1	1
83	111	1	1	1	1	1	1
84	111	1	1	1	1	1	1
85	111	1	1	1	1	1	1
86	111	1	1	1	1	1	1
87	111	1	1	1	1	1	1
88	111	1	1	1	1	1	1
89	111	1	1	1	1	1	1
90	111	1	1	1	1	1	1
91	111	1	1	1	1	1	1
92	111	1	1	1	1	1	1
93	111	1	1	1	1	1	1
94	111	1	1	1	1	1	1
95	111	1	1	1	1	1	1
96	111	1	1	1	1	1	1
97	111	1	1	1	1	1	1
98	111	1	1	1	1	1	1
99	111	1	1	1	1	1	1
100	111	1	1	1	1	1	1

If a more comprehensive view is required, select **Display Full**. If greater analysis is required, or printing in a defined layout the databases can be exported to a .csv format.



To export a Simple database, select Database/Export Simple from the menu. The following window opens. Enter a name for the new file in the File name box and click Save. A spreadsheet program can then open the file independently. Note that this exported file will not be updated by processing more data files.

8.5 Measurement Report

When a data file has been imported and viewed, a **VERSE**® Measurement Report may be produced relating to each **VERSE**® measurement. This certificate is in HTML format and it can be adjusted to the customer's requirements. Check the box in the Tools Options menu to open the certificate when created.

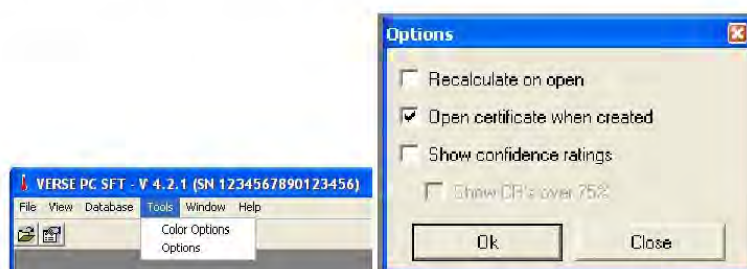


Figure 55 Option to create a certificate when accessing the run measurement

To produce the HTML file, click on the Create Certificate button. This will open an HTML page in your web browser on the PC. A certificate will be produced like that below. This will include all the inputs and values recorded during the measurement. This certificate is a true record of the measurement and it can have a name and signature box for validity and authentication purposes. A contractor may choose to present such a certificate as a record of the work carried out.

VERSE SFT Measurement Report

Location and SFT Results

Date	Time	FLR	TID	Miles	Yards	Rail	SFT
03/10/2006	12:50:53	888	111	0	0	Left	16.2C (64.7F)

Rail Details

Rail Section	Rail Height	Start Temp.	End Temp.
UIC60	152mm	10C	10C

Site Details

Location	Cant	Curve
dry	n	n

Equipment Details

Handheld Serial Number	Load Cal.	Displacement Cal.	Battery Level
55APC000C800011F4	-402.72	-32.66	94%

File Details

File Name	Runs Recorded	PC Software Version
DEV1.BIN	6	V 4.0.2

Data Recorded

Run	1	2	3	4	5	6	7	8	9	10
Time	12:51	12:52	12:54	12:55	12:56	12:57				
SFT	16.2C (64.8F)	16.4C (65.5F)	15.6C (65.1F)	17.3C (65.1F)	18.0C (64.4F)	18.3C (65.3F)	-	-	-	-
Error	0	0	0	0	0	0	20	20	20	20
User Selected	1	-	1	-	1	1	0	0	0	0

☐ User
☐ Signed
☐ Date

Figure S6 VERSE SFT Report, example

8.6 VERSE® Confidence Rating

The **Confidence Rating** is a feature which gives the **VERSE®** user an indication of the accuracy of the calculated SFT. The user has a choice as to the display of this feature, and the settings can be adjusted in the **Tools** menu by selecting **Options**.

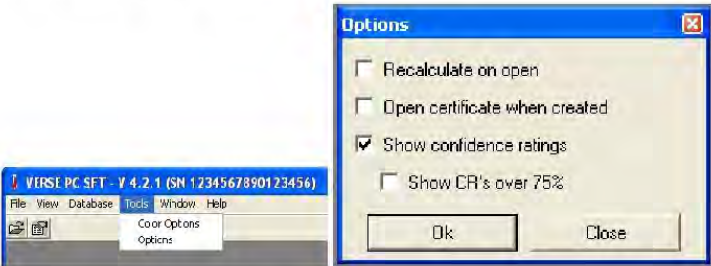


Figure S7 VERSE - Option to show confidence rating

The **Confidence Rating** will be displayed by choosing **Options**, and checking the “**Show confidence ratings**” box.

A further option exists, whereby only those ratings over 75% are displayed. This can be set by checking the “**Show CR's over 75%**” box.

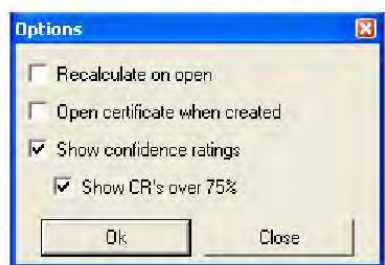


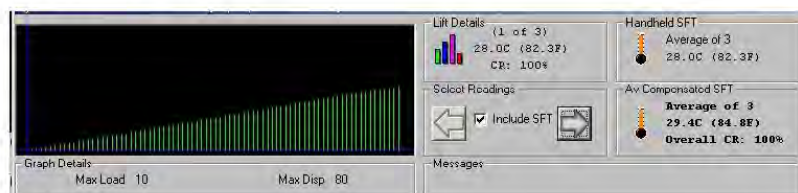
Figure 58 VERSE - Option to show confidence rating over 75%

This rating is given as a percentage and does not give any indication of the actual accuracy of the result, but is designed to warn the user when the measured data is below a certain standard.

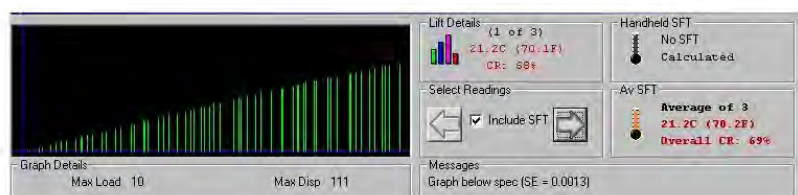
Note: It remains the responsibility of the user to decide whether the SFT calculated is valid.

The CR (confidence rating) indicates how much confidence the user can have in the calculated SFT result. It is based on how close the SFT is to the average rail temperature and also on how accurately the points fit a straight line. If a CR of less than 75% is calculated, care must be taken when interpreting the results. The software highlights this condition by displaying the SFT in red. There are some cases when a low CR can still provide a reasonable estimate of the SFT.

The following example shows a perfect fit line, which receives a CR of 100% (shown in example 1 of the supplied example files):



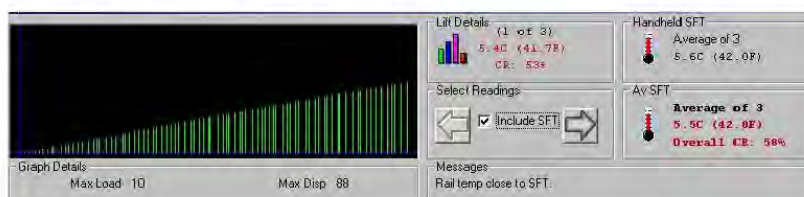
In this example the points are quite scattered but still seem to follow a straight line. This result could be used with discretion.



This example has too much scatter in the points for the result to be dependable. This is indicated in the CR of 26% compared to 68% above.



This example has a good line of fit but a poor CR still results. This is because average rail temp is 5.1°C where the SFT is 5.4°C. When operating close to the SFT the equipment becomes less accurate. Again, this result could be used with discretion.



As mentioned above, the CR is based on how well the points fit a straight line and also how close the average rail temperature is to the calculated SFT. To calculate the SFT the software uses linear regression analysis to calculate a line of best fit. The standard error (SE) of the points to this line is also calculated. A confidence rating is calculated based on the SE where 1 is a perfect fit and 0 is the worst fit. An SE of 0.001 will give a CR rating of 75%. A rating is also calculated on how close the SFT is to the average rail temperature. Above 1.5°C this rating is 1 i.e. no effect on accuracy. Below this value the rating gets worse. The two ratings are then multiplied together and converted to a percentage to give the final rating. The overall CR is the average of all the CR's of the user-selected runs.



9 Calibration and Service

Electronic system calibration should be carried out annually. This is carried out by Vortok International. On completion, a new calibration certificate will be issued.

When the complete Verse kit in a case is sent to Vortok, a technician will carry out the full inspection and service of the kit. The missing or damaged components will be replaced following quote and order from the owner/user and calibration carried out

The minimum level of equipment required by Vortok International to carry out a calibration is a content of the yellow case as shown below.



Figure 59 Basic level of equipment for calibration

⚠ CAUTION! The calibration certificate only applies to the full kit of parts supplied. If the kit components are exchanged/separated then the calibration is no longer valid.



10 Maintenance and Troubleshooting

10.1 Warning, Error Messages and Meanings

Table 2 Error table

Error/Problem	Possible cause	Correction
Error 410 - Not enough data.	Not enough data points were collected to calculate an SFT.	Repeat the run with slower lift.
		System must be returned and rechecked. Contact Vortok
Error 411 - Load too low.	The load reached was not high enough to calculate an SFT.	Repeat the run
	Leaky bottle-jack-seal damaged	Bottle jack must be replaced-contact Vortok
	Release valve not closed	Close the release valve
	Over pressure valve not tight	Tighten the valve
Error 412 - Load out of spec.	This generally means that the rail is in compression. If the data points do not create a straight line to within a certain tolerance then the SFT cannot be calculated.	Come back and repeat the run when rail temperature is lower.
Error 413 - Calc run out.	This general implies a serious input error as the calculation went through its maximum number of iterations	Repeat the run
Error 414 - Unable to calculate. (Rail temp above SFT ?).	This error occurs because there are no valid SFT results from which to calculate the average SFT value. This can be as a result of the rail temperature being above the SFT. If this is the case then it is not possible to calculate SFT values.	Repeat the run Come back and repeat the run when rail temperature is lower.
Error 415 - Check Hardware.	The handheld computer was unable to connect to the transducers. Damaged cable Loose plugs, sockets	Check connection security
		Clean all connections
		Check the cables are not trapped or damaged
	Control box not switched on	Switch on the control box
Error 416 - Version Match	The software version does not match the menu file version	Contact Vortok International
Error 417 - Key not valid.	The software key file is not valid.	New key must be issued. Contact Vortok International



10.2 Steps for a Successful **VERSE**® Measurement

- Select a measurement site and obtain a track possession
- Have received training in the use of **VERSE**® and hold a certificate of competence.
- Have a complete set **VERSE**® of equipment, in calibration and ensure the batteries are fully charged.
- Measure accurately and mark up the site as per the manual section 2
- Unclip the selected section of rail, ensuring the rail is not in compression.
- Accurately install the support blocks ($\pm 1\text{cm}$)
- Set up and record the rail temperature
- Record the rail type
- Assemble and fit the **VERSE**® equipment
- Set up and capture the data using handheld computer data logger
- Check after each lift that the data is valid and accepted by the handheld computer
- Calculate the SFT using the handheld computer data logger. (3 lifts minimum)
- Remove the **VERSE**® equipment and repack carefully in its container. Ensure that connector box is switched off.
- Depending on the result, reinstate the track or schedule a re-stressing activity.
- Remove all equipment from site and ensure the track is left in a safe condition.
- Upload data from the handheld computer to a PC and record the results in the database.
- Place the handheld computer on trickle charge and ensure that the equipment is dry and clean ready for next use.



11 Procurement Reference

11.1 VERSE Frame spare parts

Table 3 VERSE Frame Spare Parts List

Part No.	ITEM
112-1001	Case
112-1005	Lift Cable
112-1010	Plumb Bob
112-1016	Rail Hook
112-1024	Bottle Jack + Adjustable Jack Strut + Mounting Plate
112-1025	Jack Handle
112-1038	Bridge 3 rd Rail

11.2 VERSE Accessories spare parts

Table 4 VERSE spare accessories parts list

Part No.	ITEM
112-1002	Sandbag (without plate)
112-1003	Plate (without sandbag)
112-1004	Sandbag with plate
112-1007	Force Transducer Yokes
112-1009	Rail Clamp for Displacement Transducer
112-1013	Digital Thermometer (excluding thermocouple and lead)
112-1014	Thermocouple and Lead for Digital Thermometer
112-1015	Spacer Block
112-1017	Instrument Case-Radix
112- 1018	Large Rail Hook (141) - Heavy
112-1019	Linear Vernier and Extension Piece (Calibrated Set)
112-1020	Extension Piece Only (for linear vernier)
112-1021	Measuring Tape
112-1022	VERSE® Manual
112-1023	Allen Keys

11.3 VERSE electronics spare parts

Table 5 VERSE electronics spare parts

Part No.	ITEM
112-1006	Force Transducer
112-1007	Force Transducer Yokes
112-1008	Displacement Transducer
112-1009	Rail Clamp for Displacement Transducer
112-1011	Digital Control Box
112-1012	Battery Box (excluding batteries)
112-1068	Force Transducer Top Yoke VE200023
112-1069	Force Transducer Bottom Yoke VE200024
112-1070	Digital Control Box Charger

11.4 VERSE Husky spare parts (discontinued handheld)

Table 6 VERSE Husky spare parts

Part No.	ITEM	
112-1027	Husky battery charger	
112-1028	Husky NiMH battery	
112-1029	Husky data serial cable to digital box (9 pin)	
112-1043	Husky data serial cable to digital box (25 pin)	
112-1030	Husky to PC serial cable (9pin to 9pin)	
112-1044	Husky to PC serial cable (9pin to 25pin)	
112-1031	USB to serial converter	

11.5 VERSE Radix spare parts (discontinued handheld)

Table 7 VERSE Radix spare parts

Part No.	ITEM	
112-1032	Radix FW900 128Mb with software	
112-1034	Radix Charger & Cradle	
112-1035	Radix data cable to digital box	
112-1037	Radix in-car charger	
112-1040	Radix Power Module	
112-1041	Radix Keypad	

11.6 VERSE Allegro² spare parts (current handheld)

Table 8 VERSE Allegro² spare parts

Part No.	ITEM	
112-1094	Allegro ² Standard with software	
112-1092	Allegro ² 12V Vehicle Charger Cable	
112-1115	Allegro ² data cable to control box	
112-1114	Allegro ² Instrument case	
112-1102	Allegro ² Li-Ion rechargeable battery	
112-1107	Allegro ² bumper kit , orange	
112-1104	Allegro ² Replacement Hand Strap	
112-1098	Allegro ² Wall charger with international plugs kit	
112-1097	Allegro ² Micro USB PC data sync cable	

11.7 Warranty

Unless otherwise stated standard terms and conditions apply.

11.8 Training

Contact Vortok International Sales for training support.

11.9 End of Life

It is recommended that if a **VERSE**[®] unit is taken out of service at the end of its life, the unit should be returned to Vortok International for decommissioning.

Components will be recycled where possible or disposed of by Vortok International in accordance with our ISO14001 policy. The database will be maintained to reflect this.

VERSE

Issue:	A01
Date:	December 2003
eB Ref:	

1. PURPOSE

The purpose of this document is to layout the guidelines for implementation of VERSE® and to assign responsibility for the various activities within this process.

2. SCOPE

The guidelines laid out in this document are applicable to all WCMU Projects that contain track work activities. Implementation of this procedure is required when the SFT of rail is suspect either due to renewals activities or the SFT of the track is indeterminate due to the passage of time or track usage.

3. RESPONSIBILITIES

3.1 Project Track Engineer

The Project Track Engineer is responsible for:

- Identifying the track sites to be VERSE® tested. Site Assessment information is to be included in the identification of test sites.
- Reviewing the results of the VERSE® tests and identifying required follow on work or testing. Liase with the Network Rail Track Engineer as required.
- Communication of VERSE® requirements and follow on work/tests with the DPE, the Construction Manager and the Area Construction Engineer.

3.2 Responsible Field Engineer

- Verify that a site assessment has been performed, recorded on the appropriate form
- Results of the site assessment have been forwarded to the Project Track Engineer and the Contractor for consideration in the Site Specific Method Statement. Liase with the Network Rail Track Engineer as required.
- Verify work is performed in accordance with Method Statement(s), applicable Standards and this Guideline.
- Record observations and independent measurements on a Daily Report



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3.3 Construction Manager

The Construction or Site Manager is responsible for:

- Ensuring a valid purchase order is in place to supply VERSE® services.
- Scheduling the identified VERSE® and follow works if required.
- Obtaining the necessary track access to perform the VERSE® and follow on works if required.
- A Site Specific Method Statement from the Contractor must be accepted prior to site access.
- Coordinating work activities with the responsible contractor performing the works to verify understanding of work to be performed and assignment of resources.

4. ABBREVIATIONS AND DEFINITIONS

mm – millimetre(s)

DPE – Designated Project Engineer

m – metre(s)

S & C – Switches and crossings, either individual built-up components or complete layouts

Stress-free Temperature (SFT) – Is the rail temperature at which the rail is the same length as it would be in an unrestrained state and at which, therefore, there is no thermal force present.

VERSE® – Is an approved, non-destructive methodology for measuring the SFT of Continuous Welded Rail.



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5. PROCESS

5.1 Track Conditions for consideration of VERSE®

The Project Track Engineer must consider VERSE® as a method of identifying or validating track SFT if there is cause to doubt the validity of stressing records, when there are no valid stressing records or when the Project:

5.1.1 Includes track renewals and the stability of the track may have been compromised by follow on track work after the track has been stressed. (Section 12 of RT/CE/S/011 defines work that may impact track stability.)

5.1.2 Includes track maintenance where the track stability may have been compromised and may have appreciably changed the SFT. (Section 12 of RT/CE/S/011 for defines of work that may impact track stability.)

5.1.3 Track work does not include sections of the track but track work/conditions are of a nature whereby VERSE® testing is a recommended method of validating track SFT.

5.2 VERSE® Site Definition

The Project Track Engineer shall consider the following in defining the location of each VERSE® test site:

5.2.1 Randomly pick and perform one VERSE® test per mile of track, or portion of a mile (where the portion exceeds one quarter mile).

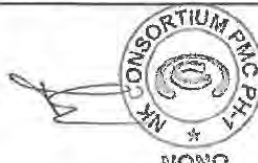
NOTE: One VERSE® test determines the SFT on up to 1500m of straight track (750m on either side of the lift point) and curved track of 3000m radius and above. For curved track between 3000m radius and 500m radius, one VERSE® test is required for every 400m (200m on either side of the lift point). This methodology is not to be used for curved track of less than 500m radius.

5.2.2 The site test limits should be no closer than 30m to any S&C.

5.2.3 The 30m test length should be free of welded joints. If this can not be achieved, then the lift point for the VERSE® should be positioned centrally between two welded joints

5.2.4 The rail within the test section (30m) shall be of uniform section.

5.2.5 Both rails of a track should be VERSE® tested to validate the 'SFT achieved' as recorded on the stressing certificate and to confirm that the SFTs of the 2 rails meet the requirements of RT/CE/S/011 in



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respect of the equality of stress in the rails.

5.2.6 If the VERSE® data meets the criteria of RT/CE/S/011, the represented track site (one mile section or portion of a mile) is considered acceptable and no further testing for that section is required.

5.2.7 If the VERSE® data from a site shows the SFT of the rail is not within the allowable of RT/CE/S/011, or if it is not within $\pm 2^{\circ}\text{C}$ of the stressing certificate for renewals sites, then an additional test site within the one quarter mile of track of either side of the failed test site should be VERSE® tested to determine if the failed site is an isolated section of track. Sites with VERSE® results having an unacceptable SFT per RT/CE/S/011, shall be re-stressed.

5.2.8 If the VERSE® data from the additional test sites is acceptable, the remainder of the represented site is considered to be acceptable.

5.2.9 If there is additional data in the second set of tests that is unacceptable, then additional testing is required. The process of unacceptable data and further testing continues until forty percent of the track site has been tested and if at that point there is still unacceptable data, then the stressing process is deemed to be out of control and one hundred percent of the track shall be re-stressed.

5.2.10 Where stressing certificates do not exist, randomly pick and perform one VERSE® per mile of track, or portion of a mile (where the portion exceeds one quarter mile). Acceptance of the data shall be per the guidelines listed in Items 5.2.5 – 5.2.9 above.

5.2.11 Where stressing certificates are more than 20 years old, randomly pick and perform one VERSE® per mile of track, or portion of a mile (where the portion exceeds one quarter mile). Acceptance of the data shall be per the guidelines listed in Items 5.2.5 – 5.2.9 above.

5.3 Environmental Conditions limiting use of VERSE®

VERSE® should not be carried out in the following conditions, as the resulting data will be indeterminate:

5.3.1.1 The rail is undergoing rapid changes of temperature (1°C in 15 minutes or greater).

5.3.1.2 Where the rail temperatures are/or could be at or above the actual SFT.

5.3.1.3 The rail temperatures are below -5°C .



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5.4 Methodology for VERSE®

The following work procedure is to be followed to ensure accurate results are obtained from VERSE®:

- 5.4.1 Prior to accessing the site, verify that the VERSE® recording equipment operates and is properly calibrated and the batteries in the apparatus are fully charged and of the correct type. Review equipment certification sheet to ensure the calibration of the equipment applies to the serial numbered components included with the kit, i.e. the computer, the transducers, and the hydraulic lifting frame are the same components that were calibrated as a unit.
- 5.4.2 At the work site, identify flash and/or alumino-thermic welds within the selected 30m test site. Finalise the lift point based on this information, ensuring that any welds are greater than 20m from the lift point. Record exact location of the lift.
- 5.4.3 Determine (by reading the brand marks on the rail web, measuring the rail height with rail gauge and the callipers) and record the rail section.
- 5.4.4 Place a calibrated thermometer on the rail web or foot as appropriate to the type of thermometer, out of direct sunlight, near the lift point and at one other location. Leave for a minimum of 15 minutes before taking a temperature reading. Record thermometer calibration numbers and temperatures.
- 5.4.5 The 30m of rail to be unclipped should be clearly marked.
- 5.4.6 Ensure excess ballast is removed from around the rail, sleepers and clips in the 30m section so the equipment can be properly set up, does not impede the results and does not obstruct re-clipping the rail.
- 5.4.7 Unclip the 30m length of rail.
- 5.4.8 Insert the 'T-pieces' at 10m on either side of the lift point.
- 5.4.9 Remove all insulators from the rail in the unclipped section.
- 5.4.10 Measure and record (to the nearest mm) each leg (from lift point to T-piece and from T-piece to anchor) of the unclipped rail.
- 5.4.11 Contractor installs the the VERSE® equipment and performs lifts, gathers data, as per their approved procedure.
- 5.4.12 Verify 'T-pieces' are removed upon completion of VERSE®.
- 5.4.13 Verify all insulators have been re-installed correctly.
- 5.4.14 Verify that the 30m of track is properly clipped before leaving the site.



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APPENDIX A

NETWORK RAIL FIELD ENGINEER'S CONTRACTOR QUALITY PLAN VERIFICATION FOR VERSE®



Date _____ Shift _____ Weather Conditions _____

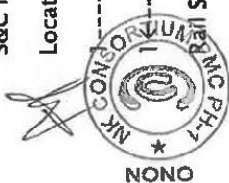
Location of lift Mile _____ Yardage _____

(Circle appropriate selection) Up/Down _____

Fast/Slow/Other (specify) _____ Field/Six Foot Rail/Ten Foot Rail _____

S&C is greater than 30m from test location _____

Locate all welded joints (X) in relation to lift point (L). Location of thermometer readings (T). Record leg lengths.



Rail Section _____

Thermometer No. _____ Reading _____ °C

2nd Reading _____ °C

Average Temperature used _____ °C SFT recorded by Verse® _____ °C

VERSE Equipment Calibration Sheet matches Serial Numbers of kit: _____

Verify rail is properly clipped up _____

Verify all tools/equipment/rubbish are removed from site _____

Comments/Slagging items: _____

Field Engineer _____ Date _____



Rete Ferroviaria Italiana
14/12/2005

RFI-DTCL
A00111P12005/0001888

Direzione Tecnica
il Direttore

Spett.le Vortok International
6 - 8 Haxter Close,
Belliver Industrial Estate
Roborough, Plymouth,
Devon, PL6 7DD England

DIREZIONE MANUTENZIONE
SEDE

DIREZIONE INVESTIMENTI
SEDE

Spett.le ITALFERR
Via Marsala, 53
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Oggetto: Autorizzazione all'uso del sistema diagnostico VERSE

Si comunica che:

visto il documento emesso dal Railway Research Institute Prague ed. aprile 2001:
"Empirical Assessment of Accuracy of Neutral temperature measurements of
continuously welded rail by means of VERSE equipment"
visto il documento emesso dal Transportation Technology Center di Pueblo ed. 2000:
"Vertical Rail Stiffness Equipment (VERSE) Trials"
visto il documento emesso da AEA Technology rail ed. novembre 1998 "Trial of the
new Verse method at Bevercotes Colliery Branch"
visto il Certificato di accettazione Railtrack No PA05/901 del 12/02/01
visto il certificato di autorizzazione all'uso emesso da DB Netz AG nel dicembre 2002
si autorizza l'uso del sistema Verse per la misura della temperatura neutra delle rotaie in
lunga rotaia saldata.

L'apparecchiatura può essere utilizzata alle seguenti condizioni:

1. l'apparecchiatura deve essere tenuta e tenuta sotto controllo



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$SFT - 4\phi - 30 = 110$
 $f_m = 35$
 $41 - 8$

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This letter is to inform you that:

- in consideration of the document issued by the Railway Research Institute Prague ed. April 2001: "Empirical Assessment of Accuracy of Neutral temperature measurements of continuously welded rail by means of VERSE equipment".
- in consideration of the document issued by the Transportation Technology Center di Pueblo ed. 2000: "Vertical Rail Stiffness Equipment (VERSE) Trials"
- in consideration of the document issued by AEA Technology rail ed. November 1998 "Trial of the new Verse method at Bevercotes Colliery Branch"
- in consideration of the homologation letter issued by Railtrack No PA05/901 dated 12/02/01
- in consideration of the homologation letter issued by DB Netz AG in December 2002

we authorize the use of the Verse system to measure the neutral temperature of the rail on long welded rails.

The equipment can be used at the following conditions:

1. the unit must be calibrated and kept under control
2. the system shall be used in accordance with what stated in the Operator's Guide to Verse serial number MC/RFI/124 issue 1, dated 11/11/2005 and issued by Vortok International, and shall only be used by trained and competent personnel who are in possession of a certificate of competence issued by Vortok International.
3. The measurement can be carried out only at rail temperatures comprised in the following range: $T_N - 30^{\circ}C < T_R < T_N - 8^{\circ}C$, where T_N is the Neutral Temperature and T_R is the rail temperature, both in $^{\circ}C$
4. For measurement in curves, the rail temperature shall not be more than ten degrees less than the neutral temperature. What stated above is related to lateral movement of the rail once it is released from the fastenings and jacked to carry out the measurements, with consequent difficulty in positioning the rail in relocated the rail into the rail seat on completion of the Verse measurement.
5. Measurements can be carried out on curves whose radius is more than 700 m
6. the thermometers for measuring the rail temperatures shall be calibrated and kept under control.

Any change to the equipment shall be communicated in advance to RFI for the appropriate evaluation.





Division / Business Unit: Corporate Services & Safety
Function: Track & Civil
Document Type: Work Instruction

Track SFT Measurement by VERSE

ETI-06-08

Applicability

ARTC Network Wide
SMS

Publication Requirement

Internal / External

Primary Source

Document Status

Version #	Date Reviewed	Prepared by	Reviewed by	Endorsed	Approved
1.2	09 Jul 18	Standards	Stakeholders	A/Manager Standards	A/General Manager Technical Standards 10/07/2018

Amendment Record

Amendment Version #	Date Reviewed	Clause	Description of Amendment
1.0	21 Nov 17		First Issue
1.1	12 Dec 17	1.3 & 1.4	Risks controlled amended; Updated responsibilities.
1.2	09 Jul 18	2.5	Additional safety requirements added following the breakage of the adjustable jack strut.

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ARTC

Track SFT Measurement by VERSE

ETI-06-08

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Track SFT Measurement by VERSE

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Introduction

1 Introduction

1.1 Purpose

The purpose of this work instruction is to facilitate the operation and maintenance of VERSE equipment for the measurement of Stress Free Temperature in rail.

1.2 Scope

This work instruction covers general information, operating instructions, and maintenance instructions for VERSE equipment. This includes instructions for post-measurement analysis of data.

1.3 Risks Controlled

This work instruction is a control for the risk of track buckle or broken rail caused by the stress free temperature out of tolerance.

1.4 Responsibilities

The user of VERSE equipment is responsible for the compliance of this work instruction.

The authority with the overall responsibility for the delivery of safe and reliable track and civil infrastructure for the business unit or their delegate is responsible for the implementation of this work.

The Manager Standards is the owner of this Work Instruction and should be contacted for any queries or suggested improvements.

1.5 Reference Documents

The following documents support this work instruction:

- ETE-00-01 Calibration of Track Inspection and Testing Equipment;
- ETM-06-08 Managing Track Stability.
- VERSE manual.

1.6 Definitions

The following terms and acronyms are used within this document:

Term or acronym	Description
ARTC	Australian Rail Track Corporation Ltd.
Hand-held computer	Radix or Husky hand held computer forming part of the VERSE equipment.
Stress Free Temperature (SFT)	The temperature at which a rail is neither in tension nor in compression.
VERSE	Proprietary device for non-destructive measurement of Stress Free Temperature in rail.

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Track SFT Measurement by VERSE

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General

2 General

2.1 Introduction

VERSE is a proprietary device used for non-destructively measuring the Stress Free Temperature in rail. VERSE is marketed by Pandrol Australia Pty Ltd.

This work instruction provides ARTC with specific guidance, and is to be read in conjunction with the manufacturer's manual.

2.2 Operating Principles

The device works on the principle that the more a rail is in tension, the greater the force required to lift it.

VERSE is essentially a lifting frame, which enables a short section of rail to be raised using a hydraulic jack, with both the lifting force and lifting height being measured. Transducers are used to determine the applied force with respect to the vertical displacement of the rail. Rail temperature at the time of lifting is also recorded.

By knowing the rail profile, rail temperature, applied lifting force, and the height through which the rail is raised, the temperature at which the rail would be stress free can be calculated. Stress Free Temperature is calculated, to the nearest whole °C, by proprietary software in a dedicated hand-held computer connected to the lifting device.

2.3 Description

A general view of an ARTC VERSE machine is shown in Figure 1.



Figure 1: VERSE

The hand operated, hydraulically activated lifting frame is placed on a sleeper and sits astride the rail being measured. The frame accommodates different rail heights, sleeper types and applied cants. The aluminium lifting frame weighs about 25 kg.

A diagram of the VERSE machine, identifying principal components, is shown in Figure 2.

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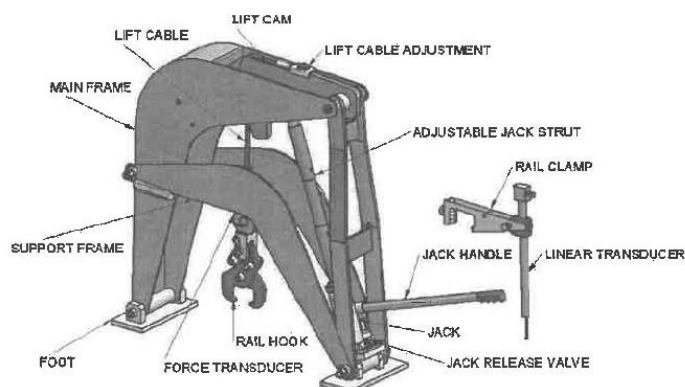


Figure 2 – Diagram of VERSE machine, showing principal components

The personal computer requirements to run the calculation software are referred to in supplier's manual.

2.4 Equipment and Tools

Additional tools and equipment required during the measurement process include:

- Tools for removing and replacing fastenings;
- Track jacks;
- Shovels and/or forks for digging out ballast
- Broom (or motorised blower) for cleaning loose material from rail seats;
- (Where applicable) lighting plant for night measurements.

And, for curved track:

- Lining bars for reinstating rail (excavator or similar on sharp curves); and
- Stringline and ruler for measuring versines.

2.5 Safety

Before all works commence, all relevant ARTC Rail Safety and WHS requirements must be in place.

When handling and positioning, use correct manual handling techniques for heavy and bulky items.

Always take particular care to ensure that hands and fingers are clear when unfolding and folding the lift frame, as a scissor action exists.

Be aware of signs of buckling, and stop the test immediately if rail is likely to buckle. Re-fasten the rail and do not perform test until rail temperature has lowered.

Operator must stop jacking when load reaches 10kN.

The adjustable jack strut should be inspected in accordance with the following steps prior to use:

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General

1. Unfold the VERSE frame in accordance with the manual.
2. Wind the adjustable jack strut from a fully contracted position to fully extended position (measure an extension of 130mm in overall Adjustable Strut Length).
3. If the Adjustable Jack Strut shows any of the three signs below, there is an indication of a bend in the Adjustable Jack Strut which could lead to premature failure.
 - a. A visible bend being detected
 - b. Difficulty in adjustment with resistance in rotation
 - c. Difficulty in adjustment with an oscillating increase and decrease in the torque required to adjust.
4. If the Adjustable Jack Strut is detected as bent, it should be replaced by the nearest VERSE service centre.

Refer to the supplier's manual for additional safety requirements.

2.6 Preliminary

Ensure all equipment is operating correctly, batteries are charged, and calibrations are current. If hand computer runs out of battery, ensure the time and date is reset correctly after recharging.

Ensure latest version of supplier's manual is available. An uncontrolled copy is available on the ARTC intranet at Engineering/Track and Civil/Manufacturer / Supplier Manuals. Contact supplier to verify version.

2.7 Site Selection

VERSE equipment shall not be used within 30 metres of turnouts, road crossings, or other fixed points in the track.

The lifted section of rail, which is 30 metres long (33 meters on AS60 rail or other similar heavy gauge rail sections), must not contain any insulated rail joints. The rail size must be the one for which the computer is programmed.

Must be of uniform rail size (i.e. must not contain a rail junction).

Should be reasonably clear of ballast around the rail foot and fastenings (to facilitate unfastening and refastening).

Should desirably contain no bolt holes or aluminothermic welds. If bolt holes or aluminothermic welds are unavoidable:

- Locate the lifting points as far away as possible from a single aluminothermic weld or set of bolt holes; and
- Locate the lifting point mid-way between two aluminothermic welds or sets of bolt holes.

Preferred site requirements, to minimise degradation of the Stress Free Temperature calculation, refer to the supplier's manual.



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General

2.8 Measuring Conditions

Ideally the VERSE lift is not carried out when sudden changes in rail temperature are occurring i.e. when the sun shines on the rail which has previously been in the shade during a measuring run.

VERSE measurements are not carried out at rail temperatures above 26°C.

8-28

VERSE measurements are not carried out at rail temperatures below 6°C on curves.

Note: If any rail movement is observed during the measuring, stop immediately and put the fasteners back. Come back and measure the SFT at a lower temperature.



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Track SFT Measurement by VERSE

ETI-06-08

Taking measurements

3 Taking measurements

All SFT measurements to be taken as per the supplier's manual and type approval.

Record and report results to the responsible ARTC Manager.

Minimum information to be recorded:

- Date
- Location details
- Rail size
- Instrument calibration details
- Rail temperature
- SFT readings

Additional information may be recorded depending on jurisdictional requirement. The information must be recorded either by written or electronic means. If written, ETM0606F-03 should be used. Electronic recording may include a computer software printout or spreadsheet.

4 VERSE kit Operation and Data Process

Refer to supplier's manual.

5 Maintenance Requirements for VERSE

Refer to supplier's manual.



ANNEXURE 5 INSTALLATION AND MAINTENANCE OF FRICTION BUFFER STOP (OLEO) (PARA.2.49)



LEADING THE WORLD IN ENERGY ABSORPTION

OLEO BUFFERS INDIA PRIVATE LIMITED Plot No. 32 and 33, KIADB Hi-Tech Hardware Park,

7th Main Road, Mahadevakodigehalli, Bengaluru, Karnataka - 562149, INDIA.

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Buffer stop Installation and maintenance manual



ARRESTORS



ELEVATOR



RAIL

OLEO Buffers India Private Limited is a company incorporated in India under company number U29268MH2010FTC210985



BUFFER STOPS

MASS OF TRAIN: 2000MT

IMPACT SPEED : 5 kmph

BUFFER STOP ASSEMBLY DRAWING NO : 7 41078 1

BUFFER STOP TYPE : PF-A1068X1956-300

PROJECT : WDFC CTP 1&2

INSTALLATION AND MAINTENANCE MANUAL

DOCUMENT REF: 9 412321

Revision	Issuer	Reason for Change/Reference	Date	Checker	Approver
A	SIDDARAJU N		10/09/2019	VIJAY KUMAR M	P. LEE
	ENGINEER - PRODUCT			PRINCIPAL ENGINEER	ENGINEERING DIRECTOR

9 41232 1 REV A1



PREPARATION FOR INSTALLATION

TRACK CONDITIONS:

- Ensure the gauge of the rail from installation drawing provided.
- The track within the installation length must be straight with no Points, Crossing or Fish Plates.
- The track should be in-line with main line or depot line.
- If installation of Buffer stop is on a site where there is a downward slope, ensure that this is stated, so allowances can be made (added sliding distance or additional friction shoes).
- If there are any thermite welds in the Buffer stop area on the rail head, both gauge and the field faces will be ground flush to the rail profile.

PRODUCT VERIFICATION:

On delivery remove all packing material and protective wrappings. Check the unit is undamaged. Before installing ensure that the design details agree with the operational requirements.

Check the following operational requirements:

1. Impact Mass m (kg)
2. Impact Speed v (m/s)
3. Max Force F (kN)
4. Impact height and Centre distance b/w impact points (mm).

The Buffer stop is designed based on the technical data received during design phase. Any deviations in values should be notified to Oleo International before installation. Never carry out unauthorized conversions or modifications on the equipment.

TYPICAL BUFFERSTOP ASSEMBLY

(Note: The Buffer stop images shown are for pictorial representation only. Please refer drawings provided for actual model)

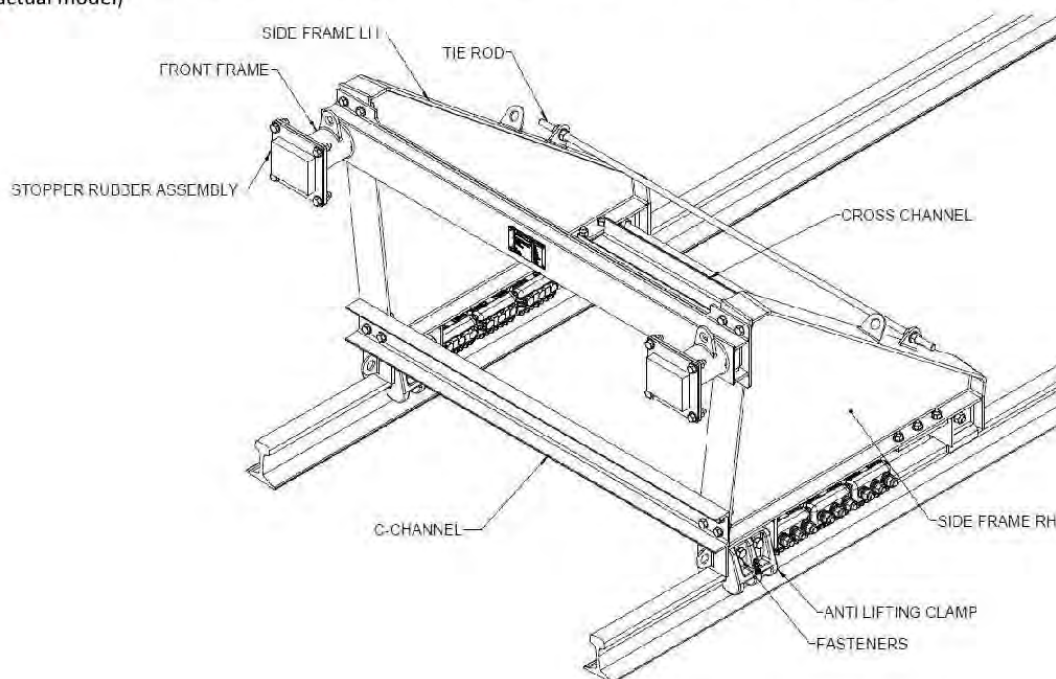


Figure-1

Note: Side frame LH and RH are defined by seeing Bufferstop from front (from train).

HANDLING

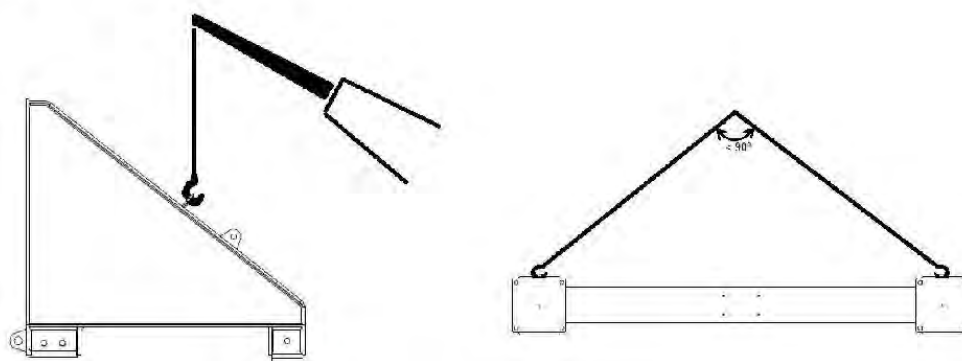


Figure-2: Lifting of Side frame and Front frame

1. The side frames (LH and RH) and front frame assembly are provided with lifting lugs for lifting hooks/shackles.
2. Ensure the lifting equipment can lift the weight of the assembly.
3. The angle between the rope/slides should not exceed 90° while lifting.
4. The frame assemblies have no symmetrical center of gravity. Pay attention to center of gravity while lifting and suspending them in air.
5. Always wear PPEs (hard hat, safety shoes and gloves) while handling the equipment.

INSTALLATION PROCEDURES FOR FRAME

Before installing frame assembly, the friction shoes under the frame should be installed. Refer the friction shoe installation manual for the standard procedure for installation, service and maintainance. Ensure that the frictions shoes on both rails are parallel with out any offsets.

INSTALLATION OF SIDE FRAME:

1. Place the frame over rail vertically (top of the friction shoes to be installed under the frame).
2. Place the anti-lift clamp and tighten with fasteners. Torque tighten with greasing after checking clearances (*Figure-5*) for complete assembly.
3. Now hold/support the side frame with a crane or suitable support method while installing the second frame.

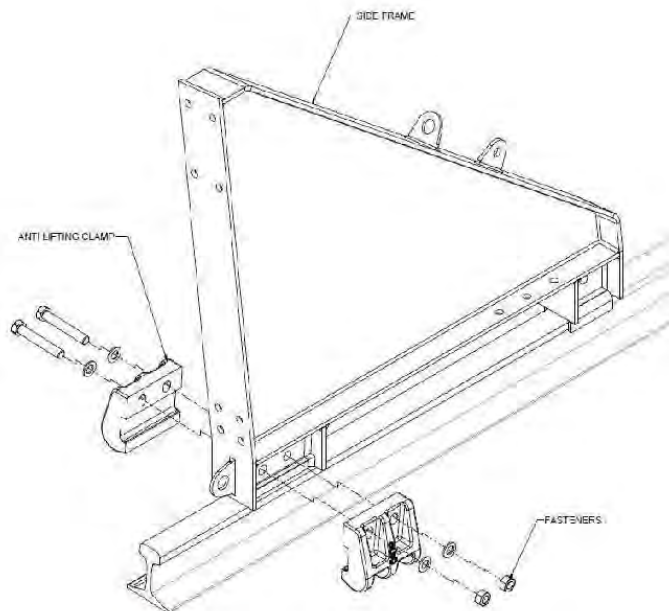


Figure-3: Side frame (Non-Insulated) installation.

Note:

For greasing anti-lifting clamps fasteners, hold anti-lifting clamp with temporary single bolt and insert other bolt. Using copper anti-seize grease, lubricate bolts (engaging threads), nuts and washers (washer nut interface) with brush. Similarly apply grease to the original temporary single bolt and re-assemble it.

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INSTALLATION OF FRONT FRAME:

1. Assemble front C-channel and rear cross channel members. Fix using bolts, washers and nuts.
2. Install the upper tie rods. Loosely position the upper rods and fix using nuts inside and outside of the side frame lugs.
3. Install the lower tie rod. Loosely position the lower rod and fix using nuts, insulation spacer and insulation washer.
4. Lift the front frame using lifting lugs provided (*Figure-2*) and assemble to the side frames using bolts, washers and nuts. Ensure the height of front frame from the rail is as per the installation drawing.
5. Assemble the stopper assemblies using screws and washers and OLEO name plate with fixings.

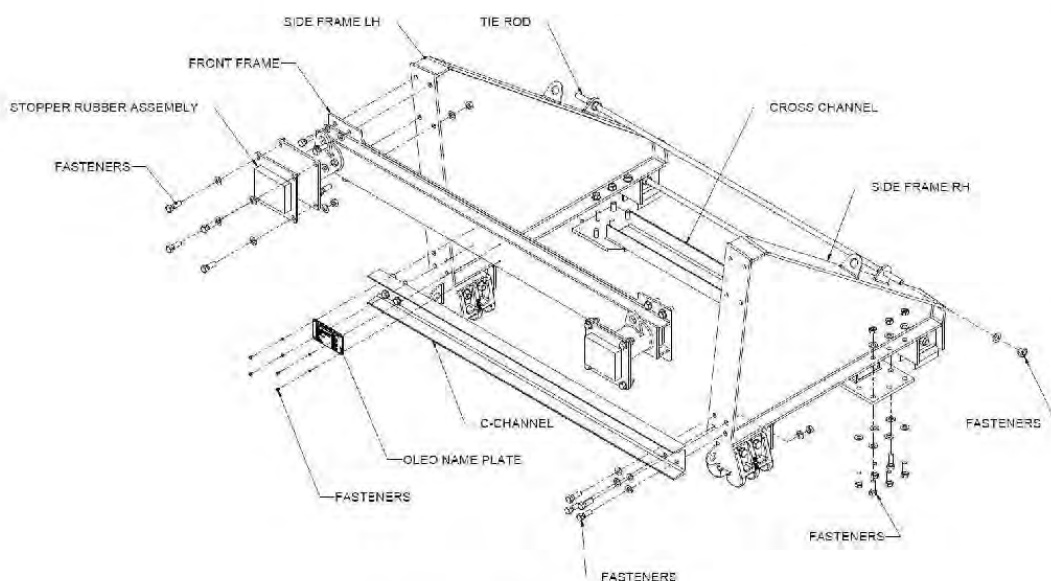


Figure-4: Front frame installation

Now adjust the whole assembly to ensure equal clearances between the rail and anti-lift clamp('S'in *Figure-5*). Once the clearances are provided, tighten all bolts to the recommended tightening torque. Ensure anti-lifting clamps fasteners are greased. Finally Install Pal Nuts to anti-lifting fasteners with recommended torque. Refer assembly drawing provided for tightening torque.

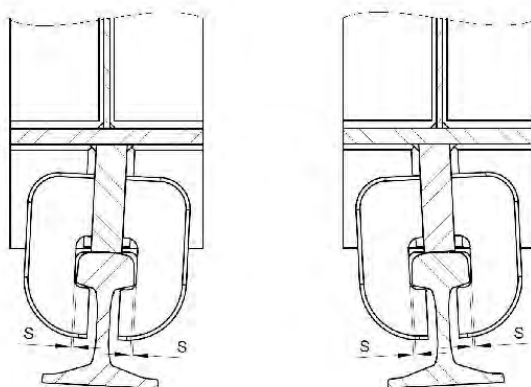


Figure-5

Note:

- The Bolt and nuts should be marked with paint pen after torquing to ensure they are torqued. It will also help during inspection to identify if initial torque is lost or loosened.



Once the installation of frame is finished, install the trailing friction shoes as per dimensions given in the installation drawing.

Gaps between the Frame and trailing friction shoes should be maintained as per the installation drawing. After installing, mark the installation position of the Buffer stop frame and all the trailing friction shoes on the rail head using paint pen for re-positioning if required.

Complete the Checklist inline with the Installation.



MAINTENANCE

The OLEO Buffer stop is essentially maintenance free, if the frame has not been in an impact and the main frame and trailing shoes have not been moved from the installed position.

It is recommended that even in the event that the frame has seen no impacts, the following inspection procedures need to be carried out.

A. ROUTINE CHECK:

1. Visually inspect the frames and other metal parts for damage (corrosion, cracks and deformation) for every six months.
2. Complete the half yearly and annual inspection checklist for maintenance.

B. AFTER IMPACT CHECK:

- C. Visually inspect the frames and other metal parts for damage like cracks and deformation.
- D. Loosen and remove anti-lift clamps and Friction Shoes inspect them for damage and wear.
- E. Check stopper rubber assembly for damage or degradation.
- F. Should there be damage or excessive wear, the damaged parts should be replaced and all bolts re-torqued according to the procedure and installation drawing.
- G. If there is no evidence of damage or excessive wear, refit the anti-lift clamps and Friction Shoes as per the installation procedure and complete the annual check list.
- H. Check the impact points' dimensions and reset to specified dimensions if changed after impact.

Note:

- If it is found any faulty during inspection or check-up, contact Oleo International for advice.

I. RE-SETTING THE BUFFER STOP AFTER IMPACT:

In the event of an impact that has caused the main frame and friction shoes to move from their starting points, please follow the below procedure to reset them.

1. Loosen and remove the anti-lift clamp and friction shoes after degreasing, check for damage and wear.
2. Refer friction shoe manual for allowable friction shoes wear, resetting and complete the installation checklist.
3. Reset the main frame and trailing shoes to their starting points as marked on the rail head and re-assemble the antilifting clamp, friction shoes according to the procedure and torque the bolts as per installation drawing.
4. Complete the annual checklist.

SERVICEABLE PARTS

1. Impact pad.
2. Anti-lift clamp.



INSTALLATION CHECKLISTS

All information about installation should be noted down along with the check points. Tick the boxes if they have been completed and record remarks if any.

Installation checklist reference: **QAD 621 Issue 1**

INSPECTION CHECK LIST

Regardless of whether the frame has been in an impact or not, inspections need to be carried out as per the checklist below.

HALF YEARLY AND ANNUAL INSPECTION

TICK	CHECK POINTS	REMARKS
	Check for damage to front frame fixings.	
	Check for damage to the impact pad and main frame.	
	Check if friction shoe bolts have lost the torque or loosened by checking the mark made during installation.	
	Check if the main frame has moved from its initial position.	
	Check if the trailing friction shoes have moved from its initial position.	
	Check for total installation length.	
	Check for visible signs of degradation in the rubber impact pad.	
	Re-mark the installation positions of all friction shoes on the rail head for re-positioning.	

Note:

- If frame is moved from its initial position, then frame should be repositioned to its original position by placing it over the reassembled Friction shoes under the frame.
- While re-assembling the frame assembly, it should be ensured that all the parts are degreased and cleaned before re-assembling it as per standard installation procedure.



SAFETY

Proper safety gear such as protective helmet, safety shoes, gloves, goggles etc, should be used during installation and assembly of Bufferstop.

Care must be taken when lifting and handling the frames and buffers to ensure that the operation is planned and that the load is both stable and safe. Proper handling equipment and methods to be followed for Buffer stops handling and assembly.

ENVIRONMENT

Oleo International is aware of the potential environmental impact of its operations and products. This concern continues beyond the supply to the end user through service and to the end of life of the product and its constituent parts.

Steel Components:- "recyclable scrap"

Capsule Fluid:- "special waste", present to a local licensed carrier

Rubber and Plastic Components:- "general waste"

Packaging materials:- please recycle where possible



OLEO INTERNATIONAL IS ACCREDITED
TO ISO 14001



ELEVATOR



BUFFER STOPS



INDUSTRIAL



RAIL

WE PROVIDE SOLUTIONS
NOT JUST PRODUCTS



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T A Savery and Co Limited is a company incorporated in England and Wales under company number 00272170 and whose registered office is at Grovelands Industrial Estate, Longford Road, Exhall, Coventry, CV7 9NE, UK



FM552731 ISO 9001



EMS552732 ISO 14001

CHECKLIST – BUFFER STOP INSTALLATION

Customer Name:		Date:	
Installation location:		Time:	
Performed by (name):		Staff No:	
Supervised by (name):		Staff No:	

PRODUCT DETAILS:

Buffer stop type	
Product code	
Installation Drawing No	

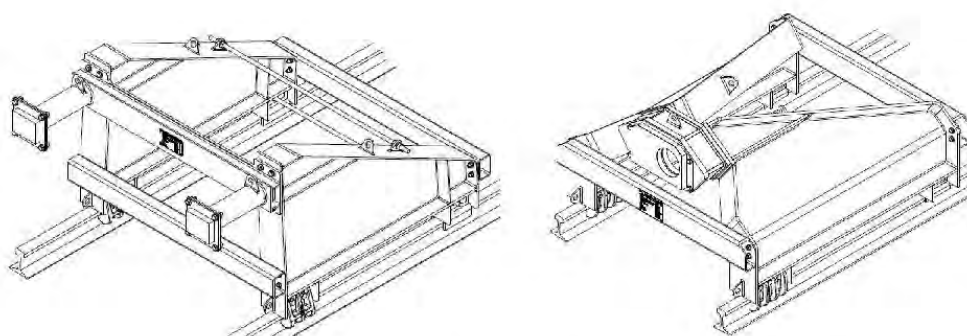
TRACK DETAILS:

Parameters	Details	Comments if any
Track gauge (mm)		
Rail Type		
Rail Condition: new or used (if it is worn out it should meet friction shoe top gap requirement)		

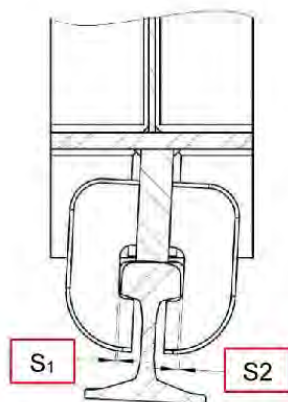
BEFORE INSTALLATION

Tick	Check points	Remarks
	Minimum installation length is available for the assembly at installation site, as per the installation drawing.	
	Track is straight and free of crossing.	
	No fish plate or un-grinded thermite welds within the sliding distance	
	No restrictions (weld burs) or unevenness on rail top, gauge or field faces, underneath and webs of the rail	
	Visual inspection is done for any damages in the parts.	
	Parts and quantities are checked as per the item list provided.	
	All required tools are available for installation as per installation drawings	
	Torque wrenches are pre set to the required torque value	
	Copy of installation manuals are available for reference.	
	The installation person/s is/are trained and qualified by Oleo	
	All parts are cleaned and degreased before installation (if reassembling/servicing)	
	No missing sleepers and Elastic clips within Buffer stop Track occupancy	

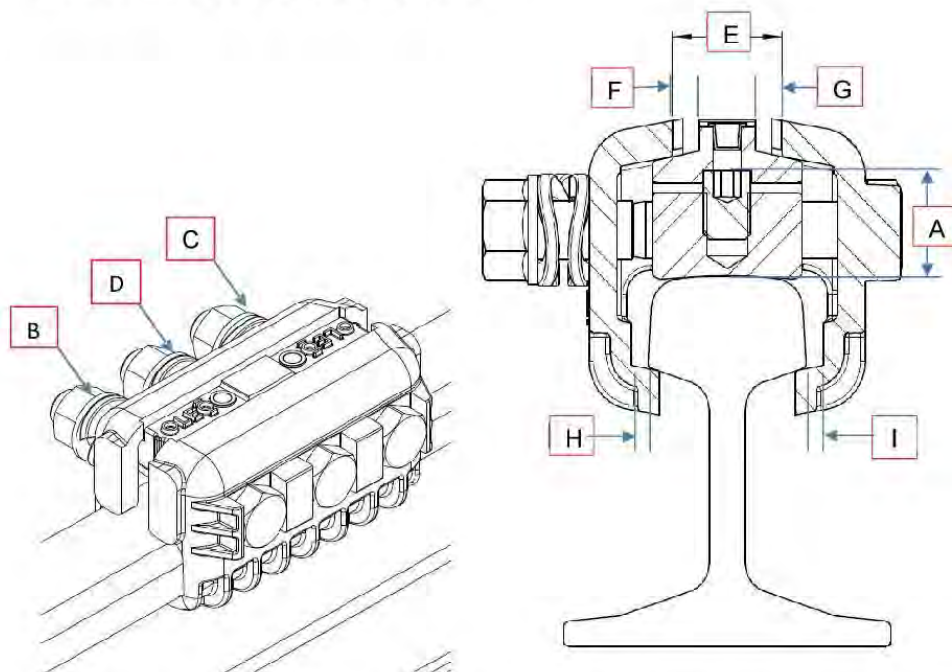
DURING INSTALLATION OF BUFFER STOP		
All bolts	Greased	
	Progressive torque	
	Final torques as per installation drawing	
S ₁		mm
S ₂		mm



(Note: The buffer stop images shown are for pictorial representation only. Please refer installation drawings provided for actual model)



DURING INSTALLATION OF FRICTION SHOE:



Note: This sheet should be completed for all the friction shoe assemblies.

FRICTION SHOE ID.					
Measurement	Value				Unit
A	G1 (Wear plate & grub screw Assy 1)				mm
	G2 (Wear plate & grub screw Assy 2)				
B	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
C	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
D	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
E	Front		Rear		mm
F	Front		Rear		mm
G	Front		Rear		mm
H	Front		Rear		mm
I	Front		Rear		mm

FRICTION SHOE ID.					
Measurement	Value				Unit
A	G1 (Wear plate & grub screw Assy 1)				mm
	G2 (Wear plate & grub screw Assy 2)				
B	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
C	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
D	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
E	Front		Rear		mm
F	Front		Rear		mm
G	Front		Rear		mm
H	Front		Rear		mm
I	Front		Rear		mm

FRICTION SHOE ID.					
Measurement	Value				Unit
A	G1 (Wear plate & grub screw Assy 1)				mm
	G2 (Wear plate & grub screw Assy 2)				
B	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
C	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
D	Greased				Yes/No
	Progressive torque				Yes/No
	Torque value				N-m
E	Front		Rear		mm
F	Front		Rear		mm
G	Front		Rear		mm
H	Front		Rear		mm
I	Front		Rear		mm



FRICITION SHOE

MASS OF TRAIN: 2000MT
 IMPACT SPEED : 5 kmph
 FRICTION SHOE ASSEMBLY DRAWING NO : 7 41078 1
 BUFFER STOP TYPE : PF-A1068X1956-300
 PROJECT : WDFC CTP 1&2

INSTALLATION AND MAINTENANCE MANUAL

DOCUMENT REF: 9 41254 1

Revision	Issuer	Reason for Change/Reference	Date	Checker	Approver
A	ARUN KUMAR H N			VIJAY KUMAR M	P. LEE
	ENGINEER - DESIGN			PRINCIPAL ENGINEER	ENGINEERING DIRECTOR

PREPARATION FOR INSTALLATION

TRACK CONDITIONS:

1. The track within the installation length must be straight with no Points, Crossing or Fish Plates.
2. The track should be in-line with main line or depot line.
3. If the rail is worn out or used, it should meet top gap requirement for installation of friction shoe.
4. If there are any thermite welds in the installation area on the rail head, both gauge and the field faces will be ground flush to the rail profile. Make sure that there are no proud welds, uneven surfaces, restrictions in rail head and web throughout the installation/sliding length.

PRODUCT VERIFICATION:

On delivery remove all packing material and protective wrappings. Check the unit is undamaged. Before installing ensure that the design details agree with the operational requirements.

TYPICAL FRICTION SHOE ASSEMBLY

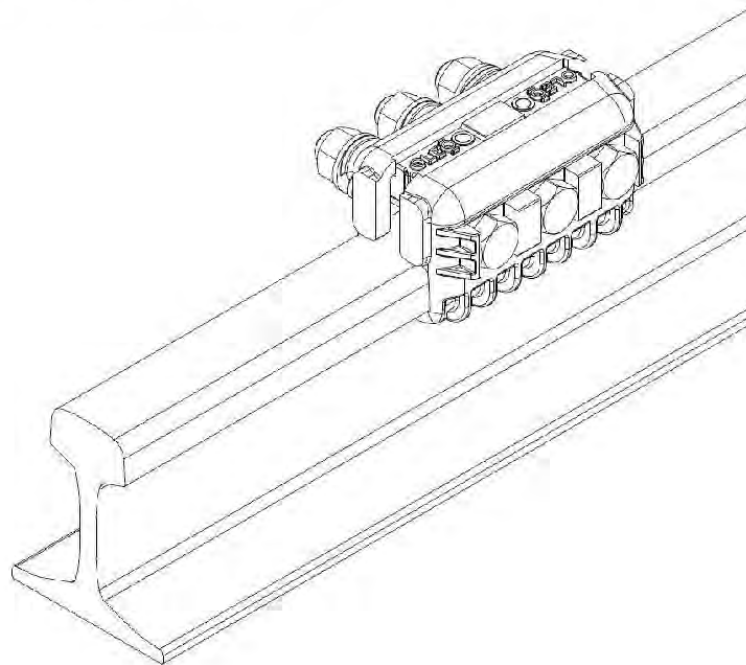


Figure-1

INSTALLATION PROCEDURE

CAUTION:

The optimum performance of friction shoe which gives correct Dynamic force with minimum oscillation is achieved by following standard installation steps with appropriate tooling and repeatable process.

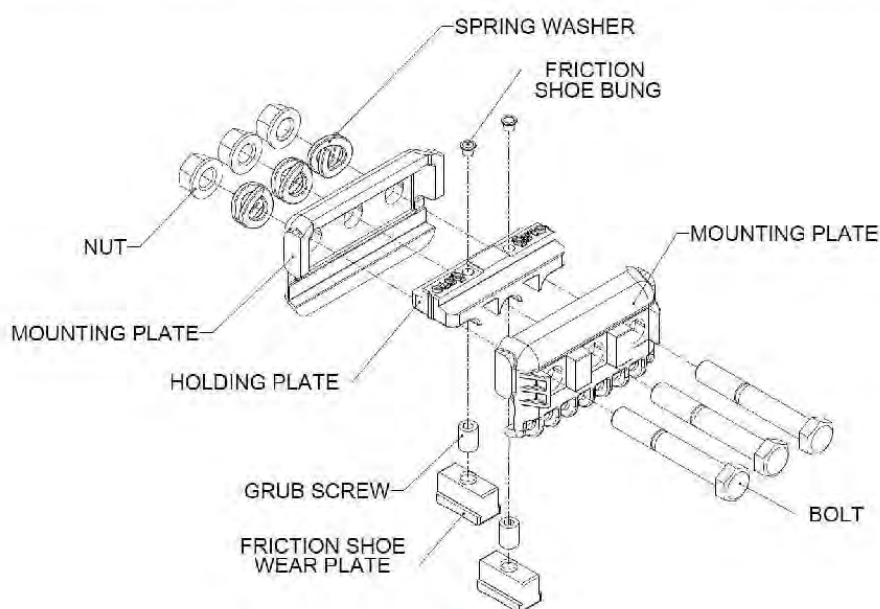


Figure-2

1. Ensure all the tools required for installation are available at site.
2. Assemble friction shoe wear plate and grub screw. Set Wear plate and grub screw height to 'C' mm (refer installation drawing) as shown in Figure-3.
3. Loosely assemble the friction shoe components with temporary single bolt as in Figure-3. Ensure that the assembly is symmetrical about the rail.

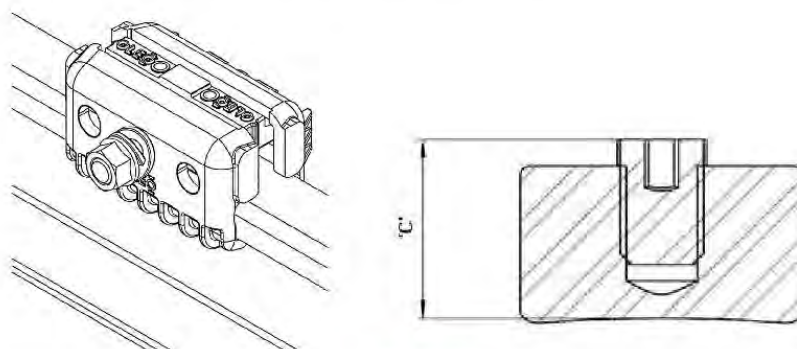


Figure-3

4. Now insert other 2 bolts. Using copper anti-seize grease, lubricate bolts (engaging threads), nuts and spring washers (washer nut interface) with brush. Similarly apply grease to the original temporary single bolt and re-assemble it.

Caution:

Grease should be applied using brush only after inserting the bolts through the mounting plates. It should be applied only to engaging threads and interface of Nut and spring washer. Must ensure that there is no over deposition.

5. Using aluminium drift make necessary alignment adjustments so that mounting plates are parallel, holding plate is centralised and parallel with the mounting plates. Remove any slack by tightening nuts. Make sure that assembly is inline with rail center after adjustments. Refer [Figure-4](#) for wrong and acceptable alignments.

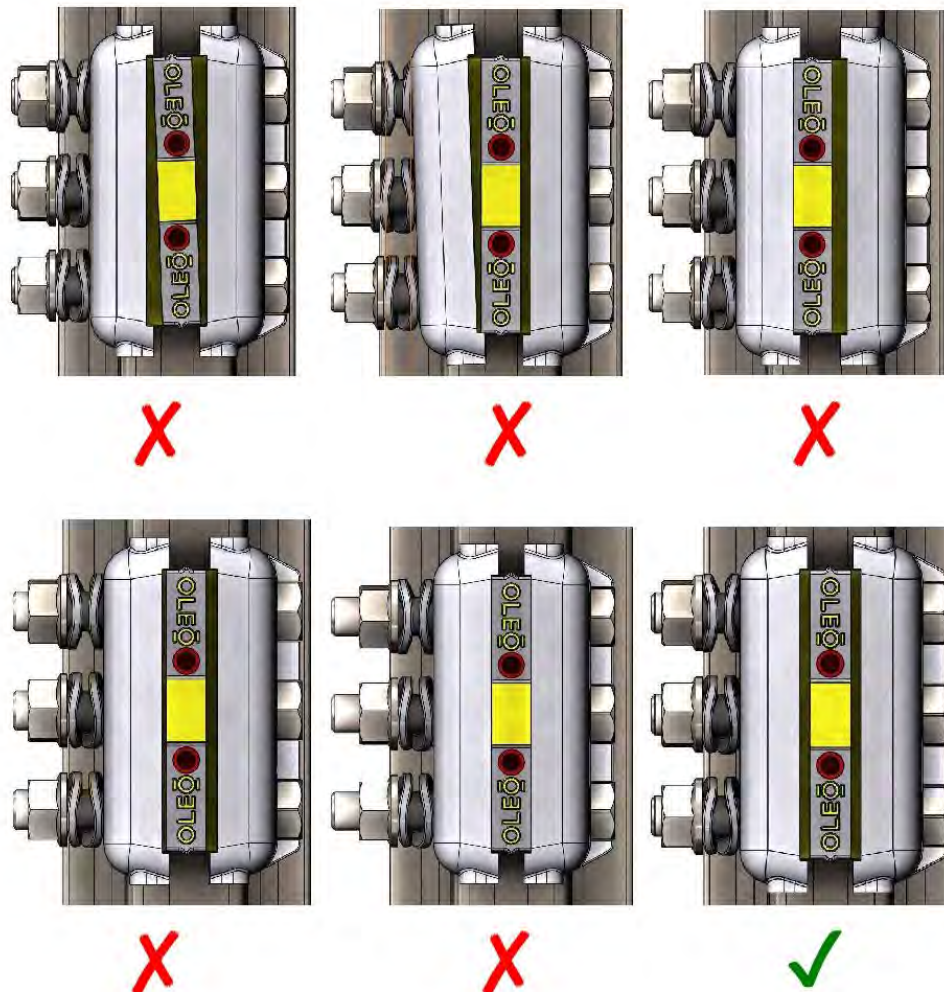
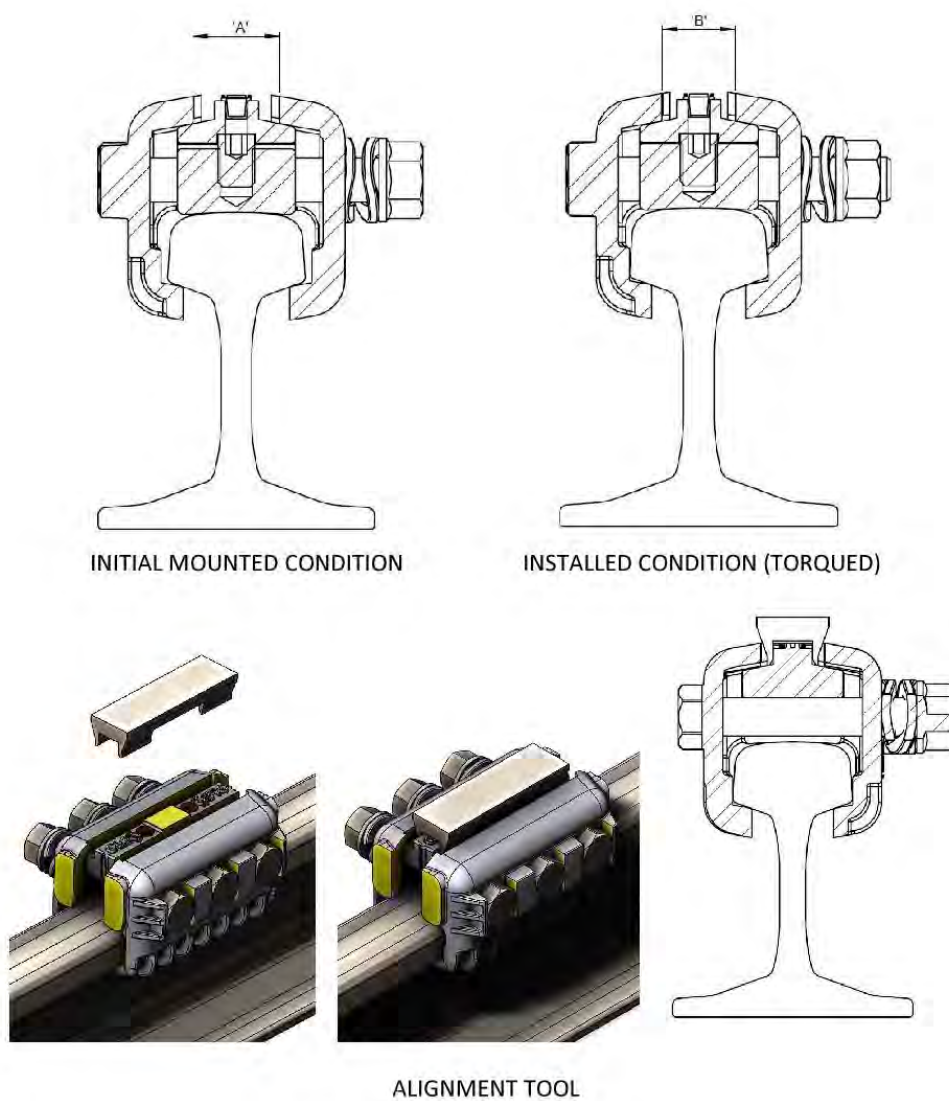


Figure-4

6. Check alignment at this stage with the aid of alignment tool. If found any misalignment repeat the above step till the correct alignment made (refer [Figure-4](#)).
7. Before tightening the bolts to torque the top gap between the two half clamps as shown in [Figure-4](#) shall be 'A' mm (refer friction shoe installation drawing).



[Figure-5](#)

8. Check that the torque tool is preset to value given in the installation drawing. The torque tool has to be preset and calibrated before sending for installation.
9. Torque the three clamping bolts in the sequence as shown in [Figure-6](#).
10. Do not torque the bolts to full torque value in a single step! Each bolt should be tightened progressively until the specified torque value is reached to maintain the alignment. For torque value refer installation drawing.

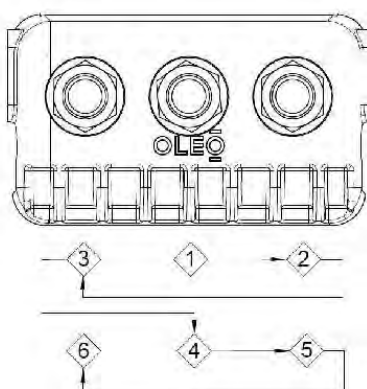


Figure-6: Bolt Torque sequence

Note:

- It is recommended that each bolt tightened in small steps. Following is the recommended torque values(T) in steps for tightening Progressively.
60% T ---> 80% T ---> 90% T ---> 95% T ---> 98% T ---> 100%T
- Special care should be taken to ensure that the holding plate and the two mounting plates remain parallel throughout the tightening process as shown in [Figure-7](#).
- The Bolt and nuts should be marked with paint pen after torquing to ensure they are torqued. It will also help during inspection to identify if initial torque is lost or loosened.



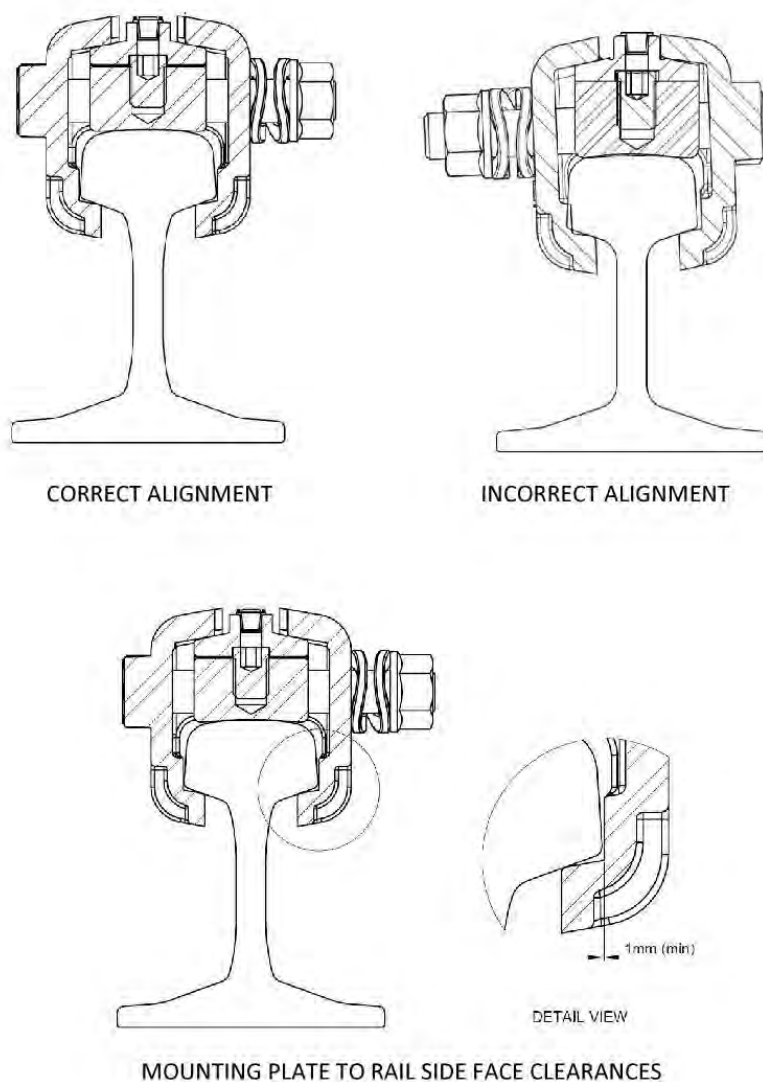


Figure-7

11. After the bolts have been torqued the top gap between on the friction shoes (Figure-5) should be 'B' mm (refer installation drawing). The friction shoes should be symmetrical with rail. Ensure clearance between the mounting plates and side faces (gauge and field faces) of rail head (Figure-7) using 1 mm feeler gauge.
12. In case of an incorrect installation as shown in Figure-7, the friction shoes shall be removed and re-installed following the above procedure.

13. WEAR:

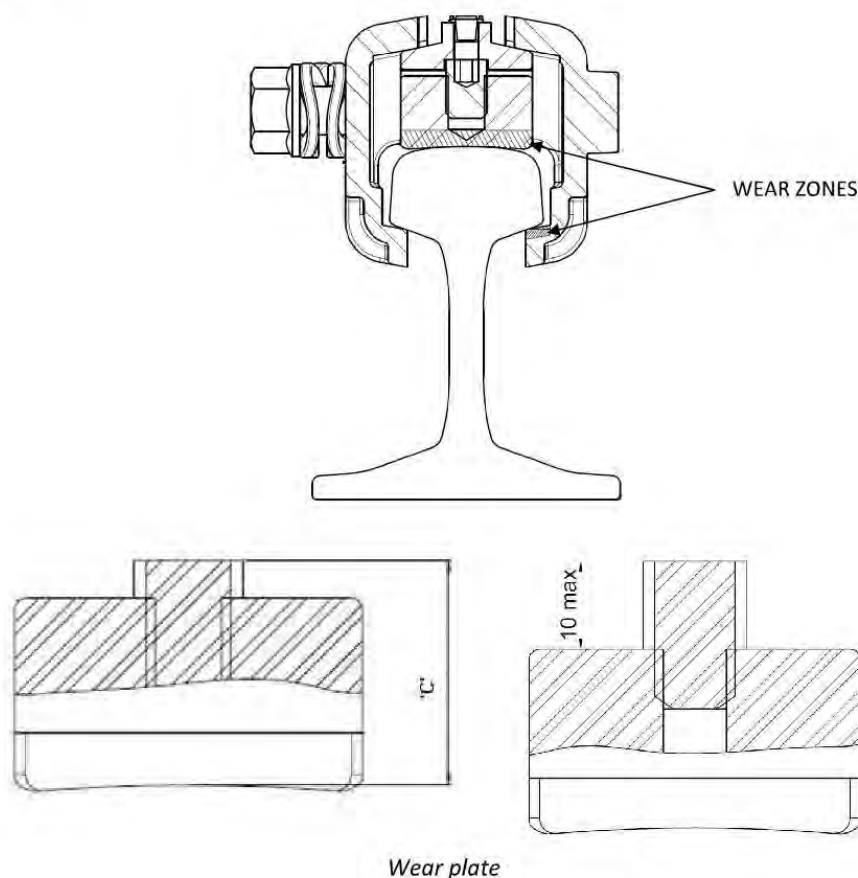


Figure-8

1. The gap between the friction shoe and the rail is adjusted by rotating grub screw in each wear plate. Maximum allowable wear plate protrusion is 10 mm to compensate wear plate and track wear.
2. In the circumstances when a significant amount of wear has occurred and the maximum permitted grub screw protrusion from wear block (10mm) has been set: if then, after assembly if the top gap on the shoe still cannot meet the required top gap 'B' mm then the wear block and mounting plates **must** be replaced. (Figure-5).
3. For rail heads with significant amount of wear, if the top gap requirement ('B' mm) is not achievable with permitted screw protrusion (10 mm), then rails **must** be replaced.
14. Gaps given for trailing friction shoes should be maintained as per the installation drawing.
15. After installing, mark the installation position of all the friction shoes on the rail head using paint pen for re-positioning.
16. Complete the installation checklist inline with the installation.



MAINTENANCE

The OLEO friction shoe is essentially maintenance free, if it has not been in an impact and has not been moved from the installed position.

It is recommended that even in the event that the friction shoe has seen no impacts, the following inspection procedures need to be carried out.

A. ROUTINE CHECK:

1. Visually inspect the all metal parts, fasteners for damage (corrosion, cracks and deformation) for every six months.
2. Refer the half yearly/ annual inspection checklist for maintenance.

B. AFTER IMPACT CHECK:

1. Loosen and remove Friction Shoes after degreasing. Inspect them for damages like crack, deformation and wear.
2. Should there be damage or excessive wear, the damaged parts should be replaced and all bolts re- torqued (refer friction shoe “**Installation procedure**” section). Refer “**Wear**” section of the manual for allowable wear of the track and friction shoes.
3. If there is no evidence of damage or excessive wear, refit the Friction Shoes as per the installation procedure. The wear plate grub screws may need adjusting to account for wear.
4. Complete the annual inspection check list.

Note:

- If it is found any faulty during inspection or check-up, contact Oleo International for advice.

INSTALLATION TOOLS REQUIRED

Refer the installation drawing .

SERVICEABLE PARTS

1. Friction shoe and its components.



INSTALLATION CHECK LIST

All information about installation should be noted down along with the check points. Tick the boxes if they have been completed and record remarks if any.

Installation checklist reference: **QAD 621 Issue 1**

INSPECTION

Regardless of whether the friction shoe has been in an impact or not, inspections need to be carried out as per the checklist below.

ANNUAL INSPECTION

TICK	CHECK POINTS	REMARKS
	Check if the friction shoes have moved from its initial position.	
	Visual inspection is done for any damages in the parts.	
	Check if friction shoe bolts have lost the torque or loosened by checking the mark made during installation.	
	Check for total installation length.	
	Check for any restrictions with in the sliding length	
	Re-mark the installation positions of all friction shoes on the rail head for re-positioning.	

Note:

- If frictions shoes found to be moved from initial positions, they must be removed and reassembled following standard installation procedure.
- While re-assembling friction shoes after impact or servicing, it should be ensured that old grease is cleaned away before attempting to remove fasteners. Also all the parts are degreased and cleaned before re-assembling it as per standard installation procedure.



SAFETY

Proper PPEs such as protective helmet, Safety shoes, gloves, goggles etc, should be used during installation and assembly of friction shoes.

Care must be taken when lifting and handling the friction shoe components. Proper handling methods to be followed for friction shoe handling and assembly.

ENVIRONMENT

Oleo International is aware of the potential environmental impact of its operations and products. This concern continues beyond the supply to the end user through service and to the end of life of the product and its constituent parts.

Steel Components:- "recyclable scrap"

Capsule Fluid:- "special waste", present to a local licensed carrier

Rubber and Plastic Components:- "general waste"

Packaging materials:- please recycle where possible



OLEO INTERNATIONAL IS ACCREDITED
TO ISO 14001



ELEVATOR



BUFFER STOPS



INDUSTRIAL



RAIL

WE PROVIDE SOLUTIONS
NOT JUST PRODUCTS



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OLEO International is a division of T A Savery and Co Limited, whose ultimate parent is Brigam Limited
T A Savery and Co Limited is a company incorporated in England and Wales under company number 00272170 and whose registered office is at Grovelands Industrial Estate, Longford Road, Exhall, Coventry, CV7 9NE, UK



FM552731 ISO 9001



EMS552732 ISO 14001

ANNEXURE 6 INSTALLATION AND MAINTENANCE OF FRICTION BUFFER STOP (RAWIE) (PARA.2.49)

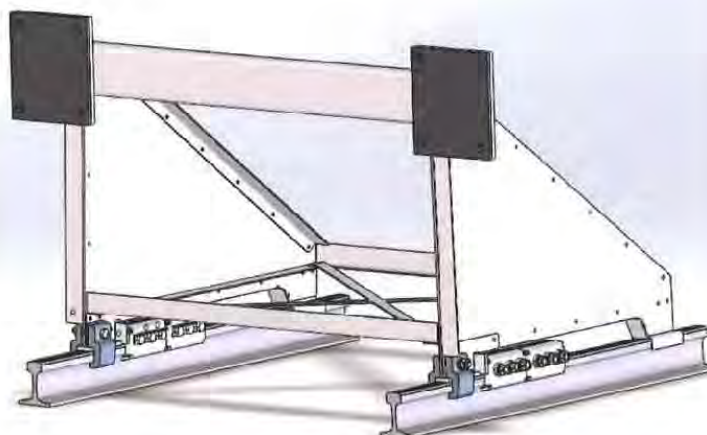
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Installation, Operating & Maintenance Manual BP351en2
2019-09-16

Friction Element Buffer Stop type 4EB

as per drawing 10-896-002a
for new flat-bottomed running rails "profile UIC60, inclined 1:20",
track gauge 1676 mm

for
WDFCC Package CTP1 & 2



Serial
Numbers
BE 12242 –
BE 12295

RAWIE-
Order No.
70996

Buyer: Startrack Fasteners-Pvt. Ltd.,
40 ,KM stone , G. T. Road,
Bahalgarh, (Vill. Asawarpur),
Haryana, India

Purchase Order:
STF18/-19/490



FR-QW51en3/2017-11-09

RAWIE® - Installation, Operating & Maintenance Manual BP351en2,
Friction Element Buffer Stop type 4EB as per drawing 10-896-002a
for WDFCC Package CTP1 & 2, RAWIE-Order No. 70996

File: BP351en3 - WDFCC Pkg CTP1-2 - 4EB - 70996 - IOM Manual.odt, printed: 2019-09-16

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Important!

This Installation, Operating & Maintenance Manual must be read before lifting, handling, transporting, storing, installing, operating, checking, resetting, maintaining or repairing the Friction Element Buffer Stop type 4EB! It must be continuously available for general use!

FR-QW51en3/2017-11-09



RAWIE® - Installation, Operating & Maintenance Manual BP351en2,
Friction Element Buffer Stop type 4EB as per drawing 10-896-002a
for WDFCC Package CTP1 & 2, RAWIE-Order No. 70996

File: BP351en3 - WDFCC Pkg CTP1-2 - 4EB - 70996 - IOM Manual.odt, printed: 2019-09-18

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 - 1.2 Function
 - 1.3 Dangers for people and environment
 - 1.4 Safeguard measures and rules of conduct
 - 1.5 Technical Specification
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 - 1.5.2 Buffer Stop Data
 - 1.6 Home Position of Buffer Stop ...
 - 1.6.1 ... for max. 247 t impacting mass, impacting at max. 10 km/h impact speed, horizontal track, Energy Safety Factor 1.25:
 - 1.7 Stopping Process ...
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 - 4.2.3 Checking worn Friction Elements and Guide Claws
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- 6 Spare Parts



7 Manufacturer
8 Enclosures

1 Basic Information

The Friction Element Buffer Stop type 4EB may be used as described in the order contract, only.

The Friction Element Buffer Stop type 4EB may be used as an emergency stop, only.

The Friction Element Buffer Stop type 4EB may be lifted, handled, transported, stored, installed, operated, checked, reset, maintained or repaired by skilled personnel, only.

Reference is made to installation drawing No. 10-896-002a "Friction Element Buffer Stop type 4LEB".

1.1 Definitions

- []: Item of parts list in section 5 "Parts List".
- The "Friction Element Buffer Stop type 4EB" [1] is called "Friction Element Buffer Stop type 4LEB", "Friction Element Buffer Stop" or "Buffer Stop" in this manual, also.
- The Buffer Stop [1] body is called "Sledge", too.
- Train, locomotive, wagon, car, vehicle, rail vehicle: Vehicle on running rails.
- Front or Face of Buffer Stop [1]: That end of the Buffer Stop [1] which is towards the impacting rail vehicle.
 - The Side Impact Plates [38b] are towards the impacting rail vehicle.
- Impact interface of Buffer Stop [1]:
 - Side Impact Plates [38b]
- Back of Buffer Stop [1]: That end of the Buffer Stop [1] body which is towards the end of track.
- Left / right: As seen from the impacting rail vehicle in driving direction.
- 'Down / Up' gradient: 'Down / Up' gradient in driving direction of the impacting rail vehicle.
- Energy Safety Factor means: The *stopping capacity* of the "Friction Element Buffer Stop type 4EB" [1] shall be equal or greater than the "*Energy Safety Factor*"-fold *impact energy*".
- The impact energy in here is the "kinetic plus potential energy" of the non-propelled rail vehicle in the moment of impact, without consideration of rotational energy.
- The "Friction Element with Anti-Theft Device" [2] is called "Friction Element" in this manual, also.
- Operating range of Buffer Stop [1]: Range of Buffer Stop [1] and stopping distance behind the Buffer Stop [1].
- A/F: Across flats.

1.2 Function

- The "Friction Element Buffer Stop type 4EB" [1] is a dormant safety device which under normal rail operating conditions does not move unless impacted by an overrunning rail vehicle.
- If an overrunning rail vehicle impacts the *impact interface* of the "Friction Element Buffer Stop [1], which is in *home position* (see section 1.6)", the Buffer Stop [1] slides along the running rails, picking up the Friction Elements [2], which are clamped on the rail heads, becomes slower via brake force of the Friction Elements [2] and stops the impacting rail vehicle safely. The brake force is generated by friction of the Friction Elements [2] at the running rails.
- The impact force acts on the rail vehicle and on the running rails.
- The Guide Claws [4] prevent the Buffer Stop [1] from flipping over at an impact. They pass the uplift forces into the running rail heads.
- The Buffer Stop [1] is non-insulated, that means: there is electrical contact between the running rails.
- **Note!** The Buffer Stop [1] will work only, if sufficient stopping distance is available! A sufficient distance from Back of Buffer Stop [1] body to End of Track is an essential precondition for the Buffer Stop [1] function!

1.3 Dangers for people and environment



- Risk of crushing between Buffer Stop [1] and impacting rail vehicle.
- Risk of crushing at the Buffer Stop [1] or inside the Buffer Stop [1] at an impact.
- Risk of crushing on the stopping distance behind the Buffer Stop [1] at an impact.

1.4 Safeguard measures and rules of conduct

- Attend Installation, Operating & Maintenance Manual!
- The Buffer Stop [1] may be lifted, handled, transported, stored, installed, operated, checked, reset, maintained or repaired by skilled personnel, only.
- Stay away from front of Buffer Stop [1]!
- Keep off the operating range of Buffer Stop [1] (at the Buffer Stop, inside the Buffer Stop and on the stopping distance behind the Buffer Stop) at rail vehicle operation!
- Use yearly maintained Buffer Stop [1], only!
- Make sure that the Buffer Stop [1] is in its Home Position! Home Position is described in section 1.6 "Home Position of Buffer Stop ...".
- Don't impact on Buffer Stop [1] with too much impact energy!
- Check Buffer Stop [1] after an impact as described in section 3.1 "Check after an impact" and reset it, in case it is not in its Home Position!



1.5 Technical Specification

1.5.1 The Friction Element Buffer Stop type 4EB is suitable for:

- | | | |
|---|--|--|
| • Impacting non-propelled rail vehicle with.... | side strike | |
| • Available track length..... | 16 m | |
| • Maximum impacting mass..... | 2000 t ... | |
| • Maximum impact speed..... | 5 km/h ... | |
| • Track gradient in the area of the impacting rail vehicle and the Buffer Stop stopping distance..... | horizontal ... |] ... on 13.7 m sliding stopping distance II |
| • Energy Safety Factor..... | 1.0 | |
| • Average side strike centre height above top of running rails..... | 1105 mm | |
| • Side strike centres distance..... | 1956 mm | |
| • Track gauge..... | 1676 mm | |
| • Running rail profile..... | UIC60 | |
| • Actual running rail height..... | new | |
| • Inclination of running rails..... | 1:20 | |
| • Straight track..... | in the area of impacting locomotive respectively of impacting car, Buffer Stop and Buffer Stop stopping distance. | |
| • Rails without joints..... | within the operating range of Buffer Stop (under the Buffer Stop and on the stopping distance behind the Buffer Stop) and 3.5 m in front of the Buffer Stop Face. | |
| • Rail heads accessible..... | within the operating range of Buffer Stop (under the Buffer Stop and on the stopping distance behind the Buffer Stop).
The required free space is shown on drawing 3156\0-47. | |



1.5.2 Buffer Stop Data

• Installation Drawing	10-896-002a
• The Buffer Stop is.....	non-insulated
• Nominal Stopping capacity on 13.7 m sliding stopping distance.....	1934 kJ
• Number of Friction Elements [2].....	4, located in Buffer Stop bottom pockets on the running rails.
• Number of Clamp Bolts [7] / [151] at each Friction Element [2].....	3
• Torque of Clamp Bolts [7] / [151] of Friction Elements [2].....	150 Nm $\pm$ 10 Nm
• Maximum impacting mass.....	2000 t ...
• Maximum impact speed.....	5 km/h ...
• Track length needed from impact front of Buffer Stop to end of track.....	15.3 m
• Nominal Impact Force (F_1).....	240 kN
• Nominal Longitudinal Force to (F_3) each running rail.....	120 kN
• Nominal Uplift Force at Guide Claws [4] to each running rail.....	93 kN
• Length of Buffer Stop [1].....	1540 mm
• Weight of Buffer Stop [1].....	550 kg

... on 13.7 m sliding stopping distance ¹ at horizontal track gradient and Energy Safety Factor 1.0

1.6 Home Position of Buffer Stop ...

- 4 Friction Elements [2]..... must be positioned in the Buffer Stop [1] bottom pockets at the running rails as shown on installation drawing 10-896-002a,
- All 4 Friction Elements [2]..... must be installed as pairs: one Friction Element of a pair at the left running rail and one at the right running rail, both absolutely at the same distance from the "Friction Element [2] pushing surface of Buffer Stop [1] body" (see installation drawing).
- and ...

1.6.1 ... for max. 2000 t impacting mass, impacting at max. 5 km/h impact speed, horizontal track, Energy Safety Factor 1.0:

- The distance from Back of Buffer Stop [1] body to End of Track must be..... 13.7 m ¹



1.7 Stopping Process ...

1.7.1 ... at 2000 t, 5 km/h, horizontal track, Energy Safety Factor 1.0

- Impacting mass: 2000 t
 - Impact speed: 5 km/h
 - Non propelled impact
 - Track gradient: horizontal
 - Energy Safety Factor: 1.0
 - Kinetic Energy ($E_{kin} = \frac{1}{2} m \cdot v^2$) plus potential energy ($E_{pot} = m \cdot g \cdot h$) = 1929 kJ
 - "Kinetic plus potential energy" times Energy Safety Factor 1.0 = 1929 kJ
 - Required sliding stopping distance: 13.7 m
 - Stopping capacity on 13.7 m sliding stopping distance: 1934 kJ
 - **The distance from Back of Buffer Stop [1] body to End of Track must be 13.7 m**
 - Track length needed from impact front of Buffer Stop to end of track: 15.3 m
- The stopping capacity on 13.7 m sliding stopping distance (1934 kJ) is greater than the "kinetic plus potential energy" times Energy Safety Factor 1.0 (1929 kJ).

1.8 Incoming Goods, Packaging

- The "Friction Element Buffer Stop type 4EB" [1] has been delivered fully assembled, unpacked. Friction Elements [2] and accessories are packed in a separate box [3].

1.9 Lifting, handling and storage

- Use proper lifting equipment and proper lifting means for lifting and handling the Buffer Stop [1].
- The weight of the unpacked "Friction Element Buffer Stop type 4EB" [1] is 550 kg.
- Use the 2 Lifting Loops [5] of the Buffer Stop [1] body for lifting the Buffer Stop [1].
- The "Friction Element Buffer Stop type 4EB" [1] and accessories shall be stored at a dry place on an even surface. Do not stack the Buffer Stops, the packages and the boxes. Do not expose components to aggressive fluids and / or chemical agents.

1.9.1 Tools needed

Following tools are needed among others for "Lifting, handling and storage":

- Proper lifting equipment (crane, ...) (The weight of the unpacked "Friction Element Buffer Stop type 4EB" [1] is 550 kg.)
- Proper lifting means.



- I Note!** The Buffer Stop [1] will stop the impacting rail vehicle only, if sufficient stopping distance is available! A sufficient distance from Back of Buffer Stop [1] body to End of Track is an essential precondition for the Buffer Stop [1] function!
- If more suitable track length is available than needed, the distance from Back of Buffer Stop [1] body to End of Track can be increased accordingly; Greater distance increases safety.

RAWIE - Installation, Operating & Maintenance Manual BP351en2,

Friction Element Buffer Stop type 4EB as per drawing 10-896-002a
for WDFCC Package CTP1 & 2, RAWIE-Order No. 70996

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1.10 Rails

Note!: Proper rail condition and rail fixation are essential preconditions for the Buffer Stop [1] function!

Make sure,

- that there is straight track in the area of impacting locomotive respectively of impacting car, Buffer Stop and Buffer Stop stopping distance,
- that track gauge and running rail profile, height and inclination are as stated in section 1.5.1 "The Friction Element Buffer Stop type 4EB is suitable for:".
- that rails and rail fixation are suitable for longitudinal and uplift forces,
- that there is no rail-joint in the running rails within the operating range of Buffer Stop [1] (under the Buffer Stop and on the stopping distance behind the Buffer Stop) and 3.5 m in front of the *Buffer Stop Face*.
- that the rail heads under the Buffer Stop [1] and on the stopping distance behind the Buffer Stop [1]
 - are accessible. The required free space is shown on drawing 3156\0-47 "Concrete cut out for embedded rails".
 - have all around an even surface,
 - are not painted,
 - are not greased,
 - are not excessively rusted.



2 Installation Procedure

Attend sections 1.3 "Dangers for people and environment" and 1.4 "Safeguard measures and rules of conduct"!

Reference is made to Buffer Stop installation drawing No. 10-896-002a which must be available during the installation.

Install the "Friction Element Buffer Stop type 4EB" [1] as follows:

2.1 Tools needed

Following tools are needed among others for installation:

- Proper lifting equipment (crane, ...) (The weight of the unpacked "Friction Element Buffer Stop type 4EB" [1] is 550 kg.)
- Proper lifting means.
- Torque wrench 100 - 250 Nm with *long* 1 9/16" socket
- Special Socket for Lock Washer [153], 36 mm A/F, to loosen Lock Nut [154] of Anti-Theft Device [150]
- Deep offset ring spanner 36 mm A/F
- Wire brush – to clean the heads of the running rails
- Bars to adjust the position of the Buffer Stop [1]
- Measuring Tape
- Chalk
- Grease
- Mark to mark Home Position of Buffer Stop [1], e. g. a Marking Post or "paint for web of the rail".
- Standard metric maintenance tool set

2.2 Preparation of Stopping Distance

- Make sure that the running rails are as stated in section 1.10 "Rails".
- Make sure there is no protuberance or obstruction other than Friction Elements [2] within the operating range of the Buffer Stop [1] (under the Buffer Stop [1] and on the stopping distance behind the Buffer Stop [1]) that would hinder the Buffer Stop [1] to slide.

2.4 Mark Home Positions of Friction Elements

Mark Home Positions of Friction Elements temporary as follows:

2.4.1 Set Buffer Buffer Stop down onto the running rails in its Home Position

- Home Position is described in section 1.6 "Home Position of Buffer Stop ...".
- Remove Guide Claws [4] from Buffer Stop [1].
- Lift the Buffer Stop [1] by crane using the 2 Lifting Loops [5]. Make sure that the *Face of Buffer Stop* is towards the impacting rail vehicle, and set it down onto the running rails in its Home Position (see section 1.6 "Home Position of Buffer Stop ...").



2.4.2 Mark Home Positions of Friction Elements

- Make sure that the distance from Back of Buffer Stop [1] body to End of Track is as stated in section 1.6 "Home Position of Buffer Stop ...".
- The 4 Friction Elements [2] of the Buffer Stop [1] bottom pockets are to be arranged as pairs: one Friction Element of the pair at the *left* running rail and one at the *right* running rail, both absolutely at the same distance from the "*Friction Element [2] pushing surface of Buffer Stop [1] body*" (see installation drawing).
- Make sure that the positions of the Friction Elements [2] are exactly as shown on installation drawing 10-896-002a.
- Mark the "Home Positions of the Friction Elements [2]", which are shown on installation drawing 10-896-002a, with chalk on running rail heads.

2.4.3 Remove Buffer Stop

- Then move Buffer Stop [1] out of the way.

2.5 Install Friction Elements

Install the Friction Elements [2] on the running rails at their as per section 2.4.2 marked home positions as follows:

- Insert 2 short Clamp Bolts [7] into outer holes of each Bolt Retainer [10] of the Friction Elements [2],
- insert 1 long Clamp Bolt [151] into centre hole of Bolt Retainer [10]
- and grease bolt threads.
- Set the Filler Block [6] on the running rail head and hold the two Supports [11] against it.
- Take Bolt Retainer [10] with Clamp Bolts [7] and [151] and insert Clamp Bolts [7] and [151] into semi-assembled Friction Element [2] so that the Bolt Threads are on field side of the track!
- Put on the Double Spring Washers [9] and Flanged Nuts [8].
- Make sure that the positions of the Friction Elements [2] are exactly as shown on installation drawing 10-896-002a.
 - **Note:** It is important that both Friction Elements [2] of each pair of Friction Elements [2] *located in Buffer Stop bottom pockets on the running rails* are absolutely at the same distance from the "*Friction Element [2] pushing surface of Buffer Stop [1] body*" (see installation drawing).
- Tighten the Flanged Nuts [8] of the Clamp Bolts [7] and [151] of the Friction Elements [2] using a Torque Wrench. It is critical that these bolts are set at a Torque of **150 Nm ±10 Nm**.

Note: Tighten the Flanged Nuts [8] of the Friction Elements [2] as one would the nuts on an auto-mobile's wheel, i. e. alternatively and not in one cycle.

2.6 Set Buffer Stop onto running rails

- Remove Guide Claws [4] from Buffer Stop [1].
- Lift the Buffer Stop [1] by crane using the 2 Lifting Loops [5]. Make sure that the *Buffer Stop Face* is towards the impacting rail vehicle, and set it down onto the running rails in its Home Position (see section 1.6 "Home Position of Buffer Stop ...").



2.7 Install Guide Claws

Install Guide Claws [4] to the Buffer Stop [1] body as shown on installation drawing 10-896-002a around both running rail heads as follows:

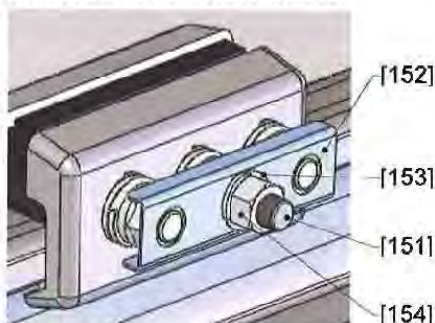
- Insert Bolt [7a] into Guide Claw [4] half. Take Guide Claw [4] half with Bolt [7a] and insert Bolt [7a] into hole of Buffer Stop [1] bottom so that the bolt thread is on field side of the track. Put on the other Guide Claw [4] half, Washer [12] and Flanged Nut [8]. Make sure that the Guide Claw [4] halves surround the corresponding running rail head.
- Tighten Flanged Nuts [8] of the Bolts [7a] of the Guide Claws [4] using a Torque Wrench: Torque has to be $230 \text{ Nm} \pm 20 \text{ Nm}$.

2.8 Intermediate check

Important!: Check again,

- that Buffer Stop [1] and all 4 Friction Elements [2] are in their correct Home Positions (defined under section 1.6 "Home Position of Buffer Stop ...") on the running rail heads,
- that the distance from Back of Buffer Stop [1] body to End of Track is as stated in section 1.6 "Home Position of Buffer Stop ...",
- that the positions of the Friction Elements [2] are exactly as shown on installation drawing 10-896-002a.
- that both Friction Elements [2] of each pair of Friction Elements [2] are absolutely at the same distance from the "Friction Element [2] pushing surface of Buffer Stop [1] body" (see installation drawing).
- that the total number of installed Friction Elements [2] is 4,
- that there are 2 short Clamp Bolts [7] and 1 long Clamp Bolt [151] in each of the 4 Friction Elements [2],
- that all Flanged Nuts [8] of the Clamp Bolts ([7] and [151]) of all 4 Friction Elements [2] are tightened to a torque of $150 \text{ Nm} \pm 10 \text{ Nm}$.
- that all 2 Guide Claws [4] are installed to the Buffer Stop [1] body and each around its corresponding running rail head as shown on installation drawing 10-896-002a and described in section 2.7 "Install Guide Claws".
- that all 2 Flanged Nuts [8] of the Bolts [7a] of the Guide Claws [4] are tightened to a torque of $230 \text{ Nm} \pm 20 \text{ Nm}$,

2.9 Install Anti-Theft Device



Anti-Theft Device [150]

Lock Nut [154]



Lock Washer [153] Special Socket

- II Note!:** The Buffer Stop [1] will stop the impacting rail vehicle only, if sufficient stopping distance is available! A sufficient distance from Back of Buffer Stop [1] body to End of Track is an essential precondition for the Buffer Stop [1] function!

Note: The Lock Nut [154] can be loosened via Special Socket for Lock Washer, only!

- Install Anti-Theft Device [150] on all 4 Friction Elements [2] as follows:
 - Put on Cover Profile [152],
 - put on Lock Washer [153] and Lock Nut [154] on the Clamp Bolts [151] in that way that saw tooth tread design of Lock Washer [153] and of Lock Nut [154] engage on each other,
 - tighten the Lock Nut [154] of the Clamp Bolts [151] of the Friction Elements [2] using a Torque Wrench. Torque has to be 100 Nm.

2.10 Mark Home Position of Buffer Stop

- Mark the Home Position of the Buffer Stop [1] (which is defined in section 1.6 "Home Position of Buffer Stop ...") weathering resistant at the longitudinal position of the impact face, e. g. with a Marking Post or with colour at the web of the rail. Place the Marking Post outside the rail vehicles kinematic envelope.

2.11 Final check

Finally check,

- that Home Position of the Buffer Stop [1] is marked as per section 2.10 "Mark Home Position of Buffer Stop",
- that there is no protuberance or obstruction other than Friction Elements [2] within the operating range of the Buffer Stop [1] (under the Buffer Stop [1] and on the stopping distance behind the Buffer Stop [1]) that would hinder the Buffer Stop [1] to slide.

3 Operating Procedure

3.1 Check after an impact

Attend sections 1.3 "Dangers for people and environment" and 1.4 "Safeguard measures and rules of conduct"!

3.1.1 Tools needed

Following tools are needed for checking after an impact:

- Measuring Tape

(Tools needed for resetting the Buffer Stop [1] are listed under section 3.2.1).

3.1.2 Procedure

When the Friction Element Buffer Stop type 4EB has been impacted, then:

- Check Buffer Stop [1] condition.
- Check that Buffer Stop [1] and Friction Elements [2] are in their Home Position. Home Position is defined in section 1.6 "Home Position of Buffer Stop ...".

When the Friction Element Buffer Stop [1] or a Friction Element [2] is not in its Home Position (e. g. due to the impact of a rail vehicle) it must be moved to its Home Position as described in section 3.2 "Resetting Instruction".

3.2 Resetting Instruction

Attend sections 1.3 "Dangers for people and environment" and 1.4 "Safeguard measures and rules of conduct"!

When the Friction Element Buffer Stop [1] or a Friction Element [2] is not in its Home Position (e. g. due to the impact of a rail vehicle) it must be moved to its Home Position, which is defined in section 1.6 "Home Position of Buffer Stop ...".

3.2.1 Tools needed

Following tools are needed among others for resetting:

- Pulling device to pull the Buffer Stop [1] (e. g. loco)
- 2-leg wire rope sling as per DIN EN 13414-1 with minimum 16 mm diameter and 10 m long, with shackles DIN 82101 nominal size 2 (WLL 2 t)
- Torque wrench 100 - 250 Nm with long 1 9/16" socket
- Special Socket for Lock Washer [153], 36 mm A/F, to loosen Lock Nut [154] of Anti-Theft Device [150]
- Deep offset ring spanner 36 mm A/F
- Bars to adjust the position of the Buffer Stop [1]
- Measuring Tape
- Vernier caliper
- Chalk
- Grease
- Standard metric maintenance tool set

3.2.2 Resetting the Buffer Stop

- Check Buffer Stop [1] condition.
- **Note:** The Lock Nut [154] can be loosened via Special Socket for Lock Washer [153], only! (see section 2.9 "Install Anti-Theft Device").
- Loosen and remove the 1 Lock Nut [154] and Lock Washer [153] of long centre Clamp Bolt [151] of the Friction Elements [2], using a Special Socket for Lock Washer [153].
- Remove Cover Profile [152] of Anti-Theft Device [150].
- Loosen all the Flanged Nuts [8] of the Clamp Bolts [7] and [151] of the Friction Elements [2].
- Loosen the Supports [11].
- Check that the Filler Blocks [6] are also loose as sometimes they will bind to the rail after an impact.
- Using a 2-leg wire rope sling as per DIN EN 13414-1 with minimum 16 mm diameter and 10 m long and shackles DIN 82101 nominal size 2 (WLL 2 t), pull the Buffer Stop [1] at the Tow Loops [14] back to its Home Position (see section 1.6 "Home Position of Buffer Stop ...").

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- Make sure that the Buffer Stop [1] is in its Home Position. Home Position is defined in section 1.6 "Home Position of Buffer Stop ...".
- Check as described under section 4.2.3 "Checking worn Friction Elements and Guide Claws" if the Friction Liners inside the Friction Elements [2] as well as the Guide Claws [4] are not worn more than allowed.
- Move Friction Elements [2] to their Home Position (see section 1.6 "Home Position of Buffer Stop ...").
- Make sure that the positions of the Friction Elements [2] are exactly as shown on installation drawing 10-896-002a.
 - **Note:** It is important that both Friction Elements [2] of each pair of Friction Elements [2] located in Buffer Stop bottom pockets on the running rails are absolutely at the same distance from the "Friction Element [2] pushing surface of Buffer Stop [1] body" (see installation drawing).
- Re-tighten the Flanged Nuts [8] of the Clamp Bolts [7] and [151] of all 4 Friction Elements [2] as described under section 2.5 "Install Friction Elements".
- Re-tighten the Flanged Nuts [8] of the Bolts [7a] of the Guide Claws [4] using a Torque Wrench: Torque has to be 230 Nm $\pm$ 20 Nm.
- Check Buffer Stop [1] as described in section 2.8 "Intermediate check".
- Install Anti-Theft Device on Friction Elements [2] as described under section 2.9 "Install Anti-Theft Device".
- Check Buffer Stop [1] as described in section 2.11 "Final check".

4 Maintenance Instruction

4.1 "Check after an impact" as per section 3.1

"Check after an impact" is described in section 3.1.

4.2 Yearly Maintenance

Attend sections 1.3 "Dangers for people and environment" and 1.4 "Safeguard measures and rules of conduct"!

4.2.1 Tools needed

Following tools are needed among others for maintenance:

- Torque wrench 100 - 250 Nm with long 1 9/16" socket
- Special Socket for Lock Washer [153], 36 mm A/F, to loosen Lock Nut [154] of Anti-Theft Device [150]
- Deep offset ring spanner 36 mm A/F
- Wire brush – to clean the heads of the running rails
- Wire brush – to clean the Friction Elements [2]
- Bars to adjust the position of the Buffer Stop [1]
- Measuring Tape
- Vernier caliper
- Chalk
- Grease
- Mark to mark Home Position of Buffer Stop [1], e. g. a Marking Post or "paint for web of the rail".
- Standard metric maintenance tool set
- Zinc paint
- Brush



RAWIE - Installation, Operating & Maintenance Manual BP351en2,
Friction Element Buffer Stop type 4EB as per drawing 10-896-002a
for WDFCC Package CTP1 & 2, RAWIE-Order No. 70996

File: BP351en3 - WDFCC Pkg CTP1-2 - 4EB - 70996 - IOM Manual.odt, printed: 2019-09-16

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- Tools needed for resetting the Buffer Stop [1] are listed under section 3.2.1.

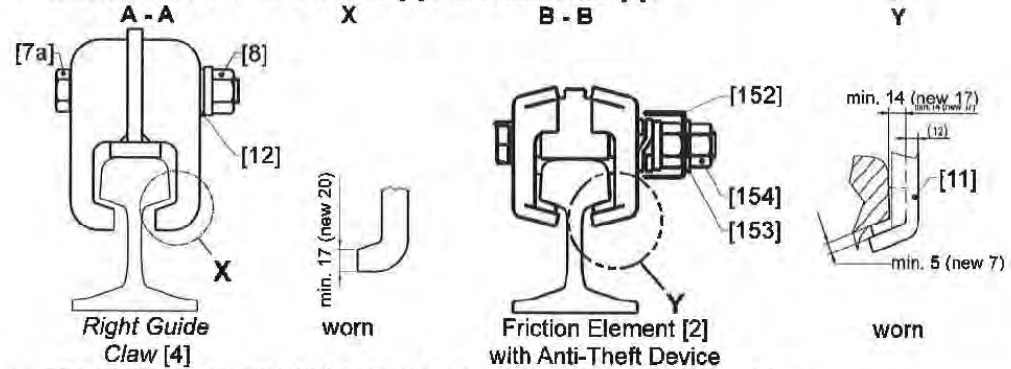
4.2.2 Procedure

- Check Buffer Stop [1] condition.
- Check that the Buffer Stop [1] is in its Home Position. Home Position is defined in section 1.6 "Home Position of Buffer Stop ...".
When the Friction Element Buffer Stop [1] is not in its Home Position it must be returned to its Home Position as described in section 3.2 "Resetting Instruction".
- Check that there is no excessive rust. If so remove and treat area with commercial rust inhibitor and repaint with zinc paint.
- **Note:** The Lock Nut [154] can be loosened via Special Socket for Lock Washer [153], only! (see section 2.9 "Install Anti-Theft Device").
- Loosen and remove the 1 Lock Nut [154] and Lock Washer [153] of long centre Clamp Bolt [151] of the Friction Elements [2], using a Special Socket for Lock Washer [153].
- Remove Cover Profile [152] of Anti-Theft Device [150].
- Loosen and clean all parts of the Friction Elements [2]. Clean and grease the threads of the Clamp Bolts [7] [151]. It is essential that the Clamp Bolt [7] [151] threads of the Friction Elements [2] are kept well greased so that the torque settings are maintained.
- Check as described under section 4.2.3 "Checking worn Friction Elements and Guide Claws" if the Friction Liners inside the Friction Elements [2] and the Guide Claws [4] are not worn more than allowed.
- Check that rail condition is as described under section 1.10 "Rails".
- Also clean the heads of the running rails within the area occupied by the Buffer Stop [1] and stopping distance.
- Mark the "Home Positions of the Friction Elements [2]" as described in section 2.4.2 "Mark Home Positions of Friction Elements".
- Install all 4 Friction Elements [2] as described under section 2.5 "Install Friction Elements" to the running rails.
- Re-tighten the Flanged Nuts [8] of the Bolts [7a] of the Guide Claws [4] using a Torque Wrench: Torque has to be 230 Nm $\pm$ 20 Nm.
- Check Buffer Stop [1] as described in section 2.8 "Intermediate check".
- Install Anti-Theft Device on Friction Elements [2] as described under section 2.9 "Install Anti-Theft Device".
- Make sure that Home Position of the Buffer Stop [1] is marked as per section 2.10 "Mark Home Position of Buffer Stop".
- Make sure that there is no protuberance or obstruction other than Friction Elements [2] within the operating range of the Buffer Stop [1] (under the Buffer Stop [1] and on the stopping distance behind the Buffer Stop [1]) that would hinder the Buffer Stop [1] to slide.



4.2.3 Checking worn Friction Elements and Guide Claws

- Check wear of Friction Elements [2] and Guide Claws [4].



- The minimum material thickness at the tip of the Guide Claw [4] is 17 mm.
- The minimum material thickness of the Friction Liners of the Supports [11] is 5 mm under the rail head and 14 mm by the sides of the rail head.
- Should due to wear the material thickness be less than minimum, the respective parts (Guide Claws [4] or Supports [11]) have to be replaced as a pair and not individually. Assemble Guide Claws [4] and Supports [11] as described under section 2 "Installation Procedure".

5 Parts List

Installation Drawing: 10-896-002a "Friction Element Buffer Stop type 4LEB"

Item	Quantity	Description
[1]	1 off	Friction Element Buffer Stop type 4EB (is called "Friction Element Buffer Stop type 4LEB", "Friction Element Buffer Stop" or "Buffer Stop" in this manual, also.)
[2]	4 off	Friction Element '3-hole with Anti-Theft Device' (consists of items [6], [7], [8], [9], [10], [11], [151], [152], [153] and [154]) (part of item [1])
[3]	1 off	Box (packing material)
[4]	2 pair	Guide Claw (consists of two claw halves and items [7a], [8] and [12]) (part of item [1])
[5]	2 off	Lifting Loop (part of item [1])
[6]	4 off	Filler Block '3-hole' (part of item [2])
[7]	8 off	Clamp Bolt ISO 4014-M24 x 180-8.8-tZn (for outside holes of item [2])
[7a]	2 off	Hexagon head bolt ISO 4014-M24 x 140-8.8-tZn (part of item [4])
[8]	14 off	Flanged Nut M24 - tZn (part of items [2] and [4])
[9]	12 off	Double Spring Washer Fe15 - tZn (part of item [2])
[10]	4 off	Bolt Retainer "3-hole" (part of item [2])
[11]	8 off	Support '3-hole' (part of item [2])
[12]	2 off	Washer ISO 7091 - 24 - 100HV - tZn (part of item [4])
[14]	2 off	Tow Loop (part of item [1])
[38b]	2 off	Side Impact Plate (part of item [1])
[150]	4 off	Anti-Theft Device (consists of items [151] [152] [153] and [154]) (part of item [2])
[151]	4 off	Clamp Bolt ISO 4016 - M24 x 200 - 4.6 - tZn (for centre hole of item [2]) (part of items [2] and [150])
[152]	4 off	Cover Profile (part of item [150])
[153]	4 off	Lock Washer (part of item [150])
[154]	4 off	Lock Nut (part of item [150])

6 Spare Parts

Use original RAWIE Spare Parts, only.

State with the order:

- Serial Number of Buffer Stop [1] (see identification plate)
- and with the order of Friction Elements [2] or Guide Claws [4]:
 - Rail profile,
 - Present rail height.

7 Manufacturer

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8 Enclosures

- Installation Drawing 10-896-002a "Friction Element Buffer Stop type 4LEB"
- Drawing 3156\0-47 "Concrete cut out for embedded rails"



RAWIE - Installation, Operating & Maintenance Manual BP351en2,
Friction Element Buffer Stop type 4EB as per drawing 10-896-002a
for WDFCC Package CTP1 & 2, RAWIE-Order No. 70996

File: BP351en3 - WDFCC Pkg CTP1-2 - 4EB - 70996 - IOM Manual.odt, printed: 2019-09-16



ANNEXURE 7 LIST OF EQUIPMENTS (PARA 1.1.2.3(H))

1. LIST OF EQUIPMENTS FOR TRACK MAINTENANCE TEAM

(A) Communication Equipment

1. Walky Talkie 4 sets

(B) Spot Tamping with Lifting Lining

2. Magnet based rail thermometer 1 No.
3. Off Track Hand Held Tamper 2 set
4. Track Lifting Jack-Hydraulic 4 sets
5. Lifting-cum-Slewing Device 2 sets

(C) Sleeper Spacing and Squaring

6. Hydraulic Sleeper Spacer 2 Nos.
7. ERC clipper and declipper 10 Nos.

(D) Inspection Gadgets

8. Inspection Kit 1 No.
9. Gauge cum Level 1 No.

(E) Ballast Dressing

10. Spade 5 Nos.
11. Crow Bar (Per member of team)
12. Rake ballast (Per member of team)

(F) Safety and Protection Equipments

13. Hooter 1 No.
14. Whistle 1 No.
15. Red Banner Flags 2 Nos.
16. Red Hand Signal Flag 2 Nos.
17. Green Hand Signals Flag 1 Nos.
18. Detonators 12 Nos.
19. Safety jackets for each team member
20. Team Number Plate

2. LIST OF EQUIPMENTS FOR RAIL MAINTENANCE TEAM

(A) Communication Equipment

1. Walky Talkie 4 sets

(B) Turnout Rail Maintenance

1. Portable Welding Generator 1 set
2. Hand Held Rail Grinder 2 sets

(C) Rail Defect Maintenance

1. Track Lifting Jacks (Hydraulic) 4 Nos.
2. Abrasive Rail Cutting Machine 1



3. Rail Cutting Disc 1 packet

4. Rail Drilling Machine 1

5. Chamfering kit 1

(D) Rail Welding Equipment

1. AT Welding Equipment 2 sets

2. Double Action Weld Trimmer 1 set

3. Rail Profile Grinder for welded joints 1 set

4. Rail welding straight edges 2 Nos.

5. Gas cutting Equipment with 1 set of spare gas cylinders 1 set

6. Rail profile weld grinder 2 Nos.

(E) Measuring Instruments

7. Magnet based rail Thermometer 2 Nos.

8. Rail Wear Gauges 2 Nos.

9. VERSE 1 set

(F) Destressing Equipment

1. Rail Tensors- Hydraulic/mechanical 2 sets

2. Rollers, wooden mallets 1 complete set for Destressing 3Km CWR

(G) Safety and Protection Equipments

1. Hooter 1 No.

2. Whistle 1 No.

3. Red Banner Flags 2 Nos.

4. Red Hand Signal Flag 2 Nos.

5. Green Hand Signals Flag 1 Nos.

6. Detonators 12 Nos.

7. Safety jackets for each team member

8. Team Number Plate

3. LIST OF EQUIPMENTS FOR CIVIL MAINTENANCE TEAM

(A) Restoration Equipments

1. Spade 10 Nos.

2. Mortar Pan (1 per person in the team)

3. Crow bar (1 per person in the team)

4. Rake Ballast (1 per person in the team)

5. Rammer 1 Nos.

(B) Safety and Protection Equipments

1. Hooter 1 No.

2. Whistle 1 No.

3. Red Banner Flags 2 Nos.

4. Red Hand Signal Flag 2 Nos.

5. Green Hand Signals Flag 1 Nos.



6. Safety jackets for each team member
7. Team Number Plate



ANNEXURE 8 RECORD OF DESTRESSING (EXAMPLE) (PARA 2.30)

Destressing details (for office use)					
Route:					
Date	Km	Line: Up/Down/ Single	Rail: Left/Right or Both	Work (Unusual occurrence / Fracture / Planned Destressing etc)	Authorised Person Executive/ Sr Executive (track)